ALAGAPPA UNIVERSITY

(A State University Accredited with 'A+ Grade by NAAC (CGPA: 3.64) in the third cycle)

Karaikudi – 630 003, Tamilnadu, India

DIRECTORATE OF DISTANCE EDUCATION

PROGRAMME PROJECT REPORT

Programme Code: 365

M.Sc., HOME SCIENCE - NUTRITION AND DIETETICS

REGULATIONS AND SYLLABUS

[From the academic year 2018-2019 onwards]

Credit Based System

(a) PROGRAMME'S MISSION & OBJECTIVES

- To provide scientifically planned, multi-dimensional, skill oriented and personality development course, which will enable the students to excel in today's context.
- To provide education not only to acquire knowledge but also to inculcate a sound philosophy for holistic living.
- To educate students towards independence and equality by inculcating a sense of self-worth, integrity and purpose.
- To empower students to respond creatively and with a sense of responsibility to the contemporary needs of the society.
- To provide quality education and at the same time inculcate a spirit of service and dedication.
- To build competent professional Nutrition & Dieticians in hospitals and specialty clinics. Thereby, the professionals can find job prospects in the field as Nutrition and Diet consultants in Food service organizations like Hotels, Hospitality services, Geriatric homes and also as administrators of Industrial canteens and other specialties.

(b) RELEVANCE OF THE PROGRAM WITH HEI'S MISSION AND GOALS

Through this program offered, the HEI will prepare students to do the learning activities independently, gaining knowledge and skills in nutrition & dietetics, and applying the knowledge they have acquired for the welfare of human.

The HEI's mission is "Affording a High Quality Higher Education to the learners so that they are transformed into intellectually competent human resources that will help in the upliftment of the Nation to Educational, Social, Technological, Environmental and Economic Magnificence".

The curriculum and syllabus, of M.Sc., Home Science - Nutrition and Dietetics course at Alagappa University, have been designed to encompass basic as well as applied knowledge of Nutrition and Dietetics to produce ready and employable Home Science postgraduates capable of availing promising professional carrier in life.

(c) NATURE OF PROSPECTIVE TARGET GROUP OF LEARNERS

The target group of learners for admission to the M.Sc., Home Science - Nutrition and Dietetics shall be graduates of a Bachelor's degree from a recognized university with minimum 55% marks in Home Science, Botany, Zoology, Biochemistry, Chemistry, Biotechnology, Microbiology, Food Science and Quality Control and Clinical Nutrition and Dietetics, M.B.B.S./ B.H.M.S./ B.A.M.S./ B. Pharmacy and any other relevant subjects in Science. The target group of learners apart from minimum prescribed qualifications is Working Professionals, Entrepreneurs, Service Personnel, Academic Faculty, Government Officials, Researchers, Home Makers and Unemployed Graduates.

(d) APPROPRIATENESS OF PROGRAMME TO BE CONDUCTED IN OPEN AND DISTANCE LEARNING MODE TO ACQUIRE SPECIFIC SKILLS AND COMPETENCE

Open and Distance Learning of M.Sc. Home Science - Nutrition and Dietetics will improve skills of the learners either in job or their pre-jobs requirements. There is a great importance and relevance in countries like India where there is an urgent need of providing, cost effective training in the field of Home Science -nutrition and dietetics to a large number of untrained work forces and need of continuing education at different levels. Thereby, the professional open and distance mode learning will improve overall knowledge of the learners and enable them to be part of the productive force in the management of quality of life in India. Through the M.Sc. Home Science - Nutrition and Dietetics program, the students will obtain knowledge in Human Physiology, Health and Nutrition, Advanced food science, Nutritional Biochemistry, Food service management, Clinical and therapeutic nutrition, Dietetics in life style diseases, Paediatric nutrition, Food microbiology and biotechnology. Over all, the course will enhance the performance of students in the field of Nutrition and Dietetics.

(e) INSTRUCTIONAL DESIGN

M.Sc., HOME SCIENCE – NUTRITION AND DIETETICS

Sl. No.	Course Code	TITLE OF THE COURSE	CIA Max.	ESE Max.	TOT Max.	C
		FIRST YEAR				
		I Semester				
1	36511	HUMAN PHYSIOLOGY	25	75	100	4
2	36512	NUTRITION AND HEALTH	25	75	100	4
3	36513	ADVANCED FOOD SCIENCE	25	75	100	4
4	36514	Lab. I: HUMAN PHYSIOLOGY, NUTRITION AND HEALTH & ADVANCED FOOD SCIENCE	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		II Semester				
5	36521	NUTRITIONAL BIOCHEMISTRY	25	75	100	4
6	36522	FUNCTIONAL FOODS AND NUTRACEUTICALS	25	75	100	4
7	36523	FOOD SERVICE MANAGEMENT	25	75	100	4
8	36524	Lab. II: NUTRITIONAL BIOCHEMISTRY, FUNCTIONAL FOODS AND NUTRACEUTICALS & FOOD SERVICE MANAGEMENT	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		SECOND YEAR				
		III Semester				
9	36531	CLINICAL AND THERAPEUTIC NUTRITION	25	75	100	4
10	36532	DIETETICS IN LIFE STYLE DISEASES	25	75	100	4
11	36533	COMMUNITY NUTRITION	25	75	100	4
12	36534	Lab. III: CLINICAL AND THERAPEUTIC NUTRITION, DIETETICS IN LIFE STYLE DISEASES, COMMUNITY NUTRITION	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		IV Semester				
13	36541	PAEDIATRIC NUTRITION	25	75	100	4
14	36542	FOOD MICROBIOLOGY AND SANITATION	25	75	100	4
15	36543	FOOD BIOTECHNOLOGY & BIOSTATISTICS	25	75	100	4
16	36544	Lab. IV: PAEDIATRIC NUTRITION, FOOD MICROBIOLOGY AND SANITATION & FOOD BIOTECHNOLOGY & BIOSTATISTICS	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		Grand Total	**400**	**1200**	**1600**	**64**

Course Code Legend:

3	6	5	Y	Z

365- M.Sc., HOME SCIENCE - NUTRITION AND DIETETICS
Y -Semester Number
Z- Course Number in the Semester.

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C: Credit Points, Max.: Maximum

36511-HUMAN PHYSIOLOGY	
Maximum marks: 100	**Credits: 4**

Objectives:

To enable the students

- ✓ To understand the basic concepts of prokaryotic and eukaryotic cell.
- ✓ To understand the basics of Human Physiology.
- ✓ To enable the students to understand the metabolic changes under various conditions.

BLOCK – I: CELL, CELL ORGANELLES, TISSUES AND MUSCLES

Unit I

Cell - Structure and organization of prokaryotic and eukaryotic cells. Cell and nuclear membrane, cell wall and cell envelope. Physiology of cytoplasm.

Unit II

Cell organelles - Structural organization and functions of intracellular organelles: nucleus, nucleolus, endoplasmic reticulum, golgi complex, mitochondria, chloroplast, lysosomes, peroxisomes and vacuoles.

Unit III

Tissues - Classification, structure and functions of epithelial, muscular, connective and nervous tissues.

Unit IV

Musculo skeletal system – structure and functions of bone, cartilage, muscle, joints, ligaments and tendons.

BLOCK – II: BLOOD, CARDIOVASCULAR, RESPIRATORY AND DIGESTIVE SYSTEM

UNIT V

Blood - Introduction to hematology, functions of blood, plasma proteins, erythrocytes, Hb, important indices of RBC & WBC, Functions of blood groups, ESR, blood viscosity, blood coagulation, Erythroblastosis foetalis and blood transfusion.

UNIT VI

Cardiovascular system - Basic properties of the heart, cardiac output, blood pressure and factors affecting it and hypertension. Nutrition and metabolism of heart. Exercise and heart Function. Techniques to identify cardiovascular disorders –angioplasty and angiogram.

UNIT VII

Respiratory system - Anatomy and physiology of respiratory organs, mechanism of respiration, gaseous exchange in lungs and tissues. Resuscitation and its methods. Respiratory disorders: dyspnoea, asphyxia, hyperpnoea, orthopnoea,

UNIT VIII

Digestive system - Anatomy, composition & functions of salivary, gastric, intestinal & pancreatic secretions. Functions of bile salts, Mechanism of secretion of digestive juices and its regulation, movements of stomach, small intestine, villi, defecation. Important of liver in digestive system, anatomy and physiology.

BLOCK – III: EXCRETORY AND REPRODUCTIVE SYSTEM, SENSE ORGANS

UNIT IX

Excretory system - Mechanism of urine formation and the role of the kidneys in water and electrolyte balance. Renal function tests. Artificial kidney, dialysis and renal transplantation.

Unit X

Reproductive system - Male and female reproductive organs: structure and functions. Menstruation, menstrual cycle, puberty, menarche, menopause, fertilization, conception, implantation. Male and female contraception's- Etiology of male and female infertility.

Unit XI

Sense organs - Physiology of vision, hearing, taste, smell and cutaneous sensations.

BLOCK – IV: ENDOCRINE, EXOCRINE AND NERVOUS SYSTEM

Unit XII

Endocrine glands - pituitary, thyroid gland, parathyroid gland, pancreas, adrenal cortex and adrenal medulla. Mechanism of action of hormones. Hormonal imbalance syndromes, hypo or hyperactivity.

Unit XIII

Exocrine glands – Structure and functions of sweat, salivary, mammary, ceruminous, lacrimal, sebaceous, and mucous glands. Gland secretion syndromes.

UNIT XIV

Nervous system - General anatomy of nervous system, functions of the different parts, reflexes, autonomic nervous system. Common test in neurological disorders- EEG, EMG, MRI, NCV.

REFERENCES

1. Leslie P. Gartner, 2016. "Textbook of Histology" Elsevier; 4th edition.
2. John E. Hall, 2015. "Guyton and Hall Textbook of Medical Physiology" W.B. Saunders & Company 13th edition.
3. Tortora G.J and Grabowski S.R, 2000. "Principles of Anatomy and Physiology". John Wiley and Sons.Inc. 9th edition.
4. Chaudhari S. K, 2000. "Concise Medical Physiology"3rd Edition.
5. West J.B. 1996. "Physiological Basis of Medical Practice". B. I. Waverly Pvt. Ltd. 12th Edition.
6. Guyton.A.C, 1991. Textbook of medical physiology, 9th edition, Philadelphia, WB Saunders.
7. Chattergee.C.C, 1987. Human Physiology (11th edition), Vol 1 & 2, Medical Allied Physiology.

OUTCOME:

This course provides the students a sound basis in human physiology to support further study in health nutrition & dietetics and medical sciences or related fields.

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36512- NUTRITION AND HEALTH	
Maximum marks: 100	**Credits: 4**

Objectives:

To enable the students

- ✓ To understand the role of nutrition in different conditions.
- ✓ To examine the techniques available for the assessment of nutritional status.
- ✓ To provide a complete understanding of the links between early nutrition and adult disease.
- ✓ To identify and overcome obstacles in the provision of healthy diets for specific age groups.
- ✓ To develop competency in planning diets to meet the nutritional requirements of different socio economic levels.

BLOCK – I: NUTRITION, DIET IN HEALTH AND PREGNANCY

UNIT I

Nutrition and diet in health - concept of adequate nutrition and malnutrition. Difference between hunger, apetite and satiety. Different food groups – major nutrients present in each group, guide in menu planning.

UNIT II

Recommended dietary allowances - Basis for requirements. ICMR Recommended Dietary Allowances (RDA) for Indians, FDA Recommendations, Basal metabolic rate (BMR) and active metabolic rate (AMR). Balanced diets.

UNIT III

Nutrition in pregnancy - Physiological changes in pregnancy. Nutritional status and general health. Importance of preconceptual nutrition. Weight gain during pregnancy and the nature of weight gain. Factors affecting maternal nutritional status.

UNIT IV

Nutrition in pregnancy - requirements, storage of nutrients in normal pregnancy, complications of pregnancy and nutritional problems in young and too old expectant mothers – causes and complications. Avoiding pregnancy associated health risk through nutrition- gestational diabetes, iron deficiency anemia and hypertensive disorders.

BLOCK–II: VACCINATION, NUTRITION IN LACTATION, INFANCY AND PRESCHOOL

UNIT V

Pregnancy and vaccination -Immunization schedule & tests during pregnancy - measles, mumps, rubella (MMR) vaccine- Tdap vaccine.

UNIT VI

Nutrition in lactation - Physiological adjustments during lactation, lactation in relation to growth and health of infants, efficiency of milk production, diet during lactation.

UNIT VII

Nutrition in infancy - Nutritional status of the infants, rate of growth as the indicator. Nutritional allowances for the infants, breast feeding Vs formula feeding, food square, weaning foods suitable for infants, feeding the premature infants and Low Birth Weight (LBW) infants, reasons for under 5 Mortality Rate (MR), interventions to prevent malnutrition.

UNIT VIII

Nutrition in preschool age - Growth and development of preschool children, food habits and nutrient intake of preschool children. Dietary allowances – supplementary foods, reasons for under 5 Mortality Rate. Interventions to prevent malnutrition among preschoolers.

BLOCK–III: NUTRITION IN SCHOOL AGE, ADOLESCENCE AND ADULTS

UNIT IX

Nutrition during school age - Physical development, nutritional status of school going children, food habits, nutritional requirements, nutrition and academic performance, interventions to prevent malnutrition.

UNIT X

Nutrition during adolescence - Changes of growth, assessment of growth – sexual maturity rating, physical, physiological and psychological changes in adolescents. Nutritional needs of the adolescents, changes needed to prevent malnutrition in adolescents.

UNIT XI

Nutrition for the adults - Nutritional requirements according to the mode of activity. Nutrition and health of women-general nutritional problems of women, anemia, osteoporosis, pre and post-menopausal syndrome, hormonal changes during menopause. Infertility – risk factors, prevention, methods of detection.

BLOCK–IV: NUTRITION IN OLD AGE, SPECIAL EVENTS AND NUTRITION MONITORING

UNIT XII

Nutrition in old age - Ageing process - physiological, metabolic, body composition changes. Nutritional & health status of elderly.

UNIT XIII

Nutrition in special events - Sports nutrition - quantity of fluids and food taken by an athlete. Space nutrition - food product created and processed for consumption by astronauts in outer space.

UNIT XIV

Nutrition monitoring and its current programmes - Nutrition Surveillance System, Integrated Child Development Services (ICDS) Programme, Nutrient Deficiency Control Programme, Supplementary Feeding Programme and Food Security Programme.

REFERENCES

1. L. Kathleen Mahan and Janice L. Raymond, 2016. Krause's Food & the Nutrition Care Process Saunders-Elsevier.
2. Park.K, 2015. Park's Textbook of Preventive and Social Medicine, 23rd ed. M/s Banarsida Bhanot, Jabalpur.
3. William's, 2009. Basic Nutrition and Diet Therapy. 13th Edition. Stacy Nix, Elsevier Mosby.
4. Tony Worsley, Basil S. Hetzel and Mark Lawrence, 2008. Public Health Nutrition: From Principles to Practice. Allen and Unwin Special Priced Titles.
5. Mahan.L.K and Stump SE, 2001. Krause's Food, Nutrition and Diet Therapy, WB Saunders Company, 10th edition.
6. FAO/WHO, 1998. Preparation and use of food based dietary guidelines. Report of a joint FAO/WHO consultation: Nicosia, Cyprus. Nutrition Programme, WHO, Geneva.

OUTCOME:

This course enables the students to understand the social relevance of nutrition and its role in health.

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36513 - ADVANCED FOOD SCIENCE	
Maximum marks: 100	**Credits: 4**

Objectives:

To enable the students

- ✓ To understand the nutritive value of foods.
- ✓ To understand the principles and chemistry of foods.
- ✓ To examine how food quality, properties, and safety are affected and controlled by handling, processing and storage practices.

BLOCK-I: FOOD PREPARATION, PROPERTIES OF FOOD AND FOOD QUALITY

UNIT I

Food in relation to health - Introduction to food science as a discipline and modern developments, different methods of cooking, functions of cooking food.

UNIT II

Functional properties of foods - Definition, structure and properties of food hydrocolloids. Hydrocolloids as gelling, emulsifying, thickening, stabilizing and coating agents. Important roles of proteins (denaturation and browning), carbohydrates (caramelization and crystallization) and fats (emulsification) in altering the functional properties of food.

UNIT III

Evaluation of food quality - Quality attributes of food – appearance factors, textural factors, and flavor factors sense of taste, texture and colour: sensory evaluation and objective evaluation. Types of sensory test. Procedures for determination and monitoring of shelf life.

BLOCK-II: CEREALS, MILLETS, PULSES, NUTS, VEGETABLES, FRUITS AND FLESH FOOD

UNIT IV

Cereals & Millets - Nutritive value, parboiling, Cereal cookery – gluten- factors affecting gluten formation, Starch granules structure and characteristics – effect of moist and dry heat, nonstarch polysaccharides- fibres, cellulose, hemicellulose, pectic substances, gums and carboxy methyl cellulose (CMC). Nutritive value of breakfast cereals and fermented products.

UNIT V

Pulses, nuts and oilseeds - Nutritive value, processing, protein foods for infants and children, soy products, protein concentrates and isolates, textured vegetable proteins.

UNIT VI

Vegetables and fruits - Nutritional importance, pigments and acids, effect of cooking on pigments and nutrients. Post-harvest changes of fruits, browning reactions- enzymatic and non-enzymatic.

UNIT VII

Flesh foods - Composition, post-mortem changes in meat, changes produced during cooking, spoilage. Effect of heat on egg proteins, egg foams, factors influencing foaming and egg products.

BLOCK-III: MARINE FOODS, MILK PRODUCTS, FATS AND BEVERAGES

Unit VIII

Nutritive value of marine foods: Fish, shrimp and sea weeds.

Unit IX

Milk and milk products – Nutritive value of milk powders, ghee, khoa, butter, paneer, cheese and ice creams - Composition, physical and functional properties.

UNIT X

Fats and oils - role of fat in cookery, rancidity, changes of fat on heating, salad dressing.

UNIT XI

Beverages – Classification, manufacture and nutritional significance and energy value.

BLOCK-III: SUGAR, FOOD ADDITIVES AND FOOD TECHNOLOGY

UNIT XII

Sugar - Properties, sugar related products, crystallization, crystalline & non- crystalline candies, stages of sugar cookery, artificial sweeteners.

Unit XIII

Food additives - Definition and needs for food additives, types of food additives and food safety, unintentional additives.

UNIT XIV

Food technology - Genetically Modified (GM) foods, Production and nutritive value of GM foods.

REFERENCES

1. Judith L. Buttriss, Ailsa A. Welch, John M. Kearney, Susan A, 2017. Lanham-New. Public Health Nutrition, 2nd Edition, ISBN: 978-1-118-66097-3.
2. B. Srilakshmi, 2015. "Food Science" New Age International Private Limited; Sixth edition.
3. Frank Lee, 2012. "Basic Food Chemistry" Springer; Softcover reprint of the original 1st edition.
4. Sunetra Roday, 2012. "Food Science and Nutrition" Oxford Press; Second edition.
5. G. L. Tandon, G. S. Siddappa, Girdhari Lal, 2009. "Preservation of Fruits and Vegetables" by Bombay Popular Prakashan.
6. Manay.N.S & Shadaksharaswamy.M, 2002. Foods-Facts & Principles. New Age International Pvt.Ltd, New Delhi.
7. Potter, N. Hotchkiss, H.J, 1996. Food Science, 5th edition, CBS publishers and distributers, New Delhi.
8. Beckhan. C.G & Graves.H.J, 1979. Foundations of food preparations, Macmillan Publishing Co, New Delhi.

OUTCOME:

This course facilitates the students to understand the basic concepts in food science

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36514 - Lab. I: HUMAN PHYSIOLOGY, NUTRITION AND HEALTH & ADVANCED FOOD SCIENCE	
Maximum marks: 100	**Credits: 4**

Objectives:

To enable the students

- ✓ To assessment of nutritional status and nutritional requirements of humans.
- ✓ To planning and preparation of therapeutic diets for various diseases and health conditions.
- ✓ To introduce the students about the tools used in health management.

HUMAN PHYSIOLOGY

1. Estimation of Glucose from blood.
2. Analysis of blood Haemoglobin.
3. Determination of Cholesterol from blood.
4. Estimation of Vitamins, Minerals, Electrolytes from blood.
5. Blood cell counts, haematocrit, blood histology/ blood smears
6. Blood typing
7. Histology: cells and tissues
8. Diffusion and osmosis
9. Urine analysis - Creatinine, Total nitrogen and Urea
10. Pregnancy test

HEALTH AND NUTRITION

1. Preparation of low cost recipes for adolescents, pregnant and lactating mothers.
2. Evaluation of the ongoing public health nutrition programmes.

ADVANCED FOOD SCIENCE

1. Database management of anthropometric indices, biochemical indices, dietary recall, energy expenditure and intake.
2. Role of portable devices in diet and health management.

REFERENCE:

1. Judith L. Buttriss, Ailsa A. Welch, John M. Kearney, Susan A. Lanham, 2017. New. Public Health Nutrition, 2nd Edition, , ISBN: 978-1-118-66097-3.
2. Paul. S., 2014. Textbook of Bio-Nutrition, Curing diseases through diet, CBS publications, first edition.
3. Maurice E Shils, James A. Olson, Moshe Shike, 2012. “Modern Nutrition in health and disease” 11th edition, Vol I & II Lea & Febiger Philadelphia, A Waverly company.
4. Srilakshmi B., 2004. Dietetics, New Age International (P) Limited Publications.
5. Sue Rodwell Williams, Eleanor D. Schlenker, 2003. “Essentials of Nutrition and Diet Therapy” C.V. Melskey Co.
6. M. Swaminathan. 2001. “Principles of Nutrition and Dietetics”, Bappeo 88, Mysore Road, Bangalore - 560 018.

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36521 - NUTRITIONAL BIOCHEMISTRY	
Maximum marks: 100	**Credits: 4**

Objectives:

To enable students to:

- ✓ To learn the biochemical role of nutrients in foods and deficiency diseases.
- ✓ To understand the metabolism of nutrients in health and diseases.

BLOCK-I: CARBOHYDRATES AND PROTEINS

Unit I

Carbohydrates - Classification, physical and chemical properties, sources, biological role.

Unit II

Carbohydrate metabolism - Glycolytic pathway, deficiency diseases, inborn errors of carbohydrate metabolism. Nutritional aspects of carbohydrate.

Unit III

Proteins - Classification, physical and chemical properties, sources, biological role and value of protein.

Unit IV

Protein metabolism – Protein synthesis, deficiency diseases and inborn errors of protein metabolism.

BLOCK-II: LIPIDS, VITAMINS AND MINERALS

Unit V

Lipids - Classification, physical and chemical properties, sources, biological role.

Unit VI

Lipid metabolism – β-oxidation. nutritional aspects of lipids, lipid based metabolic diseases, In-born errors of lipid metabolism.

Unit VII

Vitamins - Classification, characteristics, role of vitamins in metabolism, deficiency diseases.

Unit VIII

Minerals - Types, absorption & role of minerals in metabolism, minerals deficiency diseases.

BLOCK-III: NUCLEIC ACIDS AND ENZYMES

Unit IX

Nucleic acids - DNA & RNA, structure, function and metabolism, genetic disorders.

Unit X

Enzymes - Classification, nomenclature, mechanism of enzyme action, enzyme specificity, application of enzymes in clinical diagnosis.

Unit XI

Enzyme activity - Factors affecting enzyme activity, Co- enzymes and Co-factors.

BLOCK-IV: HORMONES, BUFFERS AND ELECTROLYTES

Unit XII

Hormones - Role of hormones. Interrelation between hormones and nutrients. Hormone deficiency diseases.

Unit XIII

Acid base balance - normal health, major sources of acid produced in the body, buffers, physiological role of different buffer systems.

Unit XIV

Fluid and electrolyte balance - Maintenance in normal health. Diseases of electrolytes imbalance. Role of nutrients in maintenance of fluid and electrolyte balance during disease condition.

REFERENCE:

1. Pooja Gupta. (2017) Food, Nutrition And Health, S Chand Publishing, India
2. Berg JM, Tymoczko JL and Stryer L. (2015) Biochemistry 8th ed. W.H. Freeman.
3. Murray RK, Granner DK, Mayes PA and Rodwell VW, (2015) Harper's Illustrated Biochemistry, 30th ed. McGraw-Hill (Asia).
4. Devlin Tm. (2010) Text Book of Biochemistry with clinical Correlations. 7th ed. John Wiley and sons.
5. Voet D and Voet JG. (2010) Biochemistry 4rd ed. John Wiley and Sons.
6. Nelson DL and Cox MM. (2005) Principles of Biochemistry, 4th ed. Freeman and Company.

OUTCOME:

This course provide the students a comprehensive overview about the principle, scope and applications in nutritional biochemistry

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36522 - FUNCTIONAL FOODS AND NUTRACEUTICALS	
Maximum marks: 100	**Credits: 4**

Objectives:

To enable the students

- ✓ To gain knowledge about functional foods and nutraceuticals
- ✓ To understand about the health effects
- ✓ To know the application in Industry.

BLOCK-I: FUNCTIONAL FOODS, PROBIOTICS AND PREBIOTICS

UNIT I

Definition, history, classification of functional foods: Probiotics, prebiotics and synbiotics; Nutrient vs. Non-nutrient.

UNIT II

Probiotics - Taxonomy and important features of probiotic micro- organisms. Health effects of probiotics including mechanism of action.

UNIT III

Probiotic micro- organisms in fermented milk products and non-milk products. Quality assurance of probiotics and safety.

UNIT IV

Prebiotics - chemistry, sources and bioavailability, effect of processing, effects on human health and potential applications in risk reduction of diseases, perspective for food applications for the following:

Non-digestible carbohydrates/oligosaccharides: Dietary fibre, Resistant starch, Gums.

BLOCK-II: PLANT METABOLITES AND NON- NUTRIENT EFFECT OF SPECIFIC NUTRIENTS

UNIT V

Alkaloids, Glucosinolates, Terpenoides and Phenolics- Chemistry, classes, sources, bioavailability and effects on human health

UNIT VI

Antinutrients present in food: phytate, saponin, haemagglutinins, protease, amylase and lipase inhibitors. Spices and Condiments- nutritive value and uses in cooking.

UNIT VII

Non- nutrient effect of specific nutrients: Proteins, Peptides and nucleotides, Conjugated linoleic acid and n-3 fatty acids, Vitamins and Minerals.

BLOCK-III: PROPERTIES, STRUCTURE AND FUNCTIONS OF NUTRACEUTICALS

UNIT VIII

Introduction to nutraceuticals as science - Historical perspective, classification, scope & future prospects.

UNIT IX

Applied aspects of the nutraceutical science**:** Sources of Nutraceuticals. Relation of Nutraceutical Science with other Sciences: Medicine, Human physiology, genetics, food technology, chemistry and nutrition.

UNIT X

Properties, structure and functions of various nutraceuticals - Glucosamine, Octacosanol, Lycopene, Carnitine, Melatonin and Ornithine alpha ketoglutarate. Use of proanthocyanidins, grape products, flaxseed oil as Nutraceuticals.

BLOCK-IV: NUTRACEUTICAL SUPPLEMENTS AND REMEDIES

UNIT XI

Nutraceutical rich supplements - Bee pollen, Caffeine, Green tea, grape tea, wheat grass, Lecithin, Mushroom extract, Chlorophyll, Kelp and Spirulina, *Garcinia cambogia* , *Aloe vera* and Blue Tea4.

UNIT XII

Food as remedies: Nutraceuticals bridging the gap between food and drug, Nutraceuticals in treatment for cognitive disorders. Medicinal plant derived nutraceuticals: Anti aging, anti-inflammatory compounds.

UNIT XIII

Nutraceutical remedies for Arthritis, Bronchitis, circulatory problems, hypoglycemia.

UNIT XIV

Nutraceutical remedies for Nephrological disorders, Liver disorders, Osteoporosis, Psoriasis and Ulcers.

REFERENCES

1. Dhiraj A. Vattem, Vatsala Maitin, 2016. Functional Foods, Nutraceuticals and Natural Products: Concepts and Applications, DEStech Publications, Inc.
2. Joyce I. Boye, 2015. Nutraceutical and Functional Food Processing Technology (IFST Advances in Food Science), Wiley-Blackwell.
3. Cho S. S. and Dreher, M.L, 2001. Handbook Dietary Fibre, Marcel Dekker Inc., New York.
4. Wildman, R.E.C, 2000. Handbook of Nutraceuticals and Functional Foods, CRC Press, Boca Raton.
5. Yurawecz, M.P., M.M. Mossoba, J.K.G. Kramer, M.W. Pariza and G.J. Nelson, 1999. Advances in Conjugated Linoleic Acid Research, Vol. 1. AOCS Press, Champaign.
6. Heller IR, et al. 1999. Report by CSPI. Functional foods: public health boon or 21st century quackery? Washington, DC: Center for Science in the Public Interest.
7. Thomas PL, Earl R, 1994. Opportunities in nutrition and food sciences. Washington, DC: National Academy Press; p 109.

OUTCOME:

This course empowers the students with current scenario in functional foods and neutraceuticals.

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36523 FOOD SERVICE MANAGEMENT	
Maximum marks: 100	**Credits: 4**

Objectives:

To enable the students

- ✓ To understand the objectives of different types of food service Institutions.
- ✓ To apply knowledge in space allocation of food plants
- ✓ To gain knowledge in menu planning preparation of recipes in large scale and serving and in food costing.

BLOCK-I: FOOD SERVICE INDUSTRY, INFRASTRUCTURE AND MATERIALS MANAGEMENT

UNIT I

Food service industry**:** Scope of food industry and segmentation.

UNIT II

Organization & administration of food service industry**:** types, organization structure and management.

UNIT III

Physical facilities and layout – Size and type of kitchen, design of kitchen, ventilation, lighting, flooring, carpets, wall covering and sample layout of kitchen. Storage area and equipment required.

UNIT IV

Food materials management **-** Purchasing of food materials, receiving & storing – Importance of receiving raw materials.

BLOCK-II: FOOD SERVICE MANAGEMENT IN HOSPITALS

UNIT V

Hospital food production – Menu planning for patients and process of food production. Different methods of holding foods for service.

UNIT VI

Hospital food service management - Principles and techniques of effective management, leadership and managerial abilities.

UNIT VII

Tools of food management - Organizational chart of the food service team in hospital.

UNIT VIII

Food service equipment **-** Classification, selection, purchasing, care and maintenance.

BLOCK-III: PREPARATION, SERVICE AND SANITATION OF FOOD

UNIT IX

Quantity food preparation - Types of menu, menu planning, purchasing, storage, production management, conventional and non-conventional sources of energy, Standardization and portion control.

UNIT X

Styles of service - Self-service, tray service, waiter-waitress service, vending and mobile food service system.

UNIT XI

Sanitation and hygiene - Environmental hygiene & sanitation, safe food handling practices, personal hygiene.

BLOCK-IV: HUMAN RESOURCE MANAGEMENT, MARKETING AND DIETARY ACCOUNTING

UNIT XII

Human resource management - Recruitment & selection, induction, training, performance appraisal, leadership, communication, employee benefits, laws governing food service establishment.

UNIT XIII

Marketing - Definition, marketing as a managerial function, marketing mix and promotion in food service.

Unit XIV

Dietary accounting - Definition and principles. Journal and ledger. Book of account – cash book, purchase book, sales book, purchase returns & sales returns book.

REFERENCES

1. Karen E. Drummond, Lisa M. Brefere, 2017; Nutrition for Food service and Culinary Professionals, 9th Edition (EHEP003624).
2. Ruby Parker Puckett, 2012. Food service Manual for Health Care Institutions, John Wiley & Sons.
3. Sethi M and Mahan S,, 2006. Catering Management an integrated approach, 2nd edition, John wiley & Sons, New York.
4. Tersel MC and Harger, 2005. Profession food preparation, John wiley & Sons,New York.
5. Joan C Boson, Lennox M. 2004. Hotel, hostel & hospital housekeeping, 5th edition, Book power publishers, New York.
6. Marian C Spears; 1995. Food Service Organisation; III Edition, Prentice Hall Inc., USA.
7. West and Woods, 1994. Introduction to Food Service, Macmillan Publishing Company, New York, 7 th edition.
8. Lendal. H. Kotschever, Richard Donnely, 1993 "Quantity Food Purchasing, Mac Millan Publishing Company, New York, IV Edition.

OUTCOME:

This course imparts the students about the significance of food service management, sanitation of food and dietary accounting.

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36524 - Lab. II: NUTRITIONAL BIOCHEMISTRY, FUNCTIONAL FOODS AND NUTRACEUTICALS & FOOD SERVICE MANAGEMENT	
Maximum marks: 100	**Credits: 4**

Objectives:

To enable the students

- ✓ To gain knowledge in biochemical analysis of food samples.
- ✓ To get practical experience in the laboratory and develop the skills related to functional foods and nutraceuticals.
- ✓ To provide overview about the strategies in food service management.

NUTRITIONAL BIOCHEMISTRY

1. Determination of Moisture content in Food sample.
2. Determination of Carbohydrates, Proteins and fats in Food sample.
3. Determination of Gluten content in wheat.
4. Estimation of Acidity in wheat flour.
5. Estimation of Fiber, Phosphorous and Iron content in any one food.
6. Determination of Calcium content in milk.

FUNCTIONAL FOODS AND NUTRACEUTICALS

1. Manufacturing aspects of selected nutraceuticals (Demonstration)
    a. lycopene,
    b. isoflavonoids
    c. prebiotics
    d. probiotics
    e. glucosamine
    f. phytosterols
2. Spirulina cultivation (Industrial visit)

FOOD SERVICE MANAGEMENT

1. Causes and prevention of food-borne illnesses in food service operations.
2. The levels of management and the various production and service positions in a food operation (Field visits)

REFERENCES

1. Pooja Gupta. (2017) Food, Nutrition And Health, S Chand Publishing, India
2. Paul. S., 2014.Textbook of Bio-Nutrition, Curing diseases through diet, CBS publications, first edition.
3. Maurice E Shils, James A, 2012. Olson, Moshe Shike "Modern Nutrition in health and disease" 11th edition, Vol I & II Lea & Febiger Philadelphia, A Waverly company.
4. Srilakshmi B., 2004. Dietetics, New Age International (P) limited Publications.
5. Sue Rodwell Williams, Eleanor D. Schlenker, 2003. "Essentials of Nutrition and Diet Therapy" C.V. Melskey Co.
6. M. Swaminathan, 2001. "Principles of Nutrition and Dietetics", Bappeo 88, Mysore Road, Bangalore - 560 018.
7. Messina M, Messina V, 1996.Nutritional Implications of Dietary Phytochemicals. In: Dietary Phytochemicals in Cancer Prevention and Treatment. Plenum Press. New York.

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36531 - CLINICAL AND THERAPEUTIC NUTRITION	
Maximum marks: 100	**Credit: 4**

Objectives:

To enable the students to know the

- ✓ Effect and clinical values of various diseases
- ✓ Nutritional status and dietary requirements.

BLOCK-I: ROLE OF DIETITIAN, NUTRITION PSYCHOPATHOLOGY AND HOSPITAL DIETS

UNIT I

Role of dietitian in the hospital and community- Education and personal qualifications, professional ethics and obligations. Educating the patient, diet clinics and follow up.

UNIT II

Psychology of feeding the patient, problems of feeding children, assessment of patient's needs.

UNIT III

Routine hospital diets -Regular diet, light diet, soft diet, full liquid diet, clear liquid diet and tube feeding.

UNIT IV

Enteral and parental feeding in hospitals –composition, monitoring and complications. Transitional feeding.

BLOCK-II: DIET IN FEBRILE CONDITIONS AND GASTROINTESTINAL DISORDERS

UNIT V

Modifications of diet in febrile conditions -Acute, chronic and recurrent fevers, typhoid, rheumatic fever, tuberculosis, malaria, H1N1, dengue fever and chikungunya.

UNIT VI

Gastrointestinal disorders - Esophagitis, cancer of oral cavity, ulcer, indigestion, gastritis, carcinoma of the stomach, gastric surgery and dumping syndrome.

UNIT VII

Gastrointestinal Disorders - Diarrhoea, constipation, flatulence, celiac disease, tropical sprue, steatorrhea.

UNIT VIII

Irritable bowel disease (IBD) – crohn's disease, ulcerative colitis, Irritable bowel syndrome (IBS), diverticulitis, colitis and colon cancer.

BLOCK-III: DIET IN LIVER, GALL BLADDER, PANCREAS, METABOLIC AND RENAL DISORDERS

UNIT IX

Liver, gall bladder and pancreatic disorders - Ecological factors: Dietary regimen in cirrhosis, hepatitis, hepatic coma, cholecystitis, cholelithiasis and pancreatitis

UNIT X

Metabolic disorders - Hypothyroidism, hyperthyroidism, gout, phenylketonuria and lactose intolerance

UNIT XI

Renal disorders - Contributory factors and dietary modification- acute and chronic glomerulonephritis, nephrosis, nephrosclerosis and nephrolithiasis.

BLOCK-IV: DIET IN FOOD ALLERGY, NEUROLOGICAL DISORDERS AND METABOLIC STRESS

UNIT XII

Food allergy - Definition, types, tests, dietary management and prevention.

UNIT XIII

Diet during neurological disorders - Alzheimer's disease, Parkinson's disease and epilepsy.

UNIT XIV

Diet during metabolic stress - Burns, sepsis and trauma. Surgical conditions- Cardiovascular complications, stroke and surgery, respiratory failure, hepatic failure, multi organ failure, Gastrointestinal tract (surgery and complications) and neurosurgery.

REFERENCES

1. Ellen Davis, 2017. Fight Cancer with a Ketogenic Diet: Using a Low-Carb, Fat-Burning Diet as Metabolic Therapy, 3rd edition, Ellen Davis.
2. Sylvia Escott-Stump, 2015. Nutrition and Diagnosis-Related Care. 8th edition, Wolters Kluwer.
3. Mahan.L.K and Stump SE, 2001. Krause's Food, Nutrition and Diet Therapy, WB Saunders Company, 10th edition.
4. Antia FP, Clinical Dietetics and Nutrition, 1997.Oxford University Press, New Delhi, 4th edition.
5. Garrow.JS & James W.P.T, 1993.Human Nutrition and Dieteics, Church Hill Living Stone.
6. Robinson, 1990. Normal and Therapeutic Nutrition, Oxford & LBM Publishing, Calcutta, Bombay, 17th edition.
7. Davidson, Pasmore P and Break LP, 1986. Human Nutrition and Dietetics, English language book society, Livingstone.
8. Karran, S. J. and K. G. M. M. Alberti, 1980. Practical Nutritional Support, John Wiley and Sons. Inc. New. York.

OUTCOME:

This course offers the students about the significance of diet in various health conditions.

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36532 DIETETICS IN LIFE STYLE DISEASES	
Maximum marks: 100	**Credit: 4**

Objectives:

To enable the students

- ✓ To gain knowledge about the principles of diet therapy and different therapeutic diets
- ✓ To develop aptitude for taking up dietetics as a profession.

BLOCK-I: DIETETICS IN STRESS AND WEIGHT MANAGEMENT

UNIT I

Stress – definition, types, psychosomatic disorders due to stress and functional adjustment.

UNIT II

Biological effects of stress on various systems-brain, cardiovascular system, respiratory system, non-vital organs and immune system.

UNIT III

Stress enhancing food, antistress foods and nutrients. Dietary guidelines for the management of stress.

UNIT IV

Nutrition for weight management -components of body weight, adipose tissue and regulation of body weight. Obesity-assessment, types, causes and complications.

BLOCK-II: DIETETICS IN WEIGHT REDUCTION AND DIABETES

UNIT V

Weight reduction techniques-dietary management, surgical management, lifestyle modification, Underweight-causes, complications and dietary management.

UNIT VI

Diabetes mellitus - Classification, symptoms, diagnosis, causes and complications. Management of diabetes-dietary management, artificial sweetners, diet and insulin and lifestyle management.

UNIT VII

Diabetes Insipidus & Gestational diabetes – causes, complications, dietary and life style management.

BLOCK-III: DIETETICS IN CARDIOVASCULAR DISEASES

UNIT VIII

Cardiovascular diseases - Risk factors, Blood lipids-Classification, assessment, dyslipidemia and hypercholesterolemia, Nutritional Risk Factors.

UNIT IX

Atherosclerosis - disease progression, causes, symptoms and clinical findings. Management-dietary and lifestyle.

UNIT X

Dietary management in angina pectoris, myocardial infarction and cardiac failure. Hypertension classification, causes, complications and dietary management.

BLOCK-IV: DIETETICS IN CANCER, DISEASES OF NERVOUS SYSTEM AND MUSCULO SKELETAL SYSTEM

UNIT XI

Cancer - Classification, development of cancer, risk factors-environmental, hereditary & nutritional.

UNIT XII

Nutritional effects of cancer – cachexia, energy metabolism, substrate metabolism and abnormalities in metabolism.

UNIT XIII

Cancer therapy-chemotherapy, radiation therapy, surgery, Immuno therapy and bone marrow transplantation. Nutritional problems of cancer therapy-dietary management. Role of food in the prevention of cancer.

UNIT XIV

Nutritional management in diseases of nervous system, and musculo skeletal system -Dysphagia, Epilepsy, Hyperkinetic Behaviour Syndrome, Etiology, dietary treatment in above conditions.

REFERENCES

1. Judith L. Buttriss, Ailsa A. Welch, John M. Kearney, Susan A. Lanham. 2017. -New. Public Health Nutrition, 2nd Edition, ISBN: 978-1-118-66097-3.
2. Louise Goff, Pamela Dyson. 2015. Advanced Nutrition and Dietetics in Diabetes. Wiley Blackwell.
3. Srilakshmi, B. 2006. Dietetics, New Age International (P) Ltd, Chennai.
4. Mohan, L.K. and Shump, S.E. 2001.Krause's Food Nutrition & Diet therapy, W.B.Sauders
5. Company, XII edition.
6. Shills,E.M., Olson,S.J. and Shiks,M.C. 1994. Modern Nutrition in health and disease, Lea and Febringer, Philadelphia, 8th edition.

OUTCOME:

At the end of this course, the students can interpret the role of dietetics and diet therapy.

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36533 - COMMUNITY NUTRITION	
Maximum marks: 100	**Credit: 4**

Objectives:

To enable the students

- ✓ To gain insight in to the national nutritional problems and their implications
- ✓ To understand the international contribution towards nutritional improvements in India.
- ✓ To develop skills in organizing and evaluating nutrition projects in the community.

BLOCK-I: ASSESSMENT OF NUTRITIONAL STATUS AND MALNUTRITION

UNIT I

Assessment of nutritional status - food and nutritional problems in the community, nutritional status of an individual and community.

UNIT II

Direct and indirect methods of nutritional assessment: clinical examination, nutritional anthropometry, biochemical methods, dietary survey.

UNIT III

Protein–energy malnutrition (PEM) - Aetiology, prevalence, symptoms and preventive measures.

UNIT IV

Ecology of malnutrition, nutrition and infection, Nutritional disorders: anaemia, vitamin A deficiency, iodine deficiency disorder.

BLOCK-II: MALNUTRITION IN INDIA AND NUTRITION INTERVENTION PROGRAMMES

UNIT V

Prevalence of malnutrition in India: Common nutritional problems-prevalence, morbidity and mortality rate.

UNIT VI

Strategies to overcome malnutrition in India - Need for an integrated approach to solve the problems of malnutrition.

UNIT VII

Nutrition intervention programmes. Agriculture planning, role of food technology, Environmental sanitation and Health.

UNIT VIII

Objectives and operation of nutrition intervention programmes, and other programmes organized by governmental and non-governmental agencies for the vulnerable sections of the population.

BLOCK-III: ORGANIZATIONS CONCERNED WITH MALNUTRITION AND NUTRITION EDUCATION

UNIT IX

National organizations concerned with food and nutrition- ICMR, ICARM, CHEB, CSWB and SSWB.

UNIT X

International organizations concerned with food and nutrition, FAO, WHO, UNICEF, CARE, AFPRO, CWS and World Bank.

UNIT XI

Nutrition education - nature and importance to the community, training workers in nutrition education and extension work-when to teach, whom to teach and who is to teach.

BLOCK-IV: NUTRITION EDUCATION PROGRAMMES, FOOD PRODUCTION AND FOOD SPOILAGE

UNIT XII

Principles of planning, executing and evaluating nutrition education programmes, problems of nutrition education programmes.

UNIT XIII

Food production -Objectives in agriculture planning in relation to nutrition. Green Revolution, Blue Revolution, White Revolution and Yellow Revolution.

UNIT XIV

A brief review of losses of foods in the post-harvest period and agents causing food spoilage.

REFERENCES

1. Park.K, 2017. "Park's text book of preventive and social medicine", 24th edition, M/S, Banarsidas Bhanot publishers, Jabalpur.
2. Bamji, MS, 2017. "Textbook of human Nutrition", Oxford and IBH Publishing Co, New Delhi.
3. A. K. Nigam, 2015. Statistical Aspects of Community Health and Nutrition. Woodhead Publishing India in Food Science and Nutrition.
4. Michael J. Gibney, Hester H. Vorster, Frans J. Kok, 2002. "Introduction to Human Nutrition (The Nutrition Society Textbook)" Wiley-Blackwell.
5. Jeliffee.D.B, 1996. "Assessment of Nutritional Status of the community", World Health Organisation, Geneva.
6. Swaminathan.M, 1986. "Principles of Nutrition and Dietetics", Bangalore publishing company Ltd, Bangalore

OUTCOME:

At the end of this course, the students will acquire knowledge about the basic principles and practical applications of public health and nutrition.

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36534 - Lab. III: CLINICAL AND THERAPEUTIC NUTRITION, DIETETICS IN LIFE STYLE DISEASES, COMMUNITY NUTRITION	
Maximum marks: 100	**Credit: 4**

Objectives:

To enable the students

- ✓ To obtain knowledge on different therapeutic diets and their preparation

CLINICAL AND THERAPEUTIC NUTRITION

1. Preparation of hospital diets - routine hospital diets, regular diet, soft diet, full fluid diet and tube feeding blends.
2. Diet in febrile conditions - Acute & chronic fevers – typhoid, tuberculosis.
3. Diet in - Peptic ulcer, gastritis, diarrhea
4. Diet in constipation, malabsorption syndrome.
5. Diet in Cirrhosis, hepatitis, cholelithiasis and pancreatitis.
6. Diet in hypothyroidism, hyperthyroidism, gout, phenyl ketonuria, Lactose intolerance.
7. Diet in Atherosclerosis, hypercholesterolemia, hypertension, myocardial infarction.
8. Diet in cancer.

DIETETICS IN LIFE STYLE DISEASES,

1. Diet in Diabetes mellitus and Gestational Diabetes.
2. Diet in obesity and underweight
3. Diet in Glomerulonephritis, nephrosis, nephrolithiasis & dialysis.

COMMUNITY NUTRITION

1. Diet in Anaemia, protein calorie malnutrition,
2. Diet in vitamin A, D, E, K, C and B deficiency.

REFERENCES

1. Ellen Davis, 2017. Fight Cancer with a Ketogenic Diet: Using a Low-Carb, Fat-Burning Diet as Metabolic Therapy, 3rd edition, Ellen Davis.
2. Sylvia Escott-Stump, 2015. Nutrition and Diagnosis-Related Care, Lippincott Williams and Wilkins; 8th Edition.
3. A. Catharine Ross, Benjamin Caballero, Thomas R. Ziegler M.D, 2012. Modern Nutrition in Health and Disease (Modern Nutrition in Health & Disease (Shils)) 11th Edition.
4. Mohan, L.K. and Shump, S.E, 2001. Krause's Food Nutrition & Diet therapy, W.B. Sauders Company, 12th edition.
5. Antia FP, Clinical Dietetics and Nutrition, 1997. Oxford University Press, New Delhi, 4th edition.
6. Shills, E.M., Olson, S.J. and Shiks, M.C, 1994. Modern Nutrition in health and disease, Lea and Febringer,Philadelphia,8th edition.
7. Garrow.JS & James W.P.T, 1993. Human Nutrition and Dieteics, Church Hill Living Stone.
8. Karran, S. J. and K. G. M. M. Alberti, 1980. Practical Nutritional Support, John Wiley and Sons. Inc. New York.

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36541- PAEDIATRIC NUTRITION	
Maximum marks: 100	**Credit: 4**

Objectives:

To enable the students

- ✓ To realize the importance of nutritional care and nourishment of children.
- ✓ To understand the nutritional requirements of children and the effects of paediatric diseases

BLOCK-I: NUTRITION IN INFANCY AND IMMUNIZATION SCHEDULES

UNIT I

Infancy - Physiological development, assessment of nutritional status.

UNIT II

Anthropometric measurements, biochemical parameters, clinical & dietary data of infants.

UNIT III

Nutritional and food requirements for infants.

UNIT IV

Immunization schedule during pregnancy, infancy and childhood.

BLOCK-II: NUTRITIONAL MANAGEMENT OF INFANTS AND NEWBORN SICKNESS

UNIT V

Nutritional management of premature baby, low birth weight babies and children with developmental disabilities.

UNIT VI

Infant lactation- Characteristics, causes and complications, feeding methods, growth and nutritional assessment of infant's lactation.

UNIT VII

Identification of newborn sickness-Detection of abnormal signs- cyanosis, jaundice, respiratory distress.

UNIT VIII

Bleeding, seizures, refusal and feed, abdominal distention, failure to pass meconium and urine of sick newborn.

BLOCK-III: CLINICAL NUTRITION IN INFANTS

UNIT IX

Nutritional management in malnutrition -Protein–energy malnutrition (PEM), anaemia, scurvy, rickets, vitamin A deficiency, obesity of Childhood.

UNIT X

Underweight and underweight nutrition- short term and long term consequences in infants.

UNIT XI

Nutritional management of Diarrhoea, typhoid, TB and hepatitis of infants.

BLOCK-IV: NUTRITIONAL MANAGEMENT FOR CHILDREN WITH SPECIAL CONDITIONS

UNIT XII

Lactose intolerance, celiac disease, inflammatory bowel disease, constipation and fat absorption test diet of infants. (Calculation of fluids & electrolytes-both deficit and maintenance and management of calorie intake).

UNIT XIII

Nutritional management for children with special conditions - Autism and ADH (Attention Deficit Hyperactivity disorder), epilepsy and AIDS.

UNIT XIV

Measuring, recording and plotting growth of infants.

REFERENCES

1. Pooja Gupta, 2017. Food, Nutrition And Health, S Chand Publishing, India
2. Sibal Anupam, 2015. Textbook of Pediatric Gastroenterology, Hepatology And Nutrition, Jaypee Brothers Medical Publishers; first edition.
3. A. Catharine Ross, Benjamin Caballero, Thomas R. Ziegler M.D. 2012. Modern Nutrition in Health and Disease (Modern Nutrition in Health & Disease (Shils)) Eleventh Edition.
4. Nutrition in pediatrics: Basic Sciences & clinical Applicatios, W. Allan Walker, John B Watkins & Christopher Duggan, 2003. BC Decker Inc, Hamilton, Ontario.
5. Mahan, L.K. and Escott-Stump, S. 2000. Krause's Food Nutrition and Diet Therapy, 10th Edition, W.B. Saunders Ltd.
6. Shils, M.E., Olson, J.A., Shike, M. and Ross, A.C. 1999. Modern Nutrition in Health and Disease, 9th Edition, Williams and Wilkins.
7. Fauci, S.A. et al. 1998. Harrison's Principles of Internal Medicine, 14th Edition, McGraw Hill.
8. Escott-Stump, S. 1998. Nutrition and Diagnosis Related Care, 4th Edition, Williams and Wilkins.
9. Walker, W.A. and Watkins, J.B. 1985. Nutrition in Pediatrics, Boston, Little, Brown & Co.

OUTCOME:

At the end of this course, the students will be able to know about the diet and nutrition of infants and children.

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36542 - FOOD MICROBIOLOGY AND SANITATION	
Maximum marks: 100	**Credit: 4**

Objectives:

To enable the students

- ✓ To understand the common micro-organisms associated with food borne illness
- ✓ To study about the microorganisms present in different food products.
- ✓ To study about the food packaging and labeling methods.

BLOCK-I: FUNDAMENTALS OF MICROBIOLOGY (YEAST, MOULDS AND VIRUSES)

UNIT I

Fundamentals of Microbiology – Introduction, development of microbiology. Bacteria- morphology and cultural characteristics and its importance.

UNIT II

Yeast- Morphology, culture, physiology, classification and industrial importance of yeast .

UNIT III

Moulds-morphology, physiology and multiplication, significance of moulds and common household moulds in relation to food science.

UNIT IV

Viruses and bacteriophages - discovery, morphology, reproduction and its importance.

BLOCK-II: CONTAMINATION, SPOILAGE, PRESERVATION AND MICROBES OF FRUITS, VEGETABLES AND CEREALS

UNIT V

Contamination of foods from external sources, General principles underlying spoilage - chemical changes by microorganisms.

UNIT VI

General principles of food preservation- Asepsis, Removal, Anaerobic conditions, High temperature, Low temperature, Drying, Food additives and Radiation

UNIT VII

Microbiology of fruits and vegetables - contamination, spoilage, preservation, and control of microorganisms.

UNIT VIII

Microbiology of cereals and cereal products - contamination, spoilage, preservation, and control of microorganisms.

BLOCK-III: MICROBES IN FLESHY FOODS, CANNED FOODS AND FOOD BORNE DISEASES

UNIT IX

Microbiology of fleshy foods, poultry and fish - Contamination, Spoilage , Preservation and control

UNIT X

Spoilage of Canned foods- causes of spoilage, appearance of the unopened container, grouping of canned foods on the basis of acidity, types of biological spoilage of canned foods.

UNIT XI

Food borne diseases –Food borne illness (bacteria), Food borne poisioning, infection and intoxication (non bacterial)

BLOCK-IV: FOOD SAFETY, PACKING AND FOOD STANDARDS

UNIT XII

Food Sanitation and safety – Personal hygiene-care of hands, sanitation, equipment plant, plant constructions, personal facilities, water supplies and seawage disposal.

UNIT XIII

Food packaging – Packaging methods. Moisture sorption properties of foods and selection of packaging materials. Interactions between packaging and food toxicity hazards. Bar coding - Nutrition labeling and nutrition claims, coding of food products. Packaging laws and regulations

UNIT XIV

Food laws and standards –Bureau of Indian standards - PFA, FPO, MMPO, AGMARK, CCFS, CFL, BIS & FSSAI - Consumer protection act, 1986. International standards- Codex alimentarius, ISO, WHO, FAO, WTO and HACCP.

REFERENCES

1. William C. Frazier, Dennis C. Westhoff, N.M. Vanitha, 2017. Food Microbiology, McGraw Hill Education; Fifth edition.
2. Mahendra Raj, Pal Mahendra, 2015. Sanitation in Food Establishments. LAP Lambert Academic Publishing.
3. Srilakshmi B.2008. Food Science. 4th Edition. New Age International Private Limited, New Delhi.
4. Shakuntala M.N., Shadaksharaswamy M. 2002. Foods –Facts and Principles. New Age International Publishers, New Delhi.
5. Roday,S 1999. Hygiene and Sanitation in Food Industry. Tata Mc Graw Hill Publishing Company Ltd., New Delhi.
6. Frazier,W.C & Westhoff, D.C. Food Microbiology. Tata MC Graw, 1997. Hill Publishing Company Ltd., New Delhi, 5tth Edition.
7. Potter,N.Hotchkiss, H.J. 1996. Food Science (5th edition) CBS Publishers and Distributors, New Delhi.

OUTCOME:

This course will provide the students about the role of microbes in various food products, food spoilage and food quality.

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36543 - FOOD BIOTECHNOLOGY & BIOSTATISTICS	
Maximum marks: 100	**Credit: 4**

Objectives:

To enable the Students

- ✓ To understand the application of Nutritional biotechnology.
- ✓ To obtain knowledge about the recent trends in food technology.
- ✓ To know the basic concepts of sampling and data analysis.
- ✓ To understand the applications of statistical methods in scientific research.

BLOCK-I: FOOD PROCESSING TECHNOLOGY

Unit I

Introduction to food biotechnology; Fermentation Technology – batch and continuous process, fermenter design, bioprocess control.

Unit II

Enzymes in food industry – Soluble enzymes, immobilized enzymes, amylase, invertase, isomerase – Synthesis, process and applications in food industries.

Unit III

Single cell protein – Production of microbial protein. SCP – substrates, nutritional value. Culture and process – spirulina, mushroom and yeast biomass production.

Unit IV

Regulatory aspects of biotechnology – Downstream processing, biosensors, biochips. Impact of biotechnology on the nutritional quality of foods.

BLOCK-II: FOOD TOXICANTS, ADDITIVES AND FERMENTED FOODS

Unit V

Natural food toxicants – Sources, toxicity, elimination – Lead, Mercury, Phthalates (used in plastics), Pesticides, haemagglutinins, cyanogens, saponins, gossypols, lathyrogens, favism and carcinogens.

Unit VI

Biotechnology in food industries**:** Food additives, synthesis. Acidulants – citric acid, gluconic acid, lactic acid. Sweeteners – glucose syrup and High Fructose Corn Syrup (HFCS).

Unit VII

Fermented foods – Alcoholic beverages, cheese making, fermented soya based foods, meat fermentation, vinegar, safety aspects of foods produced by biotechnology.

BLOCK-III: INTRODUCTION TO BIOSTATISTICS

Unit VIII

Introduction to biostatistics – Basic definitions and applications. Sampling – representative sample, sample size, sampling bias and sampling techniques.

Unit IX

Data collection and presentation – Types of data, methods of collection of primary and secondary data, methods of data presentation, graphical representation by histogram, polygon, ogive curves and pie diagram.

Unit X

Data classifications – categories and measurements, discrete and continuous variables. Tabulation scheme, preparation of tabular forms, methods of securing accuracy in tabulation.

Unit XI

Surveys – Graphical and diagrammatic representations. Use of computers in data processing and presentation. Choice of the sample, random samples, systematic samples, cluster samples/multistage sample and quota sample. Sources of bias and methods of reducing bias.

BLOCK-IV: APPLIED STATISTICS

Unit XII

Measures of central tendency – Mean, median, mode, their relative advantages and disadvantages, Measures of dispersion, mean deviation, coefficient of variation, percentiles and percentile ranks.

Unit XIII

Correlation – Association of attributes, contingency table, correlation, coefficient of correlation and its interpretation, rank correlation, regression equations and predictions.

Unit XIV

Probability – Rules of probability and its applications. Distribution – Normal, binomial, their properties. Importance of distributions in statistical studies. Large and small samples, X and F tests, tests for independence using contingency, analysis of variance and applications.

REFERENCES

1. B Antonisamy, Prasanna S. Premkumar, Solomon Christopher, 2017. Principles and Practice of Biostatistics, Elsevier India.
2. Ravishankar Rai V, 2015. Advances in Food Biotechnology, Wiley-Blackwell.
3. Murray, R.K. 2012. Metabolism of xenobiotic in Harphers Biochemistry, 22nd Ed, V.W. Prentice Hall Inc.
4. Dubey , R.C. 2006. "A textbook of Bio-technology", S.Chand and Company Ltd., New Delhi.
5. Kelly Anne Meckling, 2006. "Nutrient-Drug Interactions (Nutrition and Disease Prevention)" CRC Press.
6. J. Richard, Sundar P. S. S. Rao, 2006. "Introduction to Biostatistics and Research Methods".
7. Marcello Pagano, Kimberlee Gauvreau, 2000. "Principles of Biostatistics" S.Chand (G/L) & Company Ltd; 2nd edition.
8. David Moore, George P. McCabe, 1998. "Introduction to the Practice of Statistics" 3rd Edition by W. H. Freeman.
9. Parar. F.S.K. 1997. Adverse drug reactions and treatment of poisoning and drug interactions, S. Chand and Co, New Delhi.
10. Irfan A Khan, Atiya Khanum, 1994. "Fundamentals of Biostatistics" Ukaaz Publications.

OUTCOME:

This course will provides the broad overview about the food processing techniques and fundamentals of biostatistics.

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36544 - LAB. IV: PAEDIATRIC NUTRITION, FOOD MICROBIOLOGY AND SANITATION & FOOD BIOTECHNOLOGY & BIOSTATISTICS	
Maximum marks: 100	**Credit: 4**

Objectives:

To enable the students

- ✓ To study the importance of Paediatric nutrition
- ✓ To study the basic techniques of food microbiology and biotechnology
- ✓ To introduce the basics of various food processing and preservation technologies.
- ✓ To interpret the data using statistical tools.

PAEDIATRIC NUTRITION

1. Development of low cost recipes
    a. Infants
    b. Preschoolers
    c. Elementary school children.

FOOD MICROBIOLOGY AND SANITATION

1. Isolation of microorganisms from spoiled food sample.
2. Pure culture and preservation of bacteria.
3. Gram staining
4. Motility test
5. Hydrolysis of starch, gelatin and protein.
6. Identification of prepared slides
    a. Mould - mucor, rhizopus, aspergillus, penicillium, yeast
    b. Bacteria – bacilli.

FOOD BIOTECHNOLOGY

1. Determination of pH of different foods using pH meter.
2. Study quality characteristics of foods preserved by drying/dehydration/ freezing.
3. To perform pasteurization of fluids using different methods.
4. To perform blanching of different plant foods.
5. Methods of Food sampling and concept of shelf life of different foods

BIOSTATISTICS

1. Tabulation and graphical representation of data
2. Calculation of mean, median and mode
3. Calculation of standard deviation, standard error and ANOVA (One way and Two way)
4. Chi-Square test, t-Test, Regression and correlation.

REFERENCES

1. B Antonisamy, Prasanna S. Premkumar, Solomon Christopher, 2017. Principles and Practice of Biostatistics, Elsevier India.
2. Sibal Anupam, 2015. Textbook of Pediatric Gastroenterology, Hepatology And Nutrition, Jaypee Brothers Medical Publishers; first edition.
3. Bawa. A.S, O.P Chauhan et al. 2013. Food Science. New India Publishing agency.
4. A. Catharine Ross, Benjamin Caballero, Thomas R. Ziegler M.D. 2012. Modern Nutrition in Health and Disease (Modern Nutrition in Health & Disease (Shils)), 11th Edition.
5. Rao PG, 2010. Fundamentals of Food Engineering, PHI Learning Pvt Ltd, New Delhi.
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7. Meyer, 2004. Food Chemistry, New Age.
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10. Toledo Romeo T, 1999. Fundamentals of Food Process Engineering, Aspen Publishers.
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12. Potter NH, 1998. Food Science, CBS Publication, New Delhi.
13. Paine FA and Paine HY, 1992. Handbook of Food Packaging, Thomson Press India Pvt Ltd, New Delhi.

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DURATION OF THE PROGRAMME – TWO YEARS

The candidate who joins M.Sc., Home Science – Nutrition and Dietetics has to finish the course within a period of two years with four semester pattern.

FACULTY AND SUPPORT STAFF REQUIREMENT

Staff Category	Required
Core Faculty	3
Faculty -Specialization	2
Laboratory Assistant	1
Clerical Assistant	1

INSTRUCTIONAL DELIVERY MECHANISM

- ✓ Engaging learners actively in their learning
- ✓ Creating and maintaining an effective, safe learning environment
- ✓ Providing learners with information on what to expect
- ✓ Basing content on immediately applicable information and skills.
- ✓ Internet based learning and Computer-Assisted Instruction (CAI)
- ✓ Group discussion
- ✓ Webinar based intellectual support
- ✓ The Self Learning Material (SLM) will be given to the students in print form as well as in CDs (Soft copy).
- ✓ The examinations, evaluations and certificate issuing will be done by controller of examinations, Alagappa University.

CREDIT HOURS

Each semester there will be one contact programme of 168 hours duration in total comprising of both theory and practical (120 hours practical and 48 hours theory) for the Two years **M.Sc., HOME SCIENCE - NUTRITION AND DIETETICS.**

(f) PROCEDURE FOR ADMISSIONS, CURRICULUM TRANSACTION AND EVALUATION

Admission for M.Sc., HOME SCIENCE - NUTRITION AND DIETETICS shall be based on purely on merit basis. A candidate who has passed Bachelor's Degree in Science (Home Science, Botany, Zoology, Biochemistry, Chemistry, Biotechnology, Microbiology, Food Science and Quality Control and Clinical Nutrition and Dietetics, M.B.B.S./ B.H.M.S./ B.A.M.S./ B. Pharmacy and any other relevant subjects) degree with at least 55% of marks eligible for applying this programme.

The curriculum transaction will be done using advanced teaching aids and equipments. The students will be evaluated by examinations. There shall be no passing minimum for internal. For external examination, passing minimum will be 50%. In aggregate (External + Internal) the passing minimum shall be 50% for each theory and practical paper. Grading shall be based on overall marks obtained.

FEES STRUCTURE

Sl. No.	Fees Detail	Amount in Rs.		Nature of Fees
		First Year	Second Year	
1	Admission Processing Fee	**300**	**-**	Non- Refundable
2	Course Fee	**20,000**	**20,000**	Non- Refundable
3	ICT Fee	**150**	**150**	Non- Refundable
	TOTAL FEES	**20450**	**20150**	Non- Refundable

EVALUATION

The examinations shall be conducted separately for theory and practical's to assess the knowledge acquired during the study. There shall be two systems of examinations viz., internal and external examinations. The internal examinations shall be conducted as Continuous Internal Assessment test I and II (CIA Test I & II). The internal assessment shall comprise of maximum 25 marks for each subject. The following procedure shall be followed for awarding internal marks.

Theory paper (Internal Assessment) Average marks of two CIA test	10 marks
Attendance	5 marks
Seminar	5 marks
Assignment	5 marks
Total	**25 marks**

PRACTICAL'S (INTERNAL ASSESSMENT)

CIA tests	10 marks
Attendance	5 marks
Observation note book	10 marks
Total	**25 marks**

EXTERNAL EXAMINATIONS

The external theory and practical examinations shall be conducted for three hours duration at the end of each semester. The external examinations shall comprise of maximum of 75 marks for each subject. The candidate failing in any subject will be permitted to appear for each failed subject in the subsequent examination. Practical examinations and demonstration of experiments shall be conducted at the end each semester.

SCHEME OF EXTERNAL EXAMINATION

The duration of examinations for theory and practical's shall be three hours.

Question paper pattern (Theory)

- The question paper carries a maximum of 75 marks.
- The question paper consists of three sections namely Part-A, Part-B and Part-C.
- Part-A consists of 10 questions of 2 marks each (10 x 2 = 20 marks) with no choice. The candidate should answer all questions.
- Part-B consists of 5 either or choice questions. Each question carries 5 marks
- (5 x 5=25 marks).
- Part-C consists of 5 questions. Each question carries 10 marks. The candidate should answer any three questions (10 x 3 = 30 marks).

Model Question Paper (Theory)

M.Sc., DEGREE EXAMINATION

HOME SCIENCE – NUTRITION AND DIETETICS

First Semester

36511 – Human Physiology

Time : 3 hrs Maximum: 75 Marks

Part – A (10 × 2 = 20 Marks)

Answer **all** the questions

1. Power house of the cell
2. Tendons
3. Peroxisomes
4. Angioplasty
5. Orthopnoea
6. Erythroblastosis foetalis
7. Artificial kidney
8. Menarche
9. Hypothyroidsm
10. MRI

Part – B (5 × 5 = 25 Marks)

Answer **all** the questions choosing either (a) or (b)

11. (a) Write an essay about the structural organization and functions of endoplasmic reticulum and mitochondria.

Or

(b) Write an essay about Musculo skeletal system.

12. (a) Differentiate prokaryotic and eukaryotic cells.

Or

(b) Write a detailed note on fluid mosaic model of the plasma membrane.

13. (a) Discuss in detail about the functions of blood and blood proteins.

Or

(b) Write an essay about the mechanism of respiration.

14. (a) Explain in detail about the structure and functions of male reproductive organs.

Or

(b) Write an essay about the sense organs.

15. (a) Explain the principle and advantages of common test used in neurological disorders.

Or

(b) Write an essay about hormonal imbalance syndromes.

Part – C (3 × 10 = 30 Marks)

Answer **any three** questions

16. Write an essay about the classifications, structure and functions of Tissues.
17. Describe the mechanism of blood coagulation and blood coagulation disorders.
18. Write an essay about human circulatory system.
19. Explain in detail about the structure and functions of female reproductive system.
20. Write an essay about human nervous system.

Question paper pattern (Practical) (Maximum 75 marks)

Major practical	15 marks
Minor practical	10 marks
Experimental set up	5 marks
Spotters	25 marks
Viva voce	10 marks
Practical record note	10 marks
Total	75 marks

Maximum duration for completion of the course

The maximum duration for completion of M.Sc., HOME SCIENCE - NUTRITION AND DIETETICS shall not exceed five years.

MARKS AND GRADES:

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester

G_i = Grade Point obtained for course i in any semester.

n refers to the semester in which such courses were credited

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

Grade Point Average = $\frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

CGPA = $\frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses for the entire programme.}}$

CGPA	Grade	Classification of Final Result
9.5-10.0 9.0 and above but below 9.5	O+ O	First Class- Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	D++ D+ D	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	A++ A+ A	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	B+ B	Second Class
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

Commencement of this Regulation:

These regulations shall take effect from the academic year 2018-2019 (June session) i.e., for students who are to be admitted to the first year of the course during the academic year 2018-2019 (June session) and thereafter.

(g) REQUIREMENT OF THE LABORATORY SUPPORT AND LIBRARY RESOURCES

Laboratory Support

The M.Sc., HOME SCIENCE - NUTRITION AND DIETETICS curriculum of Alagappa University has strived to offer both theory courses as well as laboratory and design practice in the field of Home Science - Nutrition and Dietetics. Instruments, chemicals and glasswares are needed for conducting practicals to the better understanding of the curricula.

Library Resources

Alagappa University students and staffs have access to a world class collection of text books, reference books, conference proceedings, back volumes, non-book materials such as CD-ROMs, high quality and trusted online resources. Students need nearly 150 books and 50 practical manuals for getting fundamental knowledge in Nutrition and Dietetics.

(h) COST ESTIMATE OF THE PROGRAMME AND THE PROVISIONS:

The cost estimate of the programme and provisions for the fund to meet out the expenditure to be incurred in connection with **M. Sc., Home Science - Nutrition and Dietetics** as follows

S.No.	Expenditure Heads	Approx. Amount in Rs.
1	Programme Development	20,00,000/-
2	Programme Delivery	24,00,000/-
3	Programme Maintenance	5,00,000/-

QUALITY ASSUARANCE MECHANISM AND EXPECTED PROGRAMME OUTCOMES

Quality assurance

- Student's feedback about staffs, teaching methods and methodologies.
- Teachers review about students.
- Continuous assessment (Model practical).
- The students give feedback on teaching every semester. Feedback is also taken on their campus experience through suggestion boxes.

Expected outcome

The learners will be professionals in preparation and service of food, develop modified diets, participate in research, and educate individuals and groups on good nutritional habits. Recently, a growing number of needs for nutritional specialists in food industry, journalism, sports nutrition, corporate wellness programs, and other non-traditional dietetics settings. The outcome of the programme is in the form of job offerings in community and public-health settings, and/or in academia and research. They can be hired by sports hostels and athlete camps, health and recreational clubs and canteens. This apart, there are opportunities in catering department of star hotels and restaurants and research laboratories of food manufacturers. They can also work from home as consultants.

Minutes of the Meeting of the Board of Studies in M.Sc., Home Science - Nutrition and Dietetics (For M.Sc., Home Science - Nutrition and Dietetics Programme to be offered through ODL Mode) held at The Directorate of Distance Education, Alagappa University, Karaikudi – 630 003, on 05.09.2017, (02.00 P.M).

Members Present

1.	Dr. E. Kannapiran	-	Chairman
2.	Dr. P. Rameshthangam	-	Member
3.	Dr. K. Pandima Devi	-	Member
4.	Dr. G. Selvakumar	-	Member
5.	Dr. P. Thiruchenthilnathan	-	Member

After the deliberation and discussion the board resolved the following:

1. The board considered the curriculum design and detailed syllabi of **M.Sc., Home Science - Nutrition and Dietetics** programme, prepared as per the norms by the Chairman and the Board Members, scrutinized and suitably modified the same.
2. The board resolved to approve curriculum design, detailed syllabi and other regulations for the M.Sc., Home Science - Nutrition and Dietetics programme to be offered by the Directorate of Distance Education of Alagappa University are given in Annexure I.

Dr. E. Kannapiran

Dr. P. Rameshthangam

Dr. K. Pandima Devi

Dr. G. Selvakumar

Dr. P. Thiruchenthilnathan

ALAGAPPA UNIVERSITY

Accredited with A+ Grade by NAAC (CGPA 3.64) in the Third Cycle
Karaikudi – 630003. Tamilnadu , INDIA

Directorate of Distance Education

PROGRAMME PROJECT REPORT

for

Master of Science in Information Technology M.Sc(I.T)

Submitted to

UGC, Distance Education Bureau (DEB), New Delhi

for seeking approval to introduce programme through Distance Education Mode

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ALAGAPPA UNIVERSITY, KARAIKUDI
DIRECTORATE OF DISTANCE EDUCATION

PROGRAMMME PROJECT REPORT

Master of Science in Information Technology (M.Sc (IT))

(a) Programme's Mission and Objectives

Mission

Mission is to impart employability and creativity to the Post graduate students and lives up to the standards of Information Technology (IT) industry.

Programme Objectives:

- ✓ To facilitate students to nurture skills to practice their professions competently to meet the ever-changing needs of society such as Digital India, Safety and Privacy.
- ✓ Discover, investigate the requirements of a problem and find the solution to them using computing principles.
- ✓ Create and evaluate a computer based system, components and process to meet the specific needs of cutting edge technologies.
- ✓ Utilize current techniques and tools necessary for complex computing practices.
- ✓ Develop and integrate effectively system based components into user environment.
- ✓ Identify the need and develop the skill to employ in learning as a computing professional.
- ✓ Execute effectively in a team environment to achieve a common goal.
- ✓ Classify opportunities and use innovative ideas to create value and wealth for the betterment of the individual and society.
- ✓ Proficiency in developing application with required domain knowledge.

Programme Outcome:

- ✓ To develop in problem solving and programming skills in the various computing fields of IT industries
- ✓ To widen the ability to plan, analyze, design, code, test, implement & maintain a software product for real time system
- ✓ To support students capability to set up their own enterprise in various sectors of Computer applications
- ✓ To knowledge the students in finding solutions and developing system based applications for real time problems in various domains involving technical, managerial, economical & social constraints
- ✓ To prepare the students to pursue higher studies in computing or related disciplines and to work in the fields of teaching and research.

(b) Relevance of the program with HEI's and Alagappa University Mission and Goals

This programme is aligned with HEI's and Alagappa University mission and goals to be offered through distance mode to reach quality higher education to the unreachable and/or rural learners. Higher education in Computer Science offered through distance mode meets the mission of HEI's like digital India and e-cash transaction will enrich the Human resources for the uplift of the nation.

(c) Nature of prospective target group of learners

The nature of prospective target group of learners is graduates from various disciplines like Commerce, Mathematics, Physics, Chemistry, Biology, Electronics, and Engineering etc. It also includes the learners who want to become entrepreneurs like Web Designers, Software Developers, BPO's, KPO's etc.,

d) Appropriateness of programme to be conducted in Open and Distance Learning mode to acquire specific skills and competence;

M.Sc (IT) Programme through Distance Learning mode is developed in order to give subject-specific skills including i) Knowledge about various kinds of programming languages ii) Principles of Information Technology, Operating systems, RDBMS, Data Structure and Software Engineering ,Computer Networks, Concepts like Object Oriented Analysis and Design ii) Cutting Edge Technologies like Web Technology, .Net Frameworks, Mobile Communications.

(e) Instructional Design

e.1 Revisions of Regulation and Curriculum Design

1. The University reserves the right to amend or change the regulations, schemes of examinations and syllabi from time to time based on recent market dynamics, industrial developments, research and feedback from stakeholders and learners.

2. Each student should secure 64 credits to complete M.Sc (IT). programme.

3. Each theory and practical course carries 4 credits with 75 marks in the University End Semester Examination (ESE) and 25 marks in the Continuous Internal Assessment (CIA).

Programme code

M.Sc (Information Technology	313

M.Sc(IT) Credit Based Curriculum and Evaluation System

M.Sc(I.T) COURSE OF STUDY & SCHEME OF EXAMINATIONS

S.No	Subject Code	Title of the course	CIA Marks Max.	ESE Marks Max.	Total Marks Max.	C Max.
		FIRST YEAR				
		I SEMESTER				
1	31311	Computer Organization and Architecture	25	75	100	4
2	31312	Object Oriented Programming and Java	25	75	100	4
3	31313	Data Structures and Algorithms	25	75	100	4
4	31314	Object Oriented Programming and Java Lab	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		II SEMESTER				
5	31321	Data Mining and Warehousing	25	75	100	4
6	31322	Relational Database Management Systems (RDBMS)	25	75	100	4
7	31323	Visual Programming with •NET	25	75	100	4
8	31324	VB•NET & RDBMS Lab	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		SECOND YEAR				
		III SEMESTER				
9	31331	Open Source Software	25	75	100	4
10	31332	Operating Systems	25	75	100	4
11	31333	Computer Networks	25	75	100	4
12	31334	Open Source Lab	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		SEMESTER IV				
13	31341	Web Technology	25	75	100	4
14	31342	Software Engineering	25	75	100	4
15	31343	Cloud Computing	25	75	100	4
16	31344	Web Technology Lab	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		Grand Total	**400**	**1200**	**1600**	**64**

CIA : Continuous Internal Assessment **ESE :** End semester Examination **Max**. Maximum Marks; **C:** Credits

Course Code Legend:

3	1	3	S	C

313 – Programme code for Master of Information Technology (M.Sc(IT)
S -- Semester Number
C – Course Number in the Semester

e.2 Detailed Syllabi

The detailed Syllabi of study and shall be as shown in Appendix.

e.3 Duration of the Programme:

The M.Sc (IT) programme shall consist of a period of two years (Four Semesters).

e.3.1 Medium of Instruction

The medium of instruction is only in **English**.
The course material is also in **English.**

e.4 Faculty and Support Staff Requirements:

The following faculty and support staff is required for this programme.

S.No	Staff Category	Numbers
1	Core Faculty*	3
2	Specialization Faculty	2
3	Lab Assistant	1
4	Clerical Assistant	1

* Faculty at least in Assistant Professor level

e.5 Instructional Delivery mechanisms

The instructional delivery mechanisms of the programme includes SLM- Study materials, Lab instruction manual, Personal contact session for both theory and practical courses of the programme, e-version of the course materials in the form of CD, e-book, e-tutorials, Massive Open Online Courses (MOOC) courses, Open Educational Resources(OER) and virtual lab.

e.6 Identification of media

The printed version of SLM – study material shall be given to the learners in addition to MOOC, e-tutorial and virtual lab.

e.7 Student Support Services

The student support services will be facilitated by the Directorate of Distance Education, Alagappa University, Karaikudi and its approved learning centres located in various parts of Tamilnadu.

The pre-admission student support services like counseling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at Directorate of Distance Education or Learning centres.

The post-admission student support services like issuing Identity card, study materials will be provided thru Directorate or Learning centres. The face to face contact sessions of the programme for both theory and practical's will be held at the Directorate or Learning centres.

The student support regarding the conduct of examinations, evaluations, publication of results and certificates are done by Office of the Controller of Examinations, Alagappa University, Karaikudi.

f. Procedure for Admission:

f.1 Minimum qualification for admission

Candidates for admission to the Master of Science in Information Technology (M.Sc(I.T)) programme shall be required to have passed any degree of any Recognized University or authority accepted by the Syndicate of the Alagappa University as equivalent thereto shall be eligible.

f.2 Curriculum transaction

- The face to face contact sessions in class room teaching with the support of SLM, Power Point Presentations, web based tools, audio and animated videos.
- The practical classes are based on the respective subject study materials containing requirement for the laboratory experiments.
- Face to face contact sessions will be conducted for both theory and practical courses in the following manner.

Course Type	**Face to face contact session per semester (in Hours)**
Theory courses (3 Courses with 4 credits each)	48
Practical courses (1 Course with 4 credit)	120
Total	**168**

f.3 Evaluation

There shall be two types of evaluation systems; internal assessment and end semester examination will be conducted by the University according to the following scheme. The maximum marks for the internal assessment for both theory and practical's is 25 marks. The maximum marks for end semester examination is 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

Internal assessment

- Internal assessment of theory courses is through home assignment with workbook, case studies, review questions, quiz, multiple choice questions etc., for 25 marks.
- The internal assessment for the practical courses shall be through home assignment which includes workbook designing algorithm, preparing source code, PL/SQL coding etc., for 25 marks.
- Student should submit assignment for theory and practical courses of every course and semester.

Division of Internal Marks (Assignment)

Theory		Practical	
Assignment	**Marks**	**Assignment**	**Marks**
Review questions	15	Algorithm Design	15
Workbook, case studies, quiz, multiple choice questions	10	Workbook for preparing source code, PL/SQL coding , results	10
TOTAL	25	TOTAL	25

End Semester Examination (ESE)

The university end Semester Examinations shall be of three hours duration with maximum of 75 Marks for both theory and practical courses.

f.3.1 Minimum for a pass:

To pass in each course, a candidate is required to secure 50% marks in the End Semester Examination and 50% marks in the aggregate (marks in End Semester Examination + marks in Internal Assessment).

The student who does not secure required minimum marks for pass in a course(s) shall be required to reappear and pass the same in the subsequent examination.

f.3.2 Question Paper Pattern - Theory

The end semester examination will be conducted in the duration of 3 Hours and maximum of 75 Marks.

All the Blocks Should Be Given Equal Importance

Part – A (10 x 2 Marks: 20 Marks) Answer all questions

Part – B (5 x 5 Marks: 25 Marks) Answer all questions choosing either (a) or (b)

Part – C (3 x 10 Marks: 30 Marks) (Answer any 3 out of 5 questions)

End Semester Examination (ESE) - Practical

Students are required to prepare a separate lab record for each lab course. The practical counsellor should duly sign this lab record after each session.

Students shall prepare practical record note book which includes aim, algorithm, source code, input, expected output and result of the experiment and submit during end semester practical examination.

Division of marks in ESE – Practical (Maximum 75 marks)

The end semester practical examination will be conducted in the duration of 3 Hours and maximum of 75 Marks.

Practical details	Max. Marks
Algorithm / Flowchart	10
Source Code	20
Debugging	10
Execution	10
Results	10
Viva-Voce	5
Record	10
Total	75

f.3.3 Procedure for Completing the Course:

A student shall be permitted to continue the programme from I to IV semester irrespective of failure(s) in the courses of the earlier semesters. The candidate will qualify for the M.Sc(IT) degree only if he/she passes all the (including arrears) courses with in a period of FIVE years from the date of admission.

f.3.4 Results and Classification:

Results will be declared at the end of each semester of the University examination and the marks/grade obtained by the candidate will be forwarded to them by the Controller of Examinations, Alagappa University.

f.3.4.1 Marks and grades

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
96-100	10.00	S+	First class – Exemplary
91-95	9.5	S	
86-90	9.0	D++	First class – Distinction
81-85	8.5	D+	
76-80	8.0	D	
71-75	7.5	A++	First Class
66-70	7.0	A+	
61-65	6.5	A	
56-60	6.0	B	Second Class
50-55	5.5	C	
Below 50	0.0	F	Fail
ABSENT	0.0	AAA	Absent

For a semester

Grade Point Average[GPA] = $\sum C_i G_i / \sum C_i$

GPA = $\frac{\text{Sum of the multiplication of Grade points by the credit of the courses}}{\text{Sum of the credit of the courses in the semester}}$

= $\frac{\text{Sum of [Credit earned x Grade Points]}}{\text{Sum of the credits earned in the semester}}$

For the entire programme

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

= $\frac{\text{sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses for the entire programme}}$

Where

C_i - Credits earned for the course i in any semester

G_i – Grade Point earned for course i in any semester

n - is number of all Courses successfully cleared during the particular semester in the case of GPA and during all the semesters (programme) in the case of CGPA.

CGPA	Grade	Classification of Final Result
9.6 – 10.00	S+	First class – Exemplary*
9.1 – 9.5	S	
8.6 – 9.0	D++	First class with Distinction*
8.1 – 8.5	D+	
7.6 – 8.0	D	
7.1 – 7.5	A++	First Class
6.6 – 7.0	A+	
6.1 – 6.5	A	
5.6 – 6.0	B+	Second Class
5.0 – 5.5	C	
Below 5.0	U	Reappear

* The candidates who have passed in the first appearance and within the prescribed semester

f.4 Fees Structure:

Fee Particulars	Amount in Rs.	
	First Year	Second Year
Admission Processing Fees	300	--
Course Fees	14300	14300
ICT fees	150	150
Total Fees	**14,750**	**14,450**

The above mentioned fees structure is exclusive of examination fees.

g. Requirement of the laboratory support and library resources

g.1 Laboratory Support

A well- equipment Computer Laboratory was established in the Alagappa University, Karaikudi with necessary software's as per the practical's syllabi for conducting face to face contact sessions for practical courses of this programme. Model Practical Questions is available to the learners in the university website.

g.2 Library Resources

The Directorate of Distance Education, Alagappa University provides library facility with number of books and Self Learning materials for Computer Science programmes. The

Central library of Alagappa University provides the collection of volumes of Self Learning Materials, Printed books, Subscriptions to printed periodicals and Non-book materials in print form for the learner's references. All these library resources are meant for learner's reference purpose only.

h)Cost estimate of the programme and the provisions:

Expense details	Amount in (Rs.) Approx.
Programme development (Single time Investment)	20,00,000/-
Programme delivery (per year)	24,00,000/-
Programme maintenance (per year)	5,00,000/-

i)Quality assurance mechanism and expected programme outcomes:

i.1 University's Moto:

' Excellence in Action'

i.2 University's Vision and Mission

Vision

Achieving Excellence in all spheres of Education, with particular emphasis on ' PEARL' - Pedagogy, Extension, Administration, Research and Learning.

Mission

Affording a High Quality Higher Education to the learners so that they are transformed into intellectually competent human resources that will help in the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

i.3 University Objectives

1. Providing for instructions and training in such branches of Learning at the university may determine.
2. Fostering Research for the Advancement and Dissemination of Knowledge and Application.

i.4 Quality Policy

Attaining Benchmark Quality in every domain of 'PEARL' to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

i.5 Quality Quote

Quality Unleashes Opportunities Towards Excellence (QUOTE).

i.6. Course benchmarks

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations and number of enrolments of students. Feedback from the alumni, students, parents, stakeholders and employers will be received to analyze the benchmark qualities for the further improvement of the programme.

****

Appendix

Detailed Syllabi
FIRST YEAR
SEMESTER I

Course Code	Title of the Course
31311	COMPUTER ORGANISATION AND ARCHITECTURE

Course Objectives:

- To have a thorough understanding of the basic structure and operation of a digital computer.
- To discuss in detail the operation of the arithmetic unit including the algorithms & implementation of fixed-point and floating-point addition, subtraction, multiplication & division.
- To study the different ways of communicating with I/O devices and standard I/O interfaces.
- To study the hierarchical memory system including cache memories and virtual memory.

Course Outcomes:

Students will have thorough knowledge about

- Basic structure of a digital computer
- Arithmetic operations of binary number system
- The organization of the Control unit, Arithmetic and Logical unit, Memory unit and the I/O
- unit.

Unit No.	Contents
	BLOCK 1 : DIGITAL LOGIC CIRCUITS:
1	**Introduction :** Digital computers – Logic gates – Boolean algebra – Map simplification
2	**Combinational circuits** – Flip-flops
3	**Digital Components:** Integrated circuits – Decoders –Multiplexers – Registers.
	BLOCK 2 : DATA REPRESENTATION
4	**Introduction :** Data types – Complements – Fixed point representation – Floating point representation.
5	**Register Transfer and Microoperations:** Register transfer language – Register transfer – Bus and memory transfers –
6	**Arithmetic microoperations** – Logic Microoperations – Shift Microoperations – Arithmetic logic shift unit.

	BLOCK 3 : BASIC COMPUTER ORGANIZATION AND DESIGN:
7	**Instruction codes** – Computer registers – Computer instructions – Timing and control – Instruction cycle – Memory reference Instructions – Input-output and interrupt.
8	**Central Processing Unit:** Introduction – General register organization – Stack organization –
9	**Instruction formats** – Addressing modes – Data transfer and manipulation – Program control.
	BLOCK 4 : COMPUTER ARITHMETIC
10	**Introduction** – Addition and subtraction – Multiplication algorithms – Division algorithms – Floating-point arithmetic operations –
11	**Input-Output Organization:** Peripheral devices – Input output interface – Asynchronous data transfer
12	**Input-Output Organization:** Modes of transfer – Priority interrupt – Direct memory Access, Input-Output Processor.
	BLOCK 5 : MEMORY ORGANIZATION
13	**Memory Organization:** Memory Hierarchy – Main memory – Auxiliary memory – Associative memory –
14	**Other Memory:** Cache memory – Virtual memory – Memory management hardware.

Text Book:

M. Morris Mano, Computer System Architecture, Prentice Hall of India Pvt Ltd, Third edition, 2002. ISBN: 81-203-0855-7.

Reference Books:

1. William Stallings, Computer Organization and Architecture – Designing for Performance, 6th
2. Edition, Pearson Education, 2003.
3. Nicholas Carter, Schaum's outline of Computer Architecture, Tata McGraw Hill, 2006,
4. John L. Hennessy and David A Patterson, Computer Architecture A quantitative Approach,
5. Morgan Kaufmann / Elsevier, Fourth Edition, 2007
6. Mohammed Rafiquzzaman and Rajan Chandra, Modern Computer architecture, Galgotia Publications Pvt. Ltd., 2010
7. V.Rajaraman and T.Radhakrishnan, An Introduction to Digital computer Design, PHI Ltd, 2009.

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Course Code	Title of the Course
31312	**OBJECT ORIENTED PROGRAMMING AND JAVA**

Course Objectives:

- To provide an overview of working principles of object oriented paradigm
- To understand and apply the OOPs fundamentals

- To implement the features of OOP in real world applications

Course Outcome:

- Able to understand the object oriented programming techniques
- Able to implement, compile, test and run Java program,
- Able to make use of hierarchy of Java classes to provide a solution to a given set of requirements found in the Java API

Unit No.	Contents
	BLOCK 1 : FUNDAMENTALS OF OBJECT-ORIENTED PROGRAMMING
1	**Basic concepts** of OOP – Benefits – Applications . Java Evolution: Features – how java differs from C and C++ - java and internet- java support system – java environment
2	**Overview of Java Language** –Introduction – Simple Java Program – Comments – Java Program Structure – Tokens – Java Statements – Implementing a Java Program – JVM – Command Line Arguments. Constants – Variables – Data Types – Type Casting..
3	**Operators and Expressions:** Arithmetic Operators – Relational, Logical, Assignment, Increment and Decrement, Conditional, Bitwise, Special Operators – Arithmetic Expressions, Evaluation of Expression – Precedence of Arithmetic Operators – Type Conversions – Operator Precedence and associativity – Mathematical Functions. **Decision Making and Branching:** If –if…..else –Nesting of if……. Else – else if– switch. Decision Making and Looping: While – do – for – jump in loops – labeled loops.
	BLOCK 2 : CLASSES, OBJECTS AND METHODS
4	**class:** Defining a class –fields –methods –creating objects – accessing class members – constructors – methods overloading –static members –nesting of methods – Inheritance –overriding methods –final variables-classes –methods
5	**Arrays, Strings and Vectors** :One dimensional Arrays –creating of array – Two dimensional arrays- strings –vectors –Wrapper classes – Enumerated Types - Interfaces: Multiple Inheritance
6	**Packages:** Defining interface –Extending interfaces – Implementing Interfaces - Putting Classes Together
	BLOCK 3 MULTITHEREADED PROGRAMMING
7	**Introduction : c**reating Threads –Extending the thread class –Stopping and Blocking a thread –Life cycle of a thread –using thread methods
8	**Thread Exceptions** –Priority –Synchronization –Implementing the 'Runnable' Interface
	BLOCK 4 : MANAGING ERROR, EXCEPTION AND APPLETS
9	**Exceptions:** Types of errors –Exceptions –Syntax of Exception Handling code –

	Multiple Catch statements –using finally statement – Throwing our own Exceptions – using exceptions for Debugging -
10	**Graphics Programming:** The Graphics Class – Lines and Rectangles – Circles and Ellipses – Drawing Arcs – Drawing Polygons – Line Graphs – Using Control Loops in Applets – Drawing Bar Charts.
11	**Applet Programming:** How applets differ from Applications – preparing to write applets – Building Applet Code – Applet life cycle – creating an Executable Applet – Designing a Web Page – Applet Tag – Adding Applet to HTML file – Running the Applet – Passing parameters to Applets – Displaying Numerical values – Getting input from the user
	BLOCK 5 : MANAGING INPUT/OUTPUT FILES IN JAVA
12	**Introduction** – concept of streams –stream classes – byte stream classes – character Stream
13	**I/O classes**: –using stream –using the file class –Input / output Exceptions – creation of files – Reading / writing characters
14	**Reading writing bytes**: Random access files- Interactive input and output –Other stream classes

Text Books:

1. R. Krishnamoorthy and S. Prabhu, Internet and Java Programming, New Age International Publishers, 2004 (Unit I).
2. Programming with Java, 4e, E. Balagurusamy, Tata McGraw-Hill, 2010.

Reference books:

1. Deitel, Deitel and Nieto, Internet and World Wide Web – How to program, Pearson Education, 2000.
2. Naughton and H.Schildt, Java 2 - The complete reference, Tata McGraw-Hill, Fourth edition, 2006.
3. Elliotte Rusty Harold, Java Network Programming, O'Reilly Publishers, 2000.
4. B.Mohamal Ibrahim , Java : J2SE – A Practical Approach, Firewall media, 2006.
5. Cay S. Horstmann, Gary Cornell, Core Java, Volume I and II, 5th Edition, Pearson Education, 2003.
6. Topley, J2ME in A Nutshell, O'Reilly Publishers, 2002.
7. Hunt, Guide to J2EE Enterprise Java, Springer Publications, 2004.
8. Ed Roman, Enterprise Java Beans, Wiley Publishers, 1998.

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Course Code	Title of the Course
31313	DATA STRUCTURE AND ALGORITHMS

Course Objectives:

- The learner should be well versed with the fundamentals of Algorithms, learn various data structures, should be able to use them appropriately as per need during development of programs.

- Also, the learner should know different sorting and searching techniques so that correct techniques can be used in different programs so that the complexity of the program does not increase due the sorting/ search technique employed.

Course Outcome

After the completion of this course, the student will able to

- To write programs using structures, strings, arrays, pointers and strings for solving complex computational problem.
- Using the data structures real time applications
- Able to analyze the efficiency of Data Structures

Unit No	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction to Data Structure :** Types of Data Structure , Primitive data types
	Algorithms: –Time and space Complexity of algorithms
2	**Arrays:** Array initialization, Definition of Array, Characteristic of Array ,One-dimensional Array, Two-dimensional array and Multi dimensional array
	BLOCK 2 : LINEAR DATA STRUCTURE
3	**Stack :** Stack related terms, Operations on a stack,
4	**Representation of Stack:** Implementation of a stack – application of Stack. Expression Evaluation Polish notation.
5	**Queues:** Operations on queue Circular Queue, Representation of Queues, Application of Queues
6	**List:** Merging lists, Linked list, Single linked list, Double Linked List, Header Linked list
7	**Operation on Linked List :** Insertion and Deletion of linked list
8	**Traversal:** Traversing a linked list , Representation of linked list.
	BLOCK : 3 NON-LINEAR DATA STRUCTURE
9	**Trees:** Binary Trees, Types of Binary trees, Binary Tree Representation
10	**Binary Tree operations / Applications :** Traversing Binary Trees, Binary Search tree,
11	**Operations on Binary Tree:** Insertion and Deletion operations, Hashing Techniques.

	BLOCK 4 : SEARCHING TECHNIQUES
12	**Searching :** Introduction, Searching, Linear Search, Binary Search
	BLOCK 5 : SORTING TECHNIQUES
13	**Sorting:** Bubble sort, Insertion sort, Radix sort
14	**Other sorting Techniques:** Selection sort, Quick sort, Tree sort.

Text Books:

1. Fundamentals of Data structures, Second edition, Ellis Horowitz and Sartaj Sahini, Universities press, 2007.

2. Data Structures, Seymour Lipschutz, G.A.Vijayalakshmi Pai, Second Edition , Schaum's Outlines, Tata Mc-Graw Hill Private Ltd., 2006.

Reference Books:

1. Programming and Data Structure, Pearson Edition, Ashok N Kamthane, 2007.

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Course Code	Title of the Course
31314	**OBJECT ORIENTED PROGRAMMING AND JAVA LAB**

Course Objective:

- To understand the basic concepts of Object Oriented Programming
- To understand console and internet programming (applets) using Java

Course Requirement:

- Basic concepts of Web and Java programming

Course Outcome:

- Explore markup languages features and create interactive web pages using them
- Able to design front end web page and connect to the back end databases.
- Able to do Object oriented programming to solve the problems

Experiments based on Internet Programming Theory

Unit No.	Contents
	BLOCK 1 : JAVA FUNDAMENTAL PROBLEMS:

1	Simple Java Problems
2	class and objects
3	Conditional control using java
4	Looping using java
	BLOCK 2 : OOP CONCEPTS
5	Function overloading programs
6	Operator overloading programs
7	Inheritance programs, Packages
8	Polymorphism programs Message passing programs
	BLOCK 3 : THREAD & VIRTUAL FUNCTION
9	Threads
10	Virtual function programs
	BLOCK 4 : I/O AND EXCEPTION HANDLING
11	Exception handling programs
12	I/O manipulation programs,
	BLOCK 4 :NETWORK PROGRAMMING
13	Applet programs
14	Implementation of simple network programs using java

Reference books:

1. R. Krishnamoorthy and S. Prabhu, Internet and Java Programming, New Age International Publishers, 2004 (Unit I).
2. Programming with Java, 4e, E. Balagurusamy, Tata McGraw-Hill, 2010.
3. Deitel, Deitel and Nieto, Internet and World Wide Web – How to program, Pearson Education, 2000.
4. Naughton and H.Schildt, Java 2 - The complete reference, Tata McGraw-Hill, Fourth edition, 2006.
5. Elliotte Rusty Harold, Java Network Programming, O'Reilly Publishers, 2000.
6. B.Mohamal Ibrahim , Java : J2SE – A Practical Approach, Firewall media, 2006.
7. Cay S. Horstmann, Gary Cornell, Core Java, Volume I and II, 5th Edition, Pearson Education, 2003.
8. Topley, J2ME in A Nutshell, O'Reilly Publishers, 2002.
9. Hunt, Guide to J2EE Enterprise Java, Springer Publications, 2004.
10. Ed Roman, Enterprise Java Beans, Wiley Publishers, 1998.

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SEMESTER II

Course Code	Title of the Course
31321	DATA MINING AND WAREHOUSING

Course Objective:

- This course presents on depth of to data mining techniques; association rule, clustering, classification, web mining, temporal and sequential data mining and provide a practical exposure using data mining tool orange.
- To enable the students to learn the basic functions, principles and concepts of Data Mining
- To understand the fundamentals of Big Data Analytics

Course Requirements:

- Basic Concepts of Database

Course Outcome:

On successful completion of the course the students should have:

- Understand the data mining techniques, classification and web mining

Unit No.	Contents
	BLOCK 1 : DATA MINING AND WAREHOUSING INTRODUCTION
1	**Data Warehousing Introduction** – Definition-Architecture-Warehouse Schema-Warehouse server-OLAP operations. Data Warehouse technology – Hardware and operating system
2	**Data Mining** - Definition – DM Techniques – current trends in data mining - Different forms of Knowledge – Data selection, cleaning, Integration, Transformation, Reduction and Enrichment.
3	**Data:** Types of data - Data Quality - Data Preprocessing - Measures of similarity and dissimilarity. **Exploration:** Summary statistics – Visualization.
	BLOCK 2 : ASSOICATION RULE MINING AND CLASSIFICATION
4	**Association rules:** Introduction – Methods to discover association rule – Apriori algorithm Partition Algorithm
5	**AR Algorithms:** Pincher search algorithm – Dynamic Item set algorithm – FP Tree growth algorithm.
6	**Classification:** Decision Tree classification – Bayesian Classification – Classification by Back Propagation.
	BLOCK 3 : CLUSTERING TECHNIQUES AND MACHINE LEARNING
7	**Introduction** – Clustering Paradigms – Partitioning Algorithms – K means & K Mediod algorithms – CLARA – CLARANS – Hierarchical clustering – DBSCAN – BIRCH – Categorical Clustering algorithms – STIRR – ROCK – CACTUS.
8	**Introduction to machine learning** – Supervised learning – Unsupervised

	learning – Machine learning and data mining.
9	**Neural Networks:** Introduction – Use of NN – Working of NN Genetic Algorithm: Introduction –Working of GA.
	BLOCK 4 : WEB MINING AND VISUAL DATA MINING
10	**Introduction** –Web content mining – Web structure mining –Web usage mining –Text mining –Text clustering, Temporal mining -Spatial mining
11	**Visual data mining** – Knowledge mining – Various tools and techniques for implementation using weka, Rapidminer and Matlab.
	BLOCK 5 : INTRODUCTION TO BIG DATA ANALYTICS
12	**Big Data Characteristics**- Types of Big Data- Traditional Versus Big Data Approach
13	**Technologies** Available for Big Data
14	**Hadoop** – Introduction - What is Hadoop? - Core Hadoop Components - Hadoop Ecosystem - Physical Architecture - Hadoop Limitations

Text Books:

1. Arun K Pujari, "Data Mining Techniques", University press, 2008.
2. C S R Prabhu, "Data Warehousing – concepts, techniques and applications ", 2nd Edition, Prentice Hall of India, 2002.
3. Radha Shankarmani, M Vijayalakshmi, "Big Data Analytics", Wiley Publications, first Edition, 2016

Reference Books:

1. Jaiwei Han, Michelinne Kamber, "Data Mining: Concepts and Techniques", Harcourt India / Morgan Kauffman publishers, 2008.
2. Alex Berson, Stephen J.Smith , "Data Warehousing , Data Mining & OLAP", Tata McGraw Hill, 2004.
3. Seema Acharya, Subhashini Chellappan, "Big Data and Analytics", Wiley Publication, first edition. Reprint in 2016
4. DT Editorial Services, "Black Book- Big Data (Covers Hadoop 2, MapReduce, Hive, Yarn, PIG, R, Data visualization)", Dream tech Press edition 2016.

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Course Code	Title of the Course
31322	**RELATIONAL DATABASE MANAGEMENT SYSTEM (RDBMS)**

Course Objectives:

- To understand the fundamentals of data models
- To make a study of SQL and relational database design.

- To know about data storage techniques an query processing.
- To impart knowledge in transaction processing, concurrency control techniques and External storage

Course Requirements:

- Knowledge about the basic concepts of the database.

Course Outcome:

- Design a database using ER diagrams and map ER into Relations and normalize the relations
- Acquire the knowledge of query evaluation to monitor the performance of the DBMS.
- Develop a simple database applications using normalization.

Unit No	Contents
	BLOCK 1 INTRODUCTION
1	**Data base System Applications**, data base System VS file System – View of Data – Data Abstraction –Instances and Schemas – data Models – the ER Model
2	**Model :**Relational Model – Other Models – Database Languages – DDL – DML – database Access for applications Programs – data base Users and Administrator – Transaction Management – data base System Structure – Storage Manager – the Query Processor.
3	**History of Data base Systems** - Data base design and ER diagrams – Beyond ER Design Entities, Attributes and Entity sets – Relationships and Relationship sets – Additional features of ER Model – Concept Design with the ER Model – Conceptual Design for Large enterprises.
	BLOCK 2 : RELATIONAL MODEL
4	**Introduction**– Integrity Constraint Over relations – Enforcing Integrity constraints – Querying relational data – Logical data base Design – Introduction to Views – Destroying / altering Tables and Views.
5	**Relational Algebra** – Selection and projection set operations – renaming – Joins – Division – Examples of Algebra overviews –
6	**Relational calculus** – Tuple relational Calculus – Domain relational calculus – Expressive Power of Algebra and calculus.
	BLOCK 3 : SQL QUERY
7	**Form of Basic SQL Query** – Examples of Basic SQL Queries – Introduction to Nested Queries – Correlated Nested Queries Set – Comparison Operators – Aggregative Operators – NULL values – Comparison using Null values – Logical connectivity's – AND, OR and NOT – Impact on SQL Constructs – Outer Joins – Disallowing NULL values – Complex Integrity Constraints in SQL Triggers and Active Data bases. Schema refinement
8	**Normal forms :**Problems Caused by redundancy – Decompositions – Problem related to decomposition – reasoning about FDS – FIRST, SECOND,

	THIRD Normal forms – BCNF–
9	**Join:** Lossless join Decomposition – Dependency preserving Decomposition – Schema refinement in Data base Design – Multi valued Dependencies – FORTH Normal Form.
	BLOCK 4 TRANSACTION
10	**Introduction :**Transaction Concept- Transaction State- Implementation of Atomicity and Durability – Concurrent – Executions – Serializability- Recoverability – Implementation of Isolation – Testing for serializability
11	**Protocols :** Lock Based Protocols – Timestamp Based Protocols- Validation-Based Protocols – Multiple Granularity.
12	**Recovery and Atomicity** – Log – Based Recovery – Recovery with Concurrent Transactions – Buffer Management – Failure with loss of nonvolatile storage-Advance Recovery systems- Remote Backup systems
	BLOCK 5 STORAGE
13	**Data on External Storage** – File Organization and Indexing – Cluster Indexes, Primary and Secondary Indexes – Index data Structures – Hash Based Indexing – Tree base Indexing – Comparison of File Organizations – Indexes and
14	**Performance Tuning-** Intuitions for tree Indexes – Indexed Sequential Access Methods (ISAM) – B+ Trees: A Dynamic Index Structure.

Text Books:

1. Raghurama Krishnan, Johannes Gehrke, Data base Management Systems, 3rd Edition, TATA McGrawHill.2003.
2. Silberschatz, Korth, Data base System Concepts, 6th Edition, Tata McGraw Hill, 2011.

Reference Books:

1. Relational Database Principles 2nd Edition, Colin Ritchie, 2004
2. Sharad Maheswari and Ruchin Jain, Database management systems Complete Practical Approach,Firewall media, 2006
3. Peter Rob & Carlos Coronel, Data base Systems design, Implementation, and Management, 7th Edition.
4. Elmasri Navrate , Fundamentals of Database Systems, Pearson Education.

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Course Code	Title of the Course
31323	VISUAL PROGRAMMING WITH •NET

Course Objective:

To develop an understanding of Visual Basic .Net
To develop the skills necessary to create software solutions using VB with .Net
To learn how to analyze certain types of problems with a software solution in mind

Course Requirements:

- Basic knowledge of Visual Basic

Course Outcome:

- Able to understand and design the solution to a problem using VB. Net
- Understand and implement the features of .Net for providing programmed solutions to complex problems

Unit No	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction** - What Is Visual Studio ? - Navigating the Visual Studio - The Menu – Toolbar - Work Area
2	**Toolbox** - Solution Explorer - Status Bar - Managing VS Windows
3	**Visual Studio Project Types** - Windows Projects - Web Projects - Office Projects - SharePoint Projects - Database Projects
	BLOCK 2 : C# AND VB.NET
4	**Basic Syntax** - Code Skeleton - The Main Method - The Program Class - The First Program Namespace - VS Code Editor - Class and Member Locators – Bookmarks - Running Programs - Primitive Types and Expressions - Enums - Branching Statements - Loops
5	**Creating Classes** - Class Inheritance – Class Snippet - Writing Methods - Parameters Passing - Returning Data
6	**Method Snippets** - Coding Fields and Properties - Declaring and Using Properties - The Property Snippet
	BLOCK 3 : UNDERSTANDING DELEGATES AND EVENTS
7	Events - Delegates - Handler Code - Implementing Interfaces - The interface Snippet - Applying Arrays and Generics -
8	**Creating and Building Projects** - Constructing Solutions and Projects - Navigating the Solution Explorer - Examining Property Settings - Assembly

	Name - Default Namespace - Target Framework - Output Type
9	**Building Projects** :Startup Object - Icon and Manifest - Compiling Applications - Rebuilding Solutions/Projects - Cleaning Solutions/Projects - Managing Dependencies, Compilation Settings - Navigating with Class View - Using the Class Designer - Class Designer Code Generation
	BLOCK 4 : DEBUGGING WITH VISUAL STUDIO
10	**Debugging methods:** Breakpoints - Stepping Through Code – Inspecting Application State - Locals and Autos Windows - Watch Windows - The Immediate Window - The Call Stack Window - The Quick Watch Window - Watching Variables with Pin To Source - Working with IntelliTrace
11	**Working with Databases** - Server Explorer - Creating a Database - Adding Tables - Relating Tables with Foreign Keys - Adding Stored Procedures - Configuring Database Options
	BLOCK 5 : BUILDING PROGRAMS WITH VS 2010
12	Building Desktop Applications with WPF - Starting a WPF Project - Understanding Layout - Grid Layout - StackPanel Layout - DockPanel Layout - WrapPanel Layout - Canvas Layout
13	**Using WPF Controls** - Managing Windows for Controls - Setting Properties - Handling Events - Coding Event Handlers - Working with Data in WPF - Data Source - Configuring a ComboBox
14	**Reading and Saving Data** - Using the DataGrid - Summary -Creating Web Applications with ASP.NET MVC - Designing Silverlight Applications - Deploying Web Services with WCF

Text Book:

1. Joe Mayo, Visual Studio 2010 - A Beginner's Guide, Tata Mc Graw Hill Edition, 2010.

Reference Books:

1. Nick Randolph, David Gardner, Professional Visual Studio 2010, Wiley Publishing 2010.
2. Andrew Moore, Visual Studio 2010 All-in-One For Dummies, Weiley Publishing, 2010.

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Course Code	Title of the Course
31324	**VISUAL PROGRAMMING WITH •NET & RDBMS LAB**

Course Objective:

To develop an understanding of Visual Basic .Net
To develop the skills necessary to create software solutions using VB with .Net

To learn how to analyze certain types of problems with a software solution in mind

Course Requirements:

- Basic knowledge of Visual Basic

Course Outcome:

- Able to understand and design the solution to a problem using VB. Net
- Understand and implement the features of .Net for providing programmed solutions to complex problems

Experiments based on Visual Programming with .NET Theory

Unit No.	Contents
	BLOCK 1 : SIMPLE APPLICATIONS
1	**Simple Applications:** Developing simple applications using VB.NET a. Finding factorial Value b. Money Conversion c. Quadratic Equatin d. Temperature Conversion e. Login control
2	**Login form:** Create and Validate Login Form, Program to design Class, Program to demonstrate Inheritance, Polymorphism and Interfaces.
	BLOCK 2 : CONTROLS
3	**Controls:** Advance Controls, Common Dialog Controls. 2. Adrotator Control 3. Calendar control a. Display messages in a calendar control b. Display vacation in a calendar control c. Selected day in a calendar control using style d. Difference between two calendar dates 4. Treeview control a) Treeview control and datalist b) Treeview operations 5. Validation controls
4	**Active X Controls:** Working with intrinsic controls and ActiveX controls
	BLOCK 3 : MDI AND DATA CONTROLS
5	**MDI:** Application with multiple forms
6	**Data controls:** Application using data controls
	BLOCK 3 : DIALOGS AND MENU
7	**Dialogs:** Application with dialogs
8	**Common Dialogs:** Application using Common Dialogs
9	**Menus:** Application with Menus
	BLOCK 4 : EVENTS AND DATABASE
10	**Events and Database:** Drag and Drop Events Database Management Creating

	ActiveX Controls
11	**DataGridView:**ADO.NET Code to show records in DataGridView Control. 1. Databinding using datalist control 2. Datalist control templates 3. Databinding using datagrid 4. Datagrid control template 5. Datagrid hyperlink 6. Datagrid button column 7. Datalist event 8. Datagrid paging
12	**Database operations:**ADO.NET Code to perform Insert, Delete, Update and Select operations. Crystal Reports Web Application using ASP.NET that uses validation controls. Table creation, Renaming a Table, Copying another table, Dropping a Table Table Description: Describing Table Definitions, Modifying Tables, Joining tables, Number and Date functions.
	BLOCK 5 : SQL QUERIES AND SUB QUERIES
13	**SQL Queries:** Queries, Sub Queries, and aggregate functions DDL: Experiments using database DDL SQL statements DML: Experiment using database DML SQL statements DCL: Experiment using database DCL SQL statements
14	**APPLICATION DEVELOPMENT :** Design and Develop Application: Library information system, Students mark sheet processing, Telephone directory maintenance, Gas booking and delivering, Electricity bill processing, Bank Transaction, Pay roll processing. Personal information system, Question database and conducting Quiz and Personal diary

Reference Books:

1. Joe Mayo, Visual Studio 2010 - A Beginner's Guide, Tata Mc Graw Hill Edition, 2010.
2. Nick Randolph, David Gardner, Professional Visual Studio 2010, Wiley Publishing 2010.
3. Andrew Moore, Visual Studio 2010 All-in-One For Dummies, Weiley Publishing, 2010.

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SECOND YEAR
SEMESTER III

Course Code	Title of the Course
31331	**OPEN SOURCE SOFTWARE**

Course Objectives:

- To understand the need, advantages and applications of open source software
- To work with open source database and open source programming languages

Course Outcome

- Attained to know and work with open source software like Linux, MySql etc
- Able to do programming in open source programming languages

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	Introduction to Open sources – Need of Open Sources – Advantages of Open Sources–Application of Open Sources.
2	Open source operating systems: LINUX: Introduction– General Overview– Kernel Mode and user mode–Process–
3	Advanced Concepts–Scheduling – Personalities – Cloning – Signals – Development with Linux. .
	BLOCK 2 : OPEN SOURCE DATABASE
4	MySQL: Introduction Setting up account Starting, terminating and writing your own SQL programs
5	Record selection Technology– Working with strings – Date and Time
6	Sorting Query Results – Generating Summary – Working with metadata – Using sequences – MySQL and Web.
	BLOCK 3 :OPEN SOURCE PROGRAMMING LANGUAGE - PHP
7	PHP: Introduction – Programming in web environment – variables – constants–data types – operators –Statements
8	Functions– Arrays – OOP –String Manipulation and regular expression –File handling and data storage
9	PHP and SQL database – PHP and LDAP – PHP Connectivity –Sending and receiving E-mails –Debugging and error handling – Security – Templates.
	BLOCK 4 :OPEN SOURCE PROGRAMMING LANGUAGE - PYTHON
10	Syntax and Style – Python Objects – Numbers – Sequences – Strings –Lists and Tuples – Dictionaries –Conditionals and Loops .
11	Files – Input and Output –Errors and Exceptions – Functions – Modules – Classes and OOP – Execution Environment.
	BLOCK 5 :OPEN SOURCE PROGRAMMING LANGUAGE - PERL
12	Perl backgrounder – Perl overview– Perl parsing rules – Variables and Data – Statements
13	Control structures – Subroutines
14	Packages, and Modules-Working with Files –Data Manipulation.

Text Books:

1. Remy Card, Eric Dumas and Frank Mevel, “The Linux Kernel Book”, Wiley Publications, 2003
2. Steve Suchring, “MySQL Bible”, John Wiley, 2002

Books for Reference:

1. Rasmus Lerdorf and Levin Tatroe, “Programming PHP”, O’Reilly, 2002

2. Wesley J. Chun, "Core Phython Programming", Prentice Hall, 2001
3. Martin C. Brown, "Perl: The Complete Reference", 2nd Edition, Tata McGraw-Hill Publishing Company Limited, Indian Reprint 2009.
4. Steven Holzner, "PHP: The Complete Reference", 2nd Edition, Tata McGraw-Hill Publishing Company Limited, Indian Reprint 2009.
5. Vikram Vaswani, "MYSQL: The Complete Reference", 2nd Edition, Tata McGraw-Hill Publishing CompanyLimited, Indian Reprint 2009.

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Course Code	Title of the Course
31332	OPERATING SYSTEM

Course Objective

- Able to understand the operating system principles
- Able to know the Principles of Deadlock, processor scheduling and memory management.

Course Requirements:

- To be aware of the evolution and fundamental principles of operating system, processes and their communication

Course Outcome

- Students have acquired the knowledge about the types of operating systems
- Students have acquired the knowledge about the functions of operating system

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Introduction:** Definition of Operating Systems – Computer System Organization
2	**Computer System Architecture** – Operating System Structure – Operating System Operations
3	**System Structures:** Operating System Services – System Calls – System Programs – Operating System Design and Implementation.
	BLOCK 2 : PROCESS CONCEPT
4	**Process Concept:** Process Scheduling – Operations on Processes – Inter Process Communication
5	**Process Scheduling:** Scheduling Concepts – Scheduling Criteria – Scheduling Algorithms – Multiple Processor Scheduling
	BLOCK 3 : SYNCHRONIZATION
6	**Synchronization:** The Critical Section Problem – Synchronization Hardware –

	Semaphores – Classic Problems of Synchronization – Monitors
7	**Deadlocks:** Deadlocks Characterization – Methods for Handling Deadlocks
8	**Deadlock** Prevention – Avoidance – Detection –Recovery from Deadlock.
	BLOCK 4 : MEMORY MANAGEMENT
9	**Memory Management Strategies:** Swapping – Contiguous Memory Allocation – Paging – Segmentation
	BLOCK 5 : FILE SYSTEM
10	**File Concept** – Access Methods – Directory
11	**Structure** – File System Mounting – File Sharing – Protection.
12	**Implementing File Systems:** File System Structure – File System Implementation
13	**Directory Implementation** – Allocation Methods – Free Space Management
14	**Secondary Storage Structure:** Overview of Mass Storage Structure – Disk Structure – Disk Attachment – Disk Scheduling – Disk Management.

TEXT BOOK:

1. Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, "Operating System Principles", 7th Edition, Wiley India Edition, 2008.

REFERENCE BOOKS:

1. Andrew S.Tanenbaum, Operating Systems Design and Implementation, 3rd edition, Prentice Hall, 2006.
2. Harvey M. Deitel, An Introduction to Operating Systems, 3rd Edition, Addison Wesley 2007.

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Course Code	Title of the Course
31333	COMPUTER NETWORKS

Course Objectives:

•To understand networking concepts and basic communication model
•To understand network architectures and components required for data communication.

Course Requirements:

•Basic knowledge of networking

Course Outcome:

•Able to understand the working principles of various application protocols
•Acquire knowledge about security issues and services available

Unit No.	Contents
	BLOCK 1 : INTRODUCTION & PHYSICAL LAYER
1	**Introduction;** Computer Networks - Applications - Line configuration - Topology - Transmission Modes
2	**Categories of Network:** LAN, MAN, WAN - OSI Layer.
3	**Physical Layer:** Analog and Digital Signals Performance - Transmission Media
	BLOCK 2 : DATA LINK LAYER
4	**Data Link Layer:** Error Detection and correction – Introduction – Block Coding – Cyclic Redundancy Check – Framing – Flow and error Control –
5	**Data link layer protocols:** stop - wait protocol and sliding window protocol - ARQ, Go-back-n ARQ, selective - repeat ARQ.
6	**Multiple Access Protocols:** ALOHA – CSMA – CSMA/CD – CSMA/CA.
	BLOCK 3 : NETWORK LAYER
7	**Introduction:** Circuit switching - packet switching - message switching - Virtual circuit and Datagram subnets
8	**Routing algorithm :** Static routing -shortest path routing, Flooding, Flow based routing - Dynamic routing - distance vector routing, link state routing
9	**Other Routing Algorithms:** Hierarchical routing, Broad cast, Multi cast routing - Congestion, Control Algorithms
	BLOCK 4 : TRANSPORT LAYER
10	**Introduction:** Process to process delivery – UDP – TCP - Connection oriented Vs connectionless services.
11	**Applications and services:** Domain name system - Remote Logon – Mail Exchange - File Transfer
12	**Remote** Procedure Call - Remote File Access – WWW and HTTP – SNMP.
	BLOCK 5 : NETWORK SECURITY
13	**Introduction:** Cryptography – Encryption model – Transposition and Substitution Chipers – Cryptographic principles
14	**Symmetric key cryptography:** DES – AES – Asymmetric key cryptography: RSA – Security services.

Text Books:

1.Computer Networks, 3rd Edition, Andrew S Tanenbaum, Pearson Education, 2010.
2.Data Communications and Networking, 4th Edition, Behrouz A. Forouzan, TMH, 2009.

Reference Books:

1. Data and Computer Communications, 8th Edition, William Stallings, Prentice Hall.
2. An Engineering Approach to Computer Networks, 2nd Edition, S.Keshav, Pearson Education, 2008

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Course Code	Title of the Course
31334	OPEN SOURCE LAB

Course Objectives:

- To understand the need, advantages and applications of open source software
- To work with open source database and open source programming languages
- To develop applications in PHP using various concepts like arrays, udf's, Sessions and make the students to understand and to establish the connectivity between PHP and MySQL and develop programs to add records, retrieve records and delete records from a table.

Course Outcome

- Attained to know and work with open source software like Linux, MySql etc.,
- Able to do programming in open source programming languages.

Unit No.	Contents
	BLOCK 1 : INTRODUCTION TO LINUX
1	**Kernel** configuration, compilation and installation.
2	**Install various software** on Linux. Install and configure XAMP., Unix commands And shell programming
	BLOCK 2 : MYSQL
3	**Creating simple table** with constraints Insertion, Updation and Deletion of rows in MYSQL tables Searching of data by different criteria,Sorting of data
4	**Demonstration of joining tables** Usage of subqueries,aggregate functions Working with set operators Working with string, numeric and date functions
5	**Database connectivity** in PHP with MySQL Validating Input Formatting the Output.
	BLOCK 3 : PHP
6	**PHP Simple Programs** PHP program to perform the arithmetic operation. PHP program Adding numbers using function.
7	**PHP Web programs arrays and functions** Creating simple webpage using PHP Use of conditional statements in PHP Use of looping statements in PHP Creating different types of arrays

	Usage of array functions Creating user defined functions
8	**File manipulation using PHP** Creation of files,sessions and cookies Creating simple applications using PHP Creating simple table with constraints
	BLOCK 4 : PERL AND PYTHON PROGRMAMING
9	**PERL programs** : Simple programs using PERL
10	**Python Programming:** Use of conditional statements Use of looping statements
11	**Python Programming: Arrays** Creating different types of arrays Usage of array functions Creating user defined functions
12	**Python Programming: String**: String Handling
	BLOCK 5 : APPLICATION DEVELOPMENT
13	**Connect to a MYSQL** database with PHO, PERL and Python.
14	**Developing simple applications** using PHP and MYSQL

Books for Reference:

- Remy Card, Eric Dumas and Frank Mevel, "The Linux Kernel Book", Wiley Publications, 2003
- Steve Suchring, "MySQL Bible", John Wiley, 2002.
- Rasmus Lerdorf and Levin Tatroe, "Programming PHP", O'Reilly, 2002.
- Wesley J. Chun, "Core Phython Programming", Prentice Hall, 2001
- Martin C. Brown, "Perl: The Complete Reference", 2nd Edition, Tata McGraw-Hill Publishing Company Limited, Indian Reprint 2009.
- Steven Holzner, "PHP: The Complete Reference", 2nd Edition, Tata McGraw-Hill Publishing Company Limited, Indian Reprint 2009.
- Vikram Vaswani, "MYSQL: The Complete Reference", 2nd Edition, Tata McGraw-Hill Publishing Company Limited, Indian Reprint 2009

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SEMESTER IV

Course Code	Title of the Course
31341	**WEB TECHNOLOGY**

Course Objective:

To understand and practice markup languages
To understand and practice embedded dynamic scripting on client side Internet Programming
To understand and practice web development techniques on client-side

Course Requirement:

Basic concepts of Web and Java programming

Course Outcome:

Explore markup languages features and create interactive web pages using them
Learn and design Client side validation using scripting languages
Acquire knowledge about Open source JavaScript libraries
Able to design front end web page and connect to the back end databases.

Unit No.	Contents
	BLOCK 1 : HTML, JAVA SCRIPT and XML
1	**HTML Common tags:** List, Tables, images, forms, Frames; Cascading Style sheets.
2	**Introduction to Java Scripts**, Objects in Java Script, Dynamic HTML with Java Script.
3	**XML:** Document type definition, XML Schemas, Document Object model, Presenting XML, Using XML Processors: DOM and SAX
	BLOCK 2 : JAVA BEANS
4	**Java Beans:** Introduction to Java Beans, Advantages of Java Beans, BDK, Introspection, Using Bound properties, Bean Info Interface,
5	**Constrained properties**, Persistence, Customizes, Java Beans API, Introduction to EJB's
	BLOCK 3 : SERVLETS
6	**Web Servers and Servlets:** Tomcat web server, Introduction to Servlets: Lifecycle of a Servlet, JSDK
7	**The Servlet API,** The javax.servlet Package, Reading Servlet parameters, Reading Initialization parameters.
8	**The javax.servlet** HTTP package, Handling Http Request & Responses, Using Cookies-Session Tracking, Security Issues.
	BLOCK 4 : JAVA SERVER PAGES (JSP)
9	**Introduction to JSP:** The Problem with Servlet. The Anatomy of a JSP Page, JSP Processing. JSP Application Design with MVC Setting Up and JSP Environment: Installing the Java Software Development Kit, Tomcat Server & Testing Tomcat.
10	**JSP Application Development:** Generating Dynamic Content, Using Scripting Elements Implicit JSP Objects, Conditional Processing – Displaying Values Using an Expression to Set an Attribute, Declaring Variables and Methods
11	**Error Handling and Debugging:** Sharing Data Between JSP pages, Requests, and Users Passing Control and Date between Pages – Sharing Session and Application Data – Memory Usage Considerations
	BLOCK 5 : DATABASE ACCESS AND STRUCTS FRAMEWORK
12	**Database Access:** Database Programming using JDBC, Studying Javax.sql.* package, Accessing a Database from a JSP Page
13	**Application – Specific Database Actions**, Deploying JAVA Beans in a JSP Page
14	**Introduction to struts framework.**

TEXT BOOK:

1. Chris Bates, Web Programming, Building Internet Applications, 2nd Edition, Dreamtech
2. Patrick Naughton and Herbert Schildt, The complete Reference Java 2, 5th Edition, Tata McGraw Hill.
3. Jason Hunter Java Servlet Programming, O'Reilly
4. Hans Bergsten, Java Server Pages, O'Reilly

REFERENCE BOOK:

1. Ramesh Bangia,Web Technology, Firewall media, 2006.

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Course Code	Title of the Course
31342	SOFTWARE ENGINEERING

Course Objective:

- To know of how to do project planning for the software process.
- To learn the cost estimation techniques during the analysis of the project.
- To understand the quality concepts for ensuring the functionality of the software

Course Requirement:

- Fundamental concepts of Software Engineering

Course Outcome:

- Understand the activities during the project scheduling of any software application.
- Learn the risk management activities and the resource allocation for the projects.
- Able to create reliable, replicable cost estimation that links to the requirements of project planning and managing.

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Software:** Role of software, Software myths. Generic view of process: A layered technology, a process framework, The Capability Maturity Model Integration (CMMI)
2	**Process patterns**, Process assessment, Personal and Team process models.
3	**Process model:** The waterfall model, Incremental process models, Evolutionary process models, The Unified process.
	BLOCK 2 : REQUIREMENT ENGINEERING:
4	Design and Construction, Requirement Engineering Tasks, Requirements Engineering Process, Validating Requirements.
5	**Building the Analysis Model:** Requirement analysis, Data Modeling concepts, Object-Oriented Analysis

6	**Modeling:** Scenario-Based Modeling, Flow-Oriented Modeling Class-Based Modeling, Creating a Behavioral Model.
	BLOCK 3 : DESIGN
7	**Design Engineering:** Design process and quality, Design concepts, the design model.
8	**Architectural Design:** Software architecture, Data design, Architectural styles and patterns, Architectural Design.
9	**User interface design:** The Golden rules, User interface analysis and design, Interface analysis, Interface design steps, Design evaluation.
	BLOCK 4 : TESTING
10	**Testing Strategies:** Approach to Software Testing, Unit Testing, Integration Testing, Test strategies for Object-Oriented Software, Validation Testing, System Testing, the art of Debugging, Black-Box and White-Box testing.
11	**Product Metrics:** Software Quality, Product Metrics, Metrics for Analysis Model, Design Model, Source code and Metrics for testing, Metrics for maintenance. Metrics for Process and Projects Domains: Software Measurement, Metrics for Software Quality and Software Process.
	BLOCK 5 : RISK and QUALITY MANAGEMENT
12	Risk Strategies: Reactive vs. Proactive Risk strategies, software risks, Risk identification
13	**Risk Protection and refinement:** Risk projection, Risk refinement, Risk Mitigation, Monitoring and Management, RMMM Plan.
14	**Quality Management:** Quality concepts, Software quality assurance, Software Reviews, Formal Technical reviews, Statistical Software quality Assurance, Software reliability, The ISO 9000 quality standards.

TEXT BOOK:

1. Roger S. Pressman Software Engineering - A practitioner's Approach McGraw-Hill 6th Edition (2010)

REFERENCE BOOKS:

1. Richard Fairlay Software Engineering Concepts McGraw Hill Book Company (2005)
2. Pankaj Jalote An Integrated Approach to Software Engineering Narosa Publishing House 3rd Edition (2005)
3. Software Engineering, Somzerville, 8th Edition, Pearson Education 2007.
4. Software Engineering K.K. Agarwal & Yogesh Singh, 3rd Edition New Age International Publishers 2007.
5. Software Engineering an Engineering Approach James F. Peters, Witold Pedrycz - John Wiley & Sons 2000.
6. Software Engineering Principles and Practice Waman S Jawadekar, , Tata McGraw-Hill 2004.

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Course Code	Title of the Course
31343	CLOUD COMPUTING

Course Objective:

Lets learner to understand how to access all applications and documents from everywhere in the world, freeing from the confines of the desktop and making it easier for group members in different locations to collaborate.

Course Requirements:

Basic knowledge about internet and its application.

Course Outcome:

Understood the importance of cloud computing and its services.

Unit No.	Contents
	BLOCK 1 : INTRODUCTION
1	**Fundamentals** :Cloud Computing – History – Working of cloud computing – Cloud computing today – Pros and cons of Cloud Computing – Benefits of cloud computing
2	**Non users of Cloud computing** – Developing cloud services – Pros and Cons of Cloud service Development
3	Types of Cloud Service Development – Discovering Cloud Services development services and tools.
	BLOCK 2 : CLOUD COMPUTING FOR EVERYONE
4	**Centralizing Email Communications** – Collaborating of Grocery lists – Collaborating on To-Do lists –
5	**Collaborating** on Household budgets – Collaborating on Contact lists – Communicating across the community – Collaborating on Schedules
6	**Collaborating on group projects** and events – Cloud computing for corporation.
	BLOCK 3 : CLOUD SERVICES
7	Exploring online calendar applications – Exploring online scheduling applications – Exploring online planning and task management – Collaboration on event management –
8	Collaboration on Contact Management – Collaboration on Project Management –
9	Collaborating on Word Processing and Databases – Storing and Sharing files and other online content.
	BLOCK 4 : ISSUES IN CLOUD
10	**Federation in cloud** – Four levels of federation – Privacy in cloud
11	**Security in Cloud** –Software as a security service – Case Study: Aneka – service

	level agreements
12	**Cloud Storage:** Over view of cloud storage – Cloud storage providers – Amazon S3 – Cloud file system – Map Reduce – Hadoop
	BLOCK 5 : CLOUD DEPLOYMENT TOOLS:
13	Study of open source cloud platforms – Eucalyptus
14	Nimbus – Open Nebula

Text Books:

1. Michael Miller, "Cloud computing – Web based applications that change the way you work and collaborate online", Pearson Education Inc., 2008
2. John W.Rittinghous, James F.Ransome, "Cloud Computing: Implementation, Management and Security", CRC Press 2010.

Books for Reference:

1. Danielle Ruest and Nelson Ruest, "Virtualization: A Beginners's Guide", McGraw Hill,2009.
2. Tom White, "Hadoop: The Definitive Guide", O'RIELLY Media 2009.
3. Rajkumar Buyya, James Broberg, Andrezj Goscinski, "Cloud computing – Principles and Paradigms", John Wiley and Sons, 2011.

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Course Code	Title of the Course
31344	**WEB TECHNOLOGY LAB**

Course Objective:

- To understand and practice markup languages
- To understand and practice embedded dynamic scripting on client side Internet Programming
- To understand and practice web development techniques on client-side

Course Requirement:

- Basic concepts of Web and Java programming

Course Outcome:

- Explore markup languages features and create interactive web pages using them
- Learn and design Client side validation using scripting languages
- Acquire knowledge about Open source JavaScript libraries
- Able to design front end web page and connect to the back end databases.

Experiments based on Web Technology Theory

Unit No.	Contents
	BLOCK 1 : HTML and JAVASCRIPT
1	**HTML Common tags:** List, Tables, images, forms, Frames; Cascading Style sheets.
2	**Introduction to Java Scripts**, Objects in Java Script, Dynamic HTML with Java Script.
3	**DHTML programs**
	BLOCK 2: XML and JAVA BEANS
4	**XML:** Document type definition, XML Schemas, Document Object model, Presenting XML, Using XML Processors: DOM and SAX
5	**Simple programs using XML**
6	**Simple applications using Java Beans :** setting and getting properties of beans
	BLOCK 3 : JAVA SERVLETS PROGRAMMING
7	**Simple Server side programs:** Example. Write Servlet program to send Email message.
8	**Session Handling**
9	**Cookies :** getting and setting cookies
	BLOCK 4 : JAVA SERVER PAGES
10	Write a JSP program using JDBC. Shopping cart problem
11	Develop an Application for JSP – Servlet communication
12	Deploying JAVA Beans in a JSP Page
	BLOCK 5 : DATABASE ACCESS
13	Database Programming using JDBC
14	Accessing a Database from a JSP Page , Simple database applications with create, insert, modify and delete records. Batch Update. Stored Procedure and callable statement.

REFERENCE BOOK:

1. Chris Bates, Web Programming, Building Internet Applications, 2nd Edition, Dreamtech
2. Patrick Naughton and Herbert Schildt, The complete Reference Java 2, 5th Edition, Tata McGraw Hill.
3. Jason Hunter Java Servlet Programming, O'Reilly
4. Hans Bergsten, Java Server Pages, O'Reilly .
5. Ramesh Bangia,Web Technology, Firewall media, 2006

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MINUTES OF THE MEETING OF THE BOARD OF STUDIES FOR MASTER OF SCIENCE (INFORMATION TECHNOLGY) – M.Sc(IT) PROGRAMME

Minutes of the Meeting of the Board of Studies in Computer Science for the Master of Computer Applications (M.C.A), M.Sc(Information Technology), M.Sc. (Computer Science), Post Graduate Diploma in Computer Applications (P.G.D.C.A), Bachelor of Computer Applications (B.C.A), B.Sc (Information Technology), B.Sc. (Computer Science) Programmes to be offered through Open Distance Learning (ODL) Mode held at The Directorate of Distance Education, Alagappa University, Karaikudi – 630 003, on 04.09.2017, (11.00 A.M).

Members Present

1.	Dr. V. Palanisamy	-	Chairman
2.	Dr. E.Ramaraj	-	Member
3.	Dr. K.Kuppusamy	-	Member
4.	Dr. T.Meyyappan	-	Member
5.	Dr. S.S.Dhenakaran	-	Member
6.	Dr. K.Mahesh	-	Special Invitee
7.	Dr. A. Padmapriya	-	Special Invitee
8.	Dr. P. Prabhu	-	Member
9.	Mr.S.Balasubramanian	-	Member

After the deliberation and discussion the board resolved the following:

1. The Board considered the curriculum design and detailed syllabi of Computer Science programmes, prepared as per the norms and the Board scrutinized and necessary modifications are specified.

2. The Board resolved to approve curriculum design, detailed syllabi and other regulations for the Master of Computer Applications (M.C.A), M.Sc(Information Technology), M.Sc (Computer Science), Post Graduate Diploma in Computer Applications (P.G.D.C.A), Bachelor of Computer Applications (B.C.A), B.Sc (Information Technology), B.Sc. (Computer Science) programmes to be offered from 2018-2019 academic year onwards by the Directorate of Distance Education of Alagappa University, Karaikudi.

Dr. V. Palanisamy 4/9/17 | Dr. E.Ramaraj | Dr. K.Kuppusamy 4/9/17

Dr. T.Meyyappan 04/09/17 | Dr. S.S.Dhenakaran | Dr. K.Mahesh 4/9/17

Dr. A. Padmapriya 4/9/17 | Dr. P. Prabhu 4/9/17 | Mr.S.Balasubramanian

PROGRAMME PROJECT REPORT

M.Sc MATHEMATICS

(Distance Education Programme)

DIRECTORATE OF DISTANCE EDUCATION

ALAGAPPA UNIVERSITY

(A State University Accredited with A$^+$ Grade by NAAC (CGPA:3.64) in the Third Cycle)

KARAIKUDI- 630 003

M.Sc MATHEMATICS

Choice Based Credit System (CBCS)

(With effect from June 2018 – 2019 onwards)

a. Programme's Mission & Objectives:

To afford a High Quality Post Graduate Degree (M.Sc) Mathematics through Distance Learning mode to the students in order to nurture them in the emerging society among the young minds.

Objectives of the Course

This programme aims to develop an advanced training in mathematics with an emphasis on coursework. It offers opportunities to those who have an Honors degree or a Master degree in Mathematics to build and enhance their professional skills and qualifications in advanced mathematics in general and/or in some specialized areas of applied mathematics. The main objective of this course falls on the following aspects:

- To provide graduates with a comprehensive advanced knowledge of important areas of Mathematics
- To produce graduates of high level of analytic and technical skills required for the program
- To furnish them with the necessary background for further study in Mathematics and enhance their research capabilities
- To enable them to function effectively as teachers by giving seminar sessions in the related subjects

b. Relevance of the Programme with Alagappa University's Mission and Goals:

In order to align with the mission and goals of Alagappa University the M.Sc Mathematics is planned to deliver in Distance Learning mode which may reach the maximum number of student aspirants who are unable to thrive to spend non-elastic timings of formal conventional class room education. Such an undergraduate education in Mathematics subject with appropriate practical experiences will enrich the human resources for the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

In accordance with the mission of Alagappa University as a research-intensive institution, the teaching programme of the master's programme in psychology is based on state of the art of scientific research and maintains a strong emphasis on the acquisition of academic and research skills.

c. Nature of Prospective Target Group of Learners:

The curriculum has been designed for the learners including a class having of low level of disposable income, rural dwellers, women, unskilled men, minorities etc. to apply mathematical knowledge and problem-solving techniques to investigate a problem, generate and/or analyse information, find relationships and patterns, describe these mathematically as general rules, and justify or prove them through Distance Learning mode. Especially this curriculum will helpful to the learners, professionals who are in teaching, social workers and the people who are working in various fields.

d. Appropriateness of programme to be conducted in Distance learning mode to acquire specific skills and competence:

M.Sc Mathematics programme through Distance Learning mode is developed in order to give subject-specific skills including to:

- recognize that mathematics permeates the world around us
- appreciate the usefulness, power and beauty of mathematics
- enjoy mathematics and develop patience and persistence when solving problems
- understand and be able to use the language, symbols and notation of mathematics
- develop mathematical curiosity and use inductive and deductive reasoning when solving problems
- become confident in using mathematics to analyse and solve problems both in school and in real-life situations
- develop the knowledge, skills and attitudes necessary to pursue further studies in mathematics
- develop abstract, logical and critical thinking and the ability to reflect critically upon their work and the work of others
- develop a critical appreciation of the use of information and communication technology in mathematics
- appreciate the international dimension of mathematics and its multicultural and historical perspectives.

The programme is developed to give the students to encourage to share their thinking with teachers and peers and to examine different problem-solving strategies. Critical reflection in mathematics helps students gain insight into their strengths and weaknesses as learners and to appreciate the value of errors as powerful motivators to enhance learning and understanding.

At the end of the Programme students should be able to:

- Explain whether their results make sense in the context of the problem
- Explain the importance of their findings
- Justify the degree of accuracy of their results where appropriate
- Suggest improvements to the method when necessary.

e. Instructional Design:

e. 1. Curriculum Design:

Sl. No.	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C Max.
FIRST YEAR						
I Semester						
1.	**11311**	Algebra – I	25	75	100	4
2.	**11312**	Analysis – I	25	75	100	4
3.	**11313**	Ordinary Differential Equations	25	75	100	4
4.	**11314**	Topology – I	25	75	100	4
		Total	**100**	**300**	**400**	**16**
II Semester						
5.	**11321**	Algebra–II	25	75	100	4
6.	**11322**	Analysis–II	25	75	100	4
7.	**11323**	Topology – II	25	75	100	4
8.	**11324**	Partial Differential Equations	25	75	100	4
		Total	**100**	**300**	**400**	**16**
SECOND YEAR						
III Semester						
9.	**11331**	Differential Geometry	25	75	100	4
10.	**11332**	Optimization Techniques	25	75	100	4

11.	**11333**	Analytic Number Theory	25	75	100	4
12.	**11334**	Stochastic Processes	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		IV Semester				
13.	**11341**	Graph Theory	25	75	100	4
14.	**11342**	Functional Analysis	25	75	100	4
15.	**11343**	Numerical Analysis	25	75	100	4
16	**11344**	Probability And Statistics	25	75	100	4
		Total	**100**	**300**	**400**	**16**

Course Code Legend:

3	**1**	**1**	**X**	**Y**	**Z**

3**11-** M.Sc., Mathematics
X -Semester No
Y & Z- Course number in the semester

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C: Credit Points, Max.: Maximum

No. of Credits per Course (Theory) -4
Total No. of Credits per Semester- 16
Total No. of Credits per Programme - 16 X 4 = 64

e. 2. Detailed Syllabi:

SEMESTER-I

Course Code	Title of the Course
31111	**ALGEBRA– I**

Course Objectives:

The objective of the course is to

- introduce and study the basic properties of groups, normal sub groups and quotient groups.
- derive the notion of homomorphism, automorphism on groups and permutation groups.
- introduce the above mentioned concepts in Sylow's Theorems, direct products and finite abelian groups.
- study the structure of rings, some special classes of rings, ideals and quotient rings.
- define Euclidean rings, polynomial rings, polynomial rings over commutative rings and study their important properties and theorems.

Course Description:

UNIT – I

Set Theory - Mappings - The Integers -problems

UNIT -II

Group Theory: Definition of a group – Some examples of Groups – Some preliminary Lemmas – Subgroups

UNIT -III

A counting principle – Normal subgroups and Quotient groups

UNIT -IV

Homomorphisms – Automorphisms - Cayley's Theorem - Permutation Groups

UNIT -V

Another counting Principle – Application – Related problems

UNIT -VI

Sylow's Theorem - Direct products - Problems

UNIT -VII

Finite Abelian Groups – Supplementary problems

UNIT -VIII

Ring Theory: Definition and examples of rings – Some special classes of Rings

UNIT -IX

Ring Homomorphisms - Ideals and Quotient Rings - Problems

UNIT -X

More ideals and Quotient Rings – Related Problems

UNIT -XI

The field of quotients of an Integral Domain - Euclidean Rings – Related Problems

UNIT -XII

A Particular Euclidean Ring - Polynomial Rings

UNIT -XIII

Polynomials over the Rational Field – Related Problems

UNIT -XIV

Polynomial Rings over Commutative Rings – Supplementary Problems

REFERENCES:

1. I.N.Herstein, Topics in Algebra (2nd Edition) Wiley Eastern Limited, New Delhi, 1975.
2. M.Artin, Algebra, Prentice Hall of India, 1991.
3. John B.Fraleigh, A First Course in Abstract Algebra, Addison Wesley, Mass, 1982.
4. D.S.Malik, J.N.Mordeson and M.K.Sen, Fundamentals of Abstract Algebra, McGraw Hill (International Edition), New York, 1997.

Learning Outcomes:

After the successful completion of this course, students will be able to:

- understand the concepts of groups, normal subgroups and quotient groups.
- explain the concepts of homomorphism, automorphism on groups and permutation groups.
- analyze basic concepts about rings, ideals and quotient rings.
- demonstrate the examples of Euclidean rings, polynomial rings, polynomial rings over Commutative rings.

Course Code	Title of the Course
31112	ANALYSIS – I

Course Objectives:

The objective of the course is to:

- prove various statements by induction and emphasize the proofs' development.
- define the limit of a function at a value, a limit of a sequence, and the Cauchy criterion.
- prove various theorems about limits of sequences and functions and emphasize the proofs' development.
- prove various theorems about the series and emphasize the proofs' development.
- prove various theorems about the derivatives of functions and emphasize the proofs' development

Course Description:

UNIT -I

The Real and Complex Number Systems: Introduction- Ordered Sets –Fields- The Real Fields

UNIT -II

The Extended Real Number System- Complex field- Euclidean spaces -Problems

UNIT- III

Basic Topology: Finite- Countable and Uncountable Sets- Metric Spaces

UNIT- IV

Compact sets – Perfect sets – Connected sets- Problems

UNIT -V

Numerical sequences and series; Convergent sequences- Subsequences- Cauchy sequences- Upper and Lower limits

UNIT -VI

Special sequences- Series- Series of non–negative terms- The number e – The root and ratio tests.

UNIT -VII

Power series – Summation by parts- Absolute convergence – Addition and Multiplication of series – Rearrangements

UNIT -VIII

Continuity: Limits of Functions – Continuous Functions- Continuity and Compactness- Continuity and Connectedness

UNIT -IX

Discontinuities – Monotonic Functions - Infinite Limits and Limits at Infinity

UNIT -X

Open balls- Closed balls in R^n – Closed Sets and Adherent Points – The Bolzano-Weierstrass Theorem

UNIT -XI

The Cantor intersection theorem – The Heine - Borel covering theorem – Compactness in R^n

UNIT -XII

Derivatives – The Chain Rule- Functions with Nonzero Derivative- Zero Derivatives and Local Extrema- Rolle's theorem- The Mean-value Theorem for Derivatives

UNIT -XIII

The Mean-value Theorem for Derivatives - Intermediate value theorem for Derivatives- Taylor's Formula with Remainder

UNIT -XIV

Partial derivatives- Directional derivative- The Total derivative- The Inverse function theorem- The Implicit function theorem – Problems

REFERENCES:

1. Tom M Appostol, *Mathematical Analysis*, Second edition(1974)Addision Wesley.
2. Walter Rudin, Principles of Mathematical Analysis, III Edition, McGraw-Hill Book Company, 1976.
3. H.L.Royden, Real Analysis, Macmillan Publ.co., Inc. 4th edition, New York, 1993.
4. V.Ganapathy Iyer, Mathematical Analysis, Tata McGraw Hill, New Delhi, 1970.
5. Robert G.Bartle, Donald R.Sherbert, Introduction to *Real Analysis*, Third edition, (2000)John Wiley & Sons.

Learning Outcomes:

After the successful completion of this course, students will be able to:

- define and recognize the basic properties of the field of real numbers. Improve and outline the logical thinking.
- define and recognize the series of real numbers and convergence shown the ability of working independently and with groups.

- comprehend rigorous arguments developing the theory underpinning real analysis.
- demonstrate an understanding of limits and how they are used in sequences, series, differentiation and integration.
- appreciate how abstract ideas and rigorous methods in mathematical analysis can be applied to important practical problems.

Course Code	Title of the Course
31113	ORDINARY DIFFERENTIAL EQUATIONS

Course Objectives:

The objective of the course is to:

- formulate ordinary differential equations (ODEs) and seek understanding of their solutions, either obtained exactly or approximately by analytic or numerical methods.
- understand the concept of a solution to an initial value problem, and the guarantee of its existence and uniqueness under specific conditions.
- recognize basic types of differential equations which are solvable, and will understand the features of linear equations in particular.
- use different approaches to investigate equations which are not easily solvable. In particular, the student will be familiar with phase plane analysis.

Course Description:

UNIT-I

Linear Equations with Constant Coefficients: Introduction - The Second Order Homogeneous Equation

UNIT -II

Initial Value Problems for Second Order Equations –Related Problems

UNIT- III

Linear Dependence and Independence - Problems

UNIT-IV

A formula for the Wronskian – Non-Homogenous Equation of Order Two related Problems

UNIT- V

Homogeneous equation of order n - initial value problems for n^{th} order equations – Equations with Real Constants

UNIT -VI

Non-Homogeneous Equations of order n – Related Problems

UNIT -VII

Linear Equations with variable Coefficients: Reduction of the order of a Homogeneous Equation – Non-homogeneous Equation

UNIT -VIII

Homogeneous Equations with Analytic Coefficients – Legendre Equation.

UNIT -IX

Linear Equations with Regular Singular Points – Euler Equations - Second order Equations with Regular Singular Points – An Example

UNIT -X

Second order Equations with Regular Singular Points – General Case – Exceptional Cases

UNIT-XI

Bessel Equation – Bessel Equation (continued) – Regular Points at Infinity.

UNIT-XII

Existence and Uniqueness of Solutions to First order Equations : Equations with variables separated – Exact equations

UNIT -XIII

Method of successive approximations – Lipchitz condition – Convergence of the successive approximations.

UNIT -XIV

Nonlocal existence of solutions - Approximations to solutions and uniqueness of solutions- Existence and uniqueness of solutions to systems and n^{th} order equations - Existence and uniqueness of solutions to system.

REFERENCES:

1. Earl A.Coddington, An Introduction to Ordinary Differential Equations – Prentice Hall of India, 1987.
2. D.Somasundaram, Ordinary Differential Equations, Narosa Publishing House, Chennai, 2002.
3. M.D.Raisinghania, Advanced Differential Equations, S.Chand and Company Ltd, New Delhi, 2001.

Learning Outcomes:

After successful completion of this course, student will be able to:

- apply the fundamental concepts of ordinary differential equations and partial differential equations and the basic numerical methods for their resolution.
- solve the problems choosing the most suitable method.
- understand the difficulty of solving problems analytically and the need to use numerical approximations for their resolution.
- use computational tools to solve problems and applications of ordinary differential equations and partial differential equations.
- formulate and solve differential equation problems in the field of industrial organization engineering.
- use an adequate scientific language to formulate the basic concepts of the course.

Course Code	Title of the Course
31114	TOPOLOGY – I

Course Objectives:

The objectives of the course is to:

- explain how to distinguish spaces by means of simple topological invariants (compactness, connectedness and the fundamental group).
- explain how to construct spaces by gluing and to prove that in certain cases that the result is homeomorphic to a standard space.
- construct simple examples of spaces with given properties.
- Define Urysohn's lemma and Urysohn's metrization theorem and find the applications of them.

Course Description:

UNIT-I

Set Theory and Logic : Fundamental Concepts- Functions- Relations - The Integers and the Real Numbers

UNIT-II

The Integers and the Real Numbers - Cartesian Products- Finite Sets - Countable and Uncountable sets

UNIT-III

Infinite sets and the Axiom of choice- ordered sets- The Maximum Principle

UNIT-IV

Topological Spaces – Basis of a topology - Problems

UNIT-V

The order topology – The product topology on X x Y - Problems

UNIT-VI

The subspace topology – Closed sets and limit points – Hausdorff spaces

UNIT-VII

Continuous Functions – Continuity of a Function- Homomorphisms

The product topology

UNIT-VIII

Constructing continuous Functions -The metric topology - The quotient topology.

UNIT-IX

Connected spaces – Connected sets in the real line –component and path components

UNIT -X

Local connectedness - Compact spaces - Problems

UNIT -XI

Compact sets in the real line - Limit point compactness - Local Compactness

UNIT -XII

The Countability axioms - The separation axioms - Problems

UNIT -XIII

Normal spaces - The Urysohn's lemma

UNIT -XIV

The Uryshon's metrization theorem – Related Problems

REFERENCES:

1. James R.Munkres, Topology a first course, Prentice Hall of India Pvt. Ltd.,New Delhi (1987)
2. James Dugundji, Topology, Prentice Hall of India, New Delhi, 1975.
3. George F.Simmons, Introduction to Topology and Modern Analysis, McGraw Hill Book Co., 1963.

Learning Outcomes:

Upon successful completion of the course, students will be able to:

- define and illustrate the concept of topological spaces and continuous functions,
- prove a selection of theorems concerning topological spaces, continuous functions, product topologies, and quotient topologies.
- define and illustrate the concept of product topology
- define and illustrate the concepts of the separation axioms.
- define connectedness and compactness, and prove a selection of related theorems, and describe different examples distinguishing general, geometric, and algebraic topology.

SEMESTER-II

Course Code	Title of the Course
31121	ALGEBRA–II

Course objectives:

The main objective of the course is to:

- study the basic concepts of linear independence and bases, dual spaces and inner product spaces.
- determine roots of polynomials, extension fields and more about roots.
- introduce the concept of Galois Theory and derive the condition for the solvability by means of radicals.
- analyze the characteristic roots of linear transformation and study about Nilpotent transformations.
- define Hermitian, unitary and normal transformations.

Course Description:

UNIT-I

Vector Space: Elementary basic concepts -Problems

UNIT-II

Vector Space: Linear Independence and Bases

UNIT-III

Dual spaces – Inner product spaces – Problems

UNIT-IV

Field: Extension Fields - Roots of Polynomials – Related Problems

UNIT-V

Construction with Straight edge and Compass - More about roots.

Unit-VI

The Elements of Galois theory – Related Problems

UNIT-VII

Linear Transformations: The Algebra of linear transformations - Problems

UNIT-VIII

Linear Transformations: Characteristic roots - Matrices -Problems

UNIT-IX

Linear Transformations: Canonical Forms - Triangular Form

UNIT-X

Linear Transformations: Canonical Forms - Rational Canonical Form

UNIT-XI

Trace and Transpose – Determinants

UNIT-XII

Hermitian Transformations --Related Problems

UNIT-XIII

Unitary and Normal Transformations

UNIT-XIV

Finite Fields- Related Problems

REFERENCES:

1. I.N.Herstein, Topics in Algebra (2nd edition) John Wiley and Sons, New York.
2. P.B.Bhattacharya, S.K.Jain and S.R.Nagpaul, Basic Abstract Algebra (2nd edition) Cambridge University Press, 1997 (Indian Edition)
3. S.Lang, Algebra 3rd edition, Addison-Wesley, Mass, 1993.
4. N.Jacobson, Basic Algebra, Vol. I & II W.H.Freeman, also Published by Hindustan Publishing Company, New Delhi, 1980.

Learning Outcomes:

After successful completion of this course, students will be able to:

- analyze and demonstrate examples of linear independence and bases, dual Spaces and inner product spaces.
- assess properties implied by roots of polynomials and more about roots.
- classify and determine the trace and transpose of the matrices.
- define, illustrate and apply the concepts of unitary Hermitian and normal transformation.

Course Code	Title of the Course
31122	ANALYSIS–II

Course Objectives:

This course is aimed to provide

- an introduction to the theories for Riemann-Stieltjes Integral. It begins with the exploration of the Existence of the integral.
- the concepts of vector valued functions and Rectifiable curves are introduced.
- the notion of sequences and series is presented and to help the students to visualize the uniform convergence.
- understanding of the fundamental concepts of some special functions.
- the skill of Lebesque measure to evaluate them via examples.

Course Description:

UNIT-I

Riemann-Stieltjes Integral: Definition and Existence of the Integral

UNIT-II

Properties of the Integral- Integration and Differentiation - Problems

UNIT-III

Integration of vector-valued functions - Rectifiable curves.

UNIT-IV

Sequences and Series of functions: Discussion of main problem- Uniform convergence

UNIT-V

Uniform convergence and Continuity- Uniform convergence and Integration - Uniform convergence and Differentiation

UNIT-VI

Equicontinuous families of functions - The Stone Weierstrass theorem.

UNIT-VII

Some special functions: Power series- the Exponential- Logarithmic and Trigonometric functions

UNIT-VIII

The Algebraic completeness of the Complex field – Fourier Series – The Gamma function.

UNIT-IX

Lebesque measure - Outer measure- Measurable sets and Lebesgue measure

UNIT-X

Measurable functions- Egoroff's theorem- Lusin's theorem

UNIT-XI

Non-measurable sets – Lebesgue measurable functions – Little wood's three principles.

UNIT-XII

Lebesgue Integral: Riemann integral – Lebesgue Integral of a bounded function over a set of finite measure

UNIT-XIII

Lebesgue Integral of nonnegative measurable function – General Lebesgue integral

UNIT-XIV

Convergence theorems on measurable functions.

REFERENCES:

1. Walter Rudin, Principles of Mathematics Analysis (3rd edition), McGraw Hill 1976.
2. H.L. Royden, Real Analysis (3rd edition) Macmillan Publishing Company, New York, 1988.
3. G.De Barra, Measure Theory and Integration, Wiley Easten Ltd., New Delhi, 1987.
4. Malik S.C. and Savita Arora, Mathematical Analysis, Wiley Eastern Limited, New Delhi, 1991.
5. Robert G.Bartle, Donald R.Sherbert, Introduction to *Real Analysis*, Third edition, (2000)John Wiley & Sons.

Learning Outcomes:

Upon successful completion of this course, students will be able to:

- extend the concepts of Riemann integral
- differentiate and Integrate Complex functions.
- carry out Stone Weierstrass theorem.
- compute sequence and series of functions.
- apply techniques of measurable functions in varies fields.

Course Code	Title of the Course
31123	**TOPOLOGY – II**

Course Objectives:

The objectives of the course is to:

- explain how to distinguish regular spaces by means of simple topological invariants.
- explain how to construct Smirnov Metrization Theorem, Nagata Smirnov Metrization Theorem and to prove that in certain cases.
- Explain the concepts of complete metric spaces and open topology with suitable examples.
- construct simple examples of Baire spaces with given properties

Course Description:

UNIT-I

The Tietze Extension Theorem - Applications

UNIT-II

The Tychonoff Theorem - Problems

UNIT-III

Completely Regular Spaces - The Stone-Cech Compactification

UNIT-IV

Metrization theorems and Paracompactness: Local Finiteness - The Nagata Smirnov Metrization Theorem (Sufficiency) – The Nagata Smirnov Theorem (necessity).

UNIT- V

Paracompactness – The Smirnov Metrization Theorem - Problems

UNIT-VI

Complete Metric Spaces and Function Spaces: Complete metric spaces

UNIT-VII

A Space – Filling Curve - Compactness in Metric spaces

UNIT-VIII

Point wise and compact convergence – Related Problems

UNIT-IX

The Compact – Open Topology – Related problems

UNIT-X

Ascoli's theorem – Related problems

UNIT-XI

Baire Spaces - Applications

UNIT-XII

A Nowhere differentiable Functions.

UNIT-XIII

A Nowhere differentiable Functions - Related Problems

UNIT-XIV

Introduction to Dimension Theory

REFERENCES:

1. James R Munkres, Topology, A First Course, Prentice Hall of India, New Delhi (1984).
2. JL.Kelley, General Topology, Van Nostrnad, Reinhold Co., New York.
3. K.D.Joshi, Introduction to General Topology, Wiley Eastern Ltd., 1983.

Learning Outcomes:

Upon successful completion of the course, students will be able to:

- define and illustrate the concept of Regular spaces and Baire spaces.
- prove a selection of theorems concerning Regular spaces and Baire spaces,
- define and illustrate the concept of open topology.
- define and illustrate the concepts of the Metric spaces and Function spaces.
- define compactness and nowhere differentiable, and prove a selection of related theorems, and describe different examples

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Course Code	Title of the Course
31124	**PARTIAL DIFFERENTIAL EQUATIONS**

Course Objectives:

The objectives of this course is to:

- introduce the notion of partial differential equations.
- introduce students to how to solve linear partial differential with different methods.
- introduce some physical problems in Engineering and Biological models that results in partial differential equations.

Course Description:

UNIT-I

Ordinary differential equations in more than two variables : Surfaces and curves in three dimensions

UNIT-II

Simultaneous differential equations of the first order and the first degree in three variables

UNIT-III

Methods of solution of dx/P=dy/Q=dz/R - Orthogonal trajectories of a system of curves on a surface

UNIT-IV

Pfaffian differential forms and equations – Solution of Pfaffian differential equations - The three variables.

UNIT-V

Partial differential equations of the first order : Partial differential equations – origins of first order partial differential equations

UNIT-VI

Cauchy's problem for first order equations – Linear equations of the first order-Integral surfaces passing through a given curve

UNIT-VII

Surfaces orthogonal to a given system of surfaces - Nonlinear partial differential equations of the first order-Cauchy's method of characteristics.

UNIT-VIII

Compatible systems of first order equations – Charpits method - special types of first order equations – solutions satisfying given conditions - Jacobi's method

UNIT-IX

Partial differential equations of the second order : Origin of second order equations

UNIT-X

Linear partial differential equations with constant coefficients. Equations with variable coefficients – separation of variables

UNIT-XI

Method of integral transforms (exercise problems are excluded)

UNIT-XII

Laplace's equation : Elementary solutions of Laplace's equation – boundary value problems

UNIT-XIII

The Wave equation – Elementary solutions of the one dimensional wave equation

UNIT-XIV

The Diffusion equation : Elementary solutions of the diffusion equation – separation of variables.

REFERENCES:

1. I.N.Sneddon,Elements of Partial Differential Equations,McGraw Hill Book Company, 1986.
2. M.D.Raisinghania, Advanced Differential Equations, S.Chand&Company Ltd.,New Delhi, 2001.
3. K.Sankara Rao, Introduction to Partial Differential Equations, Second Edition, Prentice – Hall of India, New Delhi, 2006.
4. J.N.Sharma and K.Singh, Partial Differential Equations for Engineers and Scientists, Narosa Publishing House, Chennai, 2001.

Learning Outcomes:

Upon successful completion of this course, students will be able to:

- classify partial differential equations and transform into canonical form.
- solve linear partial differential equations of both first and second order.

SEMESTER-III

Course Code	Title of the Course
31131	DIFFERENTIAL GEOMETRY

Course Objectives:

The general objective of the course is to:

- introduce the concepts of a curve, arc-length, curvature, plane curves, space curves, Frenet –Serret equations.
- make the knowledge about surfaces, smooth surfaces, tangents, normals, quadratic surfaces.
- introduce the concepts of lengths of curves on surfaces, isometries of surfaces, conformal mappings of surfaces.
- understand the celebrated Gauss-Bonnet theorem, the second fundamental form, the curvature of curves on a surface, the normal and principal curvatures, geometric interpretation of principal curvatures.
- give awareness to learners about The Gaussian and mean curvatures, The surfaces of constant mean curvature, Gaussian curvature of compact surfaces.
- introduce the basic properties of Geodesics, Geodesic equations, Geodesics on surfaces of revolution.

Course Description:

UNIT-I

Introductory remark about space curves – definitions – arc length – Tangent, Normal and binormal

UNIT-II

Curvature and Torsion of a curve given as the intersection of two surfaces.

UNIT-III

Contact between curve and surfaces – Tangent surface, Involutes and evolutes

UNIT-IV

Intrinsic equations – Fundamental Existence Theorem for space curves

UNIT-V

Helices-Types of helices-Problems.

UNIT-VI

Definition of a surface – curves on a surfaces – Surfaces of revolution

UNIT-VII

Helicoids – Metric – Direction coefficients

UNIT-VIII

Families of curves – Isometric Correspondence – Intrinsic properties.

UNIT-IX

Geodesics – Canonical Geodesic equations – Normal property of Geodesics– Existence theorem.

UNIT-X

Geodesic Parallels – Problems in Geodesic parallels.

UNIT-XI

Geodesic Curvature – Gauss – Bonnet Theorem Gaussian curvature.

UNIT-XII

The Second Fundamental from – Principal curvature–Lines of Curvature

UNIT-XIII

Developables - Developables associated with space curves.

UNIT-XIV

Developables associated with curves on surfaces.

REFERENCES:

1. T.G. Willmore – An Introduction to Differential Geometry, Oxford University press (1983).

Learning Outcomes:

At the end of the module, students should be able to:

- understand the curvature and torsion of a space curve, how to compute them, and how they suffice to determine the shape of the curve.
- understand the definition of a smooth surface, and the means by which many examples may be constructed.
- understand the various different types of curvature associated to a surface, and how to compute them.
- understand the first and second fundamental forms of a surface, how to compute them, and how they suffice to determine the local shape of the surface.

- understand about Gaussian curvature, geodesics and its applications, how to compute them
- appreciate the distinction between intrinsic and extrinsic aspects of surface geometry.

Course Code	Title of the Course
31132	OPTIMIZATION TECHNIQUES

Course Objectives:

The general objective of the course is to:

- introduce the fundamental concepts of optimization techniques.
- make the learners aware of the importance of optimizations in real scenarios.
- provide the concepts of various classical and modern methods of for constrained and unconstrained problems in both single and multivariable.

Course Description:

UNIT-I

Network Models: Minimal spanning tree algorithm - Problems

UNIT-II

Shortest route algorithms -Problems

UNIT-III

Maximal flow Model-Problems

UNIT-IV

Critical path calculations - Tree and total floats - Problems

UNIT-V

Advanced Linear Programming - Simplex method using the restricted basis –

UNIT-VI

Bounded variables Algorithm - Revised Simplex method.

UNIT-VII

Game Theory - Optimal solution of Two Person Zero Sum Games

UNIT-VIII

Solution of mixed strategy Games – Related problems.

UNIT-IX

Game theory Linear programming solution of games.

UNIT-X

Classical Optimization Theory -Jacobian Method -Problems

UNIT-XI

Lagrangian Method - The Newton Raphson -Problems

UNIT-XII

Karush- Kuhn- Tucker conditions.-Problems

UNIT-XIII

Unconstrained algorithms - Non Linear Programming Algorithms.

UNIT-XIV

Separable programming – Quadratic Programming.

REFERENCES:

1. Operations Research, H.A. Taha, 8th edition, Prentice Hall, New Delhi, 2008.

Learning Outcomes:

Upon successful completion of this course, students will be able to:

- understand the theory of optimization methods and algorithms developed for solving various types of optimization problems and to formulate optimization problems.
- understand and apply the concept of optimality criteria for various types of optimization problems.
- solve various constrained and unconstrained problems in single variable as well as multivariable.
- apply the methods of optimization in real life situation.
- develop and promote research interest in applying optimization techniques in problems.

Course Code	Title of the Course
31133	ANALYTIC NUMBER THEORY

Course Objectives:

The main objective of the course is to:

- gain an understanding and appreciation of analytic number theory and some of its important applications.
- use the theory in specific examples.
- focus on the properties of prime numbers and to understand prime number theorem.
- understand the partitions of numbers and learn techniques to relate the subject with Combinatorics.

Course Description:

UNIT-I

The Fundamental Theorem of Arithmetic: Introduction – Divisibility – Greatest common divisor

UNIT-II

Prime Numbers – The series of reciprocals of the primes - The Euclidean Algorithm – The greatest common divisors of more than two numbers.

UNIT-III

Arithmetical Functions and Dirichlet Multiplication: Introduction; The Mobius function $\mu(n)$ – The Euler Totient Function $\phi(n)$ - A relation connecting ϕ and μ - A Product formula for $\phi(n)$.

UNIT-IV

The Dirichlet product of arithmetical functions: Dirichlet inverses and the mobius inversion formula - The Mangoldt function $\Lambda(n)$

UNIT-V

Multiplicative functions – Multiplicative functions and Dirichlet multiplication - The inverse of a Completely multiplicative function - Liouville's function $\lambda(n)$, The divisor functions $\sigma_\alpha(n)$

UNIT-VI

Generalized Convolutions – Formal Power Series

UNIT-VII

The Bell series of an arithmetical function - Bell series and Dirichlet Multiplication – Derivatives of arithmetical functions - The Selberg Identity.

UNIT-VIII

Averages of Arithmetical Functions: Introduction, The big oh notation. Asymptotic equality of functions

UNIT-IX

Euler's summation formula - Some elementary asymptotic formulas – The average order of d (n) – The average order of the divisor functions σ_α (n)

UNIT-X

The average order of φ (n) - An application to the distribution of lattice points, visible from the origin - The average order μ (n) and of Λ (n) - The partial sums of a Dirichlet product – Applications to μ(n) and Λ (n) Another identity for the partial sums of a Dirichlet product.

UNIT-XI

Congruences: Definition and Basic properties of congruences - Residue classes and complete residue systems - Linear congruences – Reduced residue systems and the Euler – Fermat theorem

UNIT-XII

Polynomial congruences modulo *p* Lagrange's theorem – Applications of Lagrange's theorem - Simultaneous linear congruences. The Chinese remainder theorem – Application of the Chinese remainder theorem

UNIT-XIII

Polynomial congruences with prime power moduli - The principle of cross classification - A decomposition property of reduced residue systems.

UNIT-XIV

Quadratic residuces and the Quadratic Reciprocity Law: Lagrange's symbol and its properties– Evaluation of (-1/p) and (2/P) – Gauss's Lemma – The quadratic reciprocity law - Applications of the reciprocity law - The Jacobi symbol - Applications to Diophantine Equations.

REFERENCES:

1. Tom M. Apostal, Introduction to Analytic Number theory, Springer Verlag.
2. Niven and H.S.Zuckerman, An Introduction to the Theory of Numbers, 3rd Edition, Wiley Eastern Ltd., New Delhi, 1989.
3. D.M.Burton, Elementary Number Theory, Universal Book Stall, New Delhi, 2001.

Learning Outcomes:

At the end of the course, students will be able to:

- analyze and prove results presented in analytic number theory.
- prove results similar to the ones presented in the course and apply the basic techniques, results and concepts of the course to concrete examples and exercises.
- understand the interdisciplinary nature with other mathematical branches.
- understand theoretical physics and Combinatorics with the knowledge of partition theory.

Course Code	Title of the Course
31134	STOCHASTIC PROCESSES

Course Objectives:

The objective of this course is to:

- Provide the fundamentals and advanced concepts of probability theory and random process to support graduate coursework and research in electrical, electronic and computer engineering.
- The required mathematical foundations will be studied at a fairly rigorous level and the applications of the probability theory and random processes to engineering problems will be emphasized.

Course Description:

UNIT-I

Definition of Stochastic Processes – Markov chains: definition, order of a Markov chain – Higher transition probabilities.

UNIT-II

Classification of states and chains – denumerable number of states and reducible chains

UNIT-III

Markov process with discrete state space: Poisson process – and related distributions– properties of Poisson process, Generalizations of Poisson processes –

UNIT-IV

Birth and death processes – continuous time Markov chains.

UNIT-V

Markov processes with continuous state space - Introduction, Brownian motion - Problems

UNIT-VI

Weiner process and differential equations for it, Kolmogrov equations – Problems

UNIT-VII

First passage time distribution for Weiner process – Ornstein – Uhlenbech process.

UNIT-VIII

Branching processes : Introduction – properties of generating functions of Branching process

UNIT-IX

Probability of extinction – Distribution of the total number of progency conditional limit laws due to kolmogrov and due to Yaglom,

UNIT-X

The classical Galton and Watson process – Bellman Hari's process

UNIT-XI

Stochastic processes in Queueing Systems -Concepts – Queueing model M/M/1 – transient behaviour of M/M/1 model

UNIT-XII

Birth and death process in Queueing theory: M/M/1 model and related distributions – M/M/ ∞/ - M/M/S/S – loss system – M/M/S/M.

UNIT-XIII

Non birth and death Queueing process -Bulk queues – $M^{(x)}$ / M/1 model-Problems

UNIT-XIV

Bulk service system- poisson queue with general bulk service rule – M/M(a,b)/1 model – M/M/M(K,K)/1 model – M/M(1,b)/1 – M/M (a,∞)/ 1 model.

REFERENCES:

1. J.Medhi – Stochastic Processes – New age international Private limited – Second edition –1993.
2. Gregory F.Lawler – Introduction to Stochastic Process, Chapman and Hall / CRC

Learning Outcomes

After successful completion of this course, students will be able to

- possess the basic knowledge about stochastic processes in the time domain.
- acquire more detailed knowledge about Markov processes with a discrete state space, including Markov chains, Poisson processes & birth and death processes.
- know about queuing systems and Brownian motion, in addition to mastering the fundamental principles of simulation of stochastic processes and the construction of Markov chain Monte Carlo (MCMC) algorithms.
- formulate simple stochastic process models in the time domain and provide qualitative and quantitative analyses of such models.

SEMESTER-IV

Course Code	Title of the Course
31141	GRAPH THEORY

Course Objectives:

The objective of the course is to:

- introduce students with the fundamental concepts of graph theory, with a sense of some of its modern applications.
- use these methods in subsequent courses in the design and analysis of algorithms, computability theory, software engineering, and computer systems.
- cover a variety of different problems in Graph Theory.
- come across a number of theorems and proofs.
- prove theorems which will be stated formally using various techniques.
- learn various graphs algorithms which will also be taught along with its analysis.

Course Description:

UNIT-I

Graphs – Subgraphs – Graph Isomorphism – Incidence and adjacency matrices – Vertex degrees

UNIT-II

Graphs – Walk- path - cycle – Bipartite graphs

UNIT-III

Trees – Cut Edges and Bonds – Cut vertices- Cayley's formula

UNIT-IV

Connectivity – Blocks – Euler tours – Hamiltonian cycles -Closure of a graph – Chavatal theorem for Non-Hamiltonian simple graphs.

UNIT-V

Matchings- Matchings and coverings in a Bipartite Graphs-Perfect Matchings

UNIT-VI

Independent sets – Cliques – Ramsey's numbers

UNIT-VII

Edge colourings- Edge chromatic Number – Vizing's Theorem

UNIT-VIII

Vertex colouring – Brook's theorem – Hajo's conjecture – Chromatic polynomials.

UNIT-IX

Planar graphs – Plane and Planar graphs - Dual graphs – Euler's formula

UNIT-X

Bridges – Kuratowski's Theorem – The Time table Problem

UNIT-XI

The five colour theorem – Non-Hamiltonian planar graphs

UNIT-XII

Directed graphs – Directed Path – Directed Cycles

UNIT-XIII

Networks - Flows – Cuts - Problems

UNIT-XIV

Max-Flow Min-cut theorem and Applications

REFERENCES:

1. J.A.Bondy and U.S.R Murty, *Graph Theory with Applications* Macmillan, London.
2. A Text book of Graph Theory , Balakrishnan. R, Ranganathan .K, Second Edition, Springer.
3. Invitation to Graph Theory, S.Arumugam and S.Ramachandran, Scitech Publications India

Learning Outcomes:

Upon completion of the course, students should possess the following skills:

- understand the basic concepts of graphs, directed graphs, and weighted graphs and able to present a graph by matrices.
- understand the properties of trees and able to find a minimal spanning tree for a given weighted graph.
- understand Eulerian and Hamiltonian graphs.

Course Code	Title of the Course
31142	FUNCTONAL ANALYSIS

Course Objectives

The objective of the course is to:

- the study of spaces of functions.
- introduce the students to the basic concepts and theorems

Course Description:

UNIT-I

Normed space - Banach space – Properties of Normed spaces

UNIT-I

Convex sets- Quotient spaces-Equivalent norms

UNIT-III

Finite dimensional normed spaces and subspaces- Compactness and finite dimension

UNIT-IV

Linear operators – Bounded linear operators

UNIT-V

Linear functional – Normed spaces of operators..

UNIT-VI

Continuous or bounded linear operators- Dual spaces

UNIT-VII

Inner product spaces-Definition and examples-Orthonormal sets and bases

UNIT-VIII

Annihilators-Projections

UNIT-IX

Hilbert space- Linear functionals on Hilbert spaces

UNIT-X

Reflexivity of Hilbert spaces

UNIT-XI

Riesz's theorem – Hilbert adjoint operator – Self-adjoint, unitary and normal operators.

UNIT-XII

Hahn – Banach theorem - Adjoint operator – Category theorem – Uniform boundedness theorem.

UNIT-XIII

Strong and weak convergence – Convergence of sequences of operators and functionals

UNIT-XIV

Open mapping theorem -Closed graph theorem

REFERENCES:

1. E. Kreyszig, *Introduction to Functional Analysis with Applications*, (John Wiley and Sons, 2006).
2. G. Bachman and L. Narici, *Functional Analysis*, (Academic Press, 1966)
3. F. Riesz and B. Sz. Nagay, *Functional Analysis*, (Dover Publications, Inc., 1965).

Learning Outcomes:

By the end of this course, students should be able to:

- describe the properties of normed linear spaces and construct examples of such spaces.
- extend basic notions from calculus to metric spaces and normed vector spaces.
- state and prove theorems about finite dimensionality in normed vector spaces.
- state and prove the Cauchy-Swartz inequality and apply it to the derivation of other inequalities.
- prove that a given space is a Hilbert spaces or a Banach Spaces.
- describe the dual of a normed linear space.

Course Code	Title of the Course
31143	NUMERICAL ANALYSIS

Course Objectives:

The general objective of the course is to:

- Derive appropriate numerical methods to solve algebraic and transcendental equations.
- develop appropriate numerical methods to approximate a function.
- develop appropriate numerical methods to solve a differential equation.
- derive appropriate numerical methods to evaluate a derivative at a value.
- derive appropriate numerical methods to solve a linear system of equations.
- perform an error analysis for various numerical methods.
- derive appropriate numerical methods to calculate a definite integral.
- code various numerical methods in a modern computer language.

Course Description:

UNIT-I

Transcendental and polynomial equations : Rate of convergence of iterative methods

UNIT-II

Methods for finding complex roots – Polynomial equations

UNIT-III

Birge – Vieta method, Bairstow's method, Graeffe's root squaring method.

UNIT-IV

System of Linear Algebraic equations and Eigen Value Problems : Error Analysis of direct and iteration methods

UNIT-V

Finding Eigen values and Eigen vectors – Jacobi and Power methods.

UNIT-VI

Interpolation and Approximation : Hermite Interpolations – Piecewise and Spline Interpolation - Bivariate Interpolation

UNIT-VII

Approximation – Least square approximation and best approximations.

UNIT-VIII

Differentiation and Integration : Numerical Differentiation – Optimum choice of Step – length – Extrapolation methods

UNIT-IX

Partial Differentiation – Methods based on undetermined coefficient – Gauss methods.

UNIT-X

Ordinary differential equations : Local truncation error – Problems

UNIT-XI

Euler, Backward Euler, Midpoint, -Problems

UNIT-XII

Taylor's Method –Related Problems

UNIT-XIII

Second order Runge Kutta method - Stability analysis.

REFERENCES:

1. M.K.Jain, S.R.K.Iyengar and R.K.Jain, Numerical Methods for Scientific and Engineering Computation, III Edn. Wiley Eastern Ltd., 1993.
2. Kendall E.Atkinson, An Introduction to Numerical Analysis, II Edn., John Wiley & Sons, 1983.
3. M.K.Jain, Numerical Solution of Differential Equations, II Edn., New Age International Pvt Ltd., 1983.
4. Samuel, D. Conte, Carl. De Boor, Elementary Numerical Analysis, McGraw Hill International Edn., 1983.

Learning Outcomes:

The students will become proficient in:

- understanding the theoretical and practical aspects of the use of numerical methods.
- implementing numerical methods for a variety of multidisciplinary applications.
- establishing the limitations, advantages, and disadvantages of numerical methods.
- demonstrate understanding of common numerical methods and how they are used to obtain approximate solutions.
- derive numerical methods for various mathematical operations and tasks, such as interpolation, differentiation, integration, the solution of linear and nonlinear equations, and the solution of differential equations.

Course Code	Title of the Course
31144	PROBABILITY AND STATISTICS

Course Objectives

The objective of the course is to:

- study the key concepts of probability, including discrete and continuous random variables, probability distributions, conditioning, independence, expectations, and moments.
- apply the basic rules and theorems in probability including Bayes's theorem and the Central Limit Theorem (CLT).
- apply the concepts of interval estimation and confidence intervals.
- apply the concepts of hypothesis testing t and F distributions.

Course Description:

UNIT-I

Probability and Distribution: Introduction – Set theory – The probability set function – Conditional probability and independence

UNIT-II

Random variables of the discrete type – Random variables of the continuous type –

UNIT-III

Properties of the distribution function – expectation of random variable – some special expectations – Chebyshev's Inequality.

UNIT- IV

Multivariate Distributions: Distributions of two random variables – Conditional Distributions and Expectations

UNIT-V

The correlation coefficient – Independent random variables – extension to several Random variables.

UNIT-VI

Some special Distributions: The Binomial and Related Distributions – The Poisson Distribution

UNIT-VII

The Gamma and Chi-square Distributions – The Normal Distribution – The Bivariate Normal Distribution.

UNIT-VIII

Distributions of functions of Random variables: Sampling Theory – Transformations of variables of the discrete type

UNIT-IX

Transformations of variables of the continuous type – the Beta, t and F distributions – Extensions of the change – of – variable Technique

UNIT-X

Distributions of order statistics – The Moment generating – Function, Techniques

UNIT-XI

The distributions of X and ns^2/σ^2 – Expectations of functions of Random variables

UNIT-XII

Limiting Distributions: Convergence in distribution – convergence in probability

UNIT-XIII

Limiting Moment Generating Functions – The Central Limit Theorem

UNIT-XIV

Some theorems on Limiting Distributions.

REFERENCES:

1. Introduction to Mathematical Statistics, (Fifth edition) by Robert V.Hogg and AllenT. Craig Pearson Education Asia.
2. M.Fisz, Probability, Theory and Mathematical Statistics, John Wiley and Sons, New York. 1963.
3. V.K.Rohatgi, An Introduction to Probability Theory and Mathematical Statistics, Wiley Eastern Ltd., New Delhi, 1988 (3rd Print).

Learning Outcomes:

Students who successfully complete this course should be able to demonstrate understanding of:

- basic probability axioms, rules and the moments of discrete and continuous random variables as well as be familiar with common named discrete and continuous random variables.
- how to derive the probability density function of transformations of random variables and use these techniques to generate data from various distributions.

- how to calculate probabilities, and derive the marginal and conditional distributions of bivariate random variables.
- discrete time Markov chains and methods of finding the equilibrium probability distributions.

4.4: PROJECT WORK / DISSERTATION

PROJECT WORK

- After the Completion of First Year, students are eligible to commence the Project work under the supervision of the qualified guide. The Candidates are permitted to submit the Project work on completing 18 months of the course but not later than five years after the commence of the course
- The Guide / Supervisor of the Project work shall be an approved guide of Alagappa University, Karaikudi or a person with an M.Phil Degree working with three years teaching experience in any Government or Government Aided College in Department of Mathematics with Ph.D. (Mathematics) qualification.
- The students shall submit the consent letter from the guide in the prescribed format before the commencement of the project work.
- The Project Report shall not exceed 150 Pages and be not less than 50 Pages
- The Project Report should be certified by the Approved Guide with Self Declaration of the Candidate for assuring the Quality and Originality of the work.
- There is an internal Viva-Voce examination for the Project Report submitted.

- **The Split up of marks for the project will be :**

1. Innovativeness	-	25 Marks
2. Methodology and Analysis	-	25 Marks
3. Reporting and Presentation	-	25 Marks
4. Viva – Voce examination	-	25 Marks
TOTAL :	**-**	**100 Marks**

Appendix II

(Model for wrapper and inside title page of Synopsis / thesis of the M.Sc work)

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Title of the Thesis

THESIS SUBMITTED TO ALAGAPPA UNIVERSITY IN PARTIAL FULFILMENT FOR THE AWARD OF THE DEGREE OF

MASTER OF SCIENCE

IN

MATHEMATICS

By

_____

(Name of the candidate)

(Register Number of the Candidate: _____)

Under the supervision of

_____

(Name of the Research Supervisor)

_____

DIRECTORATE OF DISTANCE EDUCATION

ALAGAPPA UNIVERSITY

[Accredited with A+ Grade by NAAC (CGPA: 3.64) in the Third Cycle]

KARAIKUDI – 630 003

INDIA

_____

Month and Year

[Note: The items in Italics as such are not to be scripted, but only the appropriate details pertaining to them need to be in the space provided]

e. 3. Duration of the Programme:

The programme for the Postgraduate degree in Mathematics shall consist of two academic years divided in to four semesters. Each semester consists of four theory Papers. Each theory course carries 4 credits and each semester consist of 16 credits.

e. 4. Faculty and Support Staff Requirements:

The programme for the Undergraduate degree in Mathematics requires the following faculty and supporting staff:

Staff Category	Required
Core Faculty*	3
Faculty – Specialization*	2
Clerical Assistant	1

* Faculty may belongs to at least Assistant Professor Level

e. 5. Instructional Delivery Mechanisms:

The instructional delivery mechanisms of the programme includes SLM – study materials, face to face contact session for theory course of the programme, e-content of the study materials in the form of CD, MOOC courses wherever applicable.

e. 6. Identification of Media:

The SLM – designed study materials will be provided in print media as well is in the form of CD which carries electronic version of the study material in addition to MOOC courses.

e. 7. Student Support Services:

The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like counselling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres. The post-admission student support services like issuance of identity card, study materials, etc. will be routed through the Learning Centres. The face to face contact sessions of the programme for both theory and practical courses will be held at the head quarter and Learning Centres. The conduct of end semester examinations, evaluation and issuance of

certificates will be done by office of the controller of examinations, Alagappa University, Karaikudi.

f. Procedure for Admission, curriculum transaction and evaluation:

f. 1. Procedure for Admission:

A candidate who has passed 12th standard (10 + 2 +3 system) of any University accepted by the Syndicate as equivalent there to shall be eligible to appear and qualify for the M.Sc Degree in Mathematics of this University after a course of study of three academic years.

f. 2. Curriculum Transactions:

The classroom teaching would be through chalk and talk method, use of OHP, Power Point presentations, web-based lessons, animated videos, etc. The face to face contact sessions would be such that the student should participate actively in the discussion. Student seminars would be conducted and scientific discussions would be arranged to improve their communicative skill.

The face to face contact sessions will be conducted in following durations;

Course Type	**Face to Face Contact Session per Semester (in Hours)**
Theory Courses (4 courses with 4 credits each)	64
Total	**64**

f. 3. Evaluation:

The evaluation shall be conducted separately for theory and practical to assess the knowledge acquired during the course of study. There shall be two systems of evaluation viz., internal assessment and end semester examinations.

In the case of theory courses the internal evaluation shall be conducted as Continuous Internal Assessment via. Student assignment preparation. The internal assessment shall comprise of maximum 25 marks for each subject. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

f. 3.1. Question Paper Pattern:

Answer all questions (one question from each unit with internal choices Time: 3 Hours Max. Marks: 75
Part A- 10 x 2 Marks = 20 Marks
Part B -5 x 5 Marks = 25 Marks
Part C- 3 x 10 Marks = 30 Marks

f. 3.2. Distribution of Marks in Continuous Internal Assessments:

The following procedure shall be followed for awarding internal marks for **theory** courses

Component	**Marks**
Assignments (5 questions per course)	25
Total	**25**

f. 3.3. Passing Minimum:

The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50% (50 marks out of 100 marks) in the University end semester examination. Then the total marks secured by the candidate will be taken and added with his/her internal marks (Maximum marks 25).

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

f. 3.4. Marks and Grades:

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester
G_i = Grade Point obtained for course i in any semester.
n refers to the semester in which such courses were credited

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

Grade Point Average = $\frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

CGPA = $\frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{}$

Sum of the credits of the courses for the entire programme

CGPA	Grad	Classification of Final Result
9.5-10.0 9.0 and above but below 9.5	O+ O	First Class- Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	D++ D+ D	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	A++ A+ A	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	B+ B	Second Class
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

f. 3.5. Maximum duration for the completion of the course:

The maximum duration for completion of M.Sc., Degree in Mathematics programme shall not exceed ten semesters from their sixth semester.

f. 3.6. Commencement of this Regulation:

These regulations shall take effect from the academic year 2018-2019 (June session) i.e., for students who are to be admitted to the first year of the course during the academic year 2018-2019 (June session) and thereafter.

f. 4. Fee Structure:

The programme has the following Fee Structure:

Sl. No.	Fees Detail	Amount in Rs.	
		First Year	**Second Year**
1	Admission Processing Fees	**300.00**	**-**
2	Tuition Fees	**4400.00**	**4400.00**
4	ICT Fees	**150.00**	**150.00**
	TOTAL	**4850.00**	**4550.00**

The above mentioned fee structure is exclusive of Exam fees.

g. Requirement of the laboratory support and Library Resources:

The students who have enrolled themselves in M.Sc., Mathematics Programme shall attend the face to face contact session for Theory Courses at their respective Learning Centres.

Directorate of Distance Education, Alagappa University, Karaikudi housing an excellent Library facility with adequate number of copies of books in relevant titles for M. Sc., Mathematics programme. The Central Library of Alagappa University also having good source of reference books. The books available at both the libraries are only for reference purpose and not for lending services.

h. Cost estimate of the programme and the provisions:

The cost estimate of the programme and provisions for the fund to meet out the expenditure to be incurred in connection with M. Sc., Mathematics programme. as follows:

S.No.	Expenditure Heads	Approx. Amount in Rs.
1	Programme Development	10,00,000/-
2	Programme Delivery	20,00,000/-
3	Programme Maintenance	3,00,000/-

i. Quality assurance mechanism and expected programme outcomes:

i. 1. University's Moto:

'Excellence in Action'

i. 2. University's Vision Statement:

Achieving Excellence in all spheres of Education, with particular emphasis on "PEARL"- Pedagogy, Extension, Administration, Research and Learning.

i. 2. University's Objectives:

1. Providing for Instructions and Training in such Branches of Learning as the University may determine.
2. Fostering Research for the Advancement and Dissemination of Knowledge

i. 3. University's Quality Policy:

Attaining Benchmark Quality in every domain of 'PEARL' to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

i. 4. University's Quality Quote:

Quality Unleashes Opportunities towards Excellence (QUOTE)

i.5. Programme's Review Mechanism:

The quality of the programme depends on scientific construction of the curriculum, strong-enough syllabi, sincere efforts leading to skillful execution of the course of the study. The ultimate achievement of M.Sc., Mathematics programme of study may reflect the gaining of knowledge and skill in the subject. And all these gaining of knowledge may help the students to get new job opportunities, upgrading in their position not only in employment but also in the society, make students feel thirsty to achieve in research in the fields associated with the discipline- Mathematics achieving in competitive examinations on the subject.

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations. Apart from the end semester examination-based review feedback from the alumni, students, parents and employers will be received and analyzed for the further improvement of the quality of the M.Sc., Mathematics Programme.

MINUTES OF THE MEETING OF THE BOARD OF STUDIES IN MATHEMATICS (DDE) HELD ON 17.06.2017 AT 2.00 p. m. IN THE DEPARTMENT OF MATHEMATICS, ALAGAPPA UNIVERSITY, KARAIKUDI.

Members Present

1. Dr. N. Anbazhagan - Chairman
2. Dr. M. Marudai - Member
3. Dr. R. Uthayakumar - Member
4. Dr. R. Asokan - Member
5. Dr. M. Mullai - Member
6. Dr. J. Vimala - Member

The chairman of the Board Dr. N. Anbazhagan welcomed the members.

1. Board of Studies in Mathematics has thoroughly discussed the B. Sc., (Mathematics), M. Sc., (Mathematics) syllabi and made necessary changes and corrections in the existing syllabi of all the above said programmes.

2. The corrected syllabi is enclosed herewith.

Dr. N. Anbazhagan | Dr. M. Marudai | Dr. R. Uthayakumar

Dr. R. Asokan | Dr. M. Mullai | Dr. J. Vimala

(PTO)

PROGRAMME PROJECT REPORT FOR M.Sc. MICROBIOLOGY

DIRECTORATE OF DISTANCE EDUCATION
ALAGAPPA UNIVERSITY
(A State University Accredited with 'A^{+}' Grade by NAAC (CGPA: 3.64) in the Third cycle)
KARAIKUDI – 630 003

Open and Distance Learning (PPR)
Course: M.Sc., Microbiology
Credit Based System (CBS)
(With effective from June 2018 – 2019 onwards)

A. Programme's mission and objectives

Microbes are omnipresent and indispensable to human life. Interactions of microbes involved in soil, environment, food, fermentation, medical, or agriculture have been studied using modern techniques. Increased attention has been directed towards use of microorganisms for wastewater treatment (bioremediation) and management of industrial effluents including distillery waste, textile industries and paper industries. Genetic improvement of strains can be explored in future for the production of biofuels. Use of recombinant microorganisms for industrial production of useful compounds has reached at commercial levels.

M.Sc. Microbiology is a two year postgraduate program. Microbiology is introduced with Microbial Physiology, Biochemistry, virology, immunology, Industrial and Agricultural microbiology are mainly grown applied sciences.

This programme is designed to provide the theoretical knowledge, understanding and practical skills to ensure a solid grounding in core and applied microbiology, along with the transferable skills which are essential to enhance your employability options. This aims to prepare graduates for a broad range of careers within industrial, commercial, governmental, environmental settings and further postgraduate study.

B. Relevance of the program with HEI's Mission and Goals

Our University Mission is "Affording a High Quality Higher Education to the learners. So that they are transformed in to intellectually competent human resources that will help in the uplift of the nation to educational, social, technological, environmental and economic magnificence" regarding that we have paved way to a tremendous rise in job opportunities for microbiologists in various industrial sectors including pharmaceuticals, agrochemicals, fertilizers, food products, government setups for environment protection, hospitals, research institutes, pathology labs to mention a few. The curriculum and syllabus, of M.Sc., (Microbiology) course at Alagappa University, have been designed to encompass basic as well as applied knowledge of microbiology to produce industry ready and employable microbiologists capable of availing promising professional carrier in life.

C. Nature of prospective target group of learners

- Working Professionals
- Entrepreneurs
- Service Personnel
- Academic Faculty
- Government Officials
- Researchers
- Home Makers
- Unemployed Graduates

D. Appropriateness of programme

The M.Sc., Microbiology programme provides opportunities for students to develop, demonstrate, understanding, skills, qualities of knowledge and other attributes in the following areas:

- Be impressed on the milestones of Microbiology and present status
- Be able to understand in depth the techniques used in Microbiology
- How fundamental chemical principles and reactions are utilized in biochemical processes
- Be impressed on the special fermentation by specific microbes
- To know the value, production, application, crop response of biofertilizers and biopesticides
- The students will be able to know the Industrial production of various products
- The students will be able to understand the Rules and regulation of industrial microbiology
- The students are able to know the alternate energy sources and biomass energy

After completion, jobs are available with a wide range of organisations in the public, private and not-for-profit sectors. Typical employers include:

- Schools, colleges, science centres, libraries and museums
- Universities and research institutes
- Government agencies and research institutions
- Chemical, pharmaceutical and petroleum companies
- Food manufacturing and processing companies
- Medical research establishments and the National Health Service
- Milk and milk product related industries
- Water treatment plants

E. Instructional design

Curriculum Design

Sem	M.Sc Microbiology		Cr.	Marks		Total
	Course Code	Title of the course		Int.	Ext.	
		FIRST YEAR				
I	36411	General Microbiology	4	25	75	100
	36412	Microbial Biochemistry	4	25	75	100
	36413	Microbial Physiology	4	25	75	100
	36414	Lab I- General Microbiology, Microbial Physiology and Biochemistry	4	25	75	100
			16	**--**	**--**	**400**
II	36421	Microbial Genetics	4	25	75	100
	36422	Molecular Biology & r DNA Technology	4	25	75	100
	36423	Food & Dairy Microbiology	4	25	75	100
	36424	Lab II-, Microbial Genetics, Molecular Biology & r DNA Technology, Food & Dairy Microbiology	4	25	75	100
			16	**--**	**--**	**400**
		SECOND YEAR				
III	36431	Immunology	4	25	75	100
	36432	Medical Microbiology	4	25	75	100
	36433	Environmental & Agriculture Microbiology	4	25	75	100
	36434	Lab III- Immunology, Medical Microbiology and Environmental & Agriculture Microbiology	4	25	75	100
			16	**--**	**--**	**400**
IV	36441	Bioprocess Technology	4	25	75	100
	36442	Microbial Biotechnology	4	25	75	100
	36443	Bioinformatics and Biostatistics	4	25	75	100
	36444	Lab IV- Bioprocess Technology & Microbial biotechnology	4	25	75	100
			16	--	--	**400**
		Grand Total	**64**	**--**	**--**	**1600**

Course Code Legend:

3	6	4	X	Y

364 - M. Sc., Microbiology Programme
X - Semester Number
Y - Course Number in the Semester

I semester	= 16 credits
II Semester	= 16credits
III semester	= 16credits
IV semester	= 16 credits
Total credits	= 64
Total Marks	= 1600

I YEAR – I SEMESTER
COURSE CODE: 36411
GENERAL MICROBIOLOGY

Objective

1. To inculcate knowledge on fundamentals of microorganisms.
2. To learn the structural organization, morphology and reproduction of microbes.
3. To know the principles of Microscopy and advancements in Microscopy.

Outcome

1. Knowledge on historical perspective of Microbiology.
2. Basic knowledge on different structure of microbes.
3. Ideas on different type of microscope.

BLOCK-1: History and Classification of Microorganisms

Unit I
Introduction to Microbiology, Haeckel's three–kingdom concept, Whittaker's Five-kingdom concept, Three-domain concept of Carl Woese.

Unit II
Classification of Bacteria according to Bergey's Manual.

Unit III
Fungi: Classification of fungi based on Alexopoulos system. - characteristics of Fungi, Industrial uses of yeast and moulds.

BLOCK-2: Microscopy, Staining techniques, Growth and Preservation methods

Unit IV
Simple, Compound, Dark-field, Phase contrast, Fluorescent and Electron microscopes. (SEM & TEM), Confocal microscopy – Principles and their applications.

Unit V
Stains and staining techniques: Simple, Differential, Structural staining methods and Imaging techniques.

Unit VI
Auxenic and synchronous, aerobic and anaerobic, Culture media and Nutritional types, Growth curve, Generation time and growth kinetics. Factors influencing microbial growth.

Unit VII
Preservation methods of microbes for storage, Sterilization and disinfection.

BLOCK-3: Prokaryotic and Eukaryotic cell structure

Unit VIII
Prokaryotic cell structure & Organization, Cell membrane, Plasma membrane, Cytoplasmic matrix, Inclusion bodies, Ribosome, Nucleiod, Prokaryotic cell wall, Capsule, Slime layers, S layers, Pili and Fimbriae, Flagella and Motility. Bacterial endospores. Archaeal cell structures.

Unit IX
General characters and classification of Blue green Algae (Cyanobacteria) Macroalgae - Biological and Economic importance of algae. Protozoa – structural characteristics, classification and reproduction.

Unit X
Eukaryotic cell structure and its organelles. Lichens and Microalgae- Structural organization and their properties.

BLOCK-4: Virology

Unit XI
Nomenclature and Classification of viruses.

Unit XII
Distinctive properties of viruses; morphology and ultrastructure. Capsids and their arrangements- types of envelopes and their composition

Unit XIII
Viral genome, their types and structures. Life cycle of virus.

Unit XIV
Virus related agents (viroids, prions).

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1. Tortora, G.J., Funke B.R, and Case C.L. (2010). **Microbiology an Introduction** (10th Edition), Benjamin Cummins, USA.
2. Dubey, R.C. and Maheswari, D.K. (2013). **A Textbook of Microbiology** (Revised Edition), S.Chand and Company Ltd., New Delhi.
3. Prescott, L.M., Harley, J.P. and Klein, D.A. (2014). **Microbiology** (9th Edition), McGraw Hill Publishers, Boston.
4. Brock, T.D., Smith, D.W. and Madigam, M.T. (2002). **Biology of Microorganisms** (Fourth Edition) Prentice Hall International, London.
5. Stanier, R.Y., Ingraham, J.L., Wheels, M.L. and Painter, P.R. (1999). **General Microbiology**, Mac Millan Educational Limited, London.
6. Boyd, R.F. (1998). **General Microbiology**, MosbyCollege Publishing, St. Louis.
7. Nester, E.W., Roberts, C.V. and Nester, M.T. (1995). **Microbiology**, A Human Perspective. IWOA, U.S.A.
8. Pelzcar, M.J., Chan, E.C.S. and Kreig, N.R. (1993). **Microbiology**, McGraw Hill Inc., New Delhi.
9. Carter, J.B and Saunders, V.A. (2007). Virology- Principles and Applications. John Wiley and Sons Ltd, UK.
10. Hull, R. (2002). **Mathews Plant Virology**. (4th Edition. Academic Press- A Harcourt Science and Technology company, New York.

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I YEAR – I SEMESTER
COURSE CODE: 36412
MICROBIAL BIOCHEMISTRY

Objective

1. To know the structural organization of bio-molecules.
2. To learn the characteristics of nucleic acids and enzymes.
3. To acquire knowledge on secondary metabolites.

Outcome

1. Knowledge on metabolism of bio – molecules.
2. General Information about nucleic acids, enzymes and vitamins.
3. Clear idea on secondary metabolites.

BLOCK-1: Carbhohydrates and Proteins

Unit I

Carbohydrates- Classification, Structure, properties of monosaccharides and disaccharides. polysaccharides - starch, cellulose, agar- agar and peptidoglycan.

Unit II

Metabolism and its regulation- Gluconeogenesis, glycolysis, kreb's cycle, pentose phosphate pathway or hexose mono phospate shunt, glyoxylate cycle and Entner Doudroff pathway.

Unit III

Amino acid and proteins- Classification based on structure, polarity, biological importance and reactivity, physical properties and chemical reactions.

Unit IV
Biosynthesis of amino acids – an overall view. Protein - Classification, physical and chemical properties. Structure – primary, secondary, tertiary and quaternary structure of proteins.

BLOCK-2: Lipids and Nucleic acids

Unit V
Lipids and fatty acids- Classification and properties. Phospholipid and cholesterol synthesis in *E.coli*.

Unit VI
Lipids and fatty acids metabolism - α, β and γ oxidation of fatty acids and lipid peroxidation.

Unit VII
Nucleic acids- Structure, synthesis and degradation of purines and pyrimidines.

BLOCK-3: Enzymes

Unit VIII
Enzymes- Classification, chemical nature and properties of enzymes. Factors affecting enzyme activity and Active site of enzyme.

Unit IX
Enzyme inhibition- Reversible, irreversible, Allosteric inhibition, Enzyme specificity and co-enzymes.

Unit X
Mechanism of enzyme action- Michaelis - Menten hypothesis, Lock and key model, induced fit theory. Isozyme, ribozyme and abzyme.

BLOCK-4: Pigments, Secondary metabolites and Vitamins

Unit XI
Microbial pigments – chlorophyll, fluorescence, phosphorescence, bacteriochlorophyll, rhodopsin, carotenoids and phycobiliproteins.

Unit XII
Secondary Metabolites- Antibiotics – Classification based upon mode of action. Biosynthesis and regulation of penicillin and streptomycin.

Unit XIII
Microbial Toxins – *Salmonella* toxin, *Cholera* toxin, Botulism toxin and Aflatoxin.

Unit XIV
Vitamins – Classification, Properties and functions, Vitamins as Co – factors and Co – enzymes.

References

1. David, A. B. (2009). **Nutritional biochemistry of Vitamins**, Cambridge.
2. Deb, A.C. (2006). **Fundamentals of Biochemistry**, New Central Book Agency Pvt. Ltd., Kolkata.
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6. Jain, J.L. (2008). **Fundamentals of Biochemistry** (5th Edition), S. Chand and Company Ltd, NewDelhi.
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8. Moat, A.G. and Foster, W. (2002). **Microbial Physiology** (4th Edition), John Wiley and Sons, New York.
9. Nelson, D.L. and Cox, M.M. (2012). **Lehingers's Principles of Biochemistry** (6th Edition), Mac Millan worth Publishers, New Delhi.
10. Srivastava, M.L. (2008). **Microbial Biochemistry**, Narosa Publishing House, New Delhi.

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I YEAR – I SEMESTER
COURSE CODE: 36413
MICROBIAL PHYSIOLOGY

Objective

1. To develop a sufficient background to students about the growth of Microbes.
2. To learn the microbial metabolism.
3. To acquire knowledge on microbial stress response.

Outcome

1. Knowledge on growth of Microbes.
2. General Information about microbial metabolism.
3. Clear idea on microbial communications.

BLOCK-1: Bacterial Growth and Nutrition

Unit I

Growth of Bacteria- Phases of growth. Growth kinetics - batch culture, continuous culture and synchronous culture. Factors affecting growth - nutrition, aeration, temperature and pH.

Unit II
Nutritional diversity in microorganisms, nutritional types - autotrophy, heterotrophy, chemotrophy, phototrophy, lithotophy and organotropy. Nutrition –essentiality of major and minor elements.

Unit III
Chemotrophism and their importance, reverse electron transport, CO_2 assimilation and reductive acetyl COA pathway.

Unit IV
Chemoheterotrophism - Acetogens, methanogens, methanogenenesis and its importance. Physiology and economic importance of methylotrophs.

BLOCK-2: Bacterial Photosynthesis, Nitrogen metabolism and Stress response

Unit V
Bacterial Photosynthesis- General types of microbial photosynthesis, oxygenic and anoxygenic. Structure of photosynthetic pigments – chlorophylls, bacteriochlorophyll, carotenoids and phycobilins.

Unit VI
Photosynthetic bacteria - green sulphur and purple. Mechanism of photosynthesis non-cyclic and cyclic electron transport. Photo phosphorylation.

Unit VII
Microbial stress responses- Osmotic stress and osmoregulation, Aerobic to anaerobic transitions, Oxidative stress, pH stress and acid tolerance, Thermal stress and heat shock response, Nutrient stress and starvation stress.

BLOCK-3: Microbial Metabolism

Unit VIII
Nitrogen metabolism- Nitrogen cycle - ammonification, nitrification, denitrification and nitrogen fixation. Nitrogenase enzyme, physiology of nitrogen fixation in symbiotic and free living bacteria.

Unit IX
Aerobic respiration- TCA cycle- intracellular location and reactions, amphibolic reactions. Glyoxalate cycle.

Unit X
Mechanisms of substrate – level phosphorylation. Respiratory electron transport in mitochondria and bacteria. Mechanism of oxidative phosphorylation.

Unit XI
Anaerobic respirations- sulphate, nitrate, carbonate respirations and their ecological significance.

BLOCK-4: Bacterial Transport and Quorum sensing

Unit XII

Bioenergetics- Entropy, enthalpy, electron carriers, artificial electron donars, inhibitors, uncouplers, energy bond and phosphorylation.

Unit XIII

Transport across membrane - diffusion, osmosis, active transport and group translocation.

Unit XIV

Quorum sensing- Mechanism and Signaling molecules.

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I YEAR – I SEMESTER
COURSE CODE: 36414
LAB I- GENERAL MICROBIOLOGY, MICROBIAL PHYSIOLOGY& BIOCHEMISTRY

Suggested Laboratory Exercises:

GENERAL MICROBIOLOGY

1. Principle and methods of sterilization.
2. Preparation of media: nutrient broth, nutrient agar plate, and soft agar.

3. Pure culture techniques: streak plate, spread plate and pour plate.
4. Motility determination – Hanging drop method.
5. Isolation and enumeration of bacteria from different environmental samples.
6. Enumeration of bacteria - viable count (plate count) and total count (Haemocytometer count).
7. Direct microscopic observation of fungal spores and mycelium.
8. Fungal slide culture.
9. Staining method: simple, negative, Gram's staining and spore staining.

BIOCHEMISTRY

10. Measurement of growth rate and generation time by turbidometry method.
11. pH metry -Preparation of buffer.
12. Spectrophotometry- Wavelength scan.
13. Chromatography - Paper and Thin layer chromatography - separation of amino acid.

MICROBIAL PHYSIOLOGY

14. Carbohydrates: Quantitative estimation of glucose, glycogen from bacterial and yeast cell.
15. Protein- Quantitative estimation of protein from bacterial yeast cell.
16. Enzyme- Estimation of alkaline phosphatase activity.
17. Environmental factor- Effect of temperature and pH on bacterial growth.
18. Physiological groupings of bacteria- Isolation of saccharophilic microorganisms (starch hydrolysis)- Proteolytic activity of microorganisms (casein and gelatin hydrolysis)- Lipolytic activity of microorganisms.
19. Utilization of Unusual compounds- Microbial degradation of azodyes.
20. Bioenergetics- Cytochrome oxidase assay - Catalase assay.
21. Nitrogen metabolism- Nitrate reduction test.

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I YEAR – II SEMESTER
COURSE CODE: 36421
MICROBIAL GENETICS

Objective

1. To extend the knowledge on molecular basis of mutation at microbial level.
2. To focus on gene regulation and expression mechanisms.
3. To understand the principles role of plasmids and gene transfer methods.

Outcome

1. Receive elaborate knowledge on mutation.
2. Better understanding about gene regulation.
3. Get thorough knowledge on gene transfer mechanisms in microbes.

BLOCK-1: Mutation, DNA damage and repair

Unit I
Mutation and its types, Mutation rate and its determination and Mutagenesis.

Unit II
Mutagens-Types, Physical mutagens, DNA reactive chemicals, base analogs, intercalating agents, metals and biological agents.

Unit III
DNA damages- Deamination of bases, alkylation, damage due to reactive oxygen, UV induced damage.

Unit IV
Repair pathways (Methyl-directed mismatch repair, Nucleotide excision repair, Base excision repair, recombinational repair, SOS inducible repair, specific repair for oxidative DNA damage, pyrimidine dimers and alkylation induced damage and adaptive response).

BLOCK-2: Bacterial Recombination

Unit V
Recombination- Types of recombinations, Models for Homologous recombination, Molecular mechanism of homologous recombination, Homologous recombination in eukaryotes, Mating type switching. Molecular mechanism for site-specific recombination. Biological roles of site specific recombination.

Unit VI
Conjugation- Conjugation by *E. coli* F factor, Structure of F-factor, Regulation of F-factor fertility, establishment of cell contact, DNA mobilization, transfer and separation of mating

pair, Hfr conjugation and chromosomal transfer, Interrupted mating and conjugational mapping.

Unit VII
Transformation- Mechanism of Natural competence and transformation in *Bacillus subtilis* and *Streptococcus pneumoniae*. Transformation by inducing artificial competence, Gene linkage and mapping by transformation.

Unit VIII
Transduction- Generalized and specialized transduction.

BLOCK-3: Operon concept and Extra Chromosomal Materials

Unit IX
Gene concept- regulation of bacterial gene expression. Lactose system - coordinate regulation, Lac components, positive, negative regulation and catabolite repression.

Unit X
Tryptophan operon - attenuation. Arabinose operon and its regulation.

Unit XI
Plasmids- Types of plasmids - F, R & Col plasmids. Properties of plasmids – sex factors, drug resistant, colicinogenic. Agrobacterium Ti and broad host range plasmid.

BLOCK-4: Transposable Elements and Epigenetics

Unit XII
Detection and purification of plasmid DNA. Transfer of plasmid DNA. Replication of plasmid. Control of copy number, plasmid amplification, curing and incompatability.

Unit XIII
Transposable elements- Types of transposable elements, Structure, genetic organization and mechanism of transposition of Tn5, Tn3 and related transposons, Bacteriophage Mu, Tn7 and IS911, Integrons, Retrotransposons.

Unit XIV
Epigenetics- Definition, Molecular basis, Mechanisms, Functions and Epigenetics in bacteria.

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I YEAR – II SEMESTER
COURSE CODE: 36422
MOLECULAR BIOLOGY & rDNA TECHNOLOGY

Objective

1. To extend the knowledge on structure and functions of genetic material
2. To focus on genome organization, transcription and translation process in Prokaryotes.
3. To understand the concepts of cloning vectors.

Outcome

1. Receive elaborate knowledge on nucleic acid
2. Better understanding of gene expression systems.
3. Get thorough knowledge on PCR and DNA sequencing techniques.

BLOCK-1: DNA and its replication, RNA and its types

Unit I

Discovery of DNA- Molecular basis of DNA as genetic material. Structure of DNA – A, B and Z form. Forms of DNA – DNA heteroduplex, circular, superhelical DNA, twisted circle. Properties of DNA - denaturation, renaturation, melting curve and hyperchromicity.

Unit II

Replication of DNA- semi conservative mode, Meselson - Stahl experiment. Enzymology of DNA replication - DNA polymerase I, II &III, topoisomerase I & II, helicase, primase, gyrase. Molecular basis of DNA replication - replication fork, origin, okazaki fragments. Types of replication - circular and theta.

Unit III

Structure of RNA - types of RNA - tRNA, mRNA and rRNA

BLOCK-2: Protein Synthesis & Cloning vectors

Unit IV
Transcription process in Prokaryotes- Initiation - promotors, upstream and downstream sequences, transcription factors. Elongation - RNA polymerase, sub units. Termination - Rho dependent and Rho independent, nus A protein. antitermination.

Unit V
RNA processing (post transcriptional modifications), inhibitors of transcription. Reverse transcription.

Unit VI
Cloning vectors – plasmids, cosmids, phasmids ,phagemids, expression vectors, plasmid vectors – $p^{BR}322$ and $p^{UC}18$, integrating shuttle vector –YAC vectors, viral vector – SV 40 and adeno virus.

BLOCK-3: Gene Cloning

Unit VII
Cloning of human insulin, interferon in *E.coli.* Recombinant vaccine development – HBs Ag in yeast.

Unit VIII
Cloning for commercial production of antibiotics (Penicillin).

Unit IX
Cloning methodologies- α complementation, sticky and blunt end cloning. Cloning from mRNA – synthesis of cDNA, cloning cDNA– cDNA library. Cloning from genomic DNA – genomic library. Shot gun cloning.

Unit X
Screening of recombinant – phenotypic expression of characters. Blotting techniques – western, northern and southern. Mapping of human genes – Human genome project.

BLOCK-4: Molecular Techniques

Unit XI
PCR- gene amplification, primer designing, optimization, variation in the PCR (RAPD and RFLP)

Unit XII
DNA sequencing – Sanger – Coulsen's method, Maxam Gilbert's method, automated sequencing and micro array.

Unit XIII
Gene silencing and antisense technology- Types and mechanism of gene silencing. Genetic factors of silencing, formation of antisense mRNA, inhibition of gene expression by antisense RNA.

Unit XIV
Plant genetic engineering- Ti plasmid, CaMV vector, Direct DNA delivery methods – micro projectile bombardment, microinjection and electroporation.

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I YEAR – II SEMESTER
COURSE CODE: 36423
FOOD & DAIRY MICROBIOLOGY

Objective
1. To learn about food spoilage factors.
2. To know the preservation methods.
3. To make aware of food borne disease.

Outcome
1. Better understanding of cause of microbes in food spoilage.
2. Get information regarding food preservation.
3. Enable them to work food fermentation industries.

BLOCK-1 Introduction to food microbiology

Unit I
General introduction of foods and related microorganisms.

Unit II
Intrinsic factors- Nutrient Content, pH and Buffering Capacity, Redox Potential, Antimicrobial Barriers and Constituents and Water Activity.

Unit III
Extrinsic factors- - Relative Humidity, Temperature and Gaseous Atmosphere.

BLOCK-2: Food contamination, Spoilage and Preservation

Unit IV
Contamination and spoilage of cereals, cereal products, fruits, vegetables, meats, meat products, fish, sea foods, eggs, poultry and canned foods.

Unit V
Food poisoning and food borne infections (bacterial, viral, fungal and protozoa), bacterial and fungal toxins.

Unit VI
General principles of food preservation- Physical and Chemical methods.

BLOCK-3: Dairy Microbiology, Microbial Products

Unit VII
Dairy microbiology- Normal flora of milk and milk products, Spoilage of milk and milk products.

Unit VIII
Fermented milk products- acidophilus milk, bifidus milk ,yoghurt manufacture of cheese and evolution of quality milk.

Unit IX
Microbial food fermentation- Fermentation in food processing, role of microorganisms in food fermentation.

Unit X
Microbial products of food; SCP, mushrooms, oriental foods Fermented beverages (fruit and cereal based).

BLOCK-4: Microbial enzymes, Quality control and Quality Assurance

Unit XI
Industrial production of enzymes- cellulases, amylases, proteases, phytases, pectinases, lipases and glucose isomerases.

Unit XII
Food borne disease outbreaks - Objectives of investigation, Field investigation, lab testing and preventive measures.

Unit XIII
Food sanitation – Microbiology of food plant sanitation, water and milk testing.

Unit XIV

Food laws and quality control – HACCP, Codex alimentarius, PFA, FPO, MFPO, BIS and AGMARK.

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I YEAR – II SEMESTER
COURSE CODE: 36424
MICROBIAL GENETICS, MOLECULAR BIOLOGY & r DNA TECHNOLOGY and FOOD & DAIRY MICROBIOLOGY

Suggested Laboratory Exercises:

MOLECULAR BIOLOGY& r DNA TECHNOLOGY

1. Single colony isolation and checking for genetic markers.
2. Measurement of growth-one step growth curve using a T even phage.
3. Titration of phages (T4).
4. Induction of Lambda Phage.
5. Induced mutagenesis – UV.
6. Isolation of antibiotic resistant mutants.
7. Isolation of auxotrophic mutants.

MICROBIAL GENETICS

8. Isolation of specialized transducing phage.
9. Bacterial conjugation – transfer of drug resistant factor (Plasmid).
10. Transposon mutagenesis of chromosomal and plasmid DNA.
11. Isolation of plasmid and chromosomal (bacterial) DNA.

12. Quality and quantity checking of DNA by UV Spectrophotometer and Submarine agarose gel electrophoresis.
13. Gene cloning – Preparation of vector and passenger – Ligation – Preparation of competent cells – Transformation of *E.coli* with recombinant plasmids.
14. Selection of recombinants by blue white selection.
15. PCR amplification – Demo.

FOOD & DAIRY MICROBIOLOGY

16. Resazurin dye reduction test.
17. Phosphatase test.
18. Litmus milk reactions.
19. Potability analysis of drinking water.
20. Bacterial examination of water (qualitative and quantitative).
21. Membrane filtration technique.

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2. Bruce Birren et al., (1997). **Genome Analysis. A Laboratory Manual. Vol I – Analysing DNA**, Cold Spring Harbor Laboratory Press.
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II YEAR – III SEMESTER
COURSE CODE: 36431
IMMUNOLOGY

Objective

1. To provide knowledge on human immunity system.
2. To understand the mechanism of antigen antibody reaction.
3. To inculcate the principles of vaccine development.

Outcome

1. Students acquire the information about immunity development.
2. Become an eminent in immunotechnology.
3. Able to understand the immunological reactions.

BLOCK-1: Immune system and Immune cells

Unit I
Basic concepts and terminologies in immunology. Haematopoiesis.

Unit II
Immune system- primary and secondary lymphoid organs: structure and functions and cells of immune system.

Unit III
Innate and acquired immune system- cells and molecules involves. Cell mediated and humoral mediated response. Role of Toll like receptors in innate immunity.

Unit IV
Maturation and Differentiation of T-cell and B-cell. T-cell and B-cell receptors.

BLOCK-2: cytokines, Antigen Antibody interactions

Unit V
Characteristics and functions of cytokines, haemokines.

Unit VI
Immunoglobulins- class, subclass, structure and fuction, Immunoglobulins genes- Organization and expression. Generation of antibody diversity.

Unit VII
Immunogenicity- Immunogens, adjuvants, epitopes, haptens and carriers.

Unit VIII
T dependent and T independent antigens. Strength of antigen-antibody interactions- affinity, avidity, valency, agglutination and precipitation.

BLOCK-3: Complement system and Major Histocompatibility Complex

Unit IX
Complement systems- mode of activation, classical and alternate pathway.

Unit X
Mechanisms of antigen processing and presentation-cytosolic and endocytic pathways. Antibody engineering.

Unit XI
Major histocompatibility complex (MHC)- structure and its interaction with peptide.

BLOCK-4: Hypersensitivity, Transplantation and Vaccines

Unit XII
Hypersensitivity reactions- Type I, II, III and IV. Autoimmune disorders.

Unit XIII
Organ transplantation and HLA tissue typing. Oncogenes and antioncogenes.

Unit XIV
Hybridoma and monoclonals. Vaccine – Introduction- types- live or attenuated, killed and DNA vaccines. Stem Cells and its clinical application- Human pluripotent stem cells (bone marrow, nerve cells, heart muscle cells and pancreatic islet cells).

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II YEAR – IV SEMESTER
COURSE CODE: 36432
MEDICAL MICROBIOLOGY

Objective

1. To inculcate on the role of normal flora and pathogenic microbes.
2. To understand the pathogenesis of various diseases.
3. To understand the various clinical microbiological techniques.

Outcome

1. Get information about various mechanisms of infection.
2. Knowledge on clinical lab techniques.
3. Acquire knowledge on control measures of diseases.

BLOCK-1: Introduction to Medical Microbiology

Unit I
Laboratory management – Safety in containment laboratory. Collection and transport of clinical samples.

Unit II
Microbiological examination of urine, blood, faeces, cerebrospinal fluid, throat swabs, sputum, pus and wound exudates.

Unit III
Normal flora of human systems – skin, respiratory tract, gastrointestinal tract and genitourinary tract. Nosocomial infections.

BLOCK-2: Bacterial diseases

Unit IV
General characters, pathogenesis, laboratory diagnosis and control measures of- Gram positive cocci disease – pharyngitis, pneumonia. Gram negative cocci– gonorrhea.

Unit V
Gram positive non spore forming bacilli – nocardiosis, diphtheria. Gram positive spore forming bacilli- aerobic- anthrax and anaerobic- Tetanus (Lockjaw).

Unit VI
General characters, pathogenesis, laboratory diagnosis and control measures of- Gram negative nonspore forming bacilli- Aerobic- pertussis. Small gram negative facultative anaerobic bacteria – Yersiniosis.

Unit VII
Enteric Gram negative bacilli – Vibriosis, *S*almonellosis. Acid fast bacteria – tuberculosis and leprosy. Cell wall less bacteria – pneumonia and Leptospirosis.

BLOCK-3: Viral and Fungal Diseases

Unit VIII
Influenza, Measles, Mumps, Chicken pox, Hepatitis A, B, C, D & E , Poliomyelitis, AIDS, Human Papilloma virus, Rabies and Yellow fever.

Unit IX
Dengue, Japanese Encephalitis, SARS and Swine Flu.

Unit X
Mechanism of pathogenesis and lab diagnosis of fungal diseases: Superficial Mycoses, Cutaneous Mycoses and Subcutaneous Mycoses.

Unit XI
Systemic mycosis, Opportunistic Mycoses and Mycotoxicosis.

BLOCK-4: Parasites, Newly Emerged Diseases and Control Mechanism

Unit XII
Mechanism of pathogenesis and lab diagnosis of Protozoology- Amoebiasis amd Malaria.

Unit XIII
Classification of antibiotics based on mode of action- antibacterial (Penicillin), antiviral (Amantidine), antifungal (Amphotericin) and antiparasitic drugs (Quinine and Metraindazole).

Unit XIV
Emerging and re-emerging infections (Chickungunya, Zika virus, H1N1 and Ebola). National programmes in prevention of infectious diseases.

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II YEAR – III SEMESTER
COURSE CODE: 36433
ENVIRONMENTAL AND AGRICULTURAL MICROBIOLOGY

Objective

1. To create awareness on Environmental Microbiology.
2. To give knowledge on plant pathogen interaction and its control.
3. To inculcate on biofertilizer and biopesticides.

Outcome

1. Acquire knowledge on agriculture microbiology.
2. Understand the biogeochemical cycles prevail in environment.
3. Able to know about global environmental problems.

BLOCK-1: Ecosystem

Unit I

Environment and Ecosystems- Definitions, biotic and abiotic environment. Environmental segments. Composition and structure of environment. Conservation and Management.

Unit II

Concept of biosphere, communities and ecosystems. Ecosystem characteristics, structure and function. Food chains, food webs and trophic structures. Ecological pyramids.

Unit III

Eutrophication- Definition, causes of eutrophication, and microbial changes in eutrophic bodies of water induced by various inorganic and organic pollutants.

BLOCK-2: Waste water treatment, Xenobiotic degradation, Environmental problems

Unit IV

Types of solid Waste, Treatment of solid wastes -composting, vermiform composting, saccharification and gasification.

Unit V

Types of liquid Waste, Treatment of liquid wastes –primary, secondary and tertiary treatment- anaerobic (methanogenesis), aerobic, trickling, activated sludge and oxidation pond.

Unit VI

Microbiology of degradation of xenobiotics (heavy metals) in the environment- ecological considerations, decay behaviour, biomagnifications, degradative plasmids and substituted hydrocarbons.

Unit VII

Global environmental problems- Ozone depletion, UV-B, green house effect and acid rain, their impact and biotechnological approaches for management. Containment of acid mine drainage applying biomining [with reference to copper extraction from low grade ores].

BLOCK-3: Soil Microbiology and Biogeochemical cycle

Unit VIII

Soil as an environment for microorganisms- Classification of soil, physical and chemical properties of soil and structure of soil.

Unit IX

Microbial interactions between plants– phyllosphere, mycorrhizae, rhizosphere and symbiotic association in root nodules.

Unit X
Biogeochemical cycles - carbon, phosphorus and sulfur.

BLOCK-4: Plant Disease Control and Management

Unit XI
Plant pathogens and classification of plant diseases- Host-pathogen recognition and specificity. Principles of plant infection - entry of pathogen in to host, colonization of host, role of enzymes, toxins and growth regulatory substances.

Unit XII
Defense mechanisms in plants - Structural and biochemical - Molecular aspects of host defense reactions - Lipoxygenase and other enzymes in the expression of disease resistance**.**

Unit XIII
Symptoms, Etiology, Epidemiology and management of the following plant diseases- Mosaic disease of tobacco, Bunchy top of banana, Bacterial blight of paddy and Grassy shoot of sugar cane.

Unit XIV
Plant disease management – exclusion, evasion, eradication and crop rotation. Sanitation - physical, chemical and biological control. Plant disease forecasting. Biotechnological approaches to disease management.

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II YEAR – III SEMESTER
COURSE CODE: 36434
IMMUNOLOGY, MEDICAL MICROBIOLOGY, ENVIRONMENTAL & AGRICULTURE MICROBIOLOGY

Suggested Laboratory Exercises:

IMMUNOLOGY

1. Collection of venous blood from human, separation and preservation of serum/plasma.
2. Blood Grouping.
3. Precipitation method- Immunodiffusion and Immunoelectrophoresis.
4. Latex Agglutination test.
5. Widal test (Tube and Slide Test).
6. ELISA.
7. Western Blotting.

MEDICAL MICROBIOLOGY

8. Isolation and identification of - Respiratory tract infections- *Pseudomonas aeruginosa,* Urinary tract infection- *Ecoli/ K. pneumonia.*
9. Fungal skin pathogens- *Dermatophytes* and *Candida.*

ENVIRONMENTAL & AGRICULTURE MICROBIOLOGY

10. Enumeration of microorganism from air.
11. Settle plate technique.
12. Air sampling technique.
13. Estimation of dissolved oxygen (DO), BOD and COD.
14. Isolation of free living nitrogen fixing bacteria from soil – *Azotobacter.*
15. Isolation of Symbiotic nitrogen fixing bacteria from root nodule – *Rhizobium.*
16. Examination of Plant Bacterial diseases- Sheath blight of rice and Wilt of potato
17. Fungal diseases – Late blight of potato and Wilt of cotton.
18. Viral diseases- Banana bunchy top virus and Tobacco Mosaic Virus.

References

1. Baily and Scott. (2017). **Diagnostic Microbiology** (14th Edition), Mosby London.
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II YEAR – IV SEMESTER
COURSE CODE: 36441
BIOPROCESS TECHNOLOGY

Objective

1. To give knowledge on strain improvement methods.
2. To learn about upstream fermentation process.
3. To understand about downstream fermentation process.

Outcome

1. Students will get knowledge on strain improvement.
2. Enable them to work in fermentation industry.
3. Students will get idea on upstream and downstream fermentation process.

BLOCK-1: Introduction to Fermentation Technology

Unit I

An overview of fermentation technology, range of fermentation processes, primary and secondary metabolites and components of fermentation process.

Unit II
Industrial microorganisms- isolation, preservation, screening, strain improvement and maintenance.

Unit III
Formulation of industrial media- Medium requirements for fermentation processes, carbon, nitrogen, mineral sources, buffers, antifoam agents and medium optimization.

Unit IV
Stoichiometry of cell growth and product formation, Sterilization of media and fementers, scale – up process and starter culture technology.

BLOCK-2: Fermentor and fermentations

Unit V
Basic design of a microbial fermentor, types of fermentation vessels. Aseptic operation and containment.

Unit VI
Body construction (stirrer glands, bearing, valves, steam traps) baffles, spargers and impellors.

Unit VII
Types of fermentations- batch, continuous, fed-batch, solid state and submerged.

Unit VIII
Aerobic and anaerobic, dual and multiple fermentations, their advantages and disadvantages.

BLOCK-3: Downstream processing

Unit IX
Importance of downstream processing in industrial fermentation processes. Problems and requirements of bio product recovery and purification.

Unit X
Downstream Processing- Recovery and purification of fermentation products- Removal of microbial cells and other solid materials, Foam separation, Precipitation, Filtration and Centrifugation.

Unit XI
Cell Disruption- Physical, Chemical methods, extractors, chromatography, membrane process, drying, crystallization and whole broth processing.

BLOCK-4: Fermentation Products and marketing

Unit XIII
A brief out lines of processes for the production of the following commercially important products – Organic acids- Citric acid, lactic acid. Amino acids- Glutamic acid, L – lysine. Solvents- Acetone, ethyl alcohol.

Unit XIV
Antibiotics- Streptomycin, penicillin and Vitamins- B12, Riboflavin.

Unit XII
Fermentation economics - Market potential, some effects of maintenance legislation on production of antibiotics and recombinant proteins.

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II YEAR – IV SEMESTER
COURSE CODE: 36442
MICROBIAL BIOTECHNOLOGY

Objective
1. To provide knowledge on application of microorganisms using technologies.
2. To understand the production and mechanism of microbial products.

Outcome
1. Students acquire the information about various uses of microorganisms.
2. Become an eminent in production of biofertilizers and pesticides.
3. Able to understand the remediation process through microbes.

BLOCK-1: Introduction to Microbial and Algal Biotechnology

Unit I
Introduction to microbial biotechnology- Scope and applications in human therapeutics, agriculture, food technology, SCP and environment.

Unit II
Algal Biotechnology- Application of cell fusion, tissue culture and hybridization techniques in algae. Algae genomics. Genetic engineering of algae- construction of transformation and expression vectors, methods of gene introduction.

Unit III
Biotechnological applications of Algae in agriculture and environment.

BLOCK-2: Biological Control

Unit IV
Microbial Pesticides- Basic principle- Antagonism, Amensalism- Siderophores, Parasitsm and Nematophagy.

Unit V
Microbial Herbicides – Formulation and their applications.

Unit VI
Microbial Insecticides- bacterial insecticide- *Pseudomaonas, Bacillus* sp., *Bacillus thuriengenesis*- toxins, BT cotton-viral insecticide- entamopathogenic fungi.

Unit VII
Pathogens- Antagonists, VAM fungi and modification of culture practices.

BLOCK-3: Commercially Important Products

Unit VIII
Cytokines, Human growth hormones, Tissue plasminogen activator and Factor VII.

Unit IX
Microbial polysaccharides and polyesters.

Unit X
Production, application of biocompost and biogas. Microbial fuel cells (Biodiesel and H_2 production).

Unit XI
Biosensor- Advantages of using microorganisms as biosensing elements, Immobilization of microorganisms, Electrochemical microbial biosensors, Optical microbial biosensor and other types.

BLOCK-4: GMO

Unit XII
Genetically Modified Microorganisms- Molecular Tools for Genetic Engineering of Microorganisms.

Unit XIII
Applications of GMM- Derived Products- Human Health, Agriculture and environment.

Unit XIV
Ethical issues raised by genetically modified microorganisms.

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1. Lodish, H., Berk A., Kaiser, C.A., Krieger, M., Scott, M. P., Brtscher A.,Ploegh, H., and Matsudaria, P. (2008). **Molecular Cell Biology** (6th Edition), W. H. Freeman, USA.
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II YEAR – IV SEMESTER
COURSE CODE: 36443
BIOINFORMATICS & BIOSTATISTICS

Objective

1. To create awareness on usage of computer aided biology.
2. To learn about individual nucleotide and protein sequence analysis.
3. To understand the concepts of data collection.

Outcome

1. Better understanding of evolutionary relationship of various species.
2. Get more knowledge on structure of proteins and nucleotides.
3. Able to understand the role of biostatistics in research.

BLOCK 1: Introduction to Bioinformatics

Unit I
Biology in the Computer age- Computational approaches to Biological questions, Basics of computers- servers and workstations.

Unit II
Operating systems- UNIX, Linux, World Wide Web. Search engines, finding scientific articles- Pubmed- Public biological databases.

Unit III
Genomics- Sequence analysis- Sequencing genomes- sequence assembly- pairwise sequence comparision- genome of the web- annotating and analyzing genome sequences.

BLOCK 2: Database

Unit IV
Genbank-sequence queries against biological databases- BLAST and FASTA-multifunctional tools for sequence analysis.

Unit V
Multiple sequence alignments, Phylogenetic alignment- profiles and motifs.

Unit VI
Proteomics- Protein data Bank, Swiss prot- biochemical pathway databases, Predicting protein structure and function from sequence-Determination of structure- feature detection-secondary structure prediction- predicting 3D structure- protein modeling.

BLOCK 3: Introduction to Biostatistics

Unit VII
Biostatistics- definition, scope, applications in biology, terminology, sampling techniques-random and non-random methods.

Unit VIII
Measures of central tendencies - Mean, mode, median, standard errors and standard deviations.

Unit IX
Probability - concepts, terminology, kinds of probabilities, theorems of probability, normal, binomial and poisson distribution. Skewness and kurtosis.

Unit X
Chi Square test- characteristics, degrees of freedom, test of goodness of fit and null hypothesis.

BLOCK 4: Statistical Analysis

Unit XI
Analysis of variance (ANOVA)- Methods of ANOVA, one way and two way classifications, F-test, steps involved in ANOVA and its importance.

Unit XII
Correlation- Definition, methods of studying the correlation, scatter diagram, Karl Pearson's efficient of correlation and rank correlation methods and types of correlations.

Unit XIII
Regression- Definition, types of regression analysis, regression equation, methods of studying regression, graphic and algebraic methods and importance of regression.

Unit XIV
Importance of statistical software in data analysis.

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1. Ewans, W.J. and Gregory, G. (2005). **Statistical methods in Bioinformatics: An Introduction** (Statistics for Biology and Health) Springer.
2. Stanton, A and Clantz, S. (2005). **Primer of Biostatistics**, The McGraw Hill Inc., New York.
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II YEAR – IV SEMESTER
COURSE CODE: 36444
INDUSTRIAL MICROBIOLOGY& MICROBIAL BIOTECHNOLOGY

Suggested Laboratory Exercises:

INDUSTRIAL MICROBIOLOGY

1. Inoculum preparation for fermentation.
2. Screening of antibiotic producing microorganisms from soil.
3. Production of extracellular metabolites from actinomycetes.
4. Production of industrially important enzymes by Submerged fermentation (Any one enzyme).
5. Production of industrially important enzymes by Solid state fermentation (Any one enzyme).
6. Assay of extracellular enzymes produced by bacteria : a) Amylase, b) Protease and c) Lipase.
7. Purification of enzymes by Filtration method / Chemical method by ammonium sulphate.
8. Wine production.
9. Microbial Production of citric acid by using *Aspergillus niger.*
10. Separation of biomass – Wet and Dry mass.
11. Immobilization of cells and enzymes.

MICROBIAL BIOTECHNOLOGY

12. Isolation of dye degrading microorganism.
13. Antibiotic sensitivity test - a) Kirby Bauer's method and b) MIC determination by filter paper assay and broth dilution assay.
14. Separation of proteins - a) Paper chromatography, b) Column chromatography.

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Duration of the Course

The course for the degree of Master of Science in Microbiology shall consist of two academic years with four semester pattern.

Faculty and Support Staff requirements

The programme for the degree of Master of Science in Microbiology requires following faculty and supporting staff:

Staff Category	Required
Faculty for Core Microbiology Subjects #	3
Faculty for Specialty Subject viz., Biochemistry, Bioinformatics, etc., #	2
Laboratory Assistant	1
Clerical Assistant	1

Faculty may be belongs to at least Assistant Professor level either permanent or part time.

Instructional Delivery Mechanisms

Each semester there will be one contact programme of 168 hours duration in total comprising of both theory and practical (120 hours practical and 48 hours theory).

Number of Assignment/Semester	Contact Sessions - Practical	Contact Sessions - Theory
4	120 hours guided experiments with support of Internal supervisor for 4 credits	48 Hours for four Assignments

The class room teaching would be through conventional lecture, use of OHP, power point presentation and novel innovative teaching ideas like television and computer aided instruction. Student seminars would be arranged to improve their awareness and communicative skill. In the laboratory, instruction would be given for the safe handling of chemicals and instruments. The practical experiments shall be conducted with special efforts to inculcate scientific knowledge among students. The students shall be trained to handle advanced instrumental facilities and shall be allowed to do experiments individually.

Identification of Media

1. Online – uses the Internet to deliver content; courses may be synchronous or asynchronous; generally widely available to learners.
    a. Web-conferencing – connecting learners through the Internet; access is often through a host established link; may use phone or computer for audio connection
    b. Digital media – includes podcasts to portable electronic devices or computers with audio, video, and documents; often involves downloading content and/or materials.
    c. Social media – connecting learners who are part of a social network; may offer text, audio and video connectivity.
2. Offline – uses physical media (e.g., DVD, USB Drive, memory card) to deliver content; asynchronous; hosted locally rather than using the Internet; generally on a single computer; uses recorded and stored text, audio, and video content

3. Teaching Mode: Applying mode of teaching such as SLM, MOOC, Virtual lab and face to face for understanding of education easily.

Student Support Service Systems

Faculty are often among the first to notice when students are experiencing problems and sometimes those are problems we are not able, nor qualified, to help them with. To find out more about your role in helping students in need, we've provided links to a number of resources the university provides:

1. Counseling
2. Academic support for student
    a. note-taking
    b. time-management
    c. exam preparation

F) Procedure for admission, curriculum transaction and evaluation

Eligibility

A candidate who has passed Bachelor's Degree in Biological Sciences (Microbiology, Biochemistry, Biotechnology, Botany, Zoology, Bioinformatics, agricultural/ Veterinary/ Fisheries Sciences/ Pharmacy) degree with at least 50% of marks and 45% marks for SC/ST candidates eligible for applying this programme. Preference in admission will be given to candidates working as teacher or scientific assistants.

Curriculum Transactions

The classroom teaching would be through chalk and talk method, use of OHP, Power Point presentations, web-based lessons, animated videos, etc. The face to face contact sessions would be such that the student should participate actively in the discussion. Student seminars would be conducted and scientific discussions would be arranged to improve their communicative skill.

For practical courses exclusive study materials containing the requirements, procedure for the experiments will be issued to the learners. In the laboratory, instruction would be given for the experiments followed by demonstration and finally the students have to do the experiments individually.

The face to face contact sessions will be conducted in following durations;

Course Type	Face to Face Contact Session per Semester (in Hours)
Theory Courses (3 courses with 4 credits each)	48
Practical Courses (1 course with 4 credits)	120
Total	**168**

Evaluation

The evaluation shall be conducted separately for theory and practical to assess the knowledge acquired during the course of study. There shall be two systems of evaluation viz., internal assessment and end semester examinations.

In the case of theory courses the internal evaluation shall be conducted as Continuous Internal Assessment via. Student's assignment preparation, seminar, etc. The internal assessment shall comprise of maximum 25 marks for each subject. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

In the case of Practical courses for M.Sc. programme in Microbiology should be conducted at every semester. The internal evaluation of practical courses will be done through continuous assessment of skill in demonstrating the experiments, record note preparation, and *viva voce*. The external evaluation consists of an end semester practical examinations which comprise of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

Scheme of External examination

The duration of examinations for theory and practical's shall be three hours.

Question paper pattern (Theory)

- The question paper carries a maximum of 75 marks.
- The question paper consists of three sections namely Part-A, Part-B and Part-C.
- Part-A consists of 10 questions of 2 marks each (10 x 2 = 20 marks) with no choice. Candidate should answer all questions.

- Part-B consists of 5 either or choice questions. Each question carries 5 marks (5 x 5=25 marks).
- Part-C consists of 5 questions. Each question carries 10 marks. The candidate should answer any three questions (10 x 3 = 30 marks).
- All blocks should be given equally importance for question setting.

Question paper pattern (Practical) (Maximum 75 marks)

Major practical	20 marks
Minor practical	10 marks
Spotters	25 marks
Viva voce	10 marks
Practical record note	10 marks
Total	**75 marks**

Distribution of Marks in Continuous Internal Assessments

The following procedure shall be followed for awarding internal marks for theory courses

Component	Marks
Student's Assignments	25
Total	**25**

The following procedure shall be followed for awarding internal marks for theory courses

Internal –Practical	Marks
Continuous Assessment (demonstration and observation of experiments)	10
Record Note Book	10
Viva	5
Total	**25**

Passing Minimum

The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50% (50 marks out of 100 marks) in the University end semester

examination. Then the total marks secured by the candidate will be taken and add with his/her internal marks (Maximum marks 25).

For a pass in the Practical paper, a candidate has to secure a minimum of 50 % (37.5 marks) marks in the University end semester practical examination (75 marks) examination. Then the total marks secured by the candidate will be taken and add with his/her internal marks (Maximum marks 25).

There is no passing minimum for the record notebook. However submission of a record notebook is a must.

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

Marks and Grades

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D^+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A^+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

$$\text{Grade Point Average} = \frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$$

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

$$CGPA = \frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses for the entire programme.}}$$

C_i = Credits earned for the course i in any semester.

G_i = Grade Point obtained for course i in any semester.

n refers to the semester in which such courses were credit.

CGPA	Grade	Classification of Final Result
9.5-10.0 9.0 and above but below 9.5	O^+ O	First Class- Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	D^{++} D^+ D	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	A^{++} A^+ A	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	B^+ B	Second Class
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

Maximum duration for the completion of the course

The maximum duration for completion of M.Sc., Degree in Microbiology programme shall not exceed ten semesters from their fourth semester.

Commencement of this Regulation

These regulations shall take effect from the academic year 2018-2019 (June session) i.e., for students who are to be admitted to the first year of the course during the academic year 2018-2019 (June session) and thereafter.

Fee Structure

The programme has the following Fee Structure:

Sl. No.	Fees Detail	Amount in Rs.		Nature of Fees
		First Year	Second Year	
1	Admission Processing Fees	**300.00**	**-**	Non- Refundable
2	Course Fees	**20000.00**	**20000.00**	Non- Refundable
5	ICT Fees	**150.00**	**150.00**	Non- Refundable
	TOTAL	**20450.00**	**20150.00**	Non- Refundable

The above mentioned fee structure is exclusive of exam fees

G. Laboratory Support

The postgraduate curriculum of Alagappa University has strived to offer both theory courses as well as laboratory and design practice in the field of Microbiology. Instruments, chemicals and glasswares are needed for conducting practicals to the better understanding of microbes.

Each semester there will be one contact programme of 168 hours duration in total comprising of both theory and practical (120 hours practical and 48 hours theory). Attendance for contact programmes is compulsory. The personal contact programme will be conducted only at Alagappa University campus, Karaikudi.

The practical books (manual) will be supplied to the students at the time admission. Hands on experience on the techniques of biological sciences will be given to the students individually during the contact sessions. The practical's will also be conducted and performed by applying virtual reality methods wherever necessary.

Library Resources

The Central Library is one of the important central facilities of Alagappa University. It has text book, reference books, conference proceedings, back volumes, standards, and non-book material such as CD-ROMs and audios. The central Library procured several e-books in different areas. The library also subscribes to about 250 current periodicals.

All routine functions of the library are automated with the help of an integrated library software package, SOUL, developed and distributed by UGC INFLIBNET. The database for the entire collection has been created and available through online Public Access Catalogue (OPAC) to the users via campus network. Now this facility is also available through Institute's Intranet as a web enabled OPAC.

H. Cost Estimate of the Programme and Provisions

Nature of Expenditure	Amount Required (Rs.)	Provision
Programme Development	**20,00,000**	University funds
Delivery	**24,00,000**	Fee collected from the students and university funds
Maintenance	**5,00,000**	Laboratory fee collected from the students

Provisions	Number
Class room + Smart class room	2
Laboratory	1
Staff room	1
Office	1

I. Quality assurance mechanism

1. Student's feedback about staffs, teaching methods and methodologies.
2. Teachers review about students.
3. Continuous assessment (Model practical).
4. The students give feedback on teaching every semester. Feedback is also taken on their campus experience through suggestion boxes.

Expected programme outcomes

1. This programme will provide higher education for rural people in this surrounding.
2. This programme will provide opportunity to study after breakup in their studies due to family circumstances especially women.

3. Gaining knowledge which helps job opportunities, research innovation, competitive exams and upgradation of their position.

Review Mechanism

We will get feedback from the students, alumni and employers from various sectors and analyze for further improvement of the quality curriculum transaction.

Minutes of the Meeting of the Board of Studies in Microbiology (For M.Sc. Microbiology Programme to be offered through ODL Mode) held at The Directorate of Distance Education, Alagappa University, Karaikudi – 630 003, on 05.09.2017, (11.00 A.M).

Members Present

1.	Dr. A. Arun	-	Chairman
2.	Dr. H. Shakila	-	Member
3.	Dr. E. Kannapiran	-	Member
4.	Dr. G. Muralitharan	-	Member
5.	Dr. P. Rameshthangam	-	Member
6.	Dr. M. JothiBasu	-	Member
7.	Dr. G. Selvakumar	-	Member

After the deliberation and discussion the board resolved the following:

1. The board considered the curriculum design and detailed syllabi of M.Sc., Microbiology programme, prepared as per the norms by the Chairman and the Board Members, scrutinized and suitably modified the same.

2. The board resolved to approve curriculum design, detailed syllabi and other regulations for the M.Sc., Microbiology programme to be offered by the Directorate of Distance Education of Alagappa University are given in Annexure I.

Dr. A. Arun | Dr. H. Shakila | Dr. E. Kannapiran | Dr. G. Muralitharan

Dr. P. Rameshthangam | Dr. M. Jothi Basu | Dr. G. Selvakumar

M.Sc. Physics

(Distance Learning Mode)

Curriculum (*2018-2019 onwards*)

for

Credit Based System

Programme Project Report

(Based on UGC-DEB 2017 guidelines)

Alagappa University

(A State University Accredited with A+ Grade by NAAC)

Karaikudi - 630 003

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Master of Science (M.Sc.) in Physics Programme
(Distance Learning Mode)

Credit Based System (CBS)
(***With effective from the academic year 2018 - 2019 onwards***)

a. Programme's Mission & Objectives:

The programme is named as Master of Science (M.Sc.) in Physics. The syllabus for this programme is framed under the rules of the Credit Based System (CBS). Physics is the natural science that involves the study of matter and its motion through space and time along with the related concepts such as energy and force. It is one of the most fundamental scientific disciplines. The main goal of Physics is to understand how the universe behaves. Physics explains the natural phenomena in the universe and often considered to be the most fundamental science. It provides a basis for all other sciences - without Physics, we could not have Biology, Chemistry, or anything else. Physics also makes significant contributions through advances in new technologies. One academic programme is necessary to create awareness to students in the emerging field and also it should teach basic concepts and developments of Physics to students to make them as scientist or technologists in this filed. Hence our task is to introduce the M.Sc. programme in Physics to educate the undergraduate students in the fascinating fields. Rigorous and comprehensive in approach, this syllabus presents essential contents in a detailed, clear and direct way. The programme is structured in such a way to impart more knowledge in science, in particular in Physics.

The major objectives of M.Sc. Physics programme are:

- To provide, thorough well designed studies of theoretical and experimental Physics, a worthwhile educational experience for all students.
- To acquire deep knowledge in the fundamental aspects of all branches of Physics.
- To acquire basic knowledge in the specialized thrust areas like Classical Mechanics, Quantum Mechanics, Mathematical Physics, Electromagnetic Theory, Electronics, Microprocessor & Electronic Instrumentation, Condensed Matter Physics, Nuclear and Particle Physics, Materials Science etc.,
- To develop abilities and skills that:
    - are relevant to the study and practice of science,

- are useful in everyday life,
- are encouraging efficient & safe practice and effective communication,
- are encouraging research and development activities.

- To develop attitudes relevant to science such as:
  - Concern for accuracy and precision,
  - Objectivity,
  - Integrity,
  - Enquiry,
  - Initiative and
  - Inventiveness.

b. Relevance of the Programme with Algappa University's Mission and Goals:

In order to align with the mission and goals of Alagappa University, the M.Sc. Physics Programme is planned to deliver in Distance Learning mode which may reach the maximum number of student aspirants who are unable to thrive to spend non-elastic timings of formal conventional class room education. Such a higher education in science subject with appropriate laboratory experiences will enrich the human resources for the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

c. Nature of Prospective Target Group of Learners:

This M.Sc. Physics Programme through Distance Learning mode is developed by keeping in mind to give opportunity to economically and socially excluded people including graduates of various socio-economic status viz., unemployed youths, employed with marginalized salary due to lack of sufficient knowledge in the subject Physics. Also, the target group of learners includes various level employees of Physics-based companies, secondary-level school teachers, research aspirants, women taking care of family - the important unit of the community, etc.

d. Appropriateness of programme to be conducted in Distance Learning mode to acquire specific skills and competence:

M.Sc. Physics Programme through Distance Learning mode is developed in order to give subject-specific skills including i) understanding of intermediate classical mechanics topics such as coordinate transformations, oscillatory motion, gravitation and other central forces, and

Lagrangian mechanics, ii) master the basic elements of mathematical physics, iii) analyze and design basic op-amp circuits, particularly various linear and non-linear circuits, iv) create and solve mathematical models of physical phenomena using analytic and numerical methods, v) develop the programming skills of microprocessor, vi) identify the fundamental models of nuclear structure that are used to describe various modes of nuclear excitation, v) provide insight into some of the steps in the production of semiconductor devices. In addition, the programme is developed to give a detailed theoretical exposure and free-hands experience on practical parts of the study in order to impart skills of Physics to the learners.

e. Instructional Design:

e. 1. Curriculum Design:

Sl. No.	Course Code No.	Title of the Course	Marks			Credit
			CIA	ESE	TOT	
FIRST YEAR						
I SEMESTER						
1	34511	Classical Mechanics	25	75	100	4
2	34512	Mathematical Physics - I	25	75	100	4
3	34513	Linear and Integrated Electronics	25	75	100	4
4	34514	Advanced Electronics and Physics Laboratory - I	25	75	100	4
Total			**100**	**300**	**400**	**16**
II SEMESTER						
5	34521	Quantum Mechanics - I	25	75	100	4
6	34522	Mathematical Physics - II	25	75	100	4
7	34523	Electromagnetic Theory	25	75	100	4
8	34524	Advanced Electronics and Physics Laboratory - II	25	75	100	4
Total			**100**	**300**	**400**	**16**
SECOND YEAR						
III SEMESTER						
9	34531	Molecular Spectroscopy	25	75	100	4
10	34532	Quantum Mechanics - II	25	75	100	4
11	34533	Microprocessor and Electronic Instrumentation	25	75	100	4
12	34534	Advanced Electronics and Physics Laboratory - III	25	75	100	4
Total			**100**	**300**	**400**	**16**
IV SEMESTER						
13	34541	Condensed Matter Physics	25	75	100	4
14	34542	Nuclear and Particle Physics	25	75	100	4
15	34543	Materials Science	25	75	100	4
16	34544	Advanced Electronics and Physics Laboratory - IV	25	75	100	4
Total			**100**	**300**	**400**	**16**

Course Code Legend:

3	4	5	X	Y

345 - M.Sc. Physics Programme

X - Semester No.

Y - Course number in the semester

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total

No. of Credits per Course (Theory):	4
No. of Credits per Course (Practical):	4
Total No. of Credits per Semester:	16
Total No. of Credits for the Programme:	64 (16 credits × 4 semesters)

e. 2. Detailed Syllabi:

I SEMESTER

34511	**CLASSICAL MECHANICS**
Objectives	*The objective of the syllabus is to impart knowledge to the students on the basic ideas of classical mechanics and its applications.*
Outcome	On successful completion of the course, a student will be able to • Explain clearly the notion of degrees of freedom and identify them for a given mechanical system • Explain clearly the notion of degrees of phase space • Demonstrate an understanding of intermediate classical mechanics topics such as coordinate transformations, oscillatory motion, gravitation and other central forces, and Lagrangian mechanics
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on Newton's laws, simple and compound pendulum, moment of inertia, relativity theory, energy and oscillator are prerequisite.
UNIT I	**NEWTON'S LAWS OF MOTION**
	Introduction – Newton's laws of motion- Kepler's laws of planetary motion- stability of orbit- Classification of a Dynamical System.
UNIT II	**LAGRANGE EQUATION**
	Lagrange's Equations for Simple Systems – Principle of Virtual Work – D'Alembert's principle – Lagrange's Equations for General Systems.
UNIT III	**HAMILTON EQUATION**
	Hamilton's Equations – Ignorable Coordinates – The Routhian Function.
UNIT IV	**HAMILTONIAN METHODS**
	Introduction – Hamilton's principle – Hamilton's Principle for a Conservative System – Principle of Least Action.
UNIT V	**HAMILTON-JACOBI THEORY**
	Characteristic Function and Hamilton-Jacobi Equation.
UNIT VI	**CANONICAL TRANSFORMATIONS**

	Phase Space and Liouville's Theorem – Special Transformations – Lagrange Brackets – Poisson Bracket – Calculus of Variations.
UNIT VII	**KINEMATICS OF RIGID BODY MOTION**
	Moments and Products of Inertia – Moment of Inertia of a Body about any Line – Through the origin of coordinate frame – The momental Ellipsoid – rotation Coordinate Axes.
UNIT VIII	**RIGID BODY EQUATIONS OF MOTION**
	Principal Axes and Principal Moments – Kinetic Energy of a Rigid Body Rotating about a Fixed Point – Angular Momentum of a Rigid Body – Eulerian Angles – The Compound Pendulum.
UNIT IX	**SPECIAL THEORY OF RELATIVITY**
	Some Fundamental Concepts: Theory of relativity, Equivalence of space and time.
UNIT X	**LORENTZ TRANSFORMATION**
	The Lorentz Transformation – Immediate Consequences of Lorentz transformations : contraction of length, time dilation, composition of velocities – The Mass of a Moving Particle – Equivalence of Mass and Energy.
UNIT XI	**ONE DIMENSIONAL OSCILLATOR**
	Potential Energy and equilibrium - one dimensional oscillator: stable, unstable and neutral equilibrium.
UNIT XII	**NORMAL MODES**
	Two coupled oscillators - normal coordinates and normal modes.
UNIT XIII	**GENERAL THEORY OF SMALL OSCILLATIONS**
	General theory of small oscillations: secular equation and eigenvalue equation.
UNIT XIV	**LINEAR TRIATOMIC MOLECULE**
	Small oscillations in normal coordinates - vibrations of a linear triatomic molecule.
Book For Study	
1. Classical Mechanics, K. Sankara Rao, PHI Learning Private Limited – 2009, New Delhi.	

2. Classical Mechanics- J.C.Upadhyaya, Himalaya Publishing House-2nd Edition, 2010.	
References	
1. Classical Mechanics- Herbert Goldstein –Pearson publishers-3rd Edition, 2011. 2. Classical mechanics – S.L. Gupta, Meenakshi Prakashan, New Delhi, 1970. 3. Introduction to Classical Mechanics - R.G. Takwala and P.S. Puranik, Tata - McGraw Hill, New Delhi, 1980.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34512	**MATHEMATICAL PHYSICS – I**
Objectives	*The main objective of this paper is to impart understanding on the fundamental thoughts of mathematical physics to the students.*
Outcome	On successful completion of the course, a student will be able to • Master the basic elements of mathematical physics and demonstrate an ability to use vector analysis, matrices and special functions in the solution of physical problems
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on vectors, matrix, integration, differentiation and Laplace transforms are prerequisite.
UNIT I	**VECTOR ANALYSIS**
	Introduction to vectors and product of vectors – Gradient, Divergence, curl.
UNIT II	**INTEGRATION OF VECTORS**
	Gauss's Theorem – Stoke's Theorem – Potential Theory – Gauss's law and Poisson's Equation – Dirac Delta function.
UNIT III	**ORTHOGONAL CURVILINEAR COORDINATES**
	Special Coordinate systems – Circular, Cylindrical Coordinates – Orthogonal coordinates – Differential Vector Operators – Spherical Polar Coordinates.
UNIT IV	**ALGEBRA OF MATRICES**
	Introduction to Matrix - Properties of matrix.
UNIT V	**RANK OF MATRIX**
	Rank of matrix and some of its theorems - Cramers rule - Characteristic equation.
UNIT VI	**EIGEN VALUES, EIGEN VECTOR, DIAGONALIZATION**
	Eigen values, eigenvectors, Orthogonal Matrices – Hermitian Matrices and Unitary Matrices – Diagonalization of Matrices - Solving differential equations.
UNIT VII	**GAMMA AND BETA FUNCTIONS**

	Gamma function - Beta function – Relation between Gamma and Beta function.
UNIT VIII	**LEGENDRE FUNCTION**
	Legendre's differential equation: Legendre polynomials - Generating functions - Recurrence relation - Rodrigue's formula – Orthogonality.
UNIT IX	**BESSEL'S FUNCTION**
	Bessel's differential equation: Bessel polynomials - Generating functions - Recurrence relation -Rodrigue's formula – Orthogonality.
UNIT X	**HERMITE FUNCTION**
	Hermite differential equation – Generating functions – Hermite polynomials - Recurrence relations – Rodrigue's formula – Orthogonality.
UNIT XI	**LAGUERRE FUNCTION**
	Laguerre differential equations – Generating functions - Laguerre polynomials - Recurrence relation - Rodrigue's formula – Orthogonality.
UNIT XII	**FOURIER TRANSFORM**
	Introduction and Definitions – Fourier Transforms – Development of the Inverse Fourier Transform – Fourier Transforms Inversion Theorem.
UNIT XIII	**FOURIER TRANSFORM OF DERIVATIVES**
	Fourier Transform of Derivatives – Convolution Theorem – Momentum Representation.
UNIT XIV	**LAPLACE TRANSFORM**
	Laplace Transforms – Laplace Transform of Derivatives – Other Properties – Convolution or Faltungs Theorem – Inverse Laplace Transform-applications.

Book For Study

1. Essential Mathematical Methods for Physicists, George B. Arfken, Hanes J.Weber, Frank E. Harris, 7th Edition, Elsevier, 2012.
2. Advanced Engineering Mathematics, Erwin Kreyszig, 9th Edition, Wiley, 2014.

References	
1. Mathematical Physics, B.D. Gupta, Vikas Publishing House Pvt. Ltd, 2010. 2. Mathematical Physics, Sathyaprakash, Sultan Chand, 2013. 3. Topics in Mathematical Physics-Parthasarathy H, Ane Books Pvt. Ltd. 2007.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34513	**LINEAR AND INTEGRATED ELECTRONICS**
Objectives	*The objective of the course is to impart in depth knowledge about Semiconductors, Diodes, Transistors, Operational amplifiers, etc to the students. The theoretical knowledge gained in the class room can be experimented in the practical classes.*
Outcome	On successful completion of the course, a student will be able to • Discuss the op-amp's basic construction, characteristics, parameter limitations, various configurations and countless applications of op-amp • Analyze and design basic op-amp circuits, particularly various linear and non-linear circuits, active filters, signal generators, and data converters
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on transistor, diode, amplifier, oscillators and semiconductor are prerequisite.
UNIT I	**SEMICONDUCTOR PHYSICS**
	Introduction to semiconductor – Intrinsic and extrinsic semiconductors – PN junction.
UNIT II	**SEMICONDUCTOR DIODE**
	Forward and Reverse bias of diode – Characteristics of forward and reverse bias of diode – Zener diode, its construction and characteristics – Zener diode as voltage regulator.
UNIT III	**SPECIAL-PURPOSE DIODES**
	Schottky diode – Tunnel diode - LED's.
UNIT IV	**TRANSISTORS**
	Transistor action – Transistor connections: CB, CE and CC configurations.
UNIT V	**TRANSISTOR AMPLIFIER**
	Transistor as an amplifier: Class A, Class B and Class C operations and their field of application.
UNIT VI	**TRANSISTOR BIASING**
	Transistor biasing and stabilization – Need for biasing – DC load line – operating point – Bias stability – Transistor biasing circuits –Fixed bias circuit – Base bias with emitter feedback – Base bias with collector feedback – Voltage

	divider bias circuit.
UNIT VII	**TRANSISTOR AUDIO POWER AMPLIFIER**
	Class B push pull amplifier-Transformer coupled audio power amplifier.
UNIT VIII	**FIELD EFFECT TRANSISTORS AND SILICON CONTROLLED RECTIFIER**
	Construction, working and I/O characteristics of FET, JFET, MOSFET and SCR.
UNIT IX	**POWER ELECTRONICS AND OPTO ELECTRONIC DEVICES**
	Construction, working and I/O characteristics of DIAC and TRIAC, Solar cells, Photo detectors.
UNIT X	**SINUSOIDAL OSCILLATORS**
	Oscillator: Wien bridge – RC phase shift, Hartley, Colpitt's oscillators.
UNIT XI	**OPERATIONAL AMPLIFIER**
	Operational amplifier – Block diagram of OP-AMP – Equivalent circuit of IC 741.
UNIT XII	**ELECTRICAL PARAMETERS OF OP-AMP**
	Electrical parameters – Input offset voltage and current input bias current – Differential input resistance – Input capacitance – Output offset voltage and nullification – CMRR – Slew rate.
UNIT XIII	**APPLICATIONS OF OP-AMP**
	Applications of OP-AMP 741: Inverting, Non-inverting Amplifiers – Adder – Subtractor – Comparator – Differentiator – Integrator – Analog computation.
UNIT XIV	**ACTIVE FILTERS**
	Active filters: First order high pass and low pass filters, Band pass filter.

Books For Study

1. Electronic Principles – Albert Malvino, David J Bates, 7th Edition, McGraw Hill. (2007).
2. Op-Amps and linear integrated circuits – Ramakant A.Gayakwad, 4th Edition, Prentice Hall. (2009).

References	
1. Principles of Electronics – V.K.Mehta, 6th Revised Edition, S.Chand and Company. (2001). 2. Electronic Devices and circuits – David A.Bell, 4th Edition, Prentice Hall. (2007).	
Mode of Evaluation	Assignment/Seminar/Written Examination

34514	**ADVANCED ELECTRONICS AND PHYSICS LABORATORY - I**
Objectives	*The main objective of this practical paper is to exercise the practical in various advanced electronics and advanced analytical experiments to the students.*
Outcome	On successful completion of the course, a student will be able to • Understand the basic operations in electronic circuits • Develop the programming skills of Microprocessor • Understand the concept of ICs manufacturing • Appreciate the applications of Microprocessor programming • Understand the basic principles of the experiments • Understand simple concepts to demonstrate an experiment
Schedule	Contact Hours: 120 hours
Prerequisite	Basic knowledge on transistor, diode, amplifier, oscillators, semiconductor, spectrometer and microscope are prerequisite.
	(Any Fifteen of the following) 1. Study the characteristics of FET. 2. Study the characteristics of transistor (CE mode). 3. Study the characteristics of Zener diode and construct regulated DC power supply. 4. Construct the logic gates using discrete components. 5. Design of FET amplifier - CS Configuration. 6. Characteristics of UJT. 7. Relaxation oscillator (UJT). 8. Transistor Astable multivibrator. 9. Monostable multivibrator (Transistor).

	10. Transistorized Hartely and Colpitt's audio oscillator. 11. Calibration of Spectrograph – Iron or Copper spectrum. 12. Michelson's Interferometer. 13. q, n, σ - Elliptical fringes. 14. q, n, σ - Hyperbolic fringes. 15. Ultrasonic Interferometer – Construction of oscillator and measurements. 16. Babinet's Compensator and study of polarized light. 17. G.M Counter – Statistical probability, Absorption measurements, Half life. 18. Any of the experiments of equal standard.

II SEMESTER

34521	**QUANTUM MECHANICS-I**
Objectives	*The main objective of this paper is to impart knowledge on the fundamental aspects of quantum mechanics to the students.*
Outcome	On successful completion of the course, a student will be able to • Know the background for the main features in the historical development of quantum mechanics • Discuss and interpret experiments displaying wavelike behaviour of matter, and how this motivates the need to replace classical mechanics by a wave equation of motion for matter (the Schrödinger equation) • Understand the central concepts and principles of quantum mechanics: the Schrödinger equation, the wave function and its physical interpretation, stationary and non-stationary states, time evolution and expectation values
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on atom, particle and wave, light and time are prerequisite.
UNIT I	**FOUNDATIONS**
	Wave particle duality – Uncertainty principle – applications – Postulates of quantum mechanics.
UNIT II	**SCHRODINGER EQUATION**
	Schrodinger equation – both time dependent and independent – Ehrenfest's theorem – eigen function and eigen vectors – probability density.
UNIT III	**ONE DIMENSIONAL PROBLEM**
	Applications to one dimensional problems: Linear harmonic oscillator and tunnel effect.
UNIT IV	**DISCRETE EIGEN VALUE PROBLEM**
	The free particle – Particle in a box.
UNIT V	**THREE DIMENSIONAL PROBLEM**

	Three dimensional harmonic oscillator – Rigid rotator.
UNIT VI	**APPLICATION TO DIATOMIC MOLECULES**
	Application to diatomic molecules – Hydrogen atom – Separation of variables and solution of R,θ,Φ equation – Discussion of bound states and parity.
UNIT VII	**REPRESENTATION THEORIES**
	Dirac's ket and bra vectors – Harmonic oscillator – Solution using ladder operator and matrix representation.
UNIT VIII	**QUANTUM DYNAMICS**
	Schrődinger, Heisenberg and interaction pictures.
UNIT IX	**TIME INDEPENDENT PERTURBATION THEORY**
	Perturbation theory (first order) – Time independent – Stark effect in hydrogen atom.
UNIT X	**VARIATIONAL PRINCIPLE**
	Variation method – Ground state of helium atom – Ground state of Deuteron.
UNIT XI	**W.K.B APPROXIMATION**
	W.K.B approximation – Application to bound states.
UNIT XII	**TIME DEPENDENT PERTURBATION THEORY**
	Time dependent perturbation theory – The golden rule and application – Spontaneous emission – Stimulated emission.
UNIT XIII	**QUANTUM THEORY OF RADIATION**
	Einstein's A & B coefficients – Semi – classical and quantum theory of radiation – Eigen value and Eigen function.
UNIT XIV	**THEORY OF SCATTERING**
	Rayleigh and Raman scattering – Selection rules.
Books For Study	
1. Quantum Mechanics I: The Fundamentals, S. Rajasekar, R. Velusamy, CRC	

Press, 2015 2. A text book of Quantum Mechanics – P.M Mathews and K.Venkatesan, McGraw Hill, New Delhi 1975. 3. Quantum Mechanics – G.Aruldhas – PHI Learning Private Limited, New Delhi 2009. 4. Quantum Mechanics - V.Devanathan, Alpha Science International, Limited, 2005.	
References	
1. Quantum Mechanics – L.Schiff, Mc Graw Hill, 1968. 2. Quantum Mechanics – J.P.Dicke and R.H.Wittke, Addison Wiley, 1978. 3. Quantum Mechanics - A.K. Ghatak and Lokanathan, McMillan, 1977. 4. Quantum Mechanics – R.Shankar,Springer, 1994.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34522	**MATHEMATICAL PHYSICS – II**
Objectives	*The main objective of this paper is to impart understanding on the fundamental thoughts of mathematical physics to the students.*
Outcome	On successful completion of the course, a student will be able to • Create and solve mathematical models of physical phenomena using analytic and numerical methods • Design, execute, and interpret experiments to test hypotheses and mathematical models
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on vectors, matrix, integration, differentiation and Laplace transforms are prerequisite.
UNIT I	**COMPLEX VARIABLE**
	Cauchy-Riemann Conditions – Cauchy's Integral Formula – Laurent Expansion – Mapping – Conformal Mapping.
UNIT II	**CALCULUS OF RESIDUES**
	Singularities – Calculus of Residues and contour integrals – Method of Steepest Descents.
UNIT III	**APPLICATION OF PARTIAL DIFFERENTIAL EQUATIONS**
	Heat equation – Laplace and Poisson equation - wave equation - separation of variables.
UNIT IV	**STURM LIOUVILLE PROBLEM**
	Green's Function – Sturm-Liouville theory - Self – Adjoint PDEs.
UNIT V	**ORTHOGONAL FUNCTIONS**
	Hermitian Operators – Gram-Schmidt Orthogonalization – Completeness of Eigen Functions.
UNIT VI	**INTRODUCTION TO TENSOR**
	Introduction – The Algebra of Tensors – Quotient law – Cartesian Tensors.
UNIT VII	**COVARIANT FORMULATION OF TENSORS**

	Four vectors in special relativity – Covariant formulation of electrodynamics.
UNIT VIII	**DUAL AND METRIC TENSORS**
	Dual tensors, irreducible tensors - Metric Tensors - Christoffel symbols - Geodesics.
UNIT IX	**INTRODUCTION TO GROUP THEORY**
	Definition of group – Homomorphism and Isomorphism.
UNIT X	**MATRIX REPRESENTATION OF GROUP THEORY**
	Matrix Representations: Reducible and Irreducible – Proof of the Orthogonality theorem.
UNIT XI	**ROTATION GROUPS**
	Rotation groups SO(2) and SO(3) – Special Unitary group SU(2) – The Character of a Representation – Construction of Character Tables.
UNIT XII	**PHYSICAL APPLICATIONS OF GROUP THEORY**
	Physical applications of group theory: Crystal – Symmetry operators – Crystallographic Point Groups.
UNIT XIII	**THEORY OF PROBABILITY**
	Probability Definitions, Simple Properties – Random Variables.
UNIT XIV	**PROBABILITY DISTRIBUTION**
	Binomial Distribution – Poisson Distribution – Gauss's Normal Distribution – Central limit theorem.

Books For Study

1. Essential Mathematical Methods for Physicists, George B. Arfken, Hanes J.Weber, Frank E. Harris, 7th Edition, Elsevier, 2012.
2. Advanced Engineering Mathematics, Erwin Kreyszig, 9th Edition, Wiley, 2014.

References

1. Mathematical Physics, B.D. Gupta, Vikas Publishing House Pvt. Ltd, 2010.
2. Mathematical Physics, Sathyaprakash, Sultan Chand, 2013.

3. Topics in Mathematical Physics-Parthasarathy H, Ane Books Pvt.Ltd 2007. 5. Chemical Applications of Group Theory, F. Albert Cotton, Addison Wiley, 2006.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34523	**ELECTROMAGNETIC THEORY**
Objectives	*The main objective of this paper is to impart the understanding on the fundamental aspects of electromagnetic theory to the students.*
Outcome	On successful completion of the course, a student will be able to • Describe the electro and magnetostatics Maxwell's equations and propagation of EM waves • Describe the reflection, refraction, dispersion and scattering of electromagnetic waves
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on reflection, refraction, dispersion, scattering and microwaves are prerequisite.
UNIT I	**ELECTRO AND MAGNETOSTATICS**
	Basics-Electrostatics and Magnetostatics. - Wave equation in terms of scalar and vector potential – Transverse nature of electromagnetic wave.
UNIT II	**FIELD EQUATIONS AND CONSERVATION LAWS**
	Maxwell's equations – Poynting theorem - Conservation of energy and momentum, continuity equation.
UNIT III	**ELECTROMAGNETIC WAVES AND WAVE PROPAGATION**
	Propagation of plane electromagnetic waves in (a) free space, (b) Isotropic and Anisotropic non- conducting medium and (c) conducting medium-skin depth.
UNIT IV	**REFLECTION AND REFRACTION OF ELECTROMAGNETIC WAVES**
	Boundary conditions at the surface of discontinuity - Reflection and refraction of electromagnetic waves at the interface of non-conducting media.
UNIT V	**FRESNEL'S EQUATION**
	Fresnel's equations – Reflection and transmission coefficients at the interface between two dielectric media.
UNIT VI	**POLARIZATION**

	Brewster's law and degree of polarization -Total internal reflection.
UNIT VII	**DISPERSION OF ELECTROMAGNETIC WAVES**
	Normal and Anomalous dispersion – Dispersion in Gases – Experimental demonstration of Anomalous dispersion in gases- Solids and Liquids.
UNIT VIII	**CLASUSIUS MOSSOTTI EQUATION**
	Clasusius-Mossotti relation – Lorentz formula.
UNIT IX	**SCATTERING OF ELECTROMAGNETIC WAVES**
	Scattering and scattering parameters - Theory of scattering of e-m waves – polarization of scattered Light – coherence and incoherence of scattered light.
UNIT X	**WAVE GUIDES**
	Wave guides: Rectangular and cylindrical waveguides.
UNIT XI	**MICROWAVES**
	Generation of microwaves – Klystron – Magnetron – Gunn diodes – Resonant cavities.
UNIT XII	**DYNAMICS OF CHARGED PARTICLES**
	Lienard-Wiechert potential-E.M fields from retarded potentials of moving point charge-e.m. fields of uniformly moving point charge- Radiation from moving charges.
UNIT XIII	**PLASMA PHYSICS**
	Introduction - Conditions for plasma existence – occurrence of plasma – charged particles in uniform constant electric field, in homogeneous magnetic fields, simultaneous homogeneous electric and magnetic fields, in nonhomogeneous magnetic fields.
UNIT XIV	**MAGNETOHYDRODYNAMICS**
	Magnetohydrodynamics – magnetic confinement -pinch effect- Instabilities- plasma waves.
Book For Study	
1. Electromagnetic theory and Electrodynamics – Satya Prakash, Kedarnath	

Ramnath & Co, 2007 2. Introduction to Electrodynamics- D.J.Griffith, Pearson Education Ltd.-4rd Edn., 2014.	
References	
1. Electromagnetics – Kraus & Carver, TMH, 1973. 2. Electromagnetic fields and waves – Paul Lorain & Dale R.Corson, CBS publishers, NewDelhi, 1986. 3. Foundations of Electromagnetic theory – Reitz, Milford & Frederick, Narosa publishing House, 1986. 4. Introduction to electromagnetic theory- Chopra and Agarwal, (1984). 5. Classical Electrodynamics, J.D.Jackson, Wiley Eastern Limited, New Delhi, 1978.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34524	**ADVANCED ELECTRONICS AND PHYSICS LABORATORY - II**
Objectives	*The main objective of this practical paper is to exercise the practical in various advanced electronics and advanced analytical experiments to the students.*
Outcome	On successful completion of the course, a student will be able to • Understand the basic operations in electronic circuits • Develop the programming skills of Microprocessor • Understand the concept of ICs manufacturing • Appreciate the applications of Microprocessor programming • Understand the basic principles of the experiments • Understand simple concepts to demonstrate an experiment
Schedule	Contact Hours: 120 hours
Prerequisite	Basic knowledge on transistor, diode, amplifier, oscillators, semiconductor, spectrometer and microscope are prerequisite.
	(Any Fifteen of the following) 1. Half adders and Full adders. 2. Integrator and Differentiator circuits using IC 741. 3. Active filters using IC 741. 4. D/A converters (a) Ladder network (b) Weighted resistor method. 5. A/D converter. 6. Encoder - Decoder circuits. 7. Square wave, Sine wave and Triangular wave generators using IC. 8. Multiplexer circuits. 9. Flip – Flop circuits using IC. 10. Powder photograph – X-ray method.

	11. Resistivity measurements of thin films. 12. Hall effect – Mobility and Hall constant determination. 13. Dielectric constant – Microwave frequency using klystron. 14. Determination of Curie point – Ferromagnetic material. 15. Susceptibility by Guoy's method. 16. Susceptibility by Quincke's method. 17. Reflection grating spectrometer. 18. Any of the experiments of equal standard.

III SEMESTER

34531	**MOLECULAR SPECTROSCOPY**
Objectives	*The main objective of this paper is to impart understanding on the fundamental thoughts of Atomic and Molecular Spectroscopy in physics to the students.*
Outcome	On successful completion of the course, a student will be able to • Appreciate the principles of spectroscopy in the different regions of the electromagnetic spectrum • Apply the concepts of group theory to molecular vibrations • Relate the theory of spectroscopy to the study of molecular structure
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on bonding in solids and spectroscopy are prerequisite.
UNIT I	**SYMMETRY ASPECTS OF MOLECULAR ORBITALS**
	Valence bond theory – Molecular orbital theory- Heitler London theory for Hydrogen molecule - Hybridization – SP – SP^2 & SP^3 Hybrids.
UNIT II	**ROTATIONAL SPECTRA OF DIATOMIC MOLECULE**
	Rotational energy of a diatomic molecule – Rigid and non-rigid rotators – isotopic substitution.
UNIT III	**STARK EFFECT**
	Stark effect – its importance in microwave spectroscopy – quadrupole hyperfine interaction.
UNIT IV	**ROTATIONAL SPECTRA OF POLYATOMIC MOLECULES**
	Rotational spectra of polyatomic molecules – pure rotational Raman spectra – diatomic linear molecule – symmetric top molecules.
UNIT V	**MOLECULAR STRUCTURE**
	Molecular structure – using IR & Raman spectroscopy.
UNIT VI	**VIBRATIONAL PROPERTIES**
	Vibrational spectra of diatomic and polyatomic molecules – Information on molecular constitution from IR studies.
UNIT VII	**VIBRATIONAL RAMAN SPECTRA**
	Vibrational Raman spectra – Vibrational course structure – Rotational course spectra.

UNIT VIII	**FRANCK-CONDON PRINCIPLE**
	Franck – Condon principle – intensity distribution – portrait parabolae – disassociation - predisassociation – mutual exclusion principle.
UNIT IX	**HYPER RAMAN EFFECT**
	Non linear Raman Phenomena-Hyper Raman effect- Classical treatment – Experimental techniques- Stimulated Raman Scattering.
UNIT X	**INVERSE RAMAN EFFECT**
	Inverse Raman Effect-Coherent Anti-Stoke's Raman Scattering-Photo acoustic Raman Scattering.
UNIT XI	**MULTIPHOTON SPECTROSCOPY**
	Multi photon spectroscopy-two photon absorption- Multiphoton absorption. X-ray spectra; rotational and vibrational spectra of diatomic molecules.
UNIT XII	**RESONANCE SPECTROSCOPY**
	Interaction between spin and magnetic field – Nuclear resonance – Bloch equations - Chemical shift – Dipole –Dipole interaction and spin lattice interaction.
UNIT XIII	**ESR AND NQR SPECTROSCOPY**
	ESR-NQR (principle only) spectroscopy and its application.
UNIT XIV	**MOSSBAUER SPECTROSCOPY**
	Mossbauer spectroscopy - applications – Electronic structure – molecular structure – crystal symmetry and molecular structures.

Book For Study

1. Handbook of Spectroscopy,1- 4 Volume, 2nd Edition., 2014, Gunter Gauglitz , David S. Moore , John Wiley & Sons, Inc.,
2. Atomic and Molecular Spectroscopy Basic Concepts and Applications, By Rita Kakkar, 2015 Cambridge University Press.
3. Condensed-Phase Molecular Spectroscopy and Photophysics, Anne Myers Kelley, 2012, Wiley.
4. Raman Spectroscopy and its Application in Nanostructures, Shu-Lin Zhang , 2012, John Wiley & Sons, Inc.,

References

1. Handbook of High-resolution Spectroscopy,Martin Quack ,Frederic Merkt, 2011, John Wiley & Sons, Inc., 2. Raman, Infrared, and Near-Infrared Chemical Imaging, Slobodan Sasic , Yukihiro Ozaki , 2010, John Wiley & Sons, Inc., 3. Molecular structure and spectroscopy, G.Aruldhass, Prentice Hall of India, New Delhi – 2001. 4. Atomic and Molecular Spectroscopy, Mool Chand Guptha, New age International Publishers, New Delhi, (2001). 5. Instrumental methods of Chemical analysis, H.Gaur, Prgati Prakasan, Ist Edition, 2001. 6. Principles of Instrumental Analysis, Skoog, Holler and Nieman, Harcourt College Publishers, 5th edition, 2003. 7. Instrumental methods of analysis, Willard et al., CBS publishers, 2005. 8. Hand book of analytical instrumentation, R.S. Khandapur, McGraw Hill, (2006). 9. Chemical Applications of Group Theory F.A.Cotton (2009) Wiley student edition.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34532	**QUANTUM MECHANICS-II**
Objectives	*The main objective of this paper is to impart in depth knowledge on the advanced theories of quantum mechanics to the students.*
Outcome	On successful completion of the course, a student will be able to • Apply principles of quantum mechanics to calculate observables on known wave functions • Grasp the concepts of spin and angular momentum, as well as their quantization- and addition rules • Explain physical properties of elementary particles, nucleons, atoms, molecules and solids (band structure) based on quantum mechanics
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on uncertainty and quantum concepts are prerequisite.
UNIT I	**THEORY OF ANGULAR MOMENTUM**
	Angular momentum of a system of particles – Commutation relations.
UNIT II	**MATRIX REPRESENTATION OF ANGULAR MOMENTUM**
	Matrix representation of angular momentum – Pauli spin matrices.
UNIT III	**ADDITION OF ANGULAR MOMENTA**
	Addition of two angular momenta – C.G. coefficients for j = ½ system only.
UNIT IV	**SELF CONSISTENT FIELD**
	Central field approximation – Thomson-Fermi Model of the Atom.
UNIT V	**IDENTICAL PARTICLES**
	Identical particles – Bosons and fermions – Symmetric and anti-symmetric wave functions.
UNIT VI	**HARTREE EQUATION**
	Hartree equation – Hartree-Fock Equation – Alkali atoms Doublet intensity and doublet separation - Periodic Table.
UNIT VII	**KLEIN-GORDAN EQUATION**

	Klein-Gordan equation – Application to Hydrogen atom.
UNIT VIII	**RELATIVISTIC QUANTUM MECHANICS**
	Dirac's Relativistic Hamiltonian – Dirac matrices and properties.
UNIT IX	**DIRAC PARTICLE**
	Spin of a Dirac particle – Negative energy states.
UNIT X	**ELEMENTS OF FIELD QUANTIZATION**
	Elements of field quantization for non-relativistic field – Quantization of relativistic field.
UNIT XI	**KLEIN-GORDAN FIELD**
	Klein Gordon field - Dirac Field – Quantization of Electromagnetic field.
UNIT XII	**SCATTERING THEORY**
	Scattering cross section – scattering amplitude.
UNIT XIII	**BORN APPROXIMATION**
	Optical Theorem – Born approximation.
UNIT XIV	**DIFFUSION SCATTERING**
	Scattering by screened Coulomb potential – Diffusion Scattering – Particle wave analysis.

Book For Study

1. Quantum Mechanics II: Advanced Topics, S.Rajasekar, R.Velusamy, CRC Press, 2015.
2. Quantum mechanics – L.Schiff, Mc-Graw Hill, 1968.
3. Quantum mechanics – B.N.Srivastava, Pragati prakashan, 1975.
4. Text book of quantum mechanics, P.M. Mathews and K.Venkatesan, McGraw Hill, New Delhi, 1975.
5. Quantum Mechanics – G.Aruldhas – PHI Learning Private Limited, New Delhi 2009.

References

1. Quantum mechanics - V.K. Thangappan, Wiley Eastern, 1985. 2. Quantum electrodynamics - P. G. Puranik, S.Chand & co, 1980. 3. A text book of quantum mechanics – A.K. Ghatak & Lokanathan, Mc Millan, 1977. 4. Quantum mechanics- Devanathan,Alpha Science International Ltd, United Kingdom (2011).	
Mode of Evaluation	Assignment/Seminar/Written Examination

34533	**MICROPROCESSOR AND ELECTRONIC INSTRUMENTATION**
Objectives	*This course is designed to provide students with the necessary foundation for entry-level industrial applications in process monitoring and controlling, with an emphasis on analysis, problem solving, exposure to open-ended problems and design methods. At the end of the course, students will be able to design an application based on microcontrollers or microprocessors.*
Outcome	On successful completion of the course, a student will be able to • Develop the programming skills of microprocessor • Appreciate the applications of microcontroller programming
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on instrumentation and programming are prerequisite.
UNIT I	**MICROPROCESSOR ARCHITECTURE (8085)**
	Introduction, Intel 8085 : Architecture, Instruction Cycle, Timing Diagram: Op-code fetch, Memory read & Memory write – Instruction Set : Instruction and Data Format, Addressing Modes, Status Flags, Instructions Set, Data Transfer, Arithmetic, Branching, and Logical group operations – Interrupts.
UNIT II	**MICROPROCESSOR ARCHITECTURE (8086)**
	Architecture of 8086, Pin diagram and pin function, Register organization, Minimum and Maximum mode operation of 8086.
UNIT III	**PROGRAMMING OF MICROPROCESSOR**
	Instructions for 8085 – Software development tools – Assembly language programs with data transfer, arithmetic, logical, bit level instructions and branch instructions.
UNIT IV	**INTERRUPTS AND STACK OPERATIONS OF 8085**
	Interrupts and interrupt service routines-Subroutine – Flow charting – Loops – Pseudo instructions – Stack Operations- Programming and applications: Traffic control system.
UNIT VI	**MICRO-CONTROLLER 8051**
	Introduction to 8 bit micro-controller, Architecture of 8051- Hardware features of 8051 - Signal description of 8051-General Purpose and Special

	Function Registers- Oscillator and clock circuit-I/O Port-Memory organization and I/O addressing by 8051, Interrupts of 8051.
UNIT VI	**8051 INSTRUCTION SET AND PROGRAMMING**
	Instructions set of 8051-Programming of 8051 (Simple Arithmetic and Logical programs).
UNIT VII	**INTRODUCTION TO INTERFACING DEVICES**
	Address space partition - Memory & I/O Interfacing – Data transfer schemes – Interrupts - I/O Ports.
UNIT VIII	**INTERFACING DEVICES (8255 AND 8259)**
	Programmable Peripheral Interface:8255 – Programmable Interrupt Controller :8259.
UNIT IX	**INTERFACING DEVICES (8257 AND 8251)**
	Programmable DMA Controllers:8257 – Programmable Communication Interface:8251.
UNIT X	**MICROPROCESSOR APPLICATIONS**
	A/D Sub systems - Applications – Temperature monitoring and Stepper motor control.
UNIT XI	**ELECTRONIC INSTRUMENTATION**
	Instrumentation amplifiers, Sample and hold circuits, Comparators, – D/A – Weighted resistor method – Resistor ladder net work method – A/D – Successive approximation method.
UNIT XII	**TEMPERATURE TRANSDUCERS**
	Classification of transducers - Temperature transducers: thermo-resistive transducers, thermoelectric, p-n junction, chemical thermometry.
UNIT XIII	**DISPLACEMENT TRANSDUCERS**
	Displacement transducers: potentiometer, resistive strain gauges, capacitive displacement transducer, LVDT transducers.
UNIT XIV	**PHOTOELECTRIC TRANSDUCERS**
	Photoelectric transducers: photovoltaic cell, photoconductive cell-Piezoelectric transducers.

Book For Study	
1. Fundamentals of Microprocessors and Microcomputers - B. Ram, 8th Edition, Dhanpat Rai Publications (P) Ltd., New Delhi, 2010. 2. Microprocessors and Microcontrollers – A.Nagoor Kani, , 2nd Edition, Mc Graw Hill Education, 2012. 3. Transducers and Instrumentation, D.V.S.Murthy, 2nd Edition, PHI Learning Private Limited -New Delhi, 2008.	
References	
1. Microprocessor Architecture, Programming and Applications- R. Gaonkar, Pearson, 1998 2. Advanced Microprocessor and Principles- A.K. Ray, K.M. Bhurchandi, 3rd Edition, Tata Mc Graw Hill Publication Co. Ltd. New Delhi. 2003.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34534	ADVANCED ELECTRONICS AND PHYSICS LABORATORY - III
Objectives	*The main objective of this practical paper is to exercise the practical in various advanced electronics and advanced analytical experiments to the students.*
Outcome	On successful completion of the course, a student will be able to • Understand the basic operations in electronic circuits • Develop the programming skills of Microprocessor • Understand the concept of ICs manufacturing • Appreciate the applications of Microprocessor programming • Understand the basic principles of the experiments • Understand simple concepts to demonstrate an experiment
Schedule	Contact Hours: 120 hours
Prerequisite	Basic knowledge on transistor, diode, amplifier, oscillators, semiconductor, spectrometer and microscope are prerequisite.
	(Any Fifteen of the following) 1. Study of Counters. 2. Monostable multivibrator using op-amp. 3. Astable multivibrator using op-amp and using IC 555. 4. Schmitt trigger using op-amp. 5. Voltage comparator. 6. Demultiplexer. 7. Logic gates using IC's. 8. Young's modulus – Cornu's method. 9. Refractive index of liquid by laser.

	10. Optical absorption studies using lasers. 11. Determination of wavelength of a laser source by diffraction grating. 12. Determination of Charge of an electron using spectrometer. 13. Thermal expansion using optical air wedge. 14. Ultrasonic interferometer. 15. Electron spin resonance spectrometer. 16. Magnetic Hystersis loop tracer. 17. Measurements and inverse square law verification. 18. Any of the experiments of equal standard.

IV SEMESTER

34541	**CONDENSED MATTER PHYSICS**
Objectives	*The objective of the course is to impart knowledge about crystalline structures, lattice vibrations, dielectrics, magnetism and superconductivity to the students. Understanding the content of this course will be useful for the students to carry out research work after the completion of Master's degree.*
Outcome	On successful completion of the course, a student will be able to • Calculate reciprocal lattice vectors for typical high symmetrical crystals and the relationship between Miller indices (hkl) and the distance between the lattice planes is to be understood • Energy band structure should be explained in terms of the periodic potential and illustrated by using Kronig-Penny model • Classification into metals, semiconductors and insulators anchored in the energy band structure
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on bonding in solids and crystal structure are prerequisite.
UNIT I	**CRYSTAL STRUCTURE**
	Crystalline solids – Crystal lattice and crystal structure, Symmetry elements, Ordered Phase of Matter.
UNIT II	**CRYSTAL SYSTEMS**
	Translation & Orientation Order, Space lattice, Unit cell and primitive cell – Bravais lattice, Crystal systems.
UNIT III	**LATTICES**
	Type of lattices: SC, BCC, FCC, HCP – Miller indices – Reciprocal lattice - Wigner-seitz cells – bonding in solids.
UNIT IV	**LATTICE VIBRATIONS**
	Lattice vibrations – Diatomic lattices – Phonons.
UNIT V	**FREE ELECTRON FERMI GAS**
	Electrical properties of metals – Free electron gas in three dimensions –

	Fermi energy – Hall effect.
UNIT VI	**BAND THEORY OF SOLIDS**
	Band theory of solids – Kronig–Penny model – Semiconductors – Classification – Fermi energies of impurity semiconductors.
UNIT VII	**DIELECTRIC PROPERTIES OF MATERIALS**
	Polarization – Local electric field, Clausius-Mosotti relation – Polarization field – Lorentz field.
UNIT VIII	**DIELECTRIC CONSTANT AND POLARIZABILITY**
	Dielectric constant – Polarizability: Electronic, Ionic- Piezo – Pyro and ferro electric properties of crystals.
UNIT IX	**PARAMAGNETISM**
	Classification of magnetic materials – Langevin's theory of paramagnetism – Quantum theory of paramagnetism.
UNIT X	**FERROMAGNETISM**
	Ferromagnetism – Weiss Molecular Field Theory – Ferromagnetic domains.
UNIT XI	**ANTIFERROMAGNETISM AND FERRIMAGNETISM**
	Antiferromagnetism – Neel's Theory – Ferrimagnetism and ferrites.
UNIT XII	**MAGNETIC MATERIALS**
	Spin waves – Hard and soft magnetic materials.
UNIT XIII	**SUPERCONDUCTIVITY – EXPERIMENTAL SURVEY**
	Introduction – Meissner effect – Isotope effect – Type I and Type II superconductors.
UNIT XIV	**SUPERCONDUCTIVITY – THEORETICAL SURVEY**
	London equations – Coherence length – BCS Theory – Cooper pair – Normal tunneling and Josephson Effect – High temperature superconductors.
Book For Study	

1. Solid State Physics – Structure and properties of Materials – M.A. Wahab, Second edition, Narosa publishers. 2005. 2. Introduction to Solid State Physics – Charles Kittel, 7th Edition, John Wiley & Sons. (1971).	
References	
1. Solid State Physics – S.O. Pillai, Wiley Eastern Ltd. (2005). 2. Solid State Physics – B.S. Saxena, R.C. Gupta and P.N. Saxena – Pragati Prakasham – Meerut. 3. Solid State Physics – A.J. Dekkar, Macmillan India Ltd., 1600. 4. Solid State Physics – S.L. Kahani, C. Hemaranjani – Sulton Chand & Sons.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34542	**NUCLEAR AND PARTICLE PHYSICS**
Objectives	*The main objective of this paper is to impart understanding on the fundamental knowledge of Nuclear & particle physics to the students.*
Outcome	On successful completion of the course, a student will be able to • Identify the fundamental models of nuclear structure that are used to describe various modes of nuclear excitation • Lay out the foundation that allows interpreting the observations obtained in typical nuclear structure experiments
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on particle, atom and nucleus are prerequisite.
UNIT I	**NECLEAR DECAY – ALPHA AND GAMMA DECAY**
	Gamow's Theory of Alpha decay – Gamma decay – Internal Conversion Nuclear Isomerism.
UNIT II	**NECLEAR DECAY – BETA DECAY**
	Fermi's theory of Beta decay – Kurie plots – Selection rules – Electron capture – Parity violation in Beta decay - Neutrinos – Measurement of neutrino helicity.
UNIT III	**NUCLEAR LIQUID DROP AND COLLECTIVE MODELS**
	Liquid Drop model – Bohr Wheeler theory – Schmidt lines – Magnetic dipole moment – Electric quadrupole moment – Collective Model.
UNIT IV	**NUCLEAR SHELL MODEL**
	Shell model – Single particle model, its validity and limitations – Rotational Spectra - Magic numbers – Spin – orbit coupling - Angular momentum of nucleus ground states – Magnetic Moments of the shell model.
UNIT V	**NUCLEAR REACTION AND MECHANISM**
	Nuclear Fission and Fusion. Nuclear reactions, reaction mechanisms, compound nuclei and direct reactions: Simple theory of deuteron – Tensor forces (qualitative).
UNIT VI	**NUCLEAR FORCE**

	Nature of nuclear force, form of nucleon-nucleon potential, charge independence and charge symmetry of nuclear forces - Normalization of deuteron wave functions.
UNIT VII	**PARTIAL WAVE ANALYSIS**
	Method of partial wave analysis and phase shifts - Effective range theory – n-p scattering at low energies– Yukawa's meson theory of nuclear forces.
UNIT VIII	**REACTION CROSS SECTIONS**
	Nuclear cross sections – Compound nuclear formation and breakup – Resonance scattering cross section.
UNIT IX	**NEUTRONS**
	Interaction of neutron with matter – Thermal neutrons – neutron cycle in a thermo nuclear reactor – Critical size.
UNIT X	**NUCLEAR REACTORS**
	Types of nuclear reactors - cylindrical and spherical- sub-nuclear particles (elementary ideas only) – source of stellar energy – controlled thermo nuclear reactions.
UNIT XI	**FUNDAMENTAL INTERACTIONS IN NATURE**
	Classification of fundamental forces – Particle Directory and quantum numbers (Charge, spin, parity, iso-spin, strangeness etc).
UNIT XII	**CLASSIFICATION OF ELEMENTARY PARTICLES**
	Leptons, Baryons and quarks. Spin and parity assignments, isospin, strangeness.
UNIT XIII	**GEL-MANN-NISHIJIMA RELATION**
	The fundamental interactions – Translations in space – Rotations in space – SU(2) and SU(3) groups – Charge conjugation – Parity – Gell-Mann-Nishijima formula.
UNIT XIV	**SYMMETRIES**
	Time reversal–CPT invariance- Applications of symmetry arguments to particle reactions, Parity non-conservation in weak interaction; Relativistic kinematics.

Book For Study	
1. Introduction to Elementary Particle Physics, Alessandro Bettin, 2nd Edition, 2014, Cambridge University Press. 2. Nuclear and Particle Physics, Claude Amsler, 2015, Iop Publishing Limited. 3. Modern Particle Physics, Mark Thomson, 2013, Cambridge University Press.	
References	
1. Nuclear and Particle Physics: An Introduction, 2nd Edition, Brian R. Martin, 2009, John Wiley & Sons, Inc. 2. Introduction to Nuclear and Particle Physics: Solutions Manual for Second Edition, C. Bromberg, Ashok Das, Thomas, 2006, Ferbel,World Scientific, Pvt Ltd 3. Modern Physics- R.Murugeshan Kiruthiga Sivaprasath, 2008, S.Chand & Company Ltd, New Delhi,30th Revised Multicolour edition. 4. Introductory Nuclear Physics-Samuel S.M. Wong, 2005, Printice Hall of India Pvt Ltd,New Delhi. 5. Nuclear Physics (Principle and applications)-J.S.Lilly, 2001, John Wiley & Sons (ASIA)Pte Ltd,Singapore. 6. Fundamentals and Nuclear Physics, Jean – Louis Basdevant, James Rich and Michel Spiro, 2005, Springer.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34543	**MATERIALS SCIENCE**
Objectives	*The main objective of this paper is to impart an interdisciplinary understanding on the fundamental scientific principles and concepts of materials that are essential for both practice and advanced study in the field of Materials Science.*
Outcome	On successful completion of the course, a student will be able to • Obtain the basis for understanding the link between different processing techniques and the characteristics of materials • Provide insight into some of the steps in the production of semiconductor devices • Provide an introduction to experimental methods that are used in parts of materials science
Schedule	Contact Hours: 16 hours
Prerequisite	Basic knowledge on material and thin films are Prerequisite.
UNIT I	**MATERIAL BEHAVIOUR**
	Elastic, Inelastic and Viscoelastic behaviour of materials – tensile strength, toughness, elongation, plastic deformation, hardness, creep and fatigue.
UNIT II	**POLYMERS**
	Polymers - Structure and Properties - Addition and condensation polymerization – Polymer types.
UNIT III	**APPLICATION OF POLYMERS**
	Application of polymers - Corrosion and Oxidation of metals – Prevention - Corrosion resistance materials.
UNIT IV	**VACUUM PUMPS**
	Kinetic Theory of gases – Gas Transport and Pumping – Vacuum Pumps – Rotary, Diffusion and Turbo molecular Pumps – Pirani and Penning Gauges.
UNIT V	**THIN FILMS BY EVAPORATION**
	Thin films by Thermal evaporation – Thickness measurement – Quartz crystal method.

UNIT VI	**THIN FILMS BY EPITAXY**
	Epitaxy – Structural Aspects – Lattice Misfit and Imperfections – Liquid Phase Epitaxy – Vapour Phase Epitaxy – Epitaxy of compound semiconductors – Applications.
UNIT VII	**PRINCIPLES OF LASER**
	Principles of Laser - Population inversion in three level and four level systems – Resonators-Q-switching and Mode locking.
UNIT VIII	**TYPES OF LASER**
	Solid state lasers – Ruby and Nd-YAG - Semiconductor lasers – GaAs/AlGaAs semiconductor laser – Gas lasers – He-Ne and CO_2 Lasers.
UNIT IX	**NON-LINEAR OPTICAL PROPERTIES OF MATERIALS**
	Electro-optic Effect – Pockel's effect, Kerr effect – Second harmonic generators – Electro-optic Modulators – Photorefractive Materials – Acousto-optic effect.
UNIT X	**COMPOSITE MATERIALS**
	Introduction – Polymer-matrix composites – Cement-matrix composites – Carbon-matrix composites – Metal-matrix composites – Ceramic-matrix composites.
UNIT XI	**APPLICATIONS OF COMPOSITE MATERIALS**
	Applications – Structural – Electronic – Thermal – Electrochemical – Environmental – Biomedical.
UNIT XII	**SMART MATERIALS**
	Amorphous and glassy materials – Structure – Preparation methods and novel properties.
UNIT XIII	**APPLICATIONS OF SMART MATERIALS**
	Shape memory alloys – Working mechanism – Pseudo elasticity – Applications – Nickel-Titanium (Nitinol) alloys – Material characteristics of Nitinol.
UNIT XIV	**MICRO ELECTRO MECHANICAL SYSTEMS**
	Introduction to Micro Electro Mechanical Systems (MEMS) – Silicon, porus Silicon and Silicon oxide based MEMS –Fabrication of

	piezoelectric and Piezo-resistive MEMS materials – Application to micro-actuators and microaccelerometers.
Book For Study	
1. Materials Science and Engineering – V. Raghavan, 6th Edition, PHI Learning, 2015. 2. Materials Science and Thin Films – Milton Ohring, 2nd Edition, ELSER, 2009. 3. Laser and Non Linear Optics- B.B. Laud, 3rd Edition, New Age International Limited, 2011. 4. Composite Materials: Science and Applications – Deborah.D.L. Chung, ANE-Books, New Delhi, 2011. 5. Smart Material Systems And MEMS: Design And Development Methodologies- Vijay K.Varadan, K.J. Vinoy, S. Gopalakrishnan, Wiley India, 2011.	
References	
1. Introduction to Materials Science for Engineers – James F. Shackelford, Madanapalli K. Muralidhara, Pearson India, 2006 2. Handbook of Thin Film Deposition: Principles, Methods, Equipment and Applications- Seshan, 2nd Edition, William Andrew Publishing, 2002.	
Mode of Evaluation	Assignment/Seminar/Written Examination

34544	ADVANCED ELECTRONICS AND PHYSICS LABORATORY - IV
Objectives	*The main objective of this practical paper is to exercise the practical in various advanced electronics and advanced analytical experiments to the students.*
Outcome	On successful completion of the course, a student will be able to • Understand the basic operations in electronic circuits • Develop the programming skills of Microprocessor • Understand the concept of ICs manufacturing • Appreciate the applications of Microprocessor programming • Understand the basic principles of the experiments • Understand simple concepts to demonstrate an experiment
Schedule	Contact Hours: 120 hours
Prerequisite	Basic knowledge on transistor, diode, amplifier, oscillators, semiconductor, spectrometer and microscope are prerequisite.
	(Any Fifteen of the following) 1. BCD to 7 segment display and BCD decoder. 2. Shift register and ring counter. 3. Operation of 7489 RAMS. 4. Arithmetic operations – Microprocessor 8085. 5. Logical operations - Microprocessor 8085. 6. Code conversion - Microprocessor 8085. 7. Determination of Plank's constant using photocell apparatus. 8. e/m by Millikan's oil drop method. 9. e/m by Thompson's oil drop method.

	10. Thermal Conductivity of a bad conductor (Lees Method). 11. Hydrogen spectrum and Rydberg's constant. 12. Determination of BH and M using magnetometers. 13. Determination of Stefan's constant. 14. Fabrey-Perot interferometer. 15. Solar cell characteristics. 16. Polarizability of liquids – Hollow prism – Spectrometer. 17. Determination of specific rotatory power of a liquid using polarimeter. 18. Any of the experiments of equal standard.

e. 3. Duration of the Programme:

The Programme for the degree of Master of Science in Physics shall consist of two academic years divided into four semesters. Each semester consists of Three Theory Papers and One Practical Paper. Each Theory course carries 4 credits and Practical course also carries 4 credits. Hence, the total credits per semester are 16 and the total credits of the programme are 64.

e. 4. Faculty and Support Staff Requirements:

The required Faculty and Supporting Staff details for the degree of Master of Science in Physics programme are given below:

Staff category	Required
Core Faculty	3
Faculty for Specialization	2
Laboratory Assistant	1
Clerical Assistant	1

e. 5. Instructional Delivery Mechanisms:

The instructional delivery mechanisms of the programme includes SLM - study materials, face to face contact session for both theory and practical courses of the programme, e-content of the study materials in the form of CD, MOOC courses and virtual laboratory, wherever applicable.

e. 6. Identification of Media:

The SLM - designed study materials will be provided in print media as well as in the form of CD which carries electronic version of the study material in addition to MOOC and virtual laboratory courses.

e. 7. Student Support Services:

The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like councelling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres. The post-admission student support services like issuance of identity card, study materials, etc. will be routed through the Learning Centres. The face to face contact sessions of the programme for both theory and practical courses will be held at the head quarter only. The conduct of end semester examinations, evaluation and issuance of certificates will be done by the Office of the Controller of Examinations, Alagappa University, Karaikudi.

f. Procedure for Admission, Curriculum Transaction and Evaluation:

f. 1. Procedure for Admission:

A candidate who has passed B.Sc., Degree Examination with Physics or Applied Physics as main course of study of any University or any of the B.Sc., degree examination with specialization such as Applied Physics, Electronics, Nuclear Physics, Biophysics, Nanoscience or any other specialization in Physics of some other University accepted by the syndicate as equivalent thereto, subject to such condition as may be prescribed therefore shall be permitted to appear and qualify for the M.Sc. Degree in Physics of this University after a course of study of two academic years.

f. 2. Curriculum Transactions:

The classroom teaching shall be through conventional lectures, use of OHP, power point presentations, web-based lessons, animated videos, etc. The lecture shall be such that the student

should participate actively in the discussion. Student seminars would be conducted and scientific discussions would be arranged to improve their communicative skill.

For practical courses exclusive study materials containing the requirements, procedure for the experiments will be issued to the learners. In the laboratory, instruction shall be given for the experiments followed by demonstration and finally the students have to do the experiments individually.

The face to face contact sessions will be conducted **only at head quarter** as per the following schedule:

Course Type	**Face to Face Contact Session per Semester (in Hours)**
Theory Courses (3 courses with 4 credits each)	48 (16 hrs per course)
Practical Courses (1 course with 4 credits)	120
Total	**168**

The above mentioned **face to face contact session (only at head quarter) is compulsory** for all the students who have enrolled in M.Sc. Physics Programme through Distance Learning Mode.

f. 3. Evaluation:

The performance of a student in each course is evaluated in terms of percentage of marks with a provision for conversion to grade points. Evaluation for each course shall be done by continuous internal assessment (CIA) as well as by an end semester examination (ESE) and will be consolidated at the end of the course.

f. 3.1. Distribution of Marks for Theory Course: (CIA 25 marks + ESE 75 marks)

The internal assessment shall comprise a maximum of 25 marks for each course. In addition to continuous evaluation component, the end semester examination, which will be a written examination of at least 3 hours duration to each course, would also form an integral component of the evaluation. The ratio of marks to be allotted to continuous internal assessment and to end semester examination is 25:75. Student must have earned 75% of attendance in each

course for appearing for the examination. Attendance need not be taken as a component for continuous assessment, although the student should put in a minimum of 75% attendance in each course.

The components for continuous internal assessment for theory course are:

Seminar (1 per course)	-	05 marks
Assignment (2 per course)	-	20 marks
Total	-	25 marks

Question paper pattern for end semester examination of theory course:

M.Sc. Physics
345XY: Course title
(2018-19 onwards)

Time: 3 Hours Max. Marks - 75

PART A: Answer all questions. All questions carry equal marks. (10 × 2 = 20 marks)
PART B: Answer all questions either (a) or (b). (5 × 5 = 25 marks)
PART C: Answer any three questions. (3 × 10 = 30 marks)

f. 3.2. Distribution of Marks for Practical Course: (CIA 25 marks + ESE 75 marks)

The evaluation of practical course, wherever applicable, will also be based on continuous internal assessment for 25 marks and an end-semester practical examination for 75 marks. The continuous assessment for practical course consists of the execution of laboratory experiments, understanding and demonstration of experiments, preparation of laboratory observation note book along with seminar/model practical, etc.

CIA	**Marks**
Execution of laboratory experiments, understanding and demonstration of experiments, preparation of laboratory observation note book along with seminar/model practical, etc.	25

ESE	**Marks**

Circuit Diagram / Diagram / Formula / Tables	10
Observation	20
Results	20
Viva - voce in practical	15
Record Note	10
Total	75

The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

f. 3.3. Passing Minimum:

The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50 % in the University external examination and 50% of the total (CIA+ESE) marks. Candidate who does not obtain the required minimum marks for a pass in a course shall be required to reappear and pass the same at a subsequent appearance.

f. 3.4. Classification of Successful Candidates

Candidates who secure not less than 60% of the aggregate marks in the whole examination shall be declared to have passed the examination in First class. All other successful candidates shall be declared to have passed in the Second class.

Candidates who obtain 75% of the marks in the aggregate shall be deemed to have passed the examination in First class with Distinction provided they pass all the examinations prescribed for the course at the first appearance.

Candidates who pass all the examinations prescribed for the programme in the first instance and within a period of two academic years from the year of admission to the programme only are eligible for University Ranking.

A candidate is deemed to have secured first rank provided he/she

(i) should have passed all the courses in first attempt itself

(ii) should have secured the highest overall grade point average (OGPA)

f. 3.5. Marks and Grading:

Once the marks of the CIA and end-semester examination for each of the courses are available, they will be added. The marks, thus obtained will then be graded as per the scheme provided in Table 1.

Table 1: Grading of the Courses

Marks	Grade Point	Letter Grade
96 and above	10	S+
91 – 95	9.5	S
86 – 90	9.0	D++
81 – 85	8.5	D+
76 – 80	8.0	D
71 – 75	7.5	A++
66 – 70	7.0	A+
61 – 65	6.5	A
56 – 60	6.0	B
50 – 55	5.5	C
Below 50	0	F

From the second semester onwards the total performance within a semester continuous performance starting from the first semester is indicated respectively **Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA)**. These two are calculated by the following formula.

$$\mathbf{GPA} = \frac{\sum_{i=1}^{n} Ci\, Gi}{\sum_{i=1}^{n} Ci}$$

Where 'Ci' is the Credit earned for the course 'i' in any semester; 'Gi' is the Grade Point obtained by the student for the course 'i' and 'n' is the number of courses **passed** in that semester. **CGPA** (Cumulative Grade Point Average) = Average Grade Point of all the Courses starting from the first semester to the current semester.

f. 3.6. Maximum Duration for the Completion of the Course:

The maximum duration for completion of M.Sc. Degree in Physics Programme shall not exceed ten semesters.

f. 3.7. Commencement of this Regulation:

These regulations shall take effect from the academic year 2018-2019 i.e., for students who are to be admitted to the first year of the programme during the academic year 2018-2019 and thereafter.

f. 4. Fee Structure:

The programme has the following Fee Structure:

Sl. No.	Fees Detail	Amount in Rs.		Nature of Fees
		First Year	Second Year	
1	Admission Processing Fee	300	-	Non- Refundable
2	Course Fee	20,000	20,000	Non- Refundable
3	ICT Fee	150	150	Non- Refundable
	TOTAL	**20,450**	**20,150**	Non- Refundable

The above mentioned fee structure is exclusive of Examination fees.

g. Requirement of the Laboratory Support and Library Resources:

A well-equipped Physics Laboratory for M.Sc. Physics (Distance Learning Mode) programme was established at the Department of Physics, Alagappa University, Karaikudi with all the necessary equipments for conducting face to face contact sessions for Practical courses. The students who have enrolled in M.Sc. Physics Programme through Distance Learning Mode have to compulsorily attend the face to face contact session for Practical Courses at the said Physics Laboratory in the head quarter **only**.

Directorate of Distance Education, Alagappa University, Karaikudi houses an exclusive Library facility with adequate number of copies of books in relevant titles for M.Sc. Physics programme. The Central Library of Alagappa University is also having good number of reference books. The books available at both the libraries are only for reference purpose and not for lending services.

h. Cost Estimate of the Programme and the Provisions:

The cost estimate of the programme and provisions for the fund to meet out the expenditure to be incurred in connection with M.Sc. Physics Programme are as follows:

Sl. No.	Expenditure Heads	Approx. Amount in Rs.
1	Programme Development (Single Time investment)	**20,00,000**
2	Programme Delivery (Per Year)	**24,00,000**
3	Programme Maintenance	**5,00,000**

i. Quality Assurance Mechanism and Expected Programme Outcomes:

i. 1. University's Moto:

'Excellence in Action'

i. 2. University's Vision Statement:

Achieving Excellence in all spheres of Education, with particular emphasis on "PEARL"- Pedagogy, Extension, Administration, Research and Learning.

i. 3. University's Objectives:

- Providing for Instructions and Training in such Branches of Learning as the University may determine.
- Fostering Research for the Advancement and Dissemination of Knowledge

i. 4. University's Quality Policy:

Attaining Benchmark Quality in every domain of 'PEARL' to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

i. 5. University's Quality Quote:

Quality Unleashes Opportunities Towards Excellence (QUOTE)

i. 6. Programme's Review Mechanism:

The quality of the programme depends on scientific construction of the curriculum, strong-enough syllabi, sincere efforts leading to skillful execution of the course of the study. The ultimate achievement of M.Sc. Physics programme of study may reflect the gaining of

knowledge and skill in Physics subject. And all these gaining of knowledge in Physics may help the students to get new job opportunities, upgrading in their position not only in employment but also in the society, make students feel thirsty to achieve in research in the fields associated with Physics discipline, achieving in competitive examinations on the subject, etc.

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations. Apart from the end semester examination-based review, feedback from the alumni, students, parents and employers will be received and analyzed for further improvement of the quality of M.Sc. Physics Programme.

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MINUTES OF THE MEETING OF THE BOARD OF STUDIES IN PHYSICS FOR M.Sc. PHYSICS (DISTANCE EDUCATION MODE) PROGRAMME HELD ON 23.06.2017

The Board of Studies Meeting for M.Sc. Physics (Distance Education Mode) programme was held on 23.06.2017 at 11:00 a.m. in the Directorate of Distance Education, Alagappa University, Karaikudi.

The following members were present:

1. Dr. G. Ravi - Chairman
   Professor and Head, Department of Physics
   Alagappa University, Karaikudi
2. Dr. R. Sivakumar - Member
   Assistant Professor, Department of Physics
   Alagappa University, Karaikudi
3. Dr. S. Sankar - Member
   Professor, Department of Physics
   Anna University, MIT Campus
   Chennai
4. Dr. S. Rajashabala - Member
   Associate Professor
   Department of Theoretical Physics
   Madurai Kamaraj University, Madurai

At the outset, the Chairman of the Board, Dr. G. Ravi, has extended a very warm welcome to all the Members of the Board and briefly narrated the need and significance of designing the regulation and syllabi for M.Sc. Physics (Distance Education mode) programme.

The board appreciated the Vice-Chancellor for taking efforts in revising the curriculum and syllabi of the said programme.

After detailed discussion and deliberations, the Board has unanimously resolved to approve the regulation and revised syllabi of the M.Sc. Physics (Distance Education mode) programme to be introduced in the Directorate of Distance Education.

The board resolved to include internal assessment mark 25 and annual examination mark 75, during the implement of this revised syllabi.

The detailed revised syllabi and the regulations are enclosed in the Appendix.

The Chairman of the Board thanked the Members of the Board for the deliberations made in the meeting.

Name of the Member	**Signature**
Dr. G. Ravi	G. Ravi 23/6/17
Dr. R. Sivakumar	R. Sivakumar 23/06/2017
Dr. S. Sankar	Sankar 23/6/17
Dr. S. Rajashabala	S. Rajashabala 23/6/17

M.Sc. (Psychology)

Programme Project Report (PPR)

a. Programme's Mission & Objectives

Mission:

The Psychology programme's mission is to contribute to the advancement of knowledge in psychological science, to share the discipline's knowledge through teaching and mentoring, and to apply that knowledge through commUNITy outreach/service and thereby tackling important social issues. Providing unique, innate talents of human resources through psychological assessment and nurturing them for individual excellence by effective training & learning.

Objectives:

- To enable the learners to understand the theoretical perspectives of psychology.
- To enable the learners to understand social influences on human behavior.
- To enable the learners to understand the dynamics of social influence on positive and negative human behavior.
- To enable the learners to understand how social psychological principles that are applied in day to day life situations.
- To describe sensational, perceptual phenomenon and its different scientific explanations.
- To impart the learners with knowledge about skills and techniques to conduct psychological assessment.
- To impart the learners with knowledge about skills and techniques relating to various approaches of Guidance & Counseling.
- To familiarize learners with new psychological therapies.
- To provide a supportive, engaging, enjoyable and challenging learning experience.

b. Relevance of the Programme with HEI's Mission and Goals:

Affording quality higher Education to the learners who are interested in pursuing higher education through distance mode, so that they are transformed into intellectually competent human resources that will help in the uplift of the nation in terms of Educational, Social, Technological, environmental and Economic magnificence (ESTEEM). This programme is very much effective in imparting quality education through flexi-timings.

Psychology helps explain how the inner-workings of the brain manifest in our everyday life. Psychology is so diverse in its focus and research, it also prepare students for a wide variety of carriers.

In accordance with the mission of Alagappa University as a research-intensive institution, the teaching programme of the master's programme in psychology is based on state of the art of scientific research and maintains a strong emphasis on the acquisition of academic and research skills.

c. Nature of prospective target group of learners:

The curriculum has been designed to fulfill the needs of diverse class of learners including a class having of low level of disposable income, rural dwellers, women, unskilled men, minorities etc. Especially this curriculum will fulfill the needs of the learners, professionals who are in teaching, social workers, medical practitioners and management professionals.

M.Sc., Psychology Employment Areas:

- Commercial and Industrial companies
- Financial organizations
- Marketing companies
- HR Departments
- Government Social Services
- Jails / Prisons
- Police Forces
- School, Colleges & Universities.

M.S c., Psychology Job Types:

- Urban planning officer
- Staff Recruiter
- Social psychologist
- Human Resources Assistant
- Counseling Psychologist
- Child support specialist, social worker
- Special education teacher
- Career counselor & career employment specialist
- Psychotherapist / Psychologist
- Psychology consultant
- Educational psychologist

d. Appropriateness of the programme to be conducted in Open and Distance Learning mode to acquire specific skills and competence

- The master's programme in psychology focuses both on in-depth study of theory and acquisition of professional and research skills.
- This programme is designed to facilitate learner's development of academic, skills, behaviour and attitudes. Content in this programme not only helps learners immediately but also prepares learners for professional success throughout their career.
- This programme provides learners with training in the skills necessary to research and enhance the discipline of psychology.
- It also provides autonomy and employability through the development of evaluative and reflective skills.
- This programme makes learners develop universally useful skills like communication, critical-thinking and proficiency.

Knowledge & Skills gained through Psychology programme:

Knowledge:

- An understanding of psychological, emotional, cognitive and social determinants of behavior.
- Basic understanding of the various disciplines in psychology as well as major theories and history of psychology.
- Familiarity and experience with the psychological research methods, including data collection and analysis.
- Knowledge of research ethics.
- Specific knowledge and abilities related to individuals concentration (i.e. Cognitive, Neuroscience, Developmental, Forensic, etc.)

Skills:

Communication skills: Learners can write and speak about issues in detail, but they are also familiar with concise writing within a pre-set format as they write practical research reports.

Problem-solving: Learners learn how to apply different strategies and approaches to understand problems and how to identify the practical steps to implement a solution.

Critical Evaluation: Learners learn to access whether evidence for a phenomenon is really what it appears to be and identify the shortcomings and pitfalls of a particular line of action.

Benefits of M.Sc., Psychology:

- The Master's degree course gives a good base for higher degree programmes like M.Phil., and Ph.D for further research studies.
- They can join schools/colleges both on private and regular basis; for regular one they should pass certain degrees / qualifications such as B.Ed., NET, SET etc.,
- They can go for social care jobs as well; here they work to meet out the needs of those coping with problems such as poverty, life crisis and emotional ill health.

M.Sc., Psychology course suitability

Those who are willing to go for teaching at higher degree level i.e., colleges and university levels both in private and government institutions are good match for it.

They should have the skills such as communication, information technology, handling of data / statistics, problem solving and ability to work in teams.

Students also should have the qualities such as human behavior and motivation, ability to critically analyze the problem, formulate a considered response, create an argument and generate new ideas, lend themselves well to careers in the creative industries, legal sector, government administration and education.

e. Instructional Design

Curriculum Design

Sl. No.	Course Code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C Max.
FIRST YEAR						
I Semester						
1.	36311	Theoretical Perspectives in Psychology	25	75	100	4
2.	36312	Life Span Psychology	25	75	100	4
3.	36313	Social Psychology	25	75	100	4
4	36314	Psychology Practical- I	25	75	100	4
		Total	**100**	**300**	**400**	**16**
II Semester						
5	36321	Theories of Personality	25	75	100	4
6	36322	Research Methodology	25	75	100	4
7.	363231 363232	**Elective(any one)** Educational Psychology Rehabilitation Psychology	25	75	100	4
8.	36324	Psychology Practical – II	25	75	100	4
		Total	**100**	**300**	**400**	**16**
SECOND YEAR						
III Semester						

9.	36331	Cognitive Neuro Psychology	25	75	100	4
10.	36332	Counseling Theories and Techniques	25	75	100	4
11.	36333	Psychopathology	25	75	100	4
12.	36334	Psychology Practical – III	25	75	100	4
		Total	**100**	**300**	**400**	**16**
		IV Semester				
16.	36341	Abnormal psychology	25	75	100	4
17.	36342	Environmental psychology	25	75	100	4
18.	36343	Positive psychology	25	75	100	4
19	363441 363442	**Elective:(any one)** Health Psychology Organisational Psychology	25	75	100	4
		Total	**100**	**300**	**400**	**16**

Course Code Legend:

3	**6**	**3**	**X**	**Y**

363 - M.Sc., Psychology Programme

X -Semester No.

Y - Course number in the semester

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C: Credit Points, Max.: Maximum

No. of Credits per Course (Theory) - 4

No. of Credits per Course (Practicals) - 4

Total No. of Credits per Semester - 16

Total No. of Credits of the Programme - 16 X 4 = 64

Detailed Syllabi

FIRST SEMESTER

Course Code	Title of the Course
36311	THEORETICAL PERSPECTIVES IN PSYCHOLOGY

Learning objectives:

- To make students get a historical perspective about the development of psychology as an independent body of knowledge.
- To emphasize the understanding human behaviour from each school's perspective in respect of human motivation, development and functioning of human personality.
- To apply the principles of each school to the development of mankind as well as its therapeutic value.

UNIT 1: Psychoanalytical perspective

History - Sigmund Freud's approach - Carl Jung, Adler, and other Neo-freudian approaches to motivation, personality, therapy and applications.

UNIT 2 : Behaviouristic perspective

Learning- Classical Conditioning (Pavlov) and Operant Conditioning (Skinner); Motivation – Drive and incentive theories (Hull), (Miller and Dollard, Rotter); Personality – Mowrer; therapeutic techniques and applications.

UNIT 3 : Humanistic & Existential perspectives

Motivation: Hierarchy of motives (Maslow) - ERG Theory (Alderfer) - Theory of needs (McClelland);

UNIT 4: Personality

Personal construct (Kelly) - Self-theory of personality (Rogers); Existential approaches - therapies and application.

UNIT 5 : Cognitive: Motivation

Cognitive balance and dissonance theory (Hieder, Festinger); Personality: Dissonance (Brehm),

UNIT 6: Social Perspectives

Social learning theory (Bandura); therapy and application.

UNIT 7 : Indigenous Perspectives

Motivation: Advaita, Buddhist and Jaina perspectives; Personality: Advaita, Upanishads, Buddhist and Jaina perspectives; Therapy (healing techniques), Applications.

UNIT 8: Psycholinguistic, Memory and Decision Making

Language structure; Theories of Language; Neurological basis of language, language Acquisition: stages in language development Memory Processes; Theories of Forgetting Models of Memory; Biological basis of memory; Strategies to improve memory Cognitive Strategies; Theories of Thought Processes, Concept formation, Creativity; Problem solving and Decision Making

UNIT 9: Attention and Perception

Attention: Definition and mechanism, determinants of attention, Selective, divided and sustained attention, Theories of Attention

UNIT 10: Approaches to study of perception:

Gestalt and physiological approaches; Perceptual Constancy; Illusion; Perception of Depth and Movements; Ecological perspective on perception

UNIT 11: Perceptual Processes

Approaches to the Study of Perception – Gestalt, Behaviouristic and Physiological Approaches, Perceptual Organization – Gestalt, Figure and Background, Laws of Perceptual Organization, Perceptual Constancy – Size, Shape, Brightness, Depth Perception, Role of Motivation and Learning in Perception, Figural After Effect.

UNIT 12: Phenomenological Theory and Cognitive Theory

Kelly's Personal Construct Theory, Roger's Self Theory, Lewin's Field Theory, Festinger's Cognitive Dissonance Theory, Mischel's Cognitive-Behavioural Theory

UNIT 13: Psychology in India

Introduction; Twentieth Century Psychology in India and future perspectives.

UNIT 14 : Approaches of Psychology

Constructivism and Social Constructivism (Vygotsky, Gregan and Bruner)

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5. Sahakian(1976) Introduction to psychology of learning. Chicoga: Rand Mcnally college publishing company.
6. Weiner B (1985) Human Motivation, New York: Springer and Verlag.

Course Code	Title of the Course
36312	LIFE SPAN PSYCHOLOGY

Learning objectives:

- To explain the importance of studying life-span development
- To describe the history of interest in the life-span perspective and indicate how contemporary concerns have arisen from previous views.
- To describe the seven basic characteristics of the life-span perspective
- To discuss the nature of development as a pattern of movement or change occurring throughout the life span
- To define and distinguish between biological processes, cognitive processes, and socioemotional processes
- To understand the major developmental periods from conception to death
- To understand the three major developmental issues (nature and nurture, continuity and discontinuity, stability and change)

UNIT 1:Introduction

An introduction to development in the lifespan- theoretical perspectives in developmental psychology and human development - Freud's psychosexual stages of development - Erikson's psychosocial stages of development - theories of learning -Piaget's cognitive stages of development - socio-cultural theories.

UNIT 2: Stages of development

Infancy, newborn and infant- prenatal diagnostic tests, genetic-environment interactions, timelines for prenatal development, teratology, and stages of childbirth. Newborn reflexes, newborn needs- patterns in infant physical development, infant temperament. EarlyChildhood- physical, cognitive, and socio emotional development- development of gross and fine motor skills- brain development. Middle childhood- physical and cognitive-growth patterns, child obesity, ADHD, and concrete operational stage of cognitive development. Social development- peer relations, divorce, and moral development.

UNIT 3: Basic Concepts

Aspects of Development, Life Span periods – Methods – Non Experimental, Experimental - Stages of Development – Principles of Development – Prenatal period – Birth – Neonatal stage – First year of Life – Early
childhood, Middle childhood – Adolescence, Adulthood and old age.

UNIT 4: Physical Development

Motor Skills – Growth rate – Physical health during Adulthood, Physical fitness & energy – Motor functions in old age. Intellectual Development

UNIT 5 Approaches

Psychometric, Piagetian and Information processing
approach – Cognitive Development – Piaget's model – Language Acquisition and Development of language, Memory, Intelligence and Moral Development.

UNIT 6: Adolescence

History of marking adolescence - puberty - the secular trend regarding menarche and social implications for pubertal timing- Eating disorders -identity status. Other adolescent problem behaviors such as juvenile delinquency, depression, behaviour disorders and suicide

UNIT 7: Early adulthood

Emerging adulthood - timeframes for physical peak and the physical declines-adult sexuality and relationships- theories on post-formal thought and theory of love.

UNIT 8: Middle adulthood

Middle adulthood section- premenopausal and menopause- hormone therapy for menopausal symptoms- midlife crisis and transitions- and family relationships- Late adulthood and the end of life, biological aging, dementia/neurocognitive disorders, stages of dying, death, hospice and palliative care, and life satisfaction in late adulthood

UNIT 9: Life Span Problems

Mental Health Problems-Sexual Problems-AIDS/HIV- Preventive Measures- Emotional imbalance-Fear and Phobic Problems- Anxiety and Stress -Coping Style- Development of adaptive and positive behavior-Human relational Problems- Mental Peace-Life satisfaction.

Block 10: Personality and Social Development

Emotions – emergence of Self – Role of parents and siblings – peer group influence – Psychoanalytic, social learning and cognitive perspectives in the personality development – Emotional problems of childhood – identity crisis in adolescence, relationship with parents and peers, sexual identity- Teenage problems.

UNIT 11: Personality and Social issues in young adulthood

Parenthood – Career planning – Intimate relationship and personal life styles – work life – personal relationship in family and work life Old age Physical changes - Psychomotor functioning – Health & fitness – Health problems – Memory changes – Work and Retirement – Adjustment to Old age - Personal Relations in Late life – Death Bereavement – Purpose and meaning of life.

UNIT 12: Social Motives, Attitudes And Learning In Social Contexts

Social motives and behaviour-The nature and measurement of attitude-Reinforcement and learning-Social learning through imitation Attitude change

UNIT 13 : Antisocial and Pro-Social Behaviour

Aggression and management- Altruism and helping behaviour

UNIT 14 : Group Processes

Group formation and maintenance Types of group Group task performance, problem solving Cooperation and competition - communication – empathy - Psycholinguistics

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2. Willem Doise (1998), Life-Span Developmental Psychology
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9. Tedeschi and Lindskold (1978) Social Psychology

Course Code	Title of the Course
36313	SOCIAL PSYCHOLOGY

Learning Objectives:

- To enable the students understand social influences on human behaviour
- To enable the students understand the dynamics of social influence on positive and negative human behaviour
- To enable students to understand how social psychological principles are applied in day to day life situations.

UNIT 1: Introduction: Nature and scope of social psychology; Overview of the history of social psychology (including development in India); Relationship with sociology and anthropology

UNIT 2 : Basic concepts of Social Psychology

Definition - Basic concepts of Social Psychology – scope – individual, society and culture – Social Psychology and related disciplines -- Social Psychology in the new millennium - Research Methods in Social Psychology: Survey, Correlation and Experimental Methods

UNIT 3: Brief history of social psychology (special emphasis on India),

Scope of social psychology, levels of social behavior, approaches towards understanding social behavior

UNIT 4 : Social Perception

Self-Perception: Self-concept, Perceived Self-control, Self-serving Bias, Self-presentation, Self-esteem, Self and Gender

Perceiving Others: Non-Verbal Communication, Attribution, Impression Formation, Impression Management

UNIT5 : Attitudes And Behavior

Attitude formation – Development of Attitudes – Attitude and its influence on behaviour - Persuasion – Change in Attitudes – Resisting persuasion – Cognitive Dissonance – Ways to manage dissonance.

UNIT 6 : Prejudice And Aggression

Prejudice -Discrimination in Action – Sources of prejudice – Social, Emotional and Cognitive - Techniques to reduce Prejudice Aggression - Theories of Aggression - Types of Aggression - Determinants ofAggression - Environmental causes - prevention and control of Aggression.

UNIT 7 : Prosocial Behavior

Motives for Prosocial behaviour - Factors that Affect Helping Behavior, External and Internal influences on helping behaviour – Long-term commitment to Prosocial Acts

UNIT 8: Understanding and evaluating

the social world: Social cognition, Social perception, Attitudes, Attitude-behaviour link; Strategies for attitude change

UNIT 9: Social interaction and Influence:

Interpersonal attraction, Pro-Social Behaviour, Aggression, Social Influence

UNIT 10: Group Dynamics and Inter-group relations: Nature of groups, Consequences of belonging (performance, decision making, cooperation and conflict), Nature of intergroup relations (prejudice, intergroup conflict, intervention techniques)

UNIT 11: Individual level processes:Person perception: attribution-theories, biases and errors Attitude: formation, change and resistance to change

UNIT 12: Interpersonal processes: Interpersonal attraction, prosocial behavior, aggression

UNIT 13: Group dynamics:

Key aspects of groups, cooperation and conflict, group decision making.

UNIT 14: Social Perception

Nonverbal Communications - Attribution: Understanding the causes of others Behaviour - Theories of Attribution- Kelley's model -Impression Formation and Management - Social Cognition: Social Information Heuristics and Automated Processing - Sources of Error. Groups: Types and formation - Theories of Group Formation - Co-ordination - Group Decision making -Group Think

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Course Code	Title of the Course
36315	PSYCHOLOGY PRACTICAL I

Tests from the following areas will be selected by the University and conducted during the I semester of the course.

1. Attention
2. Learning
3. Memory
4. Perception
5. Intelligence
6. Problem Solving
7. Creativity
8. Adjustment
9. Attitude
10. Prejudice
11. Motivation
12. Achievement Motivation

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1. Cronbach, L.J. Essentials of Psychological Testing, 1972. New Delhi, Prentice Hall Inc.
2. Woodworth R.S. and Scholsberg, 1981, Experimental Psychology, New Delhi, Taa McGraw Hill Co. Ltd.,
3. Udaipreek, T. Venkateswara Rao. Handbook of Psychological and Social Instruments. Samashti, B-2, Chamelibagh, Baoda.

SEMESTER II

Course Code	Title of the Course
36321	THEORIES OF PERSONALITY

Learning Objectives

The course will enable students to

- To understand major theoretical approaches to personality
- To understand assessment methods used in personality.
- To understand the process of personality change and development

UNIT 1: Introduction to personality and personality theory

Personality: Meaning and related concepts-Factors that contribute to personality change and personality stability - The process of personality development- The nature of theories and its functions-

UNIT 2 :Nature of personality theory

Personality theory and other psychological theories- Overview of assessments of personality-Current research focus in personality psychology.

UNIT 3: Major theoretical approaches- Psychoanalytic and Psychodynamic theories. Neo Freudians.

Sigmund Freud 's Classic Psychoanalytic theory: The structure of personality: Id, ego& super ego, The dynamics of personality: Instinct, The distribution and utilisation of psychic energy & anxiety-The development of personality-

UNIT 4:Carl Jung's Analytic theory

The structure of personality: The ego, The personal unconscious, The collective unconscious-Interactions among the systems of personality.

UNIT 5: Alfred Adler

Inferiority Feelings: The Source of Human Striving - Striving for Superiority or Perfection- The Style of Life and birth order, Erick Fromm –

UNIT 6: Freedom or Security

The Basic Human Dilemma, Personality Development in Childhood -The Basic Psychological Needs - The Productive and Non -productive Character Types , Karen Horney - The Childhood Need for Safety, Basic Anxiety: The Foundation of Neurosis The Idealized Self-Image.

UNIT 7: Major theoretical approaches: Humanistic theories

Abraham Maslow: Personality Development: The Hierarchy of Needs. The Study of Self-Actualizers. Carl Rogers: The Self and the Tendency toward Actualization, the experiential world, The Development of the Self in Childhood and Characteristics of Fully Functioning Persons.

UNIT 8: Major theoretical approaches: Trait theories

Gordon Allport: The Nature of Personality, Personality Traits, Motivation: The Functional Autonomy of Motives ,

UNIT 9: Personality Development in Childhood

The Unique Self, The Healthy Adult Personality. Raymond Cattell- Cattell's Approach to Personality Traits ,Source Traits: The Basic Factors of Personality, Dynamic Traits: The Motivating Forces-The Influences of Heredity and Environment-

UNIT 10: Stages of Personality Development

Hans Eysenck :The Dimensions of Personality- Extraversion, Neuroticism, and Psychoticism. Robert McCrae and Paul Costa: The Five-Factor Model- Arnold Buss and Robert Plomin: The Temperament Theory.

UNIT11: Major theoretical approaches: Cognitive approaches

Personal Construct Theory, Ways of Anticipating Life Events, the nature of personality.

UNIT 12: Social learning theory

Albert Bandura: Modelling: The Basis of Observational Learning -The Processes of Observational Learning,-Self-Reinforcement and Self-Efficacy ,

UNIT 13: Behaviour theories

B.F. Skinner : Reinforcement: The Basis of Behaviour, Operant Conditioning and the Skinner Box- Schedules of Reinforcement, Successive Approximation: The Shaping of Behaviour-Superstitious Behaviour, The Self-Control of Behaviour-Applications of Operant Conditioning.

UNIT14: The Psychodynamic Perspective

Classical Psychoanalysis – Freudian Psychoanalysis – Ego Psychology and Neo-Freudians –Hartman – Kohlberg – Erickson - Adler – Sullivan – Jung – Eric Berne**.**

References

1. Hall, S.C., Lindzey, G.,Campbell, B J. (2007). Theories of personality.(4th Ed). India: John Wiley & Sons, Inc.
2. Haslam, N (2007). Introduction to personality and Intelligence. London. Sage Publications Ltd
3. Schultz, P. D &Schultz, E. S (2005). Theories of personality.(8th Ed). UK: Wadsworth Publushers.
4. Freidman, H.S. and Schustack, M. W (2004). Personality. New Delhi: Pearson Education.

Course Code	Title of the Course
36322	RESEARCH METHODOLOGY

Learning Objectives

- Students will understand a general definition of research design.
- Students know why educational research is undertaken, and the audiences that profit from research studies.
- Students are able to identify the overall process of designing a research study from its inception to its report.
- Students will be familiar with ethical issues in educational research, including those issues that arise in using quantitative and qualitative research.
- Students know the primary characteristics of quantitative research and qualitative research.
- Students will be able to identify a research problem stated in a study.

UNIT 1: Meaning, Types and Process of Research

Meaning – Purpose – Types of research – Pure, applied, historical, analytical, descriptive and experimental – Significance of research in social sciences – Process of research – Meaning – Scientific method – Induction and deduction.

UNIT 2: Planning Research

Research problem – Identification, selection and formulation of research problem – Review of literature in the field of corporate management – Hypothesis – Meaning – Sources of hypothesis – Types of Hypothesis – Formulation and testing – Research design – Factors affecting research design – Evaluation of research design.

UNIT 3: Sampling Design

Census method and sampling method for investigation – Advantages and disadvantages of sampling – Principle of sampling – Essentials of a good sampling – Methods of sampling – Probability and non-probability sampling methods – Selection of a sample – Factors affecting the size of the sample – Biased sample – Sampling and non-sampling errors.

UNIT 4: Sources and Collection of Data

Sources of data – Primary and secondary data – Modes of data collection – Analytical method – Case study – Observation – Survey method – Interview – Its purpose and importance – Types of interview – Preparation for an interview – Effective interview techniques – Limitations of interview – Schedule – Its meaning and kinds – Essentials of a good schedule – Procedure for the formulation of a schedule – Questionnaire – Meaning and types – Format of a good questionnaire – Factors affecting the response to a questionnaire – Advantages and limitations of schedules and questionnaires – Pre-testing and its importance.

UNIT 5: Processing and Analysis of Data

Meaning – Importance – Process of data analysis – Editing – Coding – Tabulation – Diagrams – The process of interpretation – Guidelines for making valid interpretation – Scaling techniques – Meaning – Importance – Methods of their construction.

UNIT 6 : Foundations Of Research

Research: Meaning – objectives – Types – Research Approaches – Significance of research –Research Methods versus Methodology – Research and Scientific method – Problems encountered by researchers in India. Ethical Principles in animal research and research with human participants. Major stages in research –

UNIT 7 : Research Problems

Nature – Sources –Defining and stating a problem – Criteria of a good problem. Review of Literature: Functions – Sources – The search for the literature – Criticism.

UNIT 8 :Tools of Research

Tools of research: Criteria for selection of tools –Factors related to construction of tools – Tools of different types: Observation – Interview –Questionnaire – checklist- Rating Scales: Merits and Limitations – Writing a research proposal.

UNIT 9: Statistics

Scales of measurement - Frequency distributions and Graphs: Steps – Exact limits and midpoints of the class intervals – Graphical representation of Data: Different types of graphs – Issues to consider when preparing a graph. Measures of Central Tendency: The Mean, Median and Mode

UNIT 10 :Measures of Variability

The Range, Quartile Deviation, Average Deviation and Standard Deviation. Normal probability cure: Characteristics – Applications – Skewness and kurtosis

UNIT 11:Statistical Techniques Applied In Psychology

Correlation: Meaning – Concept of Correlation – Pearson‟s Product moment correlation – Rank order correlation – Test of Significance: t Test – Calculation and interpretations – The „t‟ ratio and its assumptions. Analysis of Variance (ANOVA): Meaning – logic – example for one-way ANOVA – interpretation – Assumptions of the ANOVA. Regression and rediction- An overview of Non-Parametric statistics.

UNIT 12 :Research Methods

Normative Survey – Experimental Research – Variables and experimental control. Experimental designs: Pre-experimental designs – True experimental designs – Quasi experimental designs – Single subject experimental designs – Ex-post Facto Designs – Interpretation and report writing.

UNIT:13 Variables

Operationally defining variables, types of variables, controlling variables. Hypothesis - Formulation, types, Research lab – Formulation of research hypotheses, Research proposal.

UNIT: 14 SPSS for Psychology

26Qantitative analysis of the data – purpose, conditions and interpretation of major

parametric and non parametric statistical techniques using SPSS

References

1. John W Best, Research in Education.
2. Anderson et-al, Thesis and Assignment Writing.
3. Goode and Hatt, Methods of Social Research.
4. Wilkinson and Bhandarkar, Methods and Techniques of Social Research.
5. ICSSR, Training in Research Methodology in Social Sciences in India.
6. Coaley, K. (2009). An Introduction to Psychological Assessment and Psychometrics.
7. New Delhi: Sage Publications.
8. Coolican, H. (2009). Research Methods in Statistics in Psychology. New Delhi:
9. Rawat Publications. M.Sc Applied Psychology, 2014-15
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11. Kerlinger, N. (1996). Foundations of behavioural research. India: Prentice Hall
12. Kothari, C.R. (2008). Research Methodology – Methods and Techniques. New Delhi: Wiley Eastern Ltd.
13. Kundu. (2010). Research Methodology. New Delhi: Pearson Publishing.
14. Myers, J. (2008). Methods in Psychological Research. New Delhi: Sage Publications.
15. Ruyon, R.P, Haber, A, Pittenger, D.J and Coleman, K.A. (2010). Fundamentals of Behavioural Statistics. New York: Mc Graw Hill.
16. Singh, A.K. (2006). Tests, Measurements and Research Methods in Behavioural Sciences. Patna: Bharati Bhavan Publishers.
17. Flick, U. (2010). Introduction to Qualitative Research (fourth edition). New Delhi: SagePublications,
18. Garret, H. E. (2005). Statistics in Psychology and Education. New Delhi: Paragon International.
19. Krishna Swamy, O. P. (1993). Methodology of research in Social Sciences. Himalaya Publishing house.
20. Kothari, C. R. (1998). Research Methods and techniques. New Delhi: Wiley Eastern.
21. Miles, J. (2001). Research Methods and Statistics. Exeter, Crucial.

Course Code	Title of the Course
363231	Elective: Educational Psychology

Learning Objectives

At the end of the Course, student will be able to:

- Analyze the different principles and theories explaining student learning
- Evaluate the effectiveness of the theories in explaining individual differences in learning
- Apply the different principles and theories of learning in the classroom
- Analyze the impact of educational psychology on the processes of teaching and learning

UNIT 1 : Psychology: An Overview

Introduction - Psychology Meaning - History of Psychology -Branches of Psychology - Research Methods in Psychology- Psychology of Learning and Education -Learning - Summary.

UNIT 2: Behavioural Learning theories

Introduction - classical conditioning by Ivan Pavlov - Classical Conditioning in Daily Life - Behaviourism - Watsan's Experiments with Little albert - Classical Conditioning in the Classroom - connectionism - Edward L.Thorndike - Implications of Thorndike's Thories - Operant Conditioning by B.F.Skinner - Schedules of Reinforcement - shaping Behaviour - Applying Operant Conditioning in the Classroom - Summary.

UNIT 3: Cognitive Learning Theories

Introduction: Cognitive - Gestalt Theories of Learning -Problem Solving by Insight - Piaget's Theory of Learning - Piaget's Theory: Application in the Classroom - Social Learning Theory: Albert Bandura -Modeling, Imitation and Reinforcement - Application of Social Learning Theory- Theory of Meaningful Learning: David Ausubel-Application of Ausubel's Learning Theory-Summary.

UNIT 4: Critical and creative Thinking

Introduction -Some Perspectives about Teaching Thinking-Definitions of Thinking-Attributes of Good Thinkers- A programme for Teaching Thinking- Critical Thinking-Components of Critical Thinking-Socratic Questioning to Enhance Critical Thinking-Creative Thinking-Definition of Creative Thinking-The Creative Process-The Creative Person-Helping Children to be More Creative-Summary.

UNIT 5: Motivation and Learning

Introduction-Some Thought on Emotion and Learning-Motivation-Expectancy-Value Theory-Valuing Task-Expecting Success-Motivating Students to Learn-Summary.

UNIT 6: Research in educational psychology

Experimental and non experimental methods

UNIT 7: Different views of learning

Developmentally based views of teaching and learning: Piaget's theory of intellectual,development and classroom application, Vygotsky's sociocultural applications,Constructivism, Language Acquisition, Development of morality, social responsibility andself control, Behaviourist views of learning.

UNIT 8: Classroom atmosphere and management

Discipline- nature and meaning. Styles. Problem behaviour.

UNIT9: Learner differences

Intelligence, Socioeconomic status, Culture, Gender, At risk students

UNIT 10 : Children with special needs

Socially disadvantaged, Disabled children, talented, gifted and creative children Mainstreaming and inclusion.

UNIT 11: Application of theories of learning in teaching

Classical conditioning, operant conditioning, Connectionism, Social cognitive learning, Cognitive Psychology perspectives to learning; David Ausubel's meaningful reception learning, Jerome Bruner's discovery learning.

UNIT 12: Effective teaching-learning and evaluation

Self regulation in learning-meaning, factors, self-regulated learning cycle; Group work and co-operation in learning-defining elements of co-operative learning, strategies for cooperative and Collaborative learning .Classroom management, creating effective learning environment, methods and technical issues in the assessment of students, Effective teaching strategies, technology based teaching strategies

UNIT 13: Use of Behaviour Modification in School setting

Rewarding Principles of Differential reinforcement Modelling Shaping Contingency management, Contracting

UNIT 14: Role of the College Counsellor

Career Counselling Individual Counselling for Personal Growth, Adjustment Problems at home or college, Interpersonal Relationship Issues, Love Failures, Attempted Suicide, Substance-Abuse Training for Life skills and Study Skills, Overall Personality Development

References

1. Snowman, J.&McCown, R.Biehler, R.F.(2012).Psychology applied to teaching (13th ed.). Wadsworth Cengage Learning.
2. Bryan, H. (2010) Education – Study and teaching (Graduate).London:SAGE
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4. Slain, R.E. (2011).Educational Psychology:Theory and Practice.(10th Edition),Pearson.
5. Ames, C. (1992). Classroom: Goals, Structure and student motivation.
6. . Eggen, P., & Kauchak, D. (1999). Educational Psychology. New Delhi: Prentice-hall of India Private Limited.
7. Kottler, J. A., &Kottler,E. (2007). Counseling Skills for Teachers, 2/E. Corwin Press
8. Woolfolk, A. (2004). Educational Psychology, 9th Edition. Delhi: Pearson Education.

Course Code	Title of the Course
363232	Elective: REHABILITATION PSYCHOLOGY

Learning Objectives

- To understand the historical perspectives, methods and functions of rehabilitation psychologist in the field of rehabilitation services.
- To become aware of psychological approach to rehabilitation in rehabilitation psychology.
- To understand the personality development among children with disabilities and their coping styles and rehabilitation process
- To be aware on the rehabilitation process in various areas.

UNIT1: Rehabilitation Psychology:

Definition- scope- methods and Functions of Rehabilitation Psychology- historical perspectives in Rehabilitation Psychology

UNIT 2: Competencies of Rehabilitation Psychologists:

Professional Competencies of rehabilitation psychologists, nature of work settings of rehabilitation psychologists, Designing training programmes for rehabilitation psychologists, Training need analysis and implementation of training programmes.

UNIT 3: Psychological Rehabilitation and Intervention:

Definition and basic principles of Psychological Rehabilitation, Assessment, diagnosis, and Intervention – Psychoanalytic therapy, Client Centred Therapy, Cognitive Behaviour therapy, Rational Emotive therapy, supportive therapy, Augmentative therapy and Behaviour therapy.

UNIT 4: Rehabilitation of Persons with Disabilities:

Lifespan development of persons with disabilities, Personality traits – Psychological problems and coping styles – Role of psychologist in disability rehabilitation.

UNIT 5: Rehabilitation Process in various areas:

Family and Marital Rehabilitation, Socio Economic Rehabilitation for Persons with disabilities, Addiction Rehabilitation, Vocational Rehabilitation, CommUNITy based rehabilitation, Disaster Rehabilitation/Reconstruction.

UNIT 6 : Disabilities

Disability - Concept and definitions, Classification of various disabilities, Incidence and prevalence Types of disabilities: Visual impairment, Hearing and speech impairment, Locomotor disability, Mental retardation, Cerebral palsy, Autism , Mental illness Learning disabilities, Multiple handicaps Etiological factors; pre-natal, natal and post-natal, chromosomal aberrations and genetic errors, Prevention of disabilities

UNIT 7 : Nature and Scope of Rehabilitation psychology

Definition, historical perspective, scope and methods, Functions of Rehabilitation Psychology: General functions and special functions, History and Philosophy of Disability Rehabilitation Goals and objectives of rehabilitation,

UNIT 8: Multi-disciplinary approach to rehabilitation:

Biological, medical, psychological, educational and social aspects.

UNIT9 : Psychological Intervention

Planning Intervention: Psychoanalytic Approach, Learning Theories and Strategies, Planning and Designing, Learning Situations, Counselling Strategies. Therapeutic services and Restorative techniques.

UNIT 10 : Designing Training Programmes for Professionals:

Training Need Analysis, Implementation of Training Programmes Monitory and Impact Studies.

UNIT 11 : Organization & Management

Evolution of Non-Government Organizations Background Characteristics of Organization Capacity Building of Non-Government Organizations

UNIT 12: Personality Development of Disabled Persons and intervention

Factors influencing personality development of disabled individuals, Life span development of people with disabilities, Assessment of personality of disabled individuals, Screening and early identification of people with developmental disabilities. Social, Psychological Perspective in Rehabilitation Psychology.

UNIT 13 : Early intervention:

Definition, assessment and strategies for intervention. Intervention packages for various disabilities. Services and programmes for disabled individuals and their families in India

UNIT 14 :Special education:

Aims, objectives and functions, Emerging trends in special education. Educational assessment and evaluation for persons with disabilities, Educational technology for disabled

References

1. Golden C.J., 1984. Current Topics in Rehabilitation Psychology: Grune & Straton, London.
2. Nirbhay N.Singh, 1998. Comprehensive Clinical Psychology: Application in Diverse Populations, Volume 9, Elsevier Science, Pergamon.
3. Zigler, E, Gates, D.B (1999). Personality development in individuals with Mental Retardation, New York: Cambridge University Press.
4. Kundu, C.L., 2000. Status of Disability in India – 2000. Rehabilitation Council of India, New Delhi.

Course Code	Title of the Course
36324	**PSYCHOLOGY PRACTICAL** – II

Tests from the following areas will be selected by the University and Conducted during the II semester of the course.

1. Anxiety Measurement
2. Anger
3. Assertiveness
4. Stress Measurement
5. Stress Coping Skills
6. Personality Measurement
7. Aptitude
8. Interest
9. Study Skills
10. Job Satisfaction
11. Organisational Climate
12. Mental Health

REFERENCES:

1. Cronbach, L.J. Essentials of Psychological Testing, 1972. New Delhi, Prentice Hall Inc.
2. Woodworth R.S. and Scholsberg, 1981, Experimental Psychology, New Delhi, Taa McGraw Hill Co. Ltd.,
3. Udaipreek, T. Venkateswara Rao. Handbook of Psychological and Social Instruments. Samashti, B-2, Chamelibagh, Baroda-2.

SECOND YEAR

III SEMESTER

Course Code	Title of the Course
36331	COGNITIVE NEURO PSYCHOLOGY

Learning Objectives

- To explain the mediatory role of cognition in behavior
- To explain the process and function of attention
- To describe sensational, perceptual phenomena and its different scientific explanations
- To elucidate how the memory system functions
- To explain the process and function of Neuropsychology
- To describe Plasticity and Restoration of brain function

UNIT 1: Cognitive Psychology

Psychological processes- Emergence of different approaches to cognitive psychology information processing- connectionism & ecological perspective;

UNIT 2: Attention

Model of attention- Functions of executive preconscious and conscious processing alerting mechanism-Selective attention: Bottom –up and top town processing automatically- division of attention-Theories of attention Bottle neck & spotlight concepts Filter model-attenuation theory- multimode theory- resources & capacity allocation model schema theory.

UNIT 3 : Introduction Cognitive Neuropsychology

Meaning- History- Assumptions of Cognitive Neuropsychology- Neuroanatomy: the nervous system, surrounding structures (skull, blood vessels, meninges, ventricles)-. Spinal cord,

brainstem, cerebellum, midbrain, thalamus, basal ganglia, cortex. The cellular foundations of the nerve function: Types of nerve cells-functions- the Four lobes and their Functions

UNIT 4 : Methods

Case Study- Animal Studies- Human Lesion Studies- Structural (X-ray, CT, MRI) and functional (SPECT, fMRI, PET, NIRS) neuroimaging. Electrophysiological methods: ectroencephalography (EEG), event-related potentials (ERP), magnetic encephalography (MEG). Controlling the brain - neurofeedback.

UNIT5 : Perception & Attention

Visual pattern recognition: Template-matching model – Feature analysis – Object recognition – Face recognition. Speech recognition – Feature analysis of speech. Context and pattern recognition – FLMP model. Attention: Auditiory attention – Filter theory – Attenuation theory. Visual attention: Neural basis – Visual search – Binding problem – Visual neglect – Object based attention. Central attention – Automaticity – Stroop effect.

UNIT 6: Memory and Cognition

Memory: Atkinson and Shiffrin Model- Neural Networks Model - Kinds of information stored in Memory: Working Memory- Episodic and Semantic Memory- Procedural MemoryMethods to Study Memory-Forgetting- Memory Distortion and Memory ConstructionMemory in Everyday Life- Memory and the Brain- Memory Improvement Techniques.

Cognition: Thinking- Basic Elements of Thought- Reasoning– Making Decisions- Problem olving and its Methods- Artificial Intelligence- Language and its Development – Language and Thought. Sensory memory: Visual – Auditory. Short-term memory – Working memory: addeley‟s theory – Long-term memory – Factors influencing memory – Techniques for studying textual material. Retention – Interference – Retrieval and inference: Plausible retrieval – Elaboration & inference – Eyewitness testimony & False memory. Associative structure and retrieval: Effect of encoding context – Encoding-specificity.

UNIT 7 : Problem Solving And Reasoning

Problem solving process – Problem solving operators: Acquisition of operators – Analogy & imitation. Operator selection: Difference-reduction model –Mean-end analysis – Tower of Hanoi –Problem representation: functional fixedness. Set effects.Reasoning about conditionals: Wason selection task – Permission schema – Probabilistic interpretation. Deductive reasoning: Categorical syllogism –Atmosphere hypothesis –Process explanation. Inductive reasoning: Hypothesis formation and testing.

UNIT 8 : Sensation & Perception

Theories of perception: top down and bottom up perspective-visuopatial sub codes pattern recognition- Perceptual phenomena Pain perception, constancies and illusions-mental imagery-

UNIT 9 : Classical and modern Psychophysics:

Fechner's contributions- Weber's law Steven's power law- signal detection theory- ROC curve Top down processes- influence of motivation & learning on perception-role of culture perceptual organization- subliminal perception and synesthesia.

UNIT 10 : Memory

Encoding theories and models of memory two store model information processing approach levels of processing levels of recall-Sensory memory- short term memory- working memory models- Storage Long – term memory episodic and semantic memory-autobiographical memory- declarative and procedural memory- Implicit and explicit memory-

UNIT 11 : Retrieval

Recall reconstruction in memory-Forgetting: Theories interference decay organic causes encoding failure- failure of reconstruction- Metacognition: Meaning and Concept and its implications.

UNIT 12: Neuropsychology

Assumptions and methods- functional modularity- anatomical functional architecture and substractivity-

UNIT 13: Methods of investigation:

Electrophysiological Single cell recording, EEG and ERP Scanning and Imaging –CAT PET MRI and FMRI; Neuropsychological Battery Luria Nebraska Neuropsychological Battery, Halstead Retain Test Battery PGI Battery of Brian Behaviour Dysfunction AIIMS neuropsychological battery; Neurodegenerative disorders Parkinson's Alzheimer, Huntington

UNIT 14 : Neuropsychological Plasticity and Restoration of Brain

Historical antecedents and approaches-Types of brain injury-Methods of localization of cognitive functions in the Brain; Function Mind and Brian relationship - computer assisted neuropsychological rehabilitation and training

References

1. Anderson, J.R. (2010). Cognitive Psychology and Its Implications. New York, NY: Worth Publishers.
2. Boller F & Grafman J (1988). Handbook of neuropsychology. New York: Elsevier
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7. Rapp, B. E. (2001). The handbook of cognitive neuropsychology: What deficits reveal about the human mind. Psychology Press.

Course Code	Title of the Course
36332	Counselling Theories and Techniques

Learning Objectives

1. To enable the students understand the nature of counselling process
2. To enable the students to understand various theoretical approaches to counselling
3. To impart the students with knowledge about skills and techniques relating to various approaches of counselling

UNIT 1 : Introduction

Counselling: Definition -Counselling as a process - Purpose and Goals of Counselling, Overview stages of counseling – Characteristics of an effective counselor - Ethics in Counselling.

UNIT 2 : Meaning and Nature

Definition - Aims and Scope of Counselling – Characteristics of effective Counselling - Application of Counselling in various areas - Diversity in Counselling - Attitude of a professional Counsellor - Personality of effective Counsellors -Values in Counselling - Ethical Considerations for a Counselor - Characteristics of a Successful Counselee - Counselee Expectations

UNIT 3 : Psychoanalytic Theory and Techniques

Key concepts – view of human nature, structure of personality, consciousness and the unconscious, Anxiety, Ego-defense mechanisms, Jung's perspectives on the Development of personality,

UNIT 4 : Contemporary trends

Self Psychology and Object-Relations Theory; Therapeutic process; Techniques – Maintaining the analytic framework- Free Association-Interpretation- Dream Analysis-

Analysis and Interpretation of Resistance- Analysis and Interpretation of Resistance; Evaluation

UNIT 5 : Humanistic Theories and Techniques

Person-Centered Theory – Key concepts – View of human nature, Basic characteristics; Therapeutic process- Application-Evaluation

UNIT 6: Gestalt Theory - Key concepts – Principles of Gestalt therapy theory, The Now, Unfinished Business, Personality as Peeling an Onion, Contact and Resistances to Contact, Energy and Blocks to Energy; Therapeutic process; Application/Techniques – The Internal Dialogue Exercise, Making the rounds, the reversal technique, the rehearsal exercise, the exaggeration exercise, staying with the feeling, the gestalt approach to dream work; Evaluation

UNIT 7: Reality Theory - Key concepts– A choice theory explanation of Behaviour, Characteristics of Reality therapy -Therapeutic process, Application – Procedures that lead to change, the "WDEP" system, Evaluation

UNIT 8 :– Behavioural Counselling: Theory And Practice

Theory Introduction – Pavlov's Classical conditioning, Watson's Conditioned Behaviourism, Skinner's Operant Behaviourism, Wolpe's Reciprocal Inhibition, Eysenck's Incubation Theory

UNIT 9: Practice

Goals for counseling - Behavioural Assessment – Relaxation Procedures – Systematic Desensitization – Behaviour Rehearsal and Assertive Training – Reinforcement Methods

UNIT 10 : Cognitive Behaviour Therapy

Introduction – Key concepts and Application – Albert Ellis's Rational Emotive Behaviour Therapy, Aaron Beck's Cognitive Therapy, Donald Meichenbaum's Cognitive Behaviour Modification

UNIT 11: Basic Concepts

Meaning of Guidance and Counseling and their differences – Approaches to counselling, person centred, Gestalt, Psychoanalytic, Cognitive, Trait factor, Behavioral and eclectic approach - Assessment Techniques - Important Factors – Tools of Assessment.

UNIT 12 : Goals of Counselling

Counselling process – characteristics of counselor –Group counselling – special areas of counselling - applied areas multicultural counselling – Ethical issues.

UNIT 13 : Cognitive Behaviour Modification

Fundamental Aspects – Cognitive Restructuring – Meichenbaum's Self Instructional training – Beck's Model –Rational Emotive Therapy (Ellis) – Thought Stopping and Variations – Problem Solving Techniques.

UNIT 14: Professional Preparation & Training

Selection, skills, counseling as a profession, desirable characteristics - Modern Trends: Career guidance, Functions of counselor, values - Assessment: Physical setting, room, length of session, group counseling, stages of counseling - Techniques: Egan's Model, Interviews, testing.

References

1. Corey, G. (1996). Theory and Practice of Counselling and Psychotherapy. 5th ed. Belmount, CA: Brook/Cole.
2. Nelson, J. (1982). The Theory and Practice of Counselling Psychology. New York: Hollt Rinehart & Winston.
3. Patterson, L. W. &Welfel, E. R. (2000). The Counselling Process. 5th ed. Belmount, CA: Brook/Cole.
4. Richard Nelson- Jones (2012). Basic Counselling Skills- a Helper's Manual, 3 rdEdition, Sage Publication India Pvt Ltd, New Delhi.
5. Brammer. L.M. and Shostrom E.L, Therapeutic Psychology, 1977, Englewood Cliffs, New Jersey.
6. D. John Antony, Skills of Counselling, 2003, Anugraha Publications.

Course Code	Title of the Course
36333	Psychopathology

Learning Objectives:

- To define psychopathology.
- To describe the criteria for abnormality.
- To explain the diathesis-stress model and how it contributes to our overall understanding of the classification and causes of psychological disorders.
- To describe the contents of the Diagnostic and Statistical Manual of Mental Disorders (DSM5)
- To explain why accurate and reliable diagnosis is important. What are possible drawbacks or disadvantages to a diagnostic system like the DSM-5?
- To define anxiety disorder. Specify what disorders are classified as anxiety disorders.

UNIT1: Introduction

Meaning of abnormal behavior-Need for classification- Historical views of abnormal behavior- Humanitarian approaches, Contemporary views of abnormal behavior- Causal factors: Biological, Psychosocial and Sociocultural

UNIT 2: Anxiety disorders

Anxiety disorders: Specific phobias, Social phobias, Panic disorder with and without agoraphobia-Generalized anxiety disorder-Obsessive-compulsive disorder- Causal factors of Anxiety disorders and treatment.

UNIT 3: Mood disorders

Unipolar mood disorders – Major depression, Dysthymia, Causal factors, Treatment and outcome, Bipolar disorders – Bipolar-I, Bipolar-II and Cyclothymic, Causal factors of Bipolar disorders, Treatment and outcome.

UNIT 4: Somatoform disorders

Somatoform disorders: Hypochondriasis- Somatization disorder,-Pain disorder,-Conversion disorder and Body dysmorphic disorder-Dissociative disorders: Depersonalization disorder,-Dissociative Amnesia, Dissociative Identity Disorder, Causal factors, Treatment and Outcome of Dissociative disorders.

UNIT 5 :Sexual Dysfunction

Sexual Desire disorders, Sexual Arousal disorders, Orgasmic disorders and Sexual pain disorders.

UNIT 6: Schizophrenia and personality disorders

Schizophrenia: Clinical picture-Subtypes of Schizophrenia: Paranoid, Disorganized, Catatonic- Undifferentiated- Residual type and other psychotic disorders- Causal factors-Treatment and Outcome.

UNIT 7: Personality disorders

Clinical Features- Categories of personality disorders: Paranoid, Schizoid, Schizotypal, Histrionic, Narcastic, Antisocial, Borderline, Avoidant, Dependent and Obsessive compulsive, Causal factors of personality disorders, Treatment and Outcome.

UNIT 8: Therapy

Psychological approaches- Behaviour therapy, Cognitive and Cognitive-Behaviour therapy-Humanistic-Experiential therapy,-Psychodynamic therapy- Marital and Family therapy-Eclecticism and Integration.

UNIT 9 : Prevention

Universal Interventions, Selective Interventions, Indicated Interventions and Deinstitutionalization.

UNIT10: Assessment and Diagnosis

Assessing psychological disorders: Clinical interview - Physical examination – Behavioural assessment – Psychological testing. Diagnosis: Classification issues - DSM IV – ICD 10

UNIT 11: Anxiety Disorders

Generalized anxiety disorder: Clinical description – Causes – Treatment Panic disorder with and without agoraphobia: Clinical description – Causes – Treatment Specific phobia: Clinical description – Causes – Treatment Post-traumatic stress disorder: Clinical description – auses – Treatment Obsessive-compulsive disorder: Clinical description – Causes – Treatment

UNIT 12 : Mood Disorders

Mood disorders: Depressive disorders – Bipolar disorder – Causes – Treatment. Sucide – Risk factors – Treatment.

UNIT 13: Eating disorders

Bulimia nervosa – Anorexia Nervosa – ingeeating disorder – Causes and treatment of eating disorders

UNIT 14 :Sleep disorders

Dys-somnias – Primary insomnia – Primary Hypersomnia – Narcolepsy –Breathing related sleep disorders – Circadian rhythm sleep disorders – Treatment: Psychological and Behavioural treatment.

References

1. Robert C. Carson & James N. Butcher.(2007) Abnormal psychology. Pearson Education Inc. New Delhi

2. Barlow and Durand.(2006).Abnormal Psychology.NewYork. Pearson India Ltd.

3. Sarason and Sarason. (2010). Abnormal Psychology: The Problem of Maladaptive Behaviour (11th Edition). New Delhi. Prentice Hall of India Pvt. Ltd.

Course Code	Title of the Course
36334	**Psychology Practical – III**

Tests from the following areas will be selected by the University and Conducted during the III semester of the course.

1. Transfer of Training
2. Self esteem
3. Interpersonal Skills
4. Communication Skills
5. Leadership
6. Group Dynamics
7. Neuro Psychological Assessment
8. Decision Making
9. Disability Assessment
10. Phobia
11. Depression
12. Obessive Compulsive Symptoms/Disorders

Course Code	Title of the Course
36341	ABNORMAL PSYCHOLOGY

Learning Objectives

- To compare, contrast, and integrate biological, psychological, and sociocultural explanations of abnormal behaviour.
- To explain the nature of stress and people's reactions to it.
- To discuss the role of psychological and behavioural factors in the causation of physical problems.
- To explain the rationale behind psychological interventions for physical conditions such as migraines.
- To describe symptoms of principal mental disorders.

UNIT 1: Introduction and Theoretical Perspective

Defining Abnormal Behaviour, Criteria of Abnormal Behaviour, Brief Mention of DSM and ICD classification systems, Causes of Abnormal Behaviour – Necessary, Predisposing, Precipitating and Reinforcing Causes.

UNIT 2 :Normality and Abnormality

Concept and Meaning of Normality- Mental Health – WHO definition- Jahoda's Healthy Personality, Allport's mature personality- Concept and meaning of Abnormality

UNIT 3: Ways of thinking about abnormality

Conception of abnormal behavior – multi dimensional view of abnormality – clinical assessment and diagnosis

UNIT 4: Disorders of childhood and adolescence

Mental Retardation - Definition, Levels of MR, Clinical Types and Causal Factors; Autism - Clinical Picture and Causal Factors;

UNIT 5: Anxiety related Disorders

Anxiety Disorders- somatoform and dissociative disorders – mood disorders – Major Depressive Disorder with Psychosocial Causal Factors.

UNIT 6: Psychoses; Personality and developmental disorders

Personality disorders – schizophrenia and psychotic disorders – cognitive disorders – developmental disorders – legal, ethical, professional and social issues.

UNIT 7: The Consumer's Decision Making Process

Models of Consumer Decision Making-Communication and Persuasion-Opinion Leadership Professes

UNIT 8: Mood disorders, **Schizophernia and other Psychotic disorders**

Depressive disorders , bipolar disorders, Cyclothymic disorders. Schizoprenia, Schizoaffective disorders, Delusionaal disorder. Brief psychotic disorder, shared psychotic disorder, Culture bound psychotic syndromes.

UNIT 9: Disorders of adult personality and behaviour : Eating disorders, Sleep disorders, impulse control disorders, personality disorders, Sexual and gender identity disorders.

UNIT 10: Classification of mental disorders & Organic Mental Disorders

ICD-10 classification of mental disorders, DSM –IV TR classification of mental disorders.

Delirium, dementia and amnestic disorders.

UNIT 11: Stress – related and somatoform disorders:

Generalized anxiety disorders, panic disorder, phobic disorders, obsessive compulsive disorder, dissociative and conversion disorder, Somatization disorder, Hypochondriasis, Body dysmorphic disorder and pain disorder.

UNIT 12: Addictive Disorders

Alcohol Dependence Syndrome – Features and Causes Other Drugs abused and harmful effects

UNIT 13: Causes and Risk factors for Abnormal Behaviour

Causality – Precipitating, Predisposing factors, Necessary, contributory and sufficient causes Feedback and Circularity in Abnormality Biological Factors –Neurotransmitters and Hormonal Imbalances, Genetic vulnerabilities, Temperament, Brain dysfunction Psychosocial factors – Role of early deprivation, Trauma, Inadequate Parenting, Marital Discord and Maladaptive peer relationships Sociocultural Factors – Sociocultural environment, Crowding Effects, Poverty, Marginalization, and other pathogenic societal influences Diathesis-Stress models for understanding human problems

UNIT 14: Adjustment Disorders

Characteristics of Adjustment Disorders – Reactions to Common Life Stressors- Characteristics of Post Traumatic Stress Disorder – Rape, Natural Disaster

References

1. Carson,R.C., Butcher,J.N and Mineka,S.(2004). Abnormal psychology. 13th Edition. New Delhi: Pearson Education.
2. Alloy,L.B.,Riskind,JH., and Manos,M.J.(2006). Abnormal Psychology – Current Perspectives.9th Edition. New Delhi: Tata McGraw-Hill Edition.
3. Barlow,D.H. and Durand,M.V. (2000). Abnormal Psychology. 2nd Edition. New Delhi:
4. Thomson Publication.
5. Bootzin,R.B.,Acocella,J.R. and Alloy,L.B. (1993). Abnormal Psychology– Current perspectives.6th Edition, International Edition, Tata Graw – Hill Inc., USA.
6. Sue,D.,Sue,,D and Sue.S. (1990). Understanding Abnormal Behaviour. 3rd Edition, Houghton Miffin Co.
7. Wright, R. (1995, August 28). The evolution of despair. Time, 146 (9), 50-57.
8. Davidson and Neal (1996). Abnormal psychology. Revised 6th Edition, John Wiley Sons
9. World Health Organization. (2008). ICD-10: International statistical classification of diseases and related health problems (10th Rev. ed.). New York, NY: Author.
10. American Psychiatric Association. (2000). Diagnostic and statistical manual of mental disorders (4th ed., text revision). Washington, DC: Author.
11. Barlow H. David and Durnad V. Mark(1999) – Abnormal Psychology; India, Brooks/Cole Publishing Company.

Course Code	Title of the Course
36342	ENVIRONMENTAL PSYCHOLOGY

Learning Objectives

- To locate, critically read, and evaluate information to solve problems;
- To critically evaluate the ideas of others;
- To apply knowledge and skills within and across the fundamental ways of knowing;
- To demonstrate an appreciation of human expression through literature and fine and performing arts;
- To demonstrate skills for effective citizenship and stewardship;

UNIT 1: Introduction

The nature and scope of environmental psychology – Role and functions of environmental psychologists – ways that environmental psychology has changed the world – understanding ordinary landscapes – psychological benefits of nature

UNIT 2 : Environmental psychology

Introduction: Nature, scope, history and focus of environmental psychology; Scientific method; models and theories of environment behavior relationship.

UNIT3: Environmental perception and cognition

Environmental perception; spatial cognition; environmental values and attitudes, attachment and identity; appraisal and assessment, personal space, privacy and territoriality.

UNIT 4: Environmental stress and health

Environmental stress, crowding, noise and overpopulation; effects of physical environment on health.

UNIT 5: Weather and climate issues

Weather and climate, seasonality, natural and technological disasters, environmental risk perception.

UNIT6 : Environmental problems and solutions

Psychology of environmental problems; sustainability, common dilemma, solution to environmental problems, sustainable design; designing more habitable environments..

UNIT 7: Theoretical Orientations

Social psychological perspective – Urie Bronfenbrenner – Baker's environmental psychology.

UNIT 8: Spatio – physical dimensions of behavior

Personal space – territoriality and crowding – urban environment and stress – noise, pollution, commuting.

UNIT 9: Towards better environment

Planning – role of media – practitioners, NGO's contribution to changing behavior to save the environment.

UNIT 10: Sustainability

Environmental degradation – resource overconsumption, pollution, climate change – human health and well being connected with environmental health – reciprocal relationship between human beings and natural world.

UNIT 11 :Public reactions to pollution

Environmental Perception, Cognition and Attitudes. Elementary Psychophysics. Perception : Theories of Environmental Perception. Environmental Cognition, Environmental Attitudes, Changing Attitudes.

UNIT 12 : Psychology of stress. Researching stress

The Environmental context. Moderators of stress response. The role of stress in understanding organism-environmental relationships

UNIT 13: Behaviours disturbing Environmental Stressors

Physical stressors : Ambient Temperature. Littering, Hummidity, Sunlight, Wind, Air and Water. Ion Concentration. The Atmospheric stressors : Carbon-dioxide, Carbon-Monoxide, Ozone. Tobacco smoke as a pollutant. Psychological Stressors : Noise :The Ubiquitous Pollutant : Noise in the work place. Noise in the living environment, noise in the laboratory Noise and social behavior : Individual Difference in response to Noise. Noise in schools and hospitals. Noise and Law.

UNIT-14: Research Methods in Environmental Psychology

Evaluating the adequacy of environmental research.

References

1. Ajzen, I. (1991). The theory of planned behavior. Organizational Behavior and Human Decision Processes. 50: 179-211.
2. Kaplan, S. & R. Kaplan (2009). Creating a larger role for environmental psychology: The Reasonable Person Model as an integrative framework. Journal of Environmental Psychology.
3. Stern, P. (2000). Toward a coherent theory of environmentally-significant behavior. Journal of Social Issues, 56(3):407-424.
4. Steg, L., G. Perlaviciute, E. van der Werff & J. Lurvink (2014). The significance of hedonic values for environmentally relevant attitudes, preferences, and actions. Environment and Behavior, 46(2): 163-192.
5. Nisbet, M. C. (2010). Study finds that fear won't don't do it: Why most efforts at climate change communication might actually backfire. Retrieved 1 December 2014 from bigthink.com

Course Code	Title of the Course
36343	POSITIVE PSYCHOLOGY

Learning Objectives

- To understand basic concepts of positive psychology and its relationship to other branches of psychology
- To gain fundamental understanding of well-being and happiness in the context of positive psychology
- To grasp basic cognitive states and processes in positive psychology
- To transfer the theoretical concepts into practical setting
- To develop an awareness of applications and implications of positive psychology concepts and theories
- To equip himself/herself with the skill and competence to apply positive psychology principles in a range of environments to increase individual and collective wellbeing

UNIT 1 : Introduction

Positive psychology: Definition; goals and assumptions; Relationship with health psychology, developmental psychology, clinical psychology

UNIT 2: Positive Psychology

Introduction and historical overview of Positive Psychology, Positive prevention and positive therapy Module

UNIT 3 : Foundations of happiness

The meaning and measure of happiness – the science of happiness – biological foundation of happiness – the happiness system – the malleable brain – the secret of smiling – positive feeling as a compass – positive traits.

UNIT 4: Emotional intelligence, Well-being and Happiness

Positive emotions: Broaden and build theory; Cultivating positive emotions; Happiness-hedonic and Euaimonic; Well- being: negative vs positive functions; Judgement and decision making; Subjective well –being: Emotional, social and psychological well-being; Model of complete mental life

UNIT 5: Positive Cognitive States and Processes

Resilience: Developmental and clinical perspectives; Sources of resilience in children; Sources of resilience in adulthood and later life; Optimism- How optimism works; variation of optimism and pessimism; Spirituality: the search for eaning(Frankl); Spirituality and well-being; Forgiveness and gratitude

UNIT 6 :Applications of Positive Psychology

Positive schooling: Components; Positive coping strategies; Gainful employment

Mental health: Moving toward balanced conceptualization; Lack of a developmental perspectives.

UNIT 7: Subjective well-being

The science of happiness and life satisfaction, Resilience in Development, Concept of flow, Positive affectivity, Social construction of self-esteem 11 Module

UNIT 8 Role pf personal control in Adaptive

Ffunctioning Optimism, Hope, Self efficacy, goal-setting for life and happiness Module

UNIT9: Interpersonal relationship

Enhancement of closeness, compassion, forgiveness and gratitude, love, empathy and altruism Module

UNIT 10: Positive response to loss Role of humour

Spirituality Module 6: Application of Positive Psychology Living well at every stage of life, Positive Psychology for children, Positive schooling, Ageing well.

UNIT11:Strategies to enhance happiness

Enhancing pleasure, engagementand meaning-making; self-related processes.

UNIT12: Character strengths and virtues

Classification, assessment and nurturance; barriers in developing strengths and virtues.

UNIT13: Meeting life challenges

Nature, type and sources of stress, individual interpretations and responses, coping strategies and their assessment, promoting healthy coping strategies and life skills.

UNIT 14: Subjective well-being

Concept and indicators, life satisfaction and happiness, determinants of happiness; theoretical frameworks.

References

1. Snyder, C.R. & Lopez, S.J. (2002). Handbook of positive psychology. (eds.). New York: Oxford University Press.
2. Baumgardner, S.R & Crothers, M.K.(2009). Positive Psychology. U.P: Dorling Kindersley Pvt Ltd.
3. Carr, A. (2004). Positive psychology, The science of happiness and human strengths.New York: Routledge.
4. Singh, A.(2013).Behavioral science: Achieving behavioral excellence for success. New Delhi: Wiley India Pvt ltd.

Course Code	Title of the Course
363441	Elective: Health Psychology

Learning Objectives

- To explain the concepts of health, illness and wellness.
- To distinguish between the biopsychosocial and biomedical models of health, including the advantages and disadvantages of each.
- To describe the various research designs employed by health psychologists.
- To define key concepts listed in the textbook.
- To explain the characteristics of the patient-centered medical home and its relationship to the biopsychosocial model of health

UNIT1: Introduction to Health Psychology

Need and significance- History- Health, disease, illness, well-being and quality of life-contributions.

UNIT 2: Health Psychology –

Introduction Define health psychology, Mind-body relationship- a brief history, Need for health psychology, Biopsychosocial model in Health Psychology.

UNIT3: Health Behaviour

The concept of Health Literacy, health behavior; factors influencing health behaviours, modification of health behavior, changing health beliefs, cognitive- behavioural approaches, Health enhancing behaviours

UNIT4: Becoming Ill and Getting Medical Treatment

Health Services – Perceiving and Interpreting systems – Using and misusing health services – The Patient-Practitioner relationship – adhering to medical advice- The hospital – Being hospitalized – Psychological problems of hospitalized patients.

UNIT5: Theoretical basis for changing health habits

Attribution Theories- Health- Locus of Control-Health Belief Model-Protection -Motivation Theory- Theory of Planned Behavior- Social Cognitive Theories- Health Action Process Approach- Trans theoretical Change Model- Models of Prevention.

UNIT 6: Stress Management:

Stress - Definitions- Models of Stress – Theories of Stress - Stress reactions – Coping and Stress Management techniques- Pain and its management - Psychological reactions of a patient to loss – Stages of Acceptance by Kubler-Ross.

UNIT 7: Management of ill and Yoga

Management of Chronic and Terminally ill – Quality of Life – Social support and rehabilitation. Role of Indian Treatment – Yoga Asanas – Principles in Yoga Practice – Pranayama.

UNIT 8 Health Psychology

Concept, Assumptions, Models (Biomedical and Biopsychosocial)

UNIT 9 Theories

Social Cognitive Theory, Theory of Planned Behavior, Health Belief model, Protection – motivation theory, Trans – theoretical model of behavior change, Self-regulatory model, latest trends.

UNIT 10 Health Promotion and Illness Prevention

Health and Behavior; Changing health habits; Cognitive behavioural approaches to health behavior change.

UNIT11: Health Care System

Indian Scenario, Attitude of Health Professionals, Burnout in health professionals, Designing health care work environment, Future challenges for health care, Growth of Health Psychology.

UNIT 12: Pain Management Control of pain

some distinctions in the clinical management of pain, pain control techniques. The management of chronic pain, the placebo effect, Psychological control and management of discomfort, Control based interventions with medical patients, individual differences in reactions to control.

UNIT 13: Psychophysiological disorders

Personality disposition. CHD, Asthmatics, Allergy, Eczema, Hding, Rheumatoid Arthritis, Peptic Ulcer, Diabetes and menstrual disorders

UNIT14: Psychoneuroimmunology

The immune system- immunocompetence/ immunocompromise.. Coping resources as moderators of the stress. Immune functioning relationship

References

1. Shelley E. Taylor. Health Psychology Third Edition. McGraw Hill International Editions, 1995.
2. Swaminathan, V.D, Latha Sathish, Psychology for Effective Living, Department of Psychology, University of Madras.
3. Brannon, J. & Feist, J. (1999). Health Psychology: An Introduction to Behavior and Health(4th ed,) Wadsworth Thomson Learning
4. Roberts, R., Towell, T. & Golding, J.F. (2001). Foundations of Health Psychology. Palgrave Houndmills, New York.
5. Taylor, E. (2006). Health Psychology. New Delthi: Mc Graw Hills Inc.

Course Code	Title of course
363442	Organisational Psychology

UNIT1: Organizational Behaviour

Meaning – Elements – Need – Approaches – Models – Global scenario; Individual Behaviour: Personality & Attitudes- Development of personality - Nature and dimensions of attitude - Organizational Commitment – Learning – Attitudes – Perception – Motivation – Ability – Their relevance to organizational behavior.

UNIT 2: Group Behaviour

Theories of Group Formation - Formal Organization and Informal Groups and their interaction- Importance of teams - Formation of teams - Team Work- Group dynamics – Group norms – Group cohesiveness – Their relevance to organizational behavior.

UNIT 3: Organizational Power and Politics

Organizational Power: Definition, Types of powers, Sources and Characteristics – Effective use of power- Organizational Politics: Factors and Impact.

UNIT 4: Organizational Stress and Conflict Management

Stress Management: Meaning – Types – Sources – Consequences – Management of stress

UNIT 5: Organizational conflict

Constructive and Destructive conflicts - Conflict Process - Strategies for encouraging constructive conflict - Strategies for resolving destructive conflict.

UNIT 6: Organizational Dynamics

Organizational Efficiency, Effectiveness and Excellence: Meaning and Approaches – Organizational Culture – Meaning, significance – Organizational Climate – Implications on organizational behavior.

UNIT 7: Communication within Organization

Meaning and Process of Communication. Models and Perspectives of Communication in organizations. Communication barriers. Types of Communication.

UNIT 8: Organizational Socialization

Organizational Socialization – Definition, Dimensions, Stages Roleof the Workplace Counsellor at each Stage

UNIT 9: Individual Behaviour in Organizations

Productive and Counterproductive Behaviour in Organizations – definitions

Job Satisfaction Employee Absenteeism Employee turnover Occupational Stress

UNIT 10 : Special groups in organisational counselling

• Chronic absentees • Accident prone • Employees with family problems • Employees with alcoholism & drug addiction • Maladjusted employees • Indisciplined employees

UNIT 11: Group Behaviour in organizations

Definition and Characteristics of Groups Stages of Groups Impact of Groups on Individual Group Effectiveness – Determinants and Enhancement Intergroup Interactions – Types and patterns Intergroup Conflicts – Reduction Strategies

UNIT 12 : Leadership Behaviour

Definition and theories Models of leadership behaviour

UNIT13: Stress and worker Well-Being

Work place Stress, Consequences of stress, theories of Stress, reducing and managing stress, Violence at work, Work Motivation theories, quality of work life, turnover, absenteeism, job involvement and commitment, Equal Employment OpportUNITy.

UNIT 14: Organisational Culture

Functions - Organisational Change and its effects –Managing Change and Resistance to change – Organisational Structure – Designs - Organizational Behaviour Modification process - Quality of Work Life – Employee Assistance Programs.

References

1. Fred Luthans, Organizational Behaviour, McGraw-Hill/Irwin, 2006.
2. Stephen P. Robbins, Organizational Behaviour, Prentice Hall; 2010
3. Keith Davis, Organizational Behavior: Human Behavior at Work, McGraw Hill, 2010
4. Griffin and Moorhead, Organizational Behavior: Managing People and Organizations, 2006.
5. Judith R. Gordon, Organizational Behavior: A Diagnostic, Prentice Hall, 2001.
6. K.Aswathappa, Organizational Behaviour, Himalaya Publishing, Mumbai, 2010
7. Judith R. Gordon, A Diagnostic Approach to Organisational Behaviour, Allyn & Bacon, 1993.

8. Jex, S. M. (2002). Organizational Psychology: A Scientist-Practitioner Approach. John Wiley and Sons. Inc, NewYork.
9. . Arnol J., & Robertson, I .T.,&Coopen, C. L. (1995). Work Psychology-Understanding human behaviour in workplace. London: Mcmillan.
10. John W. Newstrom and Keith Davis – Organizational Behaviour – Human Behaviour at Work. 10thed. Tata McGraw Hill, 2002.

Duration of the Programme (2Years)

The programme for the degree of Master of Science in Psychology shall consist of two academic years divided in to four semesters. First three semesters consist of Three Theory Papers and one Practical Paper. Fourth semester consists of four Theory papers. Theory course carry 4 credits each, Practical courses carry 4 credits, Each semester consist of 16 credits.

e. 4. Faculty and Support Staff Requirements

The programme for the degree of **Master of Science in Psychology** requires following faculty and supporting staff:

Staff Category	Required
Faculty for Core Psychology Subjects #	3
Faculty for Elective Subjects viz., Educational psychology, Rehabilitation psychology, Organizational psychology,	2
Clerical Assistant	1

e. 5. Instructional Delivery Mechanisms

The instructional delivery mechanisms of the programme includes SLM – study materials, face to face contact session for both theory and practical courses of the programme, e-content of the study materials in the form of CD, MOOC courses and virtual laboratory wherever applicable.

e. 6. Identification of Media

The SLM – designed study materials will be provided in print media as well is in the form of CD which carries electronic version of the study material in addition to MOOC and virtual laboratory courses.

e. 7. Student Support Services

The student support services will be facilitated by the head quarter i.e., Directorate of Distance Education, Alagappa University, Karaikudi and its approved Learning Centres located at various parts of Tamil Nadu. The pre-admission student support services like counseling about the programme including curriculum design, mode of delivery, fee structure and evaluation methods will be explained by the staff at head quarter and Learning Centres. The post-admission student support services like issue of identity card, study materials, etc. will be routed through the Learning Centres. The face to face contact sessions of the programme for both theory and practical courses will be held at the head quarter only. The conduct of end-semester examinations, evaluation and issue of certificates will be done by office of the controller of examinations, Alagappa University, Karaikudi.

f. Procedure for Admission, curriculum transaction and evaluation

f. 1. Procedure for Admission

A candidate who has passed any U.G degree (10 + 2+ 3 system) of any other University accepted by the Syndicate as equivalent thereto shall be eligible to appear and qualify for the M.Sc. Degree in Psychology of this University after a course of study of two academic years.

f. 2. Curriculum Transactions:

The classroom teaching would be through chalk and talk method, use of OHP, Power Point presentations, web-based lessons, animated videos, etc. The face to face contact sessions would be such that the student should participate actively in the discussion. Student seminars would be conducted and scientific discussions would be arranged to improve their communicative skill.

For practical courses exclusive study materials containing the requirements, procedure for the experiments will be issued to the learners. In the laboratory, instruction would be given for the experiments followed by demonstration and finally the students have to do the experiments individually.

The face to face contact sessions will be conducted in following durations;

Course Type	Face to Face Contact Session per Semester (in Hours)
Theory Courses Semester- I, II, & III (3 courses with 4 credits each) Semester- IV (4 courses with 4 credits)	48 64
Practical 1 Practical with 4 credits	120

f. 3. Evaluation

The evaluation shall be conducted separately for theory, practical and project work to assess the knowledge acquired during the course of study. There shall be two systems of evaluation viz., internal assessment and end semester examinations.

In the case of theory courses the internal evaluation shall be conducted as Continuous Internal Assessment viz. student assignment preparation, seminar, etc. The internal assessment shall comprise of maximum 25 marks for each subject. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

In the case of Practical courses for M.Sc. programme in Psychology should be conducted at every semester. The internal evaluation of practical courses will be done through continuous assessment of skill in demonstrating the experiments, record note preparation, and viva voce. The external evaluation consists of an end semester practical examinations which comprises of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

f. 3.1. Question Paper Pattern

Answer all questions (one question from each UNIT with internal choices Time: 3 Hours Max. Marks: 75

Part A- 10 x 2 Marks = 20 Marks

Part B -5 x 5 Marks = 25 Marks

Part C- 3 x 10 Marks = 30 Marks

f. 3.2. Distribution of Marks in Continuous Internal Assessments

The following procedure shall be followed for awarding internal marks for theory courses

Component	Marks
Seminar (1 per course)	5
Assignments (2 per course)	20
Total	**25**

The following procedure shall be followed for awarding internal marks for practical courses

Internal –Practical	Marks
Continuous Assessment (demonstration and observation of experiments)	10
Record	10
Viva	5
Total	**25**

f. 3.3. Passing Minimum

The candidate shall be declared to have passed the examination if the candidate secures a minimum of 50% (50 marks out of 100 marks) in the University end semester examination. Then the total marks secured by the candidate will be taken and added with his/her internal marks (Maximum marks 25).

For a pass in the Practical paper, a candidate has to secure a minimum of 50 % (37.5 marks) marks in the University end semester practical examination (75 marks)

examination. Then the total marks secured by the candidate will be taken and add with his/her internal marks (Maximum marks 25).

There is no passing minimum for the record notebook. However submission of a record notebook is a must.

Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

f. 3.4. Marks and Grades

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester

G_i = Grade Point obtained for course i in any semester.

n refers to the semester in which such courses were credited

For a semester

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

$$\text{Grade Point Average} = \frac{\text{Sum of the multiplication of grade points by the credits of the courses}}{\text{Sum of the credits of the courses in a semester}}$$

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

$$\text{CGPA} = \frac{\text{Sum of the multiplication of grade points by the credits of the entire programme}}{\text{Sum of the credits of the courses for the entire programme}}$$

CGPA	Grade	Classification of Final Result
9.5-10.0 9.0 and above but below 9.5	O+ O	First Class- Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	D++ D+ D	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	A++ A+ A	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	B+ B	Second Class
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

f. 3.5. Maximum duration for the completion of the course

The maximum duration for completion of M.Sc., Degree in Psychology programme shall not exceed ten semesters from their fourth semester.

f. 3.6. Commencement of this Regulation

These regulations shall take effect from the academic year 2018-2019 (June session) i.e., for students who are to be admitted to the first year of the course during the academic year 2018-2019 (June session) and thereafter.

f. 4. Fee Structure

The programme has the following Fee Structure

Sl. No.	Fees Detail	Amount in Rs.	
		First Year	Second Year
1	Admission Processing Fee	300.00	-
2	Course Fee	6100.00	6100
3	ICT Fee	150.00	150
	TOTAL	**6550.00**	**6250**

The above mentioned fee structure is exclusive of Exam fees.

g. Requirement of the laboratory support and Library Resources:

A well-equipped Psychology Laboratory exclusive for Distance Education Programme was established in the Directorate of distance Education Block, Alagappa University, Karaikudi with all the necessary equipments for conducting face to face contact sessions for Practical courses of M. Sc., Psychology programme. Directorate of Distance Education, Alagappa University, Karaikudi housing an exclusive Library facility with adequate number of copies of books in relevant titles for M. Sc., Psychology programme. The Central Library of Alagappa University also having good source of reference books. The books available at both the libraries are only for reference purpose and not for lending services.

h. Cost estimate of the programme and the provisions

The cost estimate of the programme and provisions for the fund to meet out the expenditure to be incurred in connection with M. Sc.Psychology Programme as follows:

Sl. No.	Expenditure Heads	Approx. Amount in Rs.
1	Programme Development (Single Time investment)	**10,00,000/-**
2	Programme Delivery (Per Year)	**20,00,000/-**
3	Programme Maintenance (Per Year)	**3,00,000/-**

i. Quality assurance mechanism and expected programme outcomes

i. 1. University Moto

'Excellence in Action'

i. 2. University Vision Statement

Achieving Excellence in all spheres of Education, with particular emphasis on **"PEARL"-** Pedagogy, Extension, Administration, Research and Learning.

i. 3. University Objectives

1. Providing for Instructions and Training in such Branches of Learning as the University may determine.
2. Fostering Research for the Advancement and Dissemination of Knowledge

i. 4. University Quality Policy

Attaining Benchmark Quality in every domain of **'PEARL'** to assure Stakeholder Delight through Professionalism exhibited in terms of strong purpose, sincere efforts, steadfast direction and skillful execution.

i. 5. University Quality Quote

Quality Unleashes Opportunities Towards Excellence **(QUOTE)**

i.6. Programme's Review Mechanism

The quality of the programme depends on scientific construction of the curriculum, strong-enough syllabi, sincere efforts leading to skilful execution of the course of the study. The ultimate achievement of M.Sc., Psychology programme of study may reflect the gaining of knowledge and skill in Psychology. Gaining of knowledge in Psychology may help the students to get new job opportunities, upgrading their position not only in employment but also in the society, make students feel thirsty to achieve in research in the fields associated with the discipline- Psychology, achieving in competitive examinations on the subject- Psychology, etc.

The benchmark qualities of the programme may be reviewed based on the performance of students in their end semester examinations. Also the feedback from the alumni, students, parents and employers will be received and analyzed for the further improvement of the quality of the M. Sc., Psychology Programme.

Minutes of the Meeting of the Board of Studies in Psychology (For B.Sc. Psychology & M.Sc. Psychology Programmes to be offered through ODL Mode) held at the Directorate of Distance Education, Alagappa University, Karaikudi on 01.09.2017, at 11.00 am.

Members Present

1.	Dr.P. Sivakumar	-	Chairman
2.	Dr.S. Kadhiravan	-	Member
3.	Dr.J. Sujatha Malini	-	Member
4.	Dr.AR. Saravanakumar	-	Member

The modified syllabi of B.Sc. Psychology & M.Sc. Psychology programmes were scrutinized and discussed in the meeting. The board has resolved the following:

1. The board has unanimously accepted the curriculum design and modified syllabi of B.Sc. Psychology & M.Sc. Psychology programmes are prepared as per the norms.
2. The board has approved the Regulations, Curriculum and Syllabi for B.Sc. Psychology & M.Sc. Psychology programmes to be offered by the Directorate of Distance Education, Alagappa University.
3. The Approved Syllabi are provided in the Annexure-I (B.Sc. Psychology) & Annexure-II (M.Sc. Psychology)

1/09/2017
(AR. SARAVANAKUMAR)

(J. SUJATHA MALINI) 1/9/17

01/09/17.
(S. KADHIRAVAN)

01/09/17
(P. SIVAKUMAR)

PROGRAMME PROJECT REPORT FOR

M.Sc., ZOOLOGY

DIRECTORATE OF DISTANCE EDUCATION
ALAGAPPA UNIVERSITY
KARAIKUDI – 630 003

PROGRAMME PROJECT REPORT FOR

M.Sc., ZOOLOGY

a) PROGRAMME MISSION AND OBJECTIVES

Life Sciences have fascinated humans for two reasons, namely the puzzling and the aesthetic essentials. Animals and plants are mysterious in their physiology and gorgeous in their external morphology. That is why their study is exciting and electrifying. The primary objective of this programme is to impart quality education in the subject of Zoology. Within this broad stream, students are taught varied subjects like Cell biology, Ecology, Physiology, Genetics, Immunology and Biotechnology. Knowledge of Applied Zoology paves way for self employment. It also serves to inculcate social responsibilities and serve the society by not only becoming Teachers, Scientists, Environmentalists but also responsible citizens who bridge the gap between science and common man.

b) RELEVANCE OF THE PROGRAMME WITH HEI'S MISSION AND GOALS:

The Alagappa University is functioning with following vision and Mission:

Mission: Achieving Excellence in all spheres of Education, with particular emphasis on 'PEARL"- Pedagogy, Extension, Administration, Research and Learning

Vision: Affording High Quality Higher Education to the learners so that they are transformed into intellectually competent human resources that will help in the uplift of the nation to Educational, Social, Technological, Environmental and Economic Magnificence.

Therefore, the introduction of M.Sc. Zoology programme in the Directorate of Distance Education will contribute substantially in fulfilling the mission of Alagappa University. Such a higher education in science subject with appropriate laboratory experiences will enrich the human resources for the uplift of the Nation to Educational, Social, Technological, Environmental and Economic Magnificence (ESTEEM).

NATURE OF PROSPECTIVE TARGET GROUP OF LEARNERS

- Working Professionals
- Entrepreneurs
- Service Personnel
- Academic Faculty
- Government Officials

- Researchers
- Home makers
- Unemployed Graduates

d) APPROPRIATENESS OF PROGRAMME

The M.Sc., Zoology programme has been designed to provide opportunities for students to develop and demonstrate knowledge and understanding, skills, qualities and other attributes in the following areas:

- The fundamental concepts of zoology.
- The higher-level taxonomy and diversity of animal life.
- How zoological principles can be applied to problems in conservation and applied biology.
- Practical skills in ecology and natural history.
- Undertake laboratory tasks and techniques.
- Inter-disciplinary knowledge like biophysics, biostatistics and bioinformatics .
- Analyse data using appropriate statistical methods using computer.
- It also improve the Intellectual skills of the students.
- In nutshell, these skills will enhance the performance of the students.

e) INSTRUCTIONAL DESIGN

CURRICULUM DESIGN

Sr.No	Course code	Title of the Course	CIA Max.	ESE Max.	TOT Max.	C
I YEAR						
I semester						
1	35011	Animal Diversity	25	75	100	4
2	35012	Biochemistry	25	75	100	4
3	35013	Cell and Molecular Biology	25	75	100	4
4	35014	**Practical** Lab.I: Animal Diversity, Biochemistry and Cell and Molecular Biology	25	75	100	4
Total			100	300	400	16
	II semester					
5	35021	Developmental Biology and Evolution	25	75	100	4
6	35022	Genetics	25	75	100	4
7	35023	Microbiology	25	75	100	4
8	35024	**Practical** Lab. II: Developmental Biology and Evolution, Genetics and Microbiology	25	75	100	4
Total			100	300	400	16
II YEAR						
	III semester					
9	35031	Animal Physiology	25	75	100	4
10	35032	Immunology	25	75	100	4
11	35033	Environmental Biology	25	75	100	4
12	35034	**Practical** Lab. III: Animal Physiology, Immunology, Environmental Biology	25	75	100	4
Total			100	300	400	16
	IV semester					
13	35041	Fisheries & Aquaculture	25	75	100	4
14	35042	Animal Biotechnology	25	75	100	4
15	35043	Biophysics, Biostatistics and Bioinformatics	25	75	100	4
16	35044	**Practical** Lab. IV: Fisheries and Aquaculture, Animal Biotechnology, Biophysics, Biostatistics and Bioinformatics	25	75	100	4
			100	300	400	16
Total			400	1200	1600	64

No. of Credits per Course (Theory) - 4	Total No. of Credits per Semester- 16
No. of Credits per Course (Practical) - 4	Total Credits for the Programme- 16 X 4 = 64
Total Marks: 1600	

Course Code Legend:

3	5	0	Y	Z

350- M. Sc., Zoology Programme
Y -Semester Number
Z- Course Number in the Semester

CIA: Continuous Internal Assessment, ESE: End Semester Examination, TOT: Total, C: Credit Points, Max.: Maximum

DETAILED SYLLABI

35011- ANIMAL DIVERSITY

Objectives:

- To study the classification and phylogeny of invertebrate and vertebrate animals.
- To understand the general characteristics and functions of the animals

BLOCK – I: CLASSIFICATION OF ANIMALS

Unit I
Introduction to the diversity of animals, Principles of classification, types of classification and Binomial Nomenclature; Linnaeus

Unit II
Species concept –Typological, Biological and evolutionary species concept.

Unit III
Taxonomic characters and theories of taxonomy- Numerical taxonomy, Cladistics and Molecular taxonomy.

Unit IV
Major divisions and subdivisions of the animal kingdom; Animal architecture- Cephalization Symmetry- Bilateral and Radial symmetry; Coelom in animals: Coelomata, Acoelomata and Psudocoelomata.

BLOCK – II: PROTOZOA, PORIFERA, COELENTERATA AND HELMINTHES

Unit V
Protozoa: General characteristics, classification up to class level, Protozoan parasites: Entamoeba and Plasmodium.

Unit VI
Porifera: General characters and classification, Structure of *Leucosolenia*, Canal system in sponges, Spicules in sponges

Unit VII
Coelenterata: Structure of *Obelia* colony, Polymorphism in coelenterates, Corals and coral reefs.

Unit VIII
Helminth parasites – *Taenia solium,* Nematode parasites – *Ascaris* and Parasitic adaptations.

BLOCK – III: ANNELIDA, ARTHROPODA, MOLLUSCA AND ECHINODERMATA

Unit IX
Annelida: General characters, Classification up to class - Metamerism in Annelids.

Unit X

Arthropoda: General characters, Classification up to class – Larval forms of crustaceans – Adaptive radiations in Arthropoda, Harmful and beneficial insects.

Unit XI

Mollusca and Echinodermata: General characters, Classification up to class- Cephalopod as an advanced Mollusc; Larval forms of Echinodermata -Water vascular system in Echinoderms.

BLOCK – IV: VERTEBRATES

Unit XII

General characters and classification of Prochordates and vertebrates; Pisces- classification up to orders, structural and functional adaptation of fishes

Unit XIII

Amphibians and Reptiles: Definition, general characters, classification, structural and functional adaptations of amphibians and reptiles- Mesozoic reptiles – Dinosaurs.

Unit XIV

Aves: phylogeny, flight adaptation, flightless birds and migration of birds. Mammals: General characteristics of Prototheria, metatheria and eutheria; Aquatic mammals; adaptive radiation in mammals.

Reference Books:

1. David Eisenhour, Allan Larson, Susan Keen, Larry Roberts, Cleveland Hickman Jr. 2014. Animal Diversity. McGraw Hill International, Boston.
2. Barnes, R. D., 2008. Invertebrate Zoology, *Cengage Learning (Thompson), USA*
3. Ekambaranatha Iyar, E.K. and T.N. Ananthakrishnan, 1992. A Manual of Zoology, Volume II Chordeta. Viswanathan & Co.
4. Hickman CP Jr., Roberts LS, Larson A, l'Anson H, Eisenhour DJ. 2006. Integrated Principles of Zoology. 13th ed. Boston: McGraw-Hill.
5. Jordan, E.L. and P.S. Verma, 2010, Invertebrate Zoology, S. Chand & Co Ltd., Ram Nagar, New Delhi.
6. Kotpal, R.L.,2015. Modern text book of Zoology, Vertebrates; Rastogi publications, New Delhi.
7. Kotpal, R.L., 2017. A text book of Animal Diversity; Rastogi publications, New Delhi.
8. Rajesh Karyakarle and Ajit Damle., 2005, Medical Parasitology Books & Allied (P) Ltd. Kolkata.
9. Ruppert, E.E., and Barnes, R.D., 2006. Invertebrate Zoology, VIII Edition. Holt Saunders International Edition
10. Russell- Hunter, W.D., 1979. Life of Invertebrates, Macmillan Publishing Company, New York.
11. Young, J. Z. 2004., The Life of Vertebrates. III Edition. Oxford university press.

12. Marshall, A.J., 1995. Textbook of zoology: vertebrates, AITBS publishers, New Delhi.

Outcome:

The course provides the students a comprehensive knowledge and also exhibit depth and breadth of Animal diversity.

35012 - BIOCHEMISTRY

Objectives:

- To provide an advanced understanding of the core principles and topics of biochemistry.
- To understand the structure and function of biomolecules.

BLOCK – 1: INTRODUCTION TO BIOMOLECULES

UNIT - I

Carbohydrates: Functions, classification (Mono, di and polysaccharides), Structural aspects of monosaccharide, disaccharides and polysaccharides.

UNIT - II

Lipids: Classification and functions of lipids, fatty acids, essential fatty acids, Triacylglycerols, Phospholipids, Glycolipids, lipoproteins and steroids, Properties of fats and waxes.

UNIT - III

Proteins and aminoacids: Functions, structure (Primary, Secondary, tertiary and quaternary structure), Classification and properties of proteins. General structure, classification and chemical properties of aminoacids.

UNIT - IV

Nucleic acids: Functions and components of nucleic acids. Structure and nomenclature of nucleotides. Structure of DNA (Watson and Crick model), Different forms of DNA double helix and organization of DNA in the cell.

BLOCK – II: ENZYMES, VITAMINS AND HORMONES:

UNIT – V

Enzymes: Nomenclature and classification of enzymes, Active site, factors affecting enzyme activity. Mechanism of enzyme action (Lock and key model, Induced fit model, Substrate strain model).

UNIT – VI

Isoenzymes, Regulation of enzyme activity in living system, Enzyme kinetics (MM equation, Line-Weaver and Burk plot).

UNIT – VII

Vitamins: Classification of vitamins, Chemistry, sources, biochemical functions, Recommended dietary allowances (RDA), deficiency, symptoms and hypervitaminosis.

UNIT – VIII

Hormones: General classification, mechanism of action, origin and major functions of hormones - Pituitary and Gonadal.

BLOCK – III: METABOLISM

UNIT – IX

Carbohydarte metabolism**:** Glycolysis, Citric acid cycle, glyconeogenesis, glycogenesis, glycogenolysis, hexose monophosphate shunt, Uronic acid pathway.

UNIT – X

Lipid metabolism: Fatty acid oxidation, Ketogenesis, Biosynthesis of fatty acids, metabolism of cholesterol.

UNIT – XI

Aminoacid metabolism: Amino acid pool, transamination, deamination, metabolism of ammonia, urea cycle, fate of carbon skeleton of aminoacids.

UNIT – XII

Nucleotide metabolism**:** Biosynthesis and degradation of purine and pyrimidine ribonucleotides.

BLOCK – IV: METABOLIC DISORDERS

UNIT - XII

Diabetes mellitus, Diabetes insipidus, Glycogen storage diseases, Ketoacidosis, Hyperlipoproteinemia, fatty liver.

UNIT – XIII

Antherosclerosis; phenylketonuria, maple syrup urine disease, glutaric acidemia type I, Carbamoyl phosphate synthetase I deficiency.

UNIT – XIV

Alcaptonuria; Lesch-Nyhan syndrome; Gout; lipoid congenital adrenal hyperplasia; Kearns-Sayre syndrome; Zellweger syndrome; Gaucher's disease, Niemann Pick disease.

REFERENCES

1. Berg, J.M., J.L. Tomoczko, and L. Stryer, 2008. Biochemistry, W. H. Freeman publisher, USA.
2. Lehninger, A., Nelson, D. L., and M.M. Cox 2008. Lehninger Principles of Biochemistry, W.H. Freeman, USA.
3. Voet, D. J., J.G. Voet and C. W. Pratt, 2008. Fundamental of biochemistry: Life at molecular level, Wiley publishers, USA.
4. Biochemistry 4th edition, G. Zubay, 1998. Mc Millan Publishing Co. New York.
5. Davson and Eggleton – Principles of Human Physiology, J and A. Churchill, London.
6. David Randall, 2009. Eckert Animal Physiology, W H Freeman & Co.
7. Nielsen, S, 2000. Animal Physiology, Cambridge Univ. Press, Cambridge.
8. Horton, Principles of Biochemistry, Fourth edition, Prenhall Publishers, USA.

9. RK. Murray, DK. Granner and VM. Rodwell (2006). Harper's illustrated Biochemistry Mc Graw Hill Company, Inc.
10. Rodwell (2015), Harpers Illustrated Biochemistry, 30th Edition, Mc Graw Hill Company, Inc.
11. Satyanarayana, U and Charapany, U. (2017). Biochemistry, 5th Edition, Elsevier.

Outcome:

By the end of the course, students should be able to critically discuss the core principles and topics of biochemistry with experimental knowledge.

35013 - CELL AND MOLECULAR BIOLOGY

Objectives:

- To give a firm and rigorous foundation in the principles of cell and molecular biology.
- To describe the fundamental process of gene expression and cellular functions.

BLOCK – I:CELL STRUCTURE

Unit I

Cell theory - Structural organization of Prokaryotic and Eukaryotic cells.

Unit II

Ultrastructure of Cell membrane, Nucleus, Chromosomes, Mitochondria.

Unit III

Endoplasmic reticulum, Golgi apparatus, Lysosomes, Ribosomes, Peroxisomes and their functions.

Unit IV

The cytoskeleton – Microtubules and Microfilaments - Cell cycle - Mitosis and Meiosis.

BLOCK – II: NUCLEIC ACIDS

Unit V

Structure and functions of DNA**,**Types of RNA and its function

Unit VI

Enzymes involved in Molecular Biology-DNA polymerases, RNA polymerase, Helicase, Primase, Ligase, Exonuclease and endonuclease.

Unit VII

Mechanism of prokaryotic and eukaryotic replication; machinery for replication; Synthesis of leading and lagging strands, Okazaki fragments, Difference between Prokaryotic and Eukaryotic replication.

BLOCK – III: TRANSCRIPTION AND TRANSLATION

Unit VIII

Prokaryotic transcription: Promoters, Properties of bacterial RNA polymerase, Steps: Initiation, Elongation and Termination.

Unit IX

Eukaryotic transcription: Promoters, Enhancers, Factors, properties of RNA polymerase I, II and III. Post transcriptional modification, Reverse transcription.

Unit X

Protein synthesis: Machinery, Formation of initiation complex, Translocation, Chain elongation and Termination.Post-translational modifications

Unit XI

Cell free protein synthesis, Comparison of protein biosynthesis in prokaryotes and eukaryotes.

BLOCK – IV: REGULATIONS OF GENE EXPRESSION

Unit XII

Concept of operon – Lac and trp operons, Positive and negative control, Repressor and Inducer.

Unit XIII

Hormonal regulation of gene expression, Transcription factors, Steroid receptors; DNA binding motifs in pro- and eukaryotes

Unit XIV

Analysis of Gene expression using Molecular Methodolgy

Reference Books:

1. Hunter, L. E. 2009. The Process of life- An Introduction to Molecular **B**iology, The MIT press, USA.
2. Weaver, R.F., 2008. Molecular biology, McGraw Hill higher education, USA.
3. Becker, W, L. Kleinsmith, J. Hardin, and G. Bertoni, 2008. The world of the cell, Pearson Education, London.
4. Alberts, B., A. Johnson, J. Lewis, M. Raff, K. Roberts, and P.Walter, 2007. Molecular biology of the cell, Garland publishing Inc, New York.
5. Lodish, H, A. Berk, C.A. Kaiser, M. Krieger, M. P. Scott, A. Brtscher, H. Ploegh, and P. Matsudaria, 2007. Molecular cell biology, W. H. Freeman, USA.
6. Karp, G, 2007. Cell and molecular Biology- Concepts and Experiments, John Wiley and Sons, Inc. New York.
7. Freifelder, D, 2004. Essentials of Molecular Biology, Narosa Publishing House, New Delhi.
8. De Robertis, E.D.P. and E.M.F. De Robertis, 2006, Cell & Molecular Biology, 8th Edition, Indian Reprint.
9. Brown, T.A.., 1991. Molecular Biology, Labfax, Bioscientific publishers Ltd., Oxford.
10. Benjamin Lewin, Gene VII, Oxford University Press. U.K.
11. Rastogi, S.C., 2010, Cell and Molecular Biology, Second Edition
12. Gupta, P.K., 1999, Cell and Molecular Biology, Rastogi Publications, Meerut.

13. Ajay Paul., 2011. Cell and Molecular Biology. Books and Allied Pvt, Kolkata.
14. P.S. Verma and Agarwal, 2001. Concepts of Cell Biology. S. Chand & Co.
15. Helen Kreuzer, An Introduction To Cell And Molecular Biology: Concepts And Experiments, Published October 22nd 2013 by Wiley.

Outcome:

The students will acquire fundamental ideas on organization and functions of the different cell organelles, molecular basis of cellular processes and interrelationship with special emphasize on prokaryotic and eukaryotic systems.

LAB - I: 35014: ANIMAL DIVERSITY, BIOCHEMISTRY, CELL AND MOLECULAR BIOLOGY

ANIMAL DIVERSITY

1. **Museum specimen**: Invertebrates and vertebrates: Phylum wise (at least one from each phylum)
2. **Mounting**: Earthworm – Body and pineal setae, Honey bee – sting apparatus, Cockroach – Mouth parts, Prawn – Appendages and Shark - Placoid scales
3. **Dissections:** Understanding the anatomy of frog using an appropriate software package (Carolina^{TM} Biolab^{R} – Frog), Dissection of cockroach: Digestive, reproductive & nervous systems. Dissection of available fish: General anatomy (Viscera)

BIOCHEMISTRY

1. Preparation of solutions: Molarity, Normality and Percentage.
2. Buffer preparation and Determination of pH.
3. Estimation of glucose and total protein.

CELL AND MOLECULAR BIOLOGY

1. Cell organelles from slide preparation/images.
2. Buccal mucosal epithelium – Smear preparation to detect Barr bodies.
3. Isolation and detection of DNA from gel electrophoresis (Demo only).
4. Onion root tip – Squash preparation and study of mitosis.
5. Grasshopper testis - Squash preparation and study of meiosis (Demo only).
6. *Chironomus* larva - Squash preparation of giant chromosome.
7. Separation of amino acid by paper chromatography.
8. Separation of protein by electrophoresis - SDS and Native PAGE.

Reference Books:

1. Amsath A., 2010. Practical Manual in Zoology.
2. Amsath, A. 2013. Practical manual in Zoology. MMA Publications, Adirampattinam.
3. Shivraja Sankara, 2008. Laboratory Manual for Biochemistry.
4. David A Thompson., 2011. Cell and Molecular Biology Lab Manual.
5. **Helen Kreuzer, An Introduction To Cell And Molecular Biology: Concepts And Experiments, Published October 22nd 2013 by Wiley.**
6. Jordan E.L. and Verma P.S., 2005. Invertebrate Zoology, S. Chand & co. India.
7. Joe Sambrook and David W. Rusell., 2001. Molecular Cloning: A Laboratory Manual, Cold spring harbour laboratory press, U.S.A.
8. Lundblad R. L., 2009. Practical Handbook of Biochemistry and Molecular Biology, CRC publications.

35021- DEVELOPMENTAL BIOLOGY AND EVOLUTION

Objectives:

- To understand and master in the basic concept of developmental biology and evolution of animals.

BLOCK – I: GAMETOGENESIS & FERTILIZATION

Unit I
General introduction to Developmental Biology, Spermatogenesis, Physiological maturation of sperm, Sperm structure and physiology.

Unit II
Oogenesis, Egg - size, shape, Egg membranes and organization of egg - yolk, pigments, egg cortex.

Unit III
Maturation of egg, Polarity and Symmetry, Classification of eggs.

Unit IV
Fertilization : Types and Mechanism, Monospermy and Polyspermy. Activation of egg and Egg metabolism.

BLOCK – II: CLEAVAGE AND GASTRULATION

Unit V
Types of cleavage, Factors affecting cleavage, Chemodifferentiation.

Unit VI
Blastulation, Types of blastula – Gastrulation in frog and chick, Mechanism of morphogenetic movement.

Unit VII
Metabolic and molecular changes during gastrulation; Cell motility and Differential cell affinity; Fate maps construction.

BLOCK – III: ORGANOGENESIS AND ASSISTED REPRODUCTIVE TECHNOLOGY

Unit VIII
Development of eye, brain and heart in chick. Formation of muscle and neural crest. Embryonic induction, concept of organizer.

Unit IX
Foetal membranes in chick, placenta in mammals. Origin of gene theory, Nuclear transplantation, Differential gene activation, Factors involved in teratogenesis.

Unit X
Concept of Assisted Reproductive Technology (ART) – Monitoring of ovulation phase, Super-ovulation and Cryopreservation.

Unit XI
Sperm banking, Artificial insemination, IVF, Embryo transfer and Test tube babies, Gene knock out and knock in.

BLOCK – IV: EVOLUTION

Unit XII
Lamarckism, Neolamarckism, Darwinism, Neodarwinism, Theory of natural selection, Genetic and non-genetic variations and Evolution of races to species.

Unit XIII
Evidences for evolution (Anatomical, Embryological, Physiological and Biochemical). Patterns of behavioural adaptations, Isolating mechanism and speciation.

Unit XIV
Evolution of gene families, Molecular drive, Assessment of molecular variation, human origin and migration, Phylogenetic tree at molecular level.

Reference Books:

1. Jonathan M. W. Slack, Essential Developmental Biology, 3rd Edition, December 2012, ©2012, Wiley-Blackwell.
2. Gilbert, S. F., and K. Knisely, 2009. Developmental Biology, Sinauer Associates Inc.
3. Minelli, A. 2009. Forms of Becoming: The Evolutionary Biology of Development, Princeton University Press.
4. Futuyma, D. J. (2006) Evolutionary biology, Palgrave publishers, USA.
5. Hodge, R., 2009. Developmental Biology (Genetics and Evolution). Facts on File.
6. Slack, J. M. W. 2005. Essential Developmental Biology, Wiley-Blackwell.
7. Hake S, and F. Wilt, 2003. Principles of Developmental Biology, W.W. Norton & Co.
8. Wolpert, L., R. Beddington, T. Jessell, P. Lawrence, E. Mayerowitz, and J. Smith, 2002. Principles of development, Oxford University Press, UK.
9. Twyman, R.M. Developmental Biology. Viva, New Delhi, 2008.
10. Balinsky, B.I. An Introduction to Embryology. 5th Ed., Thomas Asia Pvt. Ltd., 2004.
11. Russo, V.E.A, Brody, S., Cove, D. and Ottolenghi, S. Development: The Molecular Genetic Approach. Springer Verlag, Berlin, 1992.
12. Rao.V. Developmental Biology: A Modern Synthesis. Oxford IBH New Delhi, 1994.
13. Rastogi, V.B., Organic Evolution, 12th Ed., KedarNath Ram Nath, Meerut.
14. Brian, K. H. and Benedikt Hall, G. Strickberger's Evolution. 4th Ed., Jones and Bartlett Publishers, Inc, 2008.
15. Colbert, E.H., Morales, M. and Minkoff, E.C., 2002. Colbert's Evolution of the Vertebrates: A history of the backboned animals through time, 5th edition, John Wiley – Liss, Inc., New York.

Outcome:
On successful completion of this course students should be able to critically discuss about the concepts, principles and scope of evolution.

35022- GENETICS

Objectives:

- To study the "science of heredity" and molecular process of gene expression.
- To analyze qualitative genetic data and describe the evolution of population.

BLOCK – I: CLASSICAL GENETICS

Unit I
Definition, Terminology and Scope of Genetics - Mendel and his contribution - Hybridization techniques of Mendel.

Unit II
Mendelian principles-Monohybrid and dihybrid crosses, Simple mendelian traits in Man.

Unit III
Polygenetic inheritance, Multiple alleles – Blood group inheritance in man.

Unit IV
Interaction of genes – Allelic and Non-Allelic interaction – Complementary, Supplementary, Duplicate and Epistatic interaction.

BLCOK – II: LINKAGE, CROSSING OVER AND CHROMOSOME MAPPING

Unit V
Mechanism and theories of linkage and crossing over.

Unit VI
Chromosomal and gene mapping methods. Linkage maps, Tetrad analysis, Mapping with Molecular markers and QTL mapping.

Unit VII
Structure and types of chromosome. Sex chromosomes, Sex determination in animals-human and honey bee. Heterochromatization and Barr bodies.

Unit VIII
Sex Linked inheritance, Non disjunction – Syndromes and Pedigree analysis.

BLOCK – III: MOLECULAR GENETICS

Unit IX
Mutation – Types and its applications, Chromosomal abnormalities, Inbreeding and out breeding.

Unit X
Population genetics: Hardy-Weinberg Equilibrium-gene pool, Gene frequency, Genotypic frequency and Factors affecting Hardy-Weinberg equilibrium.

Unit XI
Twin study, Eugenics, Euthenics and Euphenics.

BLOCK – IV: GENETIC CONCEPT

Unit XII
Concept of gene – Gene expression control in prokaryotes, eukaryotes, and phages.

Unit XIII
Genetic regulation in development and Role of cell death.

Unit XIV
Differential and sequential expression of genes with reference to Drosophila.

Reference Books:

1. Emmanuel C., Ignacimuthu S., Vincent S., 2006. Applied genetics: Recent trends and Techniques, MJP Publishers.
2. Crew F.A., 2006. Animal Genetics – The Science of Animal Breeding, Lightning Source Inc.
3. Joe Bearden H., John W. Fuquay, and Scott T. Willard., 2003. Applied Animal Reproduction, 6th edition , Prentice Hall.
4. Richard M. Bourdon, 1999.Understanding Animal Breeding, 2nd Edition, Prentice Hall.
5. Terence A. Brown., 1998. Genetics: a molecular approach, Thrones Publishers.
6. *Benjamin Pierce, 2007.Genetics a conceptual approach, W.H. Freeman&Company, USA*
7. Hartwell L., 2004. Genetics from genes to genomes, McGraw-hill, USA.
8. Gahalain S. S., 2004. Fundamentals of Genetics, Anmol Publications Pvt., India.
9. *Burton S. Guttman, Anthony Griffiths, David T. Suzuki., 2002. Genetics: A Beginner's Guide,* One world Publications Epz.
10. Verma, P.S. and V.K. Agarwal, 2009, Genetics, Revised Edition, S. Chand & Co., New Delhi.
11. Gardner, E.J, Simmons, M.J. and Snustad, D.P., 2007. Principles of Genetics 7th Ed., John Wiley India.
12. Sudbery, P., 2009. Human Molecular Genetics. 2nd Ed., Dorling Kindersley (India) Pvt. Ltd.
13. Snustad, D.P. and Simmons, M.J., 2012. Genetics. VI Ed., John Wiley & Sons, Singapore.
14. Russel, P.J., 2006. iGenetics: A Molecular Approach. 2nd Ed., Pearson Education.
15. Strickberger, M.W., 2008. Genetics, MacMill Publishing.
16. Singh, B.D., 2003. Genetics, Kalyani Publishers, Ludhiana.
17. Bruce R. Korf, Mira B. Irons, **Human Genetics and Genomics, Includes Wiley E-Text, 4th Edition,** January 2013, Wiley-Blackwell

Outcome:
The students will understand the basic concepts of mendelian, molecular, evolutionary and applied genetics.

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35023- MICROBIOLOGY

Objectives:

- To inculcate knowledge on fundamentals of microorganisms.
- To learn the structural organization, morphology and reproduction of microbes.
- To know the principles of Microscopy and advancements in Microscopy.

BLOCK – I: INTRODUCTORY MICROBIOLOGY

Unit I

Introduction to Microbiology, Haeckel's three–kingdom concept, Whittaker's Five-kingdom concept, Three-domain concept of Carl Woese.

Unit II

Classification of Bacteria according to Bergey's Manual.
Fungi: Classification of fungi based on Alexopoulos system. General Characteristics of Fungi, Industrial uses of Yeast and Moulds. Lichens - Structural organization and their properties.

Unit III

Viruses: ICTV system of classification, General properties, Morphology and ultra-structure of virus (RNA, DNA).

BLOCK – II: MICROSCOPY, METHODS, NUTRITION AND MICROBIAL GROWTH

Unit IV

Principles and their applications of Simple, Compound, Fluorescent, Electron microscopes.(SEM & TEM) and Confocal microscopes.

Unit V

Stains and staining techniques: Simple, Differential and Structural staining methods, Imaging techniques.

Unit VI

Preservation methods of microbes for storage and microscopic studies, Culture collections, Sterilization and disinfection

Unit VII

Auxenic and Synchronous culture, Aerobic and Anaerobic Culture media and Nutritional types. Growth curve, Generation time and growth kinetics. Factors influencing microbial growth.

Block – III: General characteristics of bacteria,algae and Protozoa

Unit VIII

Prokaryotic cell structure & Organization: Cell membrane, Plasma membrane, Cytoplasmic matrix, Inclusion bodies, Ribosome, Nucleiod, Prokaryotic cell wall, Capsule, Slime layers, S layers, Pili and Fimbriae, Flagella and Motility.

Unit IX
Classification of Algae based on Fritsch system – General characteristicss of Micro and Macroalgae - Biological and Economic importance.

Unit X
Protozoa –General characteristics, .Impotance of *Entamoebahistolytica* and *Plasmodium* sp.

BLOCK – IV: MOLECULAR TECHNIQUES FOR IDENTIFICATION, INFECTIOUS DISEASES

Unit XI
Molecular Taxonomy, 16S/18S rRNAs and its importance in identification of microorganisms.

Unit XII
Phylogenetic tree, Types and construction of Phylogenetic tree, Molecular tools in assessing microbial diversity.

Unit XIII
Metagenomics - Sequencing methods, Data Analysis and applications.

Unit XIV
Bacterial Diseases (Tuberculosis, Typhoid, Leprosy) Viral diseases (Hepatitis, HIV, Ebola)

Reference Books:

1. Tortora, G.J., Funke B.R, and Case C.L. (2010). Microbiology an Introduction (10th Edition), Benjamin Cummins, USA.
2. Dubey, R.C. and Maheswari, D.K. (2013). A Textbook of Microbiology (Revised Edition), S.Chand and Company Ltd., New Delhi.
3. Prescott, L.M., Harley, J.P. and Klein, D.A. (2014). Microbiology (9th Edition), McGraw Hill Publishers, Boston.
4. Brock, T.D., Smith, D.W. and Madigam, M.T. (2002). Biology of Microorganisms (Fourth Edition) Prentice Hall International, London.
5. Stanier, R.Y., Ingraham, J.L., Wheels, M.L. and Painter, P.R. (1999). General Microbiology, Mac Millan Educational Limited, London.
6. Boyd, R.F. (1998). General Microbiology, MosbyCollege Publishing, St. Louis.
7. Nester, E.W., Roberts, C.V. and Nester, M.T. (1995). Microbiology, A Human Perspective. IWOA, U.S.A.
8. Pelzcar, M.J., Chan, E.C.S. and Kreig, N.R. (2001). Microbiology, McGraw Hill Inc., New Delhi.
9. Hogg, S. Essential Microbiology. John Wiley & Sons Ltd., England, 2005.
10. Madigan, M.T. and Martinko, J.M. Brock Biology of Microorganisms. 11th Ed., Prentice Hall, U.S.A, 2006.
11. Srivastava, S.H., 1994. Advances in general Microbiology, Anmol Publications, New Delhi.

Outcome:

1. Basic knowledge on historical perspectives of microbiology.
2. Knowledge on different structure, culture conditions, growth and applications of microbes in various fields.
3. Ideas on different type of microscopes.

LAB - II: 35024- DEVELOPMENTAL BIOLOGY AND EVOLUTION, GENETICS AND MICROBIOLOGY

DEVELOPMENTAL BIOLOGY AND EVOLUTION

1. Frog: Egg, blastula and yolk plug stage.
2. Chick: Egg, 24 hrs, 36 hrs, 48 hrs, 72 hrs and 96 hrs developmental stages.
3. Placental types in Mammals.
4. Animals of evolutionary importance.
5. Analogous and homologous organs.
6. Fossils.
7. Mimicry and coloration.

GENETICS

1. Klinefelter's Syndrome, Turner's Syndrome, Down's Syndrome and Cri- Du – Chat.
2. Pedigree analysis using charts and data.
3. Human karyotyping and chromosomal abnormalities.
4. Hardy- Weinberg law & Calculation of gene frequencies for dominant and recessive traits.

MICROBIOLOGY

1. Enumeration of bacteria and fungi.
2. Pure culture and preservation of bacteria.
3. Gram Staining and Negative staining.
4. Motility of bacteria.
5. Hydrolysis of starch, gelatin and protein.
6. Antibiotic susceptibility test.

Reference Books:

1. Mary S. Tyler, 1994. Developmental Biology: A guide for experiment study.
2. Bauman R. W, and N. Dolby, 2008. Microbiology Lab Manual (3rd Edition), Pearson Custom Publishing.
3. James G Cappucino, Natalie Sherman, 2007. Microbiology: A laboratory manual, Benjamin – Cummining publications, U.S.A.
4. Melissa Ann Gibbs, 2003. A Practical Guide to Developmental Biology, Oxford University Press, USA.
5. Emmanuel C., Ignacimuthu S., Vincent S., 2006. Applied genetics: Recent trends and Techniques.

35031-ANIMAL PHYSIOLOGY

Objectives:

- To study the basic physiological principles common to humans and other animals.
- To elaborate the physical and chemical functions occur in animals tissue/organ systems.

BLOCK – I: INTRODUCTION TO DIGESTIVE, RESPIRTORY AND EXCRETORY SYSTEMS

Unit I

Definition, Divisions of physiology, Relationship of physiology with other sciences, Significance of the study.

Unit II

Digestive system in man, Physiology of digestion, Absorption and Assimilation, Gastrointestinal hormones and their control in digestion.

Unit III

Respiratory system in man, Types and mechanism of respiration-Transportation of gases, Control of respiration.

Unit IV

Excretory system of human, Structure and functions of nephron, Urine formation and its regulation.

BLCOK – II: CARDIO VASCULAR SYSTEM AND NERVOUS SYSTEM

Unit V

Blood: Composition, Haemopoiesis, formed elements, Blood volume and its regulation, Haemostasis.

Unit VI

Types of heart, Structure of human heart, Heart beat and Cardiac cycle, Blood pressure, ECR and its application.

Unit VII

Types and functions of neurons, Central and Peripheral Nervous System, Synapse and its transmission, Resting and action potential, Neuro-muscular junction.

BLOCK – III: EFFECTORS AND RECEPTORS

Unit VIII

General structure and types of muscles, Sarcomere, Ultra structure of skeletal muscle, Mechanisms of muscle contraction, Chemical changes during muscle contraction, Kymograph.

Unit IX

Physiology of vision, hearing and tactile response.

Unit X
Thermoregulation in animals. Tolerance to high temperature, cold and freezing, Physiology of hibernation and aestivation.

Unit XI
Osmo-ionic regulation in freshwater and marine fishes and crustaceans – Response to hyper and hypo-osmotic media. Adaptation to pressure in high altitude – Buoyancy.

BLOCK – IV: ENDOCRINOLOGY AND ANIMAL BEHAVIOUR

Unit XII
Endocrine glands and their hormones – Mechanism in action of hormones. Hypo and Hyper secretion of hormones (Thyroid, adrenal and pancreas) and their diseases.

Unit XIII
Neuro endocrine control of hormones, Invertebrate hormones and Hormonal control of insect metamorphosis.

Unit XIV
Biological clock, Endogenous rhythm, Circadian, Circannual and Lunar periodicities.

Reference Books:

1. Richard W. Gill, Gordon A. Wyse and Margaret Anderson., 2012. Animal Physiology 3rd edition.
2. Christopher D. Moyes, Patricia M. Schulte., 2008. Principles of Animal Physiology. Pearson Education, Inc., publishing as Benjamin Cummings, San Francisco
3. Gordon A. Wyse, Margaret Anderson., 2008. Animal Physiology, 2nd edition, Richard W. Hill.
4. Neville G. Gregory., 2005. Physiology and Behavior of Animal Suffering (UFAW Animal Welfare), 1st edition, Wiley - Blackwell.
5. McGowan, C., 2016. Animal physiotherapy: assessment, treatment and rehabilitation of animals. John Wiley & Sons.
6. R.H. Williams, 2002. Text Book of Endocrinology-W.B. Saunders.
7. Campbell, A.M. and Paradise, C.J., 2016. Animal Physiology. Momentum Press.
8. Guyton Mc. and Hall, R.T. 2011. Textbook of Medical Physiology. 12th Edition Saunders Publisher, USA.
9. Sherwood L, Klandorf H and Yancey P., 2012. Animal Physiology: From genes to organisms, Cengage Learning. 5. Prosser CL, 1991. Comparative Animal Physiology, Environmental and Metabolic Animal Physiology, John Wiley and Sons.
10. Verma P.S. and V.K. Agarwal.1992. Animal Physiology. S. Chand and Co.
11. Rastogi, S.L., 1997. Essential of Animal physiology. New Age International Publishers, New Delhi

Outcome:
The course provides a comprehensive overview of animal physiology from molecular, cellular and whole animal systems approaches.

35032 - IMMUNOLOGY

Objectives

- To understand the fundamental concepts of immune systems, innate and adaptive immunity mechanism and their response.
- To provide in-depth ideas on immunotechnology and its applications.

BLOCK – I: INTRODUCTION TO IMMUNOLOGY

Unit I
Historical perspectives and scope of immunology, Types of immune cells, Tissues and organs of immune system.

Unit VII
Lymphoid organs- structure and functions of primary and secondary.

Unit II
Molecules of immune system - Antibodies, Complements, Cytokines, Interferons - Types, sources and functions. Antigen: Classification and Epitopes.

Unit III
Elements of immune system: Hematopoiesis, T- Lymphocytes, B- Lymphocytes, Generation of Lymphocyte specificity and diversity.

Unit IV
Antigen processing and presentation, Subsets of T-Cells, Memory, Helper and suppressor cells, Myeloid cells, Major histocompatibility complex (MHC)

BLOCK – II: IMMUNITY AND IMMUNE RESPONSE

Unit V
Immunity: Types of Immunity - Innate, adaptive immunity.

Unit VI
Immune Response: Types of Immune response, Effector mechanism of humoral and cell mediated immune responses.

Unit VIII
Antibody dependent cell mediated cytotoxicity, Natural killer cells. Immunity to infections – Immunoprophylaxis, Vaccines and immunization schedule.

BLOCK – III: IMMUNE DISORDRS

Unit IX
Infectious diseases; Hypersensitivity - Types I, II, III and IV.

Unit X
Autoimmune disorders and Immunodeficiency diseases. Organ Transplantation. antibody engineering

Unit XI
Cancer: Types and nature, Immunotherapy; Immune responses against tumors and transplants.

BLOCK – IV: IMMUNOLOGICAL TECHNIQUES

Unit XII
Immunocytochemistry, Antibody generation and Radioimmuno assay.

Unit XII
Detection of molecules using Immunoblot techniques, ELISA and Vaccine development

Unit X IV
Immunoprecipitation and immunofluorescence microscopy, Acquired Immuno Deficiency Syndrome (AIDS) detection and Hybridoma technology, FACS, Immunofluorescent assay.

Reference Books:

1. Judith A. Owen, Jenni Punt, Sharon A. Stranford., 2013. Kuby Immunology, 7th edition, W.H. Freeman & Company.
2. Blaine T. Smith., 2008. Concepts in Immunology and Immunotherapeutics.
3. Kannan I., 2007. Immunology.
4. Kuby, Kindt, Goldsby and Osborne, 2007. Kuby Immunology, 6th edition, W.H. Freeman & Company.
5. David Male, Jonathan Brostoff, David B Roth and Ivan Roitt, 2006. Immunology, Elsevier.
6. Helen Chapel, Mansel Haeney, Siraj Misbah and Neil Snowden., 2006. Essentials of Clinical Immunology, Blackwell Publishing.
7. Vaman Rao., 2006. Immunology, Narosa Publishing House Pvt, Ltd.
8. Peter J. Delves, Seamus J. Martin, Dennis R. Burton, Ivan M. Roitt, 2011. Roitt's Essential Immunology. 12th Ed., Wiley-Blackwell.
9. Chakravarty, A.K. Immunology and Immunotechnology. Oxford University Press. New Delhi, 2006.
10. Janeway, C.A., 2010, Immunobiology – The Immune System in Health and Disease, Churchill Livingstone, New York.

Outcome:
The course will provide basic mechanisms, distinctions and functional interplay of innate and adaptive immunity and immunological techniques.

35033-ENVIRONMENTAL BIOLOGY

Objectives:

- To understand the dynamics of ecosystem and inter-relationship between organisms.
- To quantify the biological productivity and to restore natural ecosystem.

BLOCK – I: ECOSYSTEM

Unit I

Structure, Functions and types of ecosystem - Trophic structures, Food chains, Food web, Energy flow and Ecological pyramids.

Unit II

Abiotic factors, Soil organisms, Biological effects of light, and temperature.

Unit III

Thermal stratification, Concept of limiting factors, Shelford's law of tolerance and ecotypes - Grassland and Pond ecosystem.

BLOCK – II: MARINE ECOLOGY

Unit IV

Divisions of marine environment, Physical and chemical properties of seawater, Major and minor elements.

Unit V

Primary and secondary production, Estimation and factors influencing productivity; Adaptation of plankton, Red tide, Inter tidal and deep sea ecology.

Unit VI

Unique features of Coral Reefs, Seaweeds, Seagrasses; Mangroves and estuaries.

BLOCK – III: BIOGEOCHEMICAL CYCLE AND POPULATION ECOLOGY

Unit VII

Biosphere: Types - Hydrosphere, Lithosphere and Atmosphere.

Unit VIII

General account of complete and incomplete cycle; Gaseous cycle – Carbon, Nitrogen and Oxygen cycles.

Unit IX

Sedimentary cycle: Phosphorus and Sulphur cycles.

Unit X

The population concept, Natality, Mortality, Growth rate, Population density and Age distribution, Carrying capacity, Fluctuation and Regulation.

BLOCK – IV: COMMUNITY ECOLOGY AND ENVIRONMENTAL POLLUTION

Unit XI
Community structure, Ecotone and edge effects, Ecological niche.

Unit XII
Ecological succession, Climax community - Monoclimax and polyclimax theories.

Unit XIII
Air, Water and Soil pollution - Their biological effects - Pollution control measures; Climatic changes - Green house effects, Global warming; Bioremediation and environmental awareness.

Unit XIV
Conservation of natural resources; Biodiversity hot spots of India; Endangered and threatened species, Germplasm conservation - Environmental laws.

Reference Books:

1. Henry, M., and H. Stevens, 2009. A Primer of Ecology with R (Use R), Springer
2. Odum EP, 2008. Fundamentals of Ecology, Cengage Learning (Thompson), USA.
3. Smith, T. M., and R. L. Smith, 2008. Elements of Ecology (7th Edition), Benjamin Cummings.
4. Krebs, C. J. 2008. Ecology: The Experimental Analysis of Distribution and Abundance (6th Edition), Benjamin Cummings.
5. Clark R.S. 2001. Marine Pollution, Clanderson Press Oxford, New York.
6. Strickland, J. D. and T. R. Parsons, 1972. A practical handbook of seawater analysis. Bull of Fish. Res. Bd., No. 167, pp 310.
7. James W. Nybakken and Mark D, 2000. Bertness, marine biology: An ecological approach, Benjamin-Cummings Publishing Company.
8. Saha, T. K. (2008). Ecology and Environmental Biology. Books and Allied (P) Ltd.
9. Beckman, Daniel Beckman, Marine Environmental Biology and Conservation, Published February 13th 2012 by Jones & Bartlett Publishers

Outcome:
The course provides knowledge on ecological principles/concepts and concise critical thinking to solve problems in ecology.

LAB – III: 35034-ANIMAL PHYSIOLOGY, IMMUNOLOGY, ENVIRONMENTAL BIOLOGY

ANIMAL PHYSIOLOGY

1. Estimation of salivary amylase activity.
2. Estimation of ammonia and urea.
3. Estimation of blood chloride.
4. Determination of glucose and glycogen.
5. Estimation of oxygen consumption of fish.

IMMUNOLOGY

1. Study of lymphoid organs.
2. Haemagglutination assay.
3. Study of antibody titre values.
4. Immunodiffusion – Single / Double and Immunoelectrophoresis.
5. Blood grouping
6. Human Chorionic Gonadotropin (hCG) test

ENVIRONMENTAL BIOLOGY

1. Estimation of salinity.
2. Estimation of dissolved oxygen.
3. Mounting of plankton (fresh water / marine).
4. Animal associations.
5. Intertidal fauna.
6. Construction of a food web diagram.
7. Measurement of light intensity in water bodies using Secchi disc.

Reference Books:

1. Ghai C.L., 2007. A text book of practical physiology.
2. Turgeon, M.L. 2008. Immunology & Serology in Laboratory Medicine (Immunology & Serology in Laboratory Medicine (Turgeon)), Mosby publishers.
3. Talwar, G.P., 2006. A hand book of practical & clinical immunology, CBS publishers, New Delhi
4. Hay, F.C., O.M.R. Westwood, and P.N. Nelson, 2002. Practical Immunology, Wiley-blackwell, USA.
5. Phillip L. Watson, Arlene Westhoven, Environmental Biology Laboratory Manual, Published by Hunt Publishing Company.

35041- FISHERIES AND AQUACULTURE

Objectives:

- To familiarize basic information about fishery biology, resources, management and necessary skills to identify fish species.
- To provide technical knowledge about recent aquaculture practices.

BLOCK – I: CAPTURE FISHERY AND CHARACTERISTIC ANALYSIS

Unit I

Classification of fishes, Economically important marine and freshwater fishes with regard to their fishery potential.

Unit II

Status of Indian Capture fishery, Indigenous and modern craft and gears used for capture fisheries.

Unit III

Morphometric and meristic characters of fish, Food and feeding habits, Age and growth, Spawning and reproduction of fish.

Unit IV

Endangered species and Invasive species; Conservation and Management of Fishery Resources.

BLOCK – II: AQUACULTURE

Unit V

Definition, Status of Indian aquaculture, Types of culture based on stocking density, Cultivable organisms.

Unit VI

Farm design, Structure and construction, Pond preparation, Stocking, Water quality, Feed management, harvesting, and economics; Good Management Practices.

Unit VII

Cage culture, Pen culture, Race ways culture, Poly culture, Composite fish culture and integrated fish farming. Important finfish and shell fish disease and its control measures.

BLOCK – III: HATCHERY MANAGEMENT TECHNIQUES

Unit VIII

Types of Hatchery, Brood Stock, Induced breeding, Spawning, Hatching, Larval rearing, Post larval rearing and hatchery economics

Unit IX

Live feed culture and Good management practices in hatchery.

Unit X

HACCP concept, Biosecurity and Specific pathogen free seed production and Seed packing and transportation.

BLOCK – IV: POST HARVEST TECHNOLOGY AND FISHERY BY-PRODUCTS

Unit XI
Physical and biochemical methods to examine freshness of fish.

Unit XII
Processing methods: Freezing, Canning, Smoking, Drying and Irradiation methods of preservation of fish.

Unit XIII
Quality control: HACCP, National and International standards.

Unit XIV
Fishery by-products: fish liver oil, fish ensilage, Isinglass, chitin, leather from shark skin and other value added by –products.

Reference Books:

1. Jean T. Nolan, 2009. Offshore Marine Aquaculture, Nova Science Pub Inc.
2. Michael King, 2007. Fisheries Biology, Assessment and Management, Wiley-Blackwell.
3. Pillay, T.V.R., and M. N. Kutty, 2005. Aquaculture: Principles and Practices, Wiley-Blackwell.
4. FAO, 2012. The State of World Fisheries and Aquaculture 2012 (Manuals from the Fao Training), United Nations Publications.
5. Balachandran, K. K., 2002. Post Harvest Technology of Fish and Fish Products, Daya Publishing House.
6. Bremner, H.A, 2002. Safety and Quality issues in fish processing, Publisher: CRC,(1st edition).
7. Simon Jennings, Michel Kaiser, and John D. Reynolds, 2001. Marine Fisheries Ecology, Wiley-Blackwell.
8. Chandran, K.K., 2000. Post harvest Technology of Fish and Fishery Products, Daya Publishing House, New Delhi.
9. Andy Beaumont, Pierre Boudry, Kathryn Hoare, 2010. Biotechnology and Genetics in Fisheries and Aquaculture, 2nd Edition, Wiley-Blackwell.
10. Stefano B. Longo, Rebecca Clausen, Brett Clark, The Tragedy of the Commodity: Oceans, Fisheries, and Aquaculture, Published June 25th 2015 by Rutgers University

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Outcome:
On successful completion of this course students should be able to critically discuss the fundamental concepts of fishery biology and role of aquaculture and its applications.

35042-ANIMAL BIOTECHNOLOGY

Objectives:

- To understand the principles and application of biotechnology methods in animal.
- To understand recent techniques in animal biotechnology.

BLOCK – I: GENETIC ENGINEERING

Unit I
Over view of animal biotechnology, Basic principles of genetic engineering - Genetic engineering in animal systems.

Unit II
Vectors: Plasmid, Cosmid, Phagemids - Yeast artificial chromosome (YAC) and bacterial artificial chromosome (BAC), Shuttle vectors, Yeast vectors, Minichromosomes.

Unit III
Gene transfer methods in animals –Electroporation; Microinjection; Biolistic Particle Delivery System;, Sonoporation, Optical Transfection; Protoplast Fusion, Retrovirus mediated gene transfer.

BLOCK – II: ANIMAL CELL CULTURE AND TRANSGENIC ANIMALS

Unit IV
Media requirements, preparation of media and sterilization techniques; Natural and synthetic media; Culture methods: hanging drop, suspension and monolayer culture; Primary and established cell lines.

Unit V
Characteristics of transformed cells; Methods of cell preservation- Applications of cell culture in product development and tissue repair; Bioreactors and scaling-up technologies.
Unit VI
Production and applications of transgenic animals (Fish, Mice, Sheep and Pig) and chimeras. Dolly.

Unit VII
Trangenesis; transgenic animals from foetal cells; transgenic animals in Xenotransplatation, transgenic organisms to interrupt disease cycles. Artificial insemination and embryo transfer.

BLOCK – III: MOLECULAR DIAGNOSTIC TECHNIQUES AND PHEROMONES

Unit XII
Isolation and purification of nucleic acids. Hybridization: Southern, Western and Northern hybridization.

Unit XIII
PCR, RFLP, RAPD, DNA finger-printing. DNA bar coding. Principles, methods and Instrumentation of DNA sequencing (Maxam and Gilbert method, Sanger's di-deoxy method and automated DNA sequencing).

Unit XIV
Pheromones in pest management – Insect and Rodent control; Pheromones in animal breeding – Conservation and management of indigenous Cow, Buffalo, Tiger and Elephant.

BLOCK – IV BIOTECHNOLOGY IN MEDICINE

Unit XI
Embryonic and adult, hematopoietic, epithelial and mesenchymal stem cells - applications. Knock out and Knock in Technology.

Unit VIII
Recombinant vaccines, Subunit vaccines and live vaccines - Production of Insulin and Tissue plasminogen activator.

Unit IX
Gene therapy, somatic gene therapy and germline gene therapy; Applications of advanced techniques – Positron emission tomography–computed tomography (PET-CT), Molecular imaging. Human genome project

Unit X
Ethical Issues in Genetic Engineering and Transgenics - Current Developments; social, extrinsic and intrinsic concerns, Issue of Species Boundaries, The Legal Implications of Transgenics.

Reference Books:

1. Castilho L. 2008. Animal Cell Technology: From Biopharmaceuticals to Gene Therapy, Taylor & Francis.
2. Freshney, I, 2006. Culture of Animal Cells, Publisher: John Wiley & Sons Inc (sea) Pvt. Ltd.
3. Satyanarayana, U., 2006. Biotechnology, Books and Allied (P) Ltd.
4. Animal Biotechnology, M.M. Ranga, 2000. Agrobios, India.
5. Brown, T.A, 2005. Gene cloning- An introduction, 2nd &3rd ed, Chapman &Hall. Publisher: Stanley Thornes Publishers Ltd.
6. Primrose, S.B, Richard M, Twyman, R and W. Old, 2001. Principles of gene manipulation, (6th ed), Published by Wiley-Blackwell.

7. Tsai C.S, 2001. An introduction to Computational Biochemistry, Publishers: John Wiley and Sons, Inc.,
8. Cartwright, T, 2009. Animal Cells as Bioreactors (Cambridge Studies in Biotechnology), Cambridge University Press, UK.
9. Glick, B.R. and Pasternak, J.J., 2007. Molecular Biotechnology: Principles and Applications of Recombinant DNA. 3rd Ed., ASM Press, Washington.
10. Dubey, R. C., 2009, A Text Book of Biotechnology, S. Chand Co., New Delhi.
11. R. Ian Freshney, 2016. Culture of Animal Cells: A Manual of Basic Technique and Specialized Applications. Wiley Blackwell.
12. V Kumaresan and N. Arumugam, 2014. Animal Biotechnology, Saras Publication.
13. B. Singh and S.K. Gautam, 2013. Text book of animal biotechnology. TERI press.

Outcome:

On successful completion of this course students should be able to critically discuss the applications of biotechnology in research and industry.

35043-BIOPHYSICS, BIOSTATISTICS AND BIOINFORMATICS

Objectives:

- To understand the physical laws and their role in building biomolecules.
- To provide knowledge on biophysical instruments and their application in biology.
- To provide the computational knowledge in biological applications.

BLOCK – I: BIOPHYSICS

Unit I

Introduction to Biophysics: Structure and properties of atoms and molecules-Chemical bonds – types and properties; Polymerization of organic molecules.

Unit II

Laws of thermodynamics - principle and application; Bio-energetics- coupling of chemical reactions- Redox potential- NADP/NADPH and Free energy.

Unit III

Natural radiations – Properties of light, Absorption of light, energy states of atoms, spin property of electrons- ground state and excited state of atoms and bio-molecules – their effects.

Unit IV

Spectroscopy- principle and applications; Delayed effects of radiation and measurement of radio activity – Geiger Muller counter – Isotopes as tracers - Autoradiography.

BLOCK – II: BIOSTATISTICS

Unit V

Definition and scope of biostatistics - Collection of data -Primary and secondary data.

Unit VI

Types of sampling: Random and stratified random sampling. Types of variables: continuous and discontinuous variables, qualitative and quantitative variables.

Unit VII

Presentation of data: line and bar diagram, histogram, polygon and pie diagram.

Block – III: MEASURES OF CENTRAL TENDENCY AND MEASURE OF DISPERSION

Unit VIII

Mean, Median and mode – Dispersion: Range, variance, SD, SE and CV.

Unit IX

Probability and Hypothesis testing- Normal distribution, confidence interval and P value.

Unit X

Common statistical tools: Chi-square,'t'test, – ANOVA, Correlation and Regression analysis; statistical packages.

BLOCK – IV: BIOINFORMATICS

Unit XI
Introduction to Bioinformatics, Medical-informatics, Cheminformatics and Pharmacoinformatics.

Unit XII
Current researches in Bioinformatics. Applications of Bioinformatics in cancer detection, Drug targets.

Unit -XIII
Animal genome diversity - Introduction to DNA and Protein Sequence Analysis – Introduction and Concepts to biological Databases.

Unit - XIV
Phylogenetic analysis using PHYLIP, ClustalW.

Reference Books:

1. Daniel, W.W., 2007. Biostatistics, Wiley publishers, USA.
2. Zar, 2006. Biostatistical analysis, Dorling Kindersley PvtLtd , India.
3. Nolting, B., 2006. Methods in modern biophysics, Springer, Berlin.
4. Agarwal, S.K., 2005. Advanced biophysics, APH Publishing Corporations, India
5. Daniel, M., 2004. Basic biophysics, Agrobios publications, India
6. Bailey, N.T.J., 1997. Statistical Methods in Biology, III Ed., Cam. University Press, N.Y.
7. McCleery, R.H. and Watt, T.A., 2007. Introduction to Statistics for Biology. 3rd Ed., Chapman & Hall / CRC.
8. Goutham, N, Pattabi, S., 2001. Biophysics, Narossa Publishing company, New Delhi.
9. Mount, D., 2004. "Bioinformatics: Sequence and Genome Analysis"; Cold Spring Harbor Laboratory Press, New York.
10. Lesk, A.M., 2002. "Introduction to Bioinformatics", First edition, Oxford University Press, UK.
11. Lukas, K., Buehler, Hooman, H. Rashidi, 2000. "Bioinformatics Basics: Applications in Biological Science and Medicine"; CRC Press.
12. Jean-Michel, C. Notredame, C., 2003. "Bioinformatics for Dummies"; John Wiley & Sons.
13. R. Durbin, S. Eddy, A. Krogh and G. Hitchison, 2003. Biological Sequence Analysis, Cambridge University Press, Eighth edition.
14. Andrew R. Leach, Valerie J. Gillet, 2007. An Introduction to Chemoinformatics, Springer, Revised Edition.

Outcome:
This course will provide students with the basic concepts of biophysical techniques, statistical, computational skills and their application to assess biological macromolecules.

LAB-IV: 35044- FISHERIES AND AQUACULTURE, ANIMAL BIOTECHNOLOGY, AND BIOPHYSICS, BIOSTATISTICS AND BIOINFORMATICS

FISHERIES AND AQUACULTURE

1. Identification of commercially important fin fishes, shell fishes, molluscs, lobsters and seaweed.
2. Physical, biochemical and microbiological methods to examine freshness of fish.
3. Estimation of protein, lipid, carbohydrate and salt content in fish.
4. Determination of stocking density and feed assessment.
5. Method of transportation of seeds.
6. Modern crafts and gears.

ANIMAL BIOTECHNOLOGY

1. Isolation of genomic DNA.
2. Estimation of DNA.
3. Demonstration of ELISA.
4. RAPD, RFLP (Demo).
5. Extraction and purification of Plasmid DNA.
6. Spotter: Models of PCR, Southern blotting.
7. Cloning vectors – images.

BIOPHYSICS BIOSTATISTICS AND BIOINFORMATICS

1. Spectrophotometer, pH meter, and electrophoretic unit as spotters.
2. Construction of graph and bar diagram using biological data.
3. Calculation of mean, median, mode, variance, standard deviation and standard error and Chi-Square test.
4. IN SILICO Analysis

Reference books:

1. Moyle, P. B. and J. J. Cech, Jr.,1996. Fishes an introduction to Ichthyology, Prentice Hall, New Jersey.
2. Jean T. Nolan, 2009. Offshore Marine Aquaculture, Nova Science Pub Inc.
3. Michael King, 2007. Fisheries Biology, Assessment and Management, Wiley-Blackwell.
4. Laurence Hutchinson, 2006. Ecological Aquaculture: A Sustainable Solution Permanent Publications.
5. David Scarfe, Cheng-Sheng Lee, and Patricia J. O'Bryen, 2006. Aquaculture Biosecurity: Prevention, Control, and Eradication of Aquatic Animal Disease, Wiley-Blackwell.
6. Daniel, W. W. (2007) Biostatistics, Wiley publishers, USA
7. Zar (2006) Biostatistical analysis, Dorling Kindersley Pvt. Ltd. , India.
8. Pillay, T. V. R., and M. N. Kutty, 2005. Aquaculture: Principles and Practices, Wiley-Blackwell.
9. Sam brook J., Fritsh E.F., Maniatis T., 1989. Molecular cloning, volume -3, Cold Spring Harbour Laboratory.
10. Glover D.M and Hames B.D., 1995. DNA Cloning, 2nd edition, Volume - I, II, III, IRL press at Oxford University press, New York.

DURATION OF THE PROGRAMME

The course shall consist of two academic years divided into four semesters.

Faculty and Support Staff Requirements

This programme requires the following faculty and supporting staffs

Name of the position	Required
Core Faculty	3
Faculty: Specialization	2
Laboratory Assistant	1
Clerical Assistant	1

Instructional Delivery Mechanism

Each semester there will be one contact programme of 168 hours duration in total comprising of both theory and practical (120 hours practical and 48 hours theory). The SLM (Self Learning Material) will be supplied to the students in print form as well as in CD form. The face to face contact sessions of the programme for both theory and practical courses will be held at the head quarter only. The conduct of end semester examinations, evaluation and issuance of certificates will be done by office of the controller of examinations, Alagappa University, Karaikudi.

f) PROCEDURE FOR ADMISSION, CURUCULUM TRANSACTION, AND EVALUATION

Procedure of Admission

A candidate who has passed UG degree with Zoology/Animal Sciences/Life Sciences as main subject with Chemistry / Biochemistry / Microbiology / Botany as one of the ancillary subjects from any University/colleges shall be permitted to appear and qualify for the M.Sc. Zoology programme.

Curriculum Transactions:

The class room teaching would be through conventional lecture, use of OHP, power point presentation and novel innovative teaching ideas like television and computer aided instruction. Student seminars would be arranged to improve their awareness and

communicative skill. In the laboratory, instruction would be given for the safe handling of chemicals and instruments. For practical courses exclusive study materials containing the requirements, procedure for the experiments will be issued to the learners. In the laboratory, instruction would be given for the experiments followed by demonstration and finally the students have to do the experiments individually.

Face to face contact session will be conducted as given in below table.

Course Type	**Face to Face Contact Session/semester (in Hours)**
Theory courses (3 Courses with four credits each)	48
Practical courses (1 course with 4 credits)	120
Total	168

Evaluation

The examinations shall be conducted separately for theory and practical's to assess the knowledge acquired during the study. There shall be two systems of examinations viz., internal and external examinations. In the case of theory courses, the internal evaluation shall be conducted as Continuous Internal Assessment via. Student assignments preparation and seminar, etc. The internal assessment shall comprise of maximum 25 marks for each course. The end semester examination shall be of three hours duration to each course at the end of each semester. The end semester examinations shall comprise of maximum of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

In the case of Practical courses, the internal evaluation of practical courses will be done through continuous assessment of skill in demonstrating the experiments, record note preparation, and *viva voce*. The external evaluation consists of an end semester practical examinations which comprise of 75 marks for each course. The candidate failing in any course(s) will be permitted to appear for each failed course(s) in the subsequent examination.

Question paper pattern (Theory)

- The question paper carries a maximum of 75 marks.
- The question paper consists of three sections namely Part-A, Part-B and Part-C.

- Part-A consists of 10 questions of 2 marks each (10 x 2 = 20 marks) with no choice. The candidate should answer all questions.
- Part-B consists of 5 either or choice questions. Each question carries 5 marks (5 x 5=25 marks).
- Part-C consists of 5 questions. Each question carries 10 marks. The candidate should Answer any three questions (10 x 3 = 30 marks).
- All Blocks shall be given equal importance..

Passing minimum

- There shall be no Passing Minimum for Internal.
- For External Examination, Passing Minimum shall be of 50% (Fifty Percentage) of the maximum marks (75) prescribed for the paper.
- In the aggregate (External + Internal) the passing minimum shall be of 50% for each Paper and Practical.
- Grading shall be based on overall marks obtained (internal + external).

There is no passing minimum for the record notebook. However submission of a record notebook is a must. Candidate who does not obtain the required minimum marks for a pass in a course shall be required to appear and pass the same at a subsequent appearance.

f. 3.4. Marks and Grades:

The following table gives the marks, grade points, letter, grades and classification to indicate the performance of the candidate.

Range of Marks	Grade Points	Letter Grade	Description
90-100	9.0-10.0	O	Outstanding
80-89	8.0-8.9	D+	Excellent
75-79	7.5-7.9	D	Distinction
70-74	7.0-7.4	A+	Very Good
60-69	6.0-6.9	A	Good
50-59	5.0-5.9	B	Average
00-49	0.0	U	Re-appear
ABSENT	0.0	AAA	ABSENT

C_i = Credits earned for the course i in any semester
G_i = Grade Point obtained for course i in any semester.
n refers to the semester in which such courses were credited

For a semester;

Grade Point Average [GPA] = $\sum_i C_i G_i / \sum_i C_i$

Grade Point Average = Sum of the multiplication of grade points by the credits of the courses

Sum of the credits of the courses in a semester

For the entire programme;

Cumulative Grade Point Average [CGPA] = $\sum_n \sum_i C_{ni} G_{ni} / \sum_n \sum_i C_{ni}$

CGPA = Sum of the multiplication of grade points by the credits of the entire programme

Sum of the credits of the courses for the entire programme

CGPA	Grad	Classification of Final Result
9.5-10.0 9.0 and above but below 9.5	O+ O	First Class- Exemplary*
8.5 and above but below 9.0 8.0 and above but below 8.5 7.5 and above but below 8.0	D++ D+ D	First Class with Distinction*
7.0 and above but below 7.5 6.5 and above but below 7.0 6.0 and above but below 6.5	A++ A+ A	First Class
5.5 and above but below 6.0 5.0 and above but below 5.5	B+ B	Second Class
0.0 and above but below 5.0	U	Re-appear

*The candidates who have passed in the first appearance and within the prescribed semester of the PG Programme are eligible.

Maximum duration for completion of the course

The maximum duration for completion of M.Sc. degree in Zoology shall not exceed five years.

Commencement of this regulation

These regulations shall come into effect from the academic year 2018-19 for students who are admitted to the first year of the course during the academic year 2018-19.

Fee structure

Sl. No.	Fees Detail	Amount in Rs.		Nature of Fees
		First Year	Second Year	
1	Admission Processing Fees	300.00	-	Non-Refundable
2	Course Fees	20000		
5	ICT Fees	150.00	150.00	Non-Refundable
	TOTAL	20450.00	20150.00	Non-Refundable

g) REQUIREMENT OF THE LABORATORY SUPPORT AND LIBRARY RESOURCES:

LABORATORY SUPPORT

The Directorate of Distance Education of Alagappa University has established the well equipped laboratory exclusively for conducting face to face contact sessions for M.Sc., Zoology programme at Science Campus, Alagappa University, Karaikudi. The personal contact programme will be conducted only at Alagappa University campus, Karaikudi. The practical books (manual) will be supplied to the students at the time admission. Hands on experience on the techniques of biological sciences will be given to the students individually during the contact sessions. The practical's will also be conducted and performed by applying virtual reality methods wherever necessary.

LIBRARY RESOURCES

The Central Library is one of the important central facilities of Alagappa University. It has text book, reference books, conference proceedings, back volumes, standards, and non-book material such as CD-ROMs and audios. The central Library procured several e-books in different areas. The library also subscribes to about 250 current periodicals. The Directorate

of Distance Education of Alagappa University has adequate number of copies of books related to Zoology Programme.

h) COST ESTIMATE OF THE PROGRAMME AND THE PROVISIONS:

Sl. No.	Nature of Expenditure	Amount in Rs. (Approx.)
1	Programme Development	20,00,000
2	Programme Delivery	24,00,000
3	Programme Maintenance	5,00,000

i) QUALITY ASSURANCE MECHANISM AND EXPECTED PROGRAMME OUTCOMES

- The feedback from students on teaching will be collected every semester using standard formats.
- Feedback on the curriculum will also be collected from the experiences of the students which help teachers in fine tuning of deliverables in the classroom as well as in laboratories.
- It helps in improving the standard of teaching as expected by the students.
- Exit survey feedback on various parameters to improve and quality of the programme and support services like course material, library, laboratories and infrastructure.
- It helps to Strengthen the contents of the program to meet the requirements of the employment market and keep the curriculum as a treasure of knowledge.
- This programme provides opportunities for students to develop and demonstrate knowledge and understanding, skills, qualities and other attributes.

Minutes of the Meeting of the Board of Studies in Zoology (For M.Sc. Zoology Programme to be offered through ODL Mode) held at The Directorate of Distance Education, Alagappa University, Karaikudi – 630 003, on 01.09.2017, (11.00 A.M).

Members Present

1.	Dr. G. Archunan	-	Member
2.	Dr. E. Kannapiran	-	Chairman
3.	Dr. P. Thiruchenthilnathan	-	Member
4.	Dr. P. Srinivasan	-	Member
5.	Dr. P. Rameshthangam	-	Member
6.	Dr. G. Selvakumar	-	Member

After the deliberation and discussion the board resolved the following:

1. The board considered the curriculum design and detailed syllabi of M.Sc., Zoology programme, prepared as per the norms by the Chairman and the Board Members, scrutinized and suitably modified the same.
2. The board resolved to approve curriculum design, detailed syllabi and other regulations for the M.Sc., Zoology programme to be offered by the Directorate of Distance Education of Alagappa University are given in Annexure I.

Dr. G. Archunan	Dr. E. Kannapiran	Dr. P. Thiruchenthilnathan
Dr. P. Srinivasan	Dr. P. Rameshthangam	Dr. G. Selvakumar